

TEMPORARY REVISION NOTICE No. 8A

April 06, 2023

CAGE Code: 63367

TO: HOLDERS OF COMPONENT MAINTENANCE MANUAL 25-33-20, ENDURA BEVERAGE MAKER
PN 4510-1 SERIES AND 4510-2 SERIES.

DESCRIPTION OF TR 8A, DATED APRIL 06, 2023 HIGHLIGHTS

This temporary revision replaces the specified pages in the manual that you currently have.

Insert the Temporary Revision No. 8A pages in B/E AEROSPACE CMM 25-33-20, Revision 07, in accordance with the instructions on each temporary revision page. Record the Temporary Revision on the CMM Temporary Revision page.

Internal Reference: DCR C23-150007

<u>TR PAGE</u>	<u>PAGE</u>	<u>CHANGE</u>
3008A	3008	Revised Figure 3006 to delete items 235 and 240 and to add items 235A, 240A, and 242.
3009A	3009	Revised paragraph 2.B.(5)(e) to add item 242.
7037A	7037	Revised Figure 7036 to delete items 235 and 240 and to add items 235A, 240A, and 242.
7038A	7038	Revised paragraph 2.N.(12)(g) to add item 242.
10008A	10008	Revised Alpha/Numerical Index to update item 235 as non illustrated and to add items 165A and 235A for IPL Figures 2, 2A, 2B, and 2C.
10011A and 10012A	10011 and 10012	Revised Alpha/Numerical Index to update the quantity of item 555 as 4 for IPL Figures 2, 2A, 2B, and 2C. Revised Alpha/Numerical Index to add items 242 and 555A for IPL Figures 2, 2A, 2B, and 2C.
10014A	10014	Revised Alpha/Numerical Index to update the quantity of item 190 as 8 for IPL Figures 2, 2A, 2B, and 2C.
10015A	10015	Revised Alpha/Numerical Index to add item 240A for IPL Figures 2, 2A, 2B, and 2C.
10018A	10018	Revised Alpha/Numerical Index to update item 240 as non illustrated for IPL Figures 2, 2A, 2B, and 2C.
10034A	10034	Revised Alpha/Numerical Index to add item 747A for IPL Figures 2, 2A, 2B, and 2C.
10052A	10052	Revised IPL Figure 2 to delete items 235 and 240 and to add items 235A, 240A, and 242.
10055A	10055	Revised IPL Figure 2 to add superseded by statement for item 165 and to add item 165A and to update the quantity of item 190 as 8.

<u>TR PAGE</u>	<u>PAGE</u>	<u>CHANGE</u>
10056A	10056	Revised IPL Figure 2 to update item 235 as non illustrated, to update the effectivity and to add superseded by statement. Revised IPL Figure 2 to update item 240 as non illustrated and to add superseded by statement. Revised IPL Figure 2 to add items 235A, 242, and 243.
10061A	10061	Revised IPL Figure 2 to update the quantity of item 555 as 4 and to add superseded by statement. Revised IPL Figure 2 to add item 555A.
10064A/10065A	10064/10065	Revised IPL Figure 2 to add superseded by statement for item 747 and to add item 747A.
10066A	10066	Revised IPL Figure 2A to delete items 235 and 240 and to add items 235A, 240A, and 242.
10069A	10069	Revised IPL Figure 2A to add superseded by statement for item 165 and to add item 165A and to update the quantity of item 190 as 8.
10070A	10070	Revised IPL Figure 2A to update item 235 as non illustrated, to update the effectivity and to add superseded by statement. Revised IPL Figure 2A to update item 240 as non illustrated and to add superseded by statement. Revised IPL Figure 2A to add items 235A, 242, and 243.
10075A	10075	Revised IPL Figure 2A to update the quantity of item 555 as 4 and to add superseded by statement. Revised IPL Figure 2A to add item 555A.
10078A/10079A	10078/10079	Revised IPL Figure 2A to add superseded by statement for item 747 and to add item 747A.
10080A	10080	Revised IPL Figure 2B to delete items 235 and 240 and to add items 235A, 240A, and 242.
10083A	10083	Revised IPL Figure 2B to add superseded by statement for item 165 and to add item 165A and to update the quantity of item 190 as 8.
10084A	10084	Revised IPL Figure 2B to update item 235 as non illustrated, to update the effectivity and to add superseded by statement. Revised IPL Figure 2B to update item 240 as non illustrated and to add superseded by statement. Revised IPL Figure 2B to add items 235A, 242, and 243.
10089A	10089	Revised IPL Figure 2B to update the quantity of item 555 as 4 and to add superseded by statement. Revised IPL Figure 2B to add item 555A.
10092A/10093A	10092/10093	Revised IPL Figure 2B to add superseded by statement for item 747 and to add item 747A.
10094A	10094	Revised IPL Figure 2C to delete items 235 and 240 and to add items 235A, 240A, and 242.
10097A	10097	Revised IPL Figure 2C to add superseded by statement for item 165 and to add item 165A and to update the quantity of item 190 as 8.

<u>TR PAGE</u>	<u>PAGE</u>	<u>CHANGE</u>
10098A	10098	Revised IPL Figure 2C to update item 235 as non illustrated, to update the effectivity and to add superseded by statement. Revised IPL Figure 2C to update item 240 as non illustrated and to add superseded by statement. Revised IPL Figure 2C to add items 235A, 242, and 243.
10103A	10103	Revised IPL Figure 2C to update the quantity of item 555 as 4 and to add superseded by statement. Revised IPL Figure 2C to add item 555A.
10106A/10107A	10106/10107	Revised IPL Figure 2C to add superseded by statement for item 747 and to add item 747A.

COMPONENT MAINTENANCE MANUAL

Endura Beverage Maker

Component Maintenance Manual
(with Illustrated Parts List)

25-33-20

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September 21, 2022

TO: HOLDERS OF COMPONENT MAINTENANCE MANUAL 25-33-20, ENDURA BEVERAGE MAKER.

DESCRIPTION OF REVISION 07, DATED SEPTEMBER 21, 2022 HIGHLIGHTS

This revision replaces the manual that you currently have in its entirety.'

All changed pages keep data necessary to do maintenance on all equipment models. Black bars on the side of the page identify the changes of this revision.

This revision does not include copies of service bulletins (SB) or service information letters (SIL) applicable to this equipment. This manual includes data from applicable SBs and SILs. For this revision, move the SBs and SILs from your manual to this manual.

Page numbers, figure numbers, table numbers, etc. have been allowed to increment normally. Also, identification of previous deletions have been removed. Pages where only figure references, table reference, etc. have changed or text has been removed by previous revisions are not listed below. Pages with relocated content are not listed below. The table below shows all other pages of the manual where content has been added, changed, or removed.

<u>PAGE</u>	<u>CHANGE</u>
T-1	Revised title page to update address.
T-3/T-4	Revised to update contact information.
RR-1/RR-2	Revised to add revision 07 and revision date.
RTR-2	Revised to add TR 7A, 7B, 7C, and 7D.
SBL-1	Revised SB List to update SB 4510-25-16 from Rev 01 to Rev 02 and to add SB 4X1X-25-002.
SBL-2	Revised SB List to add SILs SIL-MISC-3201-1, 4XXX-SIL-001, 4X1X-EN-SIL-02 / Rev 001, 4X10EN-SIL-001, and 4510-SIL-07.
LEP-1 thru LEP-4	Revised to reflect the current revision.
1002	Updated paragraph 2.A. and added Note as per TR 7C.
1017/1018	Revised Table 1002 to add fault isolation for resistance check.
3009	Added Note to step 2.B.(5)(f) as per TR 7D.
3031	Revised Figure 3025 to add illustration for PN 3520-1529-01.
3032	Revised steps 2.L.(6), (7), and (8) to add disassembly procedure for PN 3530-0279-01, PN 3530-0279-23, and PN 3520-1529-01 respectively.
3033/3034	Deleted steps 2.L.(9) thru (13).
4001	Revised Table 4001 to delete Sodium Bicarbonate as per TR 7B.
4003/4004	Revised paragraph 2.E to add reference to UG-4XXX-25-002. Deleted paragraphs 2.E.(1) thru 2.E.(7).
5002	Revised paragraph 2.C.(4) to add Figure 5001 reference.
5004	Added note for paragraph 3.
6002	Revised Caution to correct figure reference from 9005 to 9003 as per TR 7B.

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

<u>PAGE</u>	<u>CHANGE</u>
7001	Revised Note in paragraph 1.E as per TR 7C.
7002	Revised Table 7001 to add Clamp, Loctite-5900, Soft-Cloth, and Syringe as per TR 7D and to add FDA Lubricant and Sealant.
7005 thru 7034 and 7037 thru 7044	Added Figure 7003 and paragraph 2.C. for PN 3520-1529-01. Renumbered subsequent paragraphs and figures.
7006	Revised Figure 7004 and figure title as per TR 7B.
7007	Revised Figure 7005 and figure title as per TR 7B.
7008	Revised Figures 7006 and 7007 and figure titles as per TR 7B.
7019	Revised paragraph 2.L.(7)(b) to update torque units from in-lb to ft-lbs as per TR 7B.
7034 thru 7038	Added paragraph 2.N.(12)(a) for the procedure to install platen inlay and added Figures 7034 and 7035 as per TR 7D.
7037	Revised Figure 7036 to add note for Black Adhesive Sealant as per TR 7D.
7038	Added paragraph 2.N.(12)(k) for the application of black adhesive sealant to the chassis assembly as per TR 7D.
7042 and 7043	Added paragraph 2.N.(15)(b) for the installation of o-ring as per TR 7C.
9001	Added Note for step 1.B. and revised Table 9001 to add Clamp as per TR 7D.
9002	Revised Table 9001 to add Loctite-5900 as per TR 7D and FDA Lubricant and Sealant.
9003	Revised Table 9001 to delete Sodium Bicarbonate and to add Soft-Cloth and Syringe as per TR 7D.
10002	Revised table 10001 to update Vendor information.
10017	Revised Alpha/Numerical Index to add PN TE50442. Revised Alpha/Numerical Index to update item numbers from -350 to 350 for IPL Figure 2, 2A, 2B and 2C.
10018	Revised Alpha/Numerical Index to add PN 0R50099.
10024	Revised Alpha/Numerical Index to add PN 3520-1529-01, PN 3520-1616-01, and PN 3520-1620-01.
10029	Revised Alpha/Numerical Index to add PN 3575-1269-01 as per TR 7D.
10030	Revised Alpha/Numerical Index to add PN 3575-3037-01.
10052	Revised IPL Figure 2 to update item number from 350A to 350 as per TR 7C.
10056	Revised IPL Figure 2 to add PN 3575-1269-01 as item 205 as per TR 7D.
10058	Revised IPL Figure 2 to show item 350 as illustrated as per TR 7C.
10066	Revised IPL Figure 2A to update item number from 350A to 350 as per TR 7C.
10070	Revised IPL Figure 2A to add PN 3575-1269-01 as item 205 as per TR 7D.
10072	Revised IPL Figure 2A to show item 350 as illustrated as per TR 7C.
10080	Revised IPL Figure 2B to update item number from 350A to 350 as per TR 7C.
10084	Revised IPL Figure 2B to add PN 3575-1269-01 as item 205 as per TR 7D.

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<u>PAGE</u>	<u>CHANGE</u>
10086	Revised IPL Figure 2B to show item 350 as illustrated as per TR 7C.
10094	Revised IPL Figure 2C to update item number from 350A to 350 as per TR 7C.
10098	Revised IPL Figure 2C to add PN 3575-1269-01 as item 205 as per TR 7D.
10100	Revised IPL Figure 2C to show item 350 as illustrated as per TR 7C.
10130/10131	Revised IPL Figure 11 to add replaced by and PRE SIL statements to items 30B and 30D. Added PN 3520-1529-01 as items 30E and 30F and PN TE50442 as item 32.
10133	Revised IPL Figure 12 to add PRE SIL statement to item 1B.
10134 and 10035	Added new IPL Figure 12A for PN 3520-1529-01.

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Endura Beverage Maker

Component Maintenance Manual (with Illustrated Parts List)

This manual includes coverage of the following equipment:

<u>Unit</u>	<u>Collins Part Number</u>
Endura Beverage Maker	4510-11UF-00
	4510-11LF-00
	4510-12UF-00
	4510-12LF-00
	4510-13UF-00
	4510-13LF-00
	4510-14UF-00
	4510-14LF-00
	4510-15UF-00
	4510-15LF-00
	4510-16UF-00
	4510-16LF-00
	4510-17UF-00
	4510-17LF-00
	4510-18UF-00
	4510-18LF-00
	4510-21UF-00
	4510-21LF-00
	4510-22UF-00
	4510-22LF-00
	4510-23UF-00
	4510-23LF-00
	4510-24UF-00
	4510-24LF-00
	4510-25UF-00
	4510-25LF-00

B/E AEROSPACE, a part of Collins Aerospace

15701 W. 95th Street

Lenexa, KS 66219

USA

CAGE Code: 63367

25-33-20

T-1

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

<u>Unit</u>	<u>Collins Part Number</u>
Endura Beverage Maker	4510-26UF-00
	4510-26LF-00
	4510-27UF-00
	4510-27LF-00
	4510-28UF-00
	4510-28LF-00
	4510-11UG-00
	4510-11LG-00
	4510-12UG-00
	4510-12LG-00
	4510-13UG-00
	4510-13LG-00
	4510-14UG-00
	4510-14LG-00
	4510-15UG-00
	4510-15LG-00
	4510-16UG-00
	4510-16LG-00
	4510-17UG-00
	4510-17LG-00
	4510-18UG-00
	4510-18LG-00
	4510-21UG-00
	4510-21LG-00
	4510-22UG-00
	4510-22LG-00
	4510-23UG-00
	4510-23LG-00
	4510-24UG-00
	4510-24LG-00
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RECORD OF REVISIONS

REV. NO.	ISSUE DATE	INSERT DATE	BY	REV. NO.	ISSUE DATE	INSERT DATE	BY
000	May 15/04						
001	Mar 31/06						
002	May 10/13						
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07	Sep 21/22						

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RECORD OF TEMPORARY REVISIONS

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A	Feb 14/07					LEP 1 and 1009
2B	May 14/10					3027, 7007, 10088, 10089, and 10103
3A	Jul 10/13					1002, 1004, 1008, 1010, 1011, and 9001
3B	Jul 12/13					6002
3C	Nov 14/13					10032, 10034, 10121, 10138, and 10140
4A	Oct 01/14					HL-1
4B	Mar 25/15					HL-1, 2002, 7003, 7004, 7005, 10015, 10026, 10120, 10121, and 10123
4C	Mar 31/16					3021, 7014, 10008, 10010, 10015, 10026, 10048, 10049, 10062, 10063, 10076, 10077, 10090, 10091, and 10105
5A	Sep 07/16					10030, 10058, 10072, 10086, and 10100
5B	Oct 14/16					10030, 10058, 10072, 10086, and 10100
5C	Mar 08/17					7016, 7017, 10009, 10029, and 10110
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5F	Aug 02/17					3012, 3013, 7022, 10027, 10038, 10042, 10046, 10050, 10052, 10060, 10066, 10074, 10080, 10088, 10094, and 10102
6A	Feb 23/18	Sep 12/19				7016A and 9001A
6B	Mar 28/18	Sep 12/19				10008A and 10113A
6C	Oct 24/18	Sep 12/19				1004A and 7002A
6D	Nov 05/18	Sep 12/19				1004A and 7002A
6E	Jan 15/19	Sep 12/19				1004A
6F	Apr 29/19	Sep 12/19				1002A and 1015A
7A	Jan 22/20	Sep 21/22				6002A, 7005A, 7006A, and 7007A
7B	Feb 11/20	Sep 21/22				4001A, 7005A, 7006A, 7007A, and 9003A
7C	Aug 03/20	Sep 21/22				1002A, 7001A, 7018A, 7039A, 7040A, 9001A, 10052A, 10058A, 10066A, 10072A, 10080A, 10086A, 10094A, and 10100A
7D	Mar 03/22	Sep 21/22				3009A, 7002A, 7033A, 7033B, 7033C, 7034A, 7035A, 9001A, 9002A, 9003A, 10029A, 10056A, 10070A, 10084A, and 10098A

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SERVICE BULLETIN LIST

SERVICE BULLETIN NO / REV	ISSUE DATE	SUBJECT	MANUAL REVISION NUMBER	MANUAL REVISION DATE
3530-0121-25-01 / Rev 04	May 18/18	Printed Circuit Board Assembly Replacement	No Effect	Sep 12/19
3520-0534-25-001 / Rev 003	Aug 16/16	New Screws for Brew Head Assembly, PN 3520-0534-01	05	Oct 13/17
3520-0594-25-001 / Rev 004	Nov 06/17	Beverage Maker/Coffee Maker Vent Valve Ball Replacement	No Effect	Sep 12/19
3520-0594-25-002 / Rev 001	Oct 19/16	Beverage Maker/Coffee Maker Vent Valve Ball Replacement	05	Oct 13/17
3520-0594-25-003 / Rev 003	Jul 18/17	Improved Durability Vent Valve Components	05	Oct 13/17
4510-25-01	Nov 19/04	Handle Decal Installation	01	Mar 31/06
4510-25-02 / Rev 002	Dec 18/17	Beverage Makers - 4510 Series - Tea Tube Replacement	No Effect	Sep 12/19
4510-25-03	Aug 16/05	Server Retainer Assy Replacement	01	Mar 31/06
4510-25-06	Aug 31/06	Vent Valve Replacement	02	May 10/13
4510-25-07 / Rev 005	Oct 21/14	Vent Valve Replacement	04	Jun 30/16
4510-25-08 / Rev 002	Oct 21/13	Low Water Sensor Replacement	03	Sep 08/14
4510-25-009 / Rev 002	Oct 21/13	Pressure Relief Valve Replacement	03	Sep 08/14
4510-25-13 / Rev 001	Apr 25/16	Beverage Maker Disable Platen Heater Function	04	Jun 30/16
4510-25-14 / Rev 002	Jul 25/17	New Tea Funnel, PN 3530-0879-01	05	Oct 13/17
4510-25-16 / Rev 002	Jan 11/21	Modify Beverage Maker PN 4510-12UF-00 to Beverage Maker 4510-22UF-00	No Effect	Sep 21/22
4X1X-25-002 / Rev 0	Jun 01/22	Equipment/Furnishings - Beverage Maker - Updates Manifold Assemblies and New Solenoids	07	Sep 21/22
CM-50016-SIL-001 / Rev 002	Jun 01/16	Application of Never Seez (CM-50016) on Heater Threads	04	Jun 30/16
SIL-4510-2500-101	Apr 04/17	Beverage Maker Installation & Operating Instructions	No Effect	Oct 13/17

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SERVICE BULLETIN NO / REV	ISSUE DATE	SUBJECT	MANUAL REVISION NUMBER	MANUAL REVISION DATE
3520-0579-SIL-001	Apr 23/14	Improved Reliability for IR Sensor PN 3520-0579-01	No Effect	Jun 30/16
3520-0579-SIL-002	Mar 30/17	Improved Reliability for IR Sensor PN 3520-0579-01 with new LED, PN LP-50098	No Effect	Oct 13/17
3530-0261-SIL-001 / Rev 001	Jan 14/15	Platen Heater PN 3530-0261 Product Improvement	04	Jun 30/16
3575-0650-SIL-001 / Rev 003	Jan 15/15	Clarification of Secondary Retention Device on Units Shown In Effectivity Table	No Effect	Jun 30/16
4510-SIL-01	Dec 23/04	Optional Replacement of Microcontroller PN 3530-0270-01 from Rev. A01 to Rev. B00 for all 4510 Series mfg before 1/1/2005.	No Effect	Mar 31/06
4510-SIL-02	Nov 06/06	Overtemperature Thermostat Replacement	02	May 10/13
4510-SIL-03	Oct 02/09	Configuration Change - Platen Assy	02	May 10/13
4510-SIL-05 / Rev 001	Jan 26/12	Informational SIL to advise the microcontroller PN 3530-0270-01 used in the Endura PC Board 3530-0121-01 is at Rev D00.	No Effect	May 10/13
4510-SIL-06	Mar 26/13	Suggested Special Tools and Fixtures for Endura 4510 Products	02	May 10/13
SIL-MISC-3201-1 / Rev 00	Apr 23/20	New Locking Patch Screws Available for Servers, PN MISC-3201	No Effect	Sep 21/22
4XXX-SIL-001 / Rev 001	May 18/20	Vent Valve Ball, PN MISC-50509	07	Sep 21/22
4X1X-EN-SIL-02 / Rev 03	Aug 11/22	Solenoid Component and Sub-component Interchangability Guidance	07	Sep 21/22
4X10-EN-SIL-001 / Rev 00	Oct 01/20	Manufacturing Improvement for IR Sensor Assembly PN 3520-0579-01	No Effect	Sep 21/22
4510-SIL-007 / Rev 01	Jul 01/22	Fault Isolation for Resistance Check	07	Sep 21/22

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4510-1 Series and 4510-2 Series

LIST OF EFFECTIVE PAGES

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	10005	Sep 21/22	10045	Sep 21/22	
	10006	Sep 21/22	10046	Sep 21/22	
	10007	Sep 21/22	10047	Blank	
	10008	Sep 21/22	10048	Sep 21/22	
	10009	Sep 21/22	10049	Sep 21/22	
	10010	Sep 21/22	10050	Sep 21/22	
	10011	Sep 21/22	10051	Blank	
	10012	Sep 21/22	* 10052	Sep 21/22	
	10013	Sep 21/22	10053	Sep 21/22	
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	10022	Sep 21/22	10062	Sep 21/22	
	10023	Sep 21/22	10063	Sep 21/22	
	* 10024	Sep 21/22	10064	Sep 21/22	
	10025	Sep 21/22	10065	Blank	
	10026	Sep 21/22	* 10066	Sep 21/22	
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	10078	Sep 21/22		10119	Sep 21/22
	10079	Blank		10120	Sep 21/22
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	10088	Sep 21/22		10129	Sep 21/22
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	10099	Sep 21/22			
	* 10100	Sep 21/22			
	10101	Sep 21/22			
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	10105	Sep 21/22			
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INTRODUCTION

1. **General**

- A. This manual has been prepared to fulfill the requirements of ATA iSpec 2200 for Component Maintenance Manuals.
- B. All procedures established in this manual have been performed and verified acceptable by trained personnel at B/E Aerospace, Inc. A competent shop mechanic who is unfamiliar with these units should, with a minimum amount of effort and tools, be able to isolate the problems of a nonfunctional unit and restore it to service.
- C. Procedures are provided for the full range of maintenance and include all the work functions that could or may be done to the unit throughout its entire service life.
- D. A competent shop mechanic should use this manual first to test and isolate a faulty component. The mechanic should then go directly to the applicable manual subject to effect a repair in the specific problem area without disturbing other areas.

2. **Abbreviations**

The following abbreviations may be used in this manual:

ABBREVIATION	MEANING
AR	As Required
° C	Degree Centigrade
CAGE	Commercial and Government Entity
CM	Centimeter
DIA	Diameter
EFF	Effectivity
ESD	Electro Static Devices
° F	Degree Fahrenheit
FIG	Figure
GA	Gauge
GPM	Gallons Per Minute
ID	Inner Diameter
IN	Inch
IN-LBS	Inch-Pounds
IPL	Illustrated Parts List
KPAG	Kilo Pascals Gauge
LBS	Pound

ABBREVIATION	MEANING
LPM	Liter Per Minute
MM	Millimeter
NHA	Next Higher Assembly
NO	Number
OD	Outer Diameter
PN	Part Number
PSIG	Pounds per Square Inch Gauge
REQ	Required
RF	Reference
RTD	Resistive Temperature Detector
SB	Service Bulletin
SIL	Service Information Letter
SUPSD	Superseded
SUPSDS	Supersedes
TTL	Total
V	Vendor

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DESCRIPTION AND OPERATION

1. General

- A. Beverage Makers PNs 4510-1 and 4510-2 series are manufactured by B/E Aerospace, Inc., Commercial Aircraft Products Group.
- B. The Beverage Maker (refer to Figure 1) is designed for use in aircraft galleys. The Beverage Maker is easily removed from the permanently installed rail assembly after a locking pin is released.

2. Description

- A. The Beverage Maker is of modular design, utilizing cast aluminum-magnesium-titanium construction for maintainability and reliability. The front of the casting serves as a framework for the housing platen, electrical controls, brew assembly, and water faucet. The rear of the casting serves as a framework for the hot water tank, power module, plumbing and controls, and provides close dimensional control for accurate connection to the aircraft electrical system, water supply, and drain.

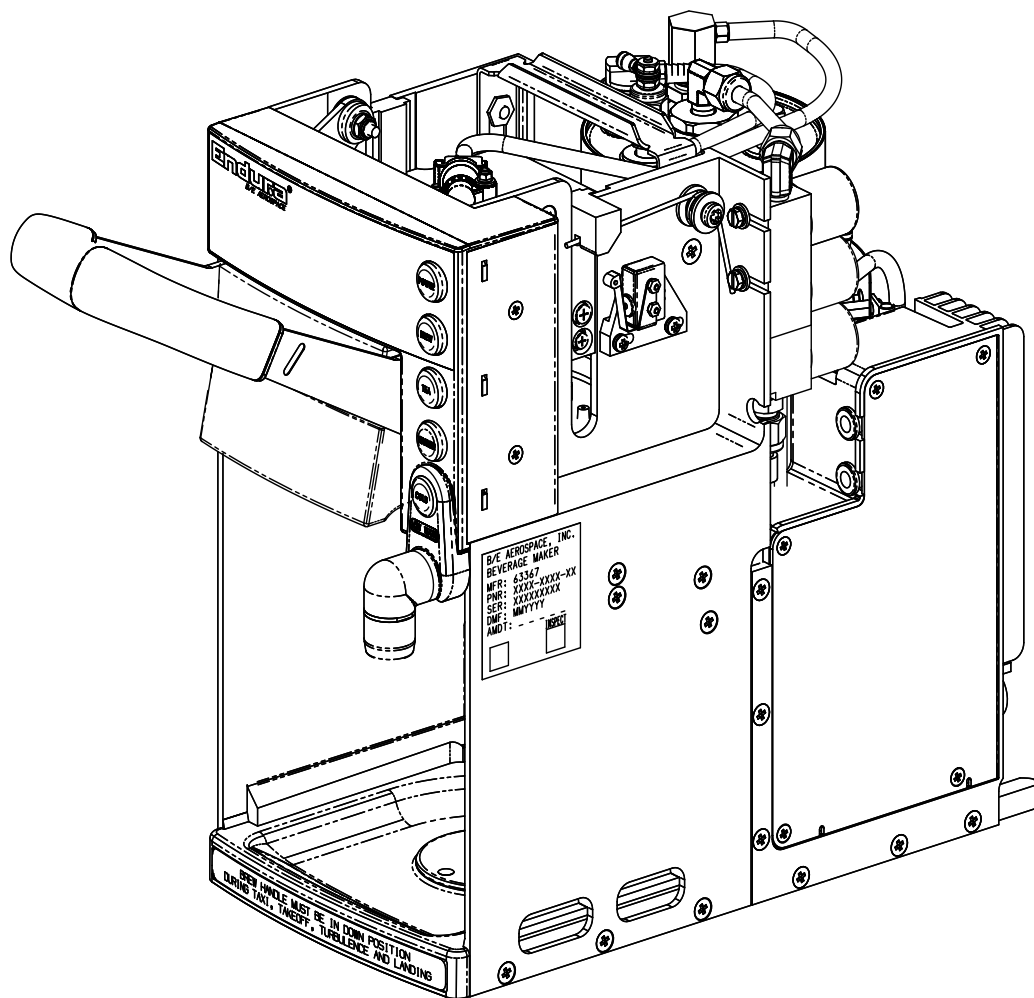


Figure 1
Beverage Maker

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- B. The Beverage Maker heats water for a full server of coffee. Optional controls are available to heat a server full of hot water for tea or other uses. Optional controls are available to provide hot or cold water from a faucet on the front of the beverage maker. The faucet aerator decreases likelihood of splashing.
- C. The water is heated in a single tank by three 900 watt immersion type heaters mounted in the bottom of the tank.
- D. An optional 15 watt WARMER, mounted in the bottom of the platen will maintain liquid temperature between 165° F (74° C) to 195° F (91° C) for approximately 30 minutes.
- E. An insulated server is available. It should not be used when the WARMER is on.
- F. The brew chamber uses the standard size 4.25 in (10.79 cm) x 5.25 in (13.33 cm) coffee bag (pillow pack). The brew chamber is sealed during the brew cycle to retain grounds in the event that the pillow pack is torn or broken.
- G. The server and brew cup are restrained under all flight conditions while the brew handle is in the down position. The server is also covered by the server retainer when the brew handle is down.
- H. The Beverage Maker provides a vent valve to allow automatic filling or draining of the Beverage Maker when interfaced with an aircraft water system.

3. Operation (refer to Figure 1)

NOTE: The server must be in place and the brew handle in the down position before the BREW button or TEA button can be pressed to start a brew cycle.

- A. Three-phase 115 VAC, 400 Hz is required at the power connection for operation of the Beverage Maker.
- B. When the Beverage Maker is placed on an energized rail, power is supplied to the controller board and the electrical system.
- C. If the water level in the tank is low, the LOW WATER indicator light will come on until the tank is filled.
- D. Pressing the POWER button starts the preheat cycle so water is heated for brewing.
- E. When the POWER button is pressed, with the brew handle in the down position and the server in place in the server cavity, brewing may be started by pressing the BREW button or the optional TEA button.
- F. When the temperature in the water tank reaches the set point [approximately 196° F (91° C)], the brew solenoid valve opens and lets water flow. The water flow will continue until the level sensors detect the liquid in the server or the brew handle is lifted. Completion of an electrical circuit across the level sensor will signal the controller board to close the brew solenoid valve.
- G. A time delay causes the BREW button light to stay on for 10-20 seconds after the brew solenoid valve closes. This time lets most hot liquid drain into the server.
- H. If the brew cycle is interrupted, it may be reset by pressing the POWER button OFF and ON again.

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CAUTION: DO NOT USE THE WARMER WHEN USING AN INSULATED SERVER. THE WARMER CAN MAKE THE OUTER SURFACE OF THE INSULATED SERVER VERY HOT.

- J. Power may be applied to the optional WARMER by pressing the POWER button and the WARMER button to the ON positions. With the server in the server cavity, the WARMER keeps coffee temperature between 165° F (74° C) to 195° F (91° C) for approximately 30 minutes.
- K. In the event of failures disabling the electronic safety features, control circuits prevent Beverage Maker operation when internal tank temperature is below 32° F (0° C) or above 235° F (113° C).
- L. Additional safety features provided in the Beverage Maker design include:
 - (1) If the water tank external temperature exceeds 225° F (107° C), power to the heaters will automatically be removed by three resettable over temperature thermostats.
 - (2) A pressure relief valve is set at the factory to open in the event that the water tank internal pressure exceeds 80 to 85 psig (5.52 to 5.86 barg).
 - (3) A low water sensor prevents operation of tank heaters and the LOW WATER indicator light comes ON any time the tank water level drops below the sensor mounted in the tank.
 - (4) A coffee level sensor provides accurate control of liquid level in the server and stops a brew cycle with a full server.
 - (5) Over temperature detection trips the safety latch if the tank temperature goes above 225° F (107° C). The POWER button light goes into double-blink mode while the Beverage Maker is connected to a power supply until the temperature goes down and/or the required circuits are repaired, and the safety latch is reset.
 - (6) Phase loss detection detects the loss of one of the phases in the circuit. The POWER button light goes into triple-blink mode while the Beverage Maker is connected to a circuit where one or more phases is not open or disconnected. Once all phases are restored, cycling the POWER button OFF and ON will eliminate the blinking POWER light.

4. Operating Instructions

NOTE: The Beverage Maker is designed to consistently brew coffee or tea of uniform extraction. Control of the coffee blend and water quality are beyond the control of this equipment and remain the responsibility of the user.

A. Brew Coffee

- (1) To prepare Beverage Maker to brew coffee:
 - (a) Press the POWER button. This starts operation of the tank heaters.
 - (b) Remove brew cup from the brew chamber above the server cavity.

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- (c) Put a pillow pack of coffee, with its seam down, on top of the brew cup's metal base plate.

NOTE: Make sure all of the pillow pack is in the brew cup. Do not compress or fold the pillow pack. This could cause the pack to break open during the brew cycle.

- (d) Put brew cup into brew chamber.
- (e) Put an empty server into server cavity.
- (f) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.

- (2) To brew a server of coffee (after doing the steps above):

CAUTION: DO NOT USE THE WARMER WHEN USING AN INSULATED SERVER. THE WARMER CAN MAKE THE OUTER SURFACE OF THE INSULATED SERVER VERY HOT.

- (a) Press the WARMER button (where option is installed). This warmer or platen heater is used to keep coffee hot in the (non-insulated) server.
- (b) Press the BREW button. Brewing will start when the water comes to brew temperature. From a cold start, this will take approximately three minutes. The Beverage Maker will then brew a server of coffee in approximately three minutes.
- (c) When the brew cycle is complete (shown by the BREW button light going OFF) lift the brew handle.
- (d) Remove the brew cup and discard the used pillow pack. If there are any coffee grounds in the brew cup, clean the brew cup with clean water.
- (e) Put a new pillow pack in the brew cup and put the brew cup in the brew chamber.
- (f) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.

B. Brew Tea

- (1) To prepare Beverage Maker to brew tea:

- (a) Press the POWER button.
- (b) Put tea bag(s) in empty server.
- (c) Put server into server cavity.
- (d) Put an empty brew cup into brew chamber.
- (e) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.
- (f) Press TEA button.
- (g) When TEA button light goes OFF, the tea is ready.
- (h) Lift the brew handle to access the server.

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5. Leading Particulars

A. Refer to Table 1 for leading particulars of the Beverage Maker.

Table 1
Leading Particulars

OPERATING CHARACTERISTICS			
Preheat Time	3 minutes 30 seconds (maximum)		
Brew Cycle Time (50 oz ± 3 oz)	3 minutes		
Tea Cycle Time (50 oz ± 3 oz)	3 minutes		
Unit Recovery Time	60 seconds (maximum)		
Initial or Subsequent Cycle, Server Temperature	175° F (79° C) to 195° F (91° C)		
ELECTRICAL DATA			
Operating Voltage	115 VAC, 400 Hz, 3-phase		
Wattage (Phase A, B, C)	2760 Watts Total (Phase A - 960 Watts, Phase B - 900 Watts and Phase C - 900 Watts)		
OVERALL DIMENSIONS			
Length	Width	Height (with rail)	Weight (Dry)
15.76 in (400 mm)	6.31 in (160 mm)	12.03 in (306 mm)	17 lbs (7.7 kg)

NOTE: Overall dimensions vary with options installed.

NOTE: The (dry) weight shown is an upper limit. Some installed options may weigh less.

Table 2
Part Number and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-11UF-00	Sump	Hot Water from Faucet	Upper Plate	Standard
4510-11LF-00	Sump	Hot Water from Faucet	Lower Plate	Standard
4510-12UF-00	Sump	Hot Water from Faucet, Tea Brew	Upper Plate	Standard
4510-12LF-00	Sump	Hot Water from Faucet, Tea Brew	Lower Plate	Standard
4510-13UF-00	Sump	Hot or Cold Water from Faucet	Upper Plate	Standard
4510-13LF-00	Sump	Hot or Cold Water from Faucet	Lower Plate	Standard
4510-14UF-00	Sump	Hot Water from Faucet, Platen Heater	Upper Plate	Standard
4510-14LF-00	Sump	Hot Water from Faucet, Platen Heater	Lower Plate	Standard
4510-15UF-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Standard

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Table 2 (Cont.)
Part Number and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-15LF-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Standard
4510-16UF-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-16LF-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-17UF-00	Sump	Cold Water from Faucet, Tea Brew	Upper Plate	Standard
4510-17LF-00	Sump	Cold Water from Faucet, Tea Brew	Lower Plate	Standard
4510-18UF-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-18LF-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-21UF-00	Rear Connection	Hot Water from Faucet	Upper Plate	Standard
4510-21LF-00	Rear Connection	Hot Water from Faucet	Lower Plate	Standard
4510-22UF-00	Rear Connection	Hot Water from Faucet, Tea Brew	Upper Plate	Standard
4510-22LF-00	Rear Connection	Hot Water from Faucet, Tea Brew	Lower Plate	Standard
4510-23UF-00	Rear Connection	Hot or Cold Water from Faucet	Upper Plate	Standard
4510-23LF-00	Rear Connection	Hot or Cold Water from Faucet	Lower Plate	Standard
4510-24UF-00	Rear Connection	Hot Water from Faucet, Platen Heater	Upper Plate	Standard
4510-24LF-00	Rear Connection	Hot Water from Faucet, Platen Heater	Lower Plate	Standard
4510-25UF-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Standard
4510-25LF-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Standard
4510-26UF-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-26LF-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-27UF-00	Rear Connection	Cold Water from Faucet, Tea Brew	Upper Plate	Standard
4510-27LF-00	Rear Connection	Cold Water from Faucet, Tea Brew	Lower Plate	Standard
4510-28UF-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-28LF-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-11UG-00	Sump	Hot Water from Faucet	Upper Plate	Swivel
4510-11LG-00	Sump	Hot Water from Faucet	Lower Plate	Swivel
4510-12UG-00	Sump	Hot Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-12LG-00	Sump	Hot Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-13UG-00	Sump	Hot or Cold Water from Faucet	Upper Plate	Swivel
4510-13LG-00	Sump	Hot or Cold Water from Faucet	Lower Plate	Swivel
4510-14UG-00	Sump	Hot Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-14LG-00	Sump	Hot Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-15UG-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-15LG-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Swivel

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Table 2 (Cont.)
Part Number and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-16UG-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-16LG-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-17UG-00	Sump	Cold Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-17LG-00	Sump	Cold Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-18UG-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-18LG-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-21UG-00	Rear Connection	Hot Water from Faucet	Upper Plate	Swivel
4510-21LG-00	Rear Connection	Hot Water from Faucet	Lower Plate	Swivel
4510-22UG-00	Rear Connection	Hot Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-22LG-00	Rear Connection	Hot Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-23UG-00	Rear Connection	Hot or Cold Water from Faucet	Upper Plate	Swivel
4510-23LG-00	Rear Connection	Hot or Cold Water from Faucet	Lower Plate	Swivel
4510-24UG-00	Rear Connection	Hot Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-24LG-00	Rear Connection	Hot Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-25UG-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-25LG-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-26UG-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-26LG-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-27UG-00	Rear Connection	Cold Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-27LG-00	Rear Connection	Cold Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-28UG-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-28LG-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel

6. Airworthiness Limitations

- A. The Airworthiness Limitation Section is FAA approved and specifies maintenance required under 14 CFR 43.16 and 91.403, unless an alternative program has been FAA approved.
- B. The Beverage Maker PN 4510-1 and 4510-2 series and its components have no life limited parts, no mandatory replacement times for parts, and no defined structural inspection intervals.

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TESTING AND FAULT ISOLATION

1. General

WARNING: THE BEVERAGE MAKER AND WATER IN THE TANK MAY BE VERY HOT.

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY FOR TESTING PURPOSES.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY-FREE ENVIRONMENT.

NOTE: The following tests are to be performed with non-insulated server, PN MISC-3201. Use of an insulated server requires that the WARMER is not used. An insulated server may affect water temperature tests.

A. Perform a functional test on the Beverage Maker at every shop visit.

NOTE: After replacing components, a functional test should be performed to make sure the Beverage Maker operates correctly. Refer to the ILLUSTRATED PARTS LIST Section for item descriptions and locations.

B. Completely read and thoroughly understand the entire TESTING AND FAULT ISOLATION section. Become thoroughly familiar with the test sequences and their interaction before performing any tests.

C. Perform a visual inspection to detect obvious problems or defects. Refer to the INSPECTION/CHECK, DISASSEMBLY, and ASSEMBLY Sections.

D. The work area for TESTING should be clean and well organized.

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2. Test Equipment

- A. Refer to Table 1001 for test equipment and materials. See the SPECIAL TOOLS, FIXTURES AND EQUIPMENT Section for more information.

NOTE: Equivalent substitutes may be used for listed items.

Table 1001
Test Equipment and Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
1/8 inch NPT plug	AN933-1D or MIT-022P	Commercially Available or FasTest, Inc. 1646 Terrace Drive Roseville, MN 55113-1317
Electrical connector cable between Beverage Maker and power supply	998552	B/E Aerospace, Inc. (V63367)
High Low Temperature RTD wiring harness with 4-position connector	997112	B/E Aerospace, Inc. (V63367)
Megohmmeter	None	Commercially Available
Milliohmmer	QuadTech Model 1880	QuadTech Inc (V0PK96)
Ohmmeter	None	Commercially Available
Power Supply, including: 3-phase 115 ± 3 VAC, 400 Hz, 10 amps per phase; 3 ammeters (0-25 amp); and 3 voltmeters (0-250 volt) (refer to Figure 1001)	None	Commercially Available
Server	MISC-3201	B/E Aerospace, Inc. (V63367)
Stopwatch	None	Commercially Available
Thermometer, 50-250° F (10-121° C)	None	Commercially Available
Water Supply, including: pressure regulator (0-150 psig, 0-10.34 barg) and flow meter (0-0.5 GPM, 0-1.9 LPM) (refer to Figure 1001)	None	Commercially Available

- B. Set-up for Functional Tests (refer to Table 1001 and Figure 1001)

- (1) Provide a power supply that meets the requirements described in Table 1001.
- (2) Provide a water supply that meets the requirements described in Table 1001.
- (3) Provide a server and Beverage Maker for TESTING.

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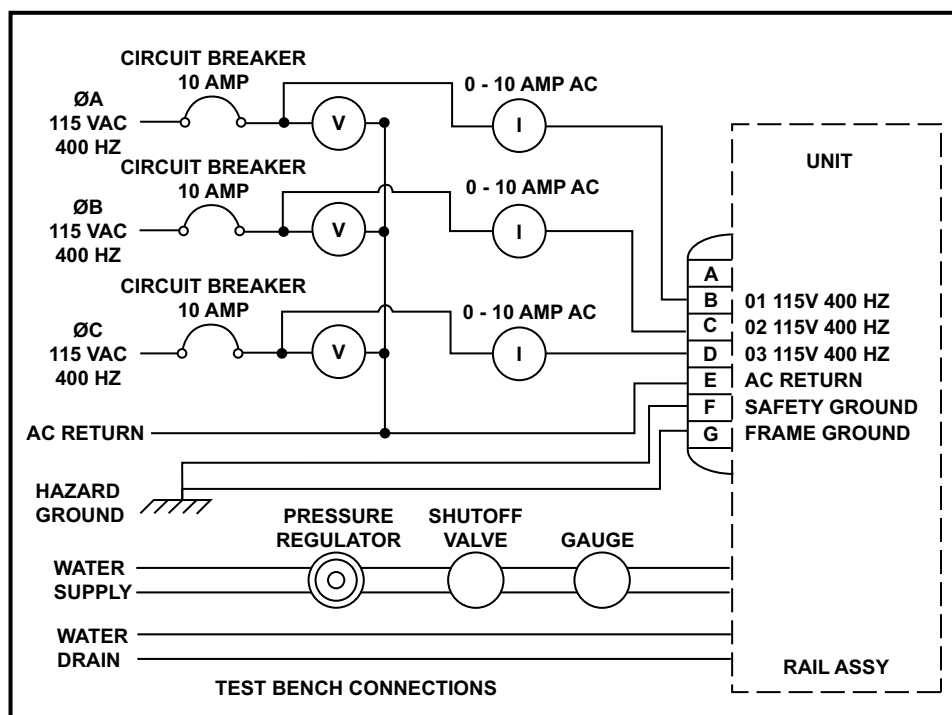


Figure 1001
Functional Test Set-up

3. Functional Tests (refer to Figure 1001 and Table 1002)

NOTE: When testing a Beverage Maker, each test should be performed in the sequence listed below.

NOTE: If a Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

A. Dielectric Test

NOTE: Do a dielectric test only after a major rewiring has been done. A dielectric test is not needed every time the unit comes into the shop for repair.

- (1) Make sure that the P1 connector is removed from J1 connector on circuit board.
- (2) Install shorting connector into PP1.
 - (a) Connect one lead of the Hi Pot tester to the PP1 pigtail (B,C,D and E) and the other lead to the PP1 pigtail (F & G). Gradually apply, in increments of 250 to 500 volts per second, a test voltage of 1500 volts, 50/60 Hz. Maintain the test voltage for one full minute and make sure it should not exceed more than 2 milliamps.
- (3) Use Megohmmeter with output voltage of 500 volts DC across same wires as mentioned in paragraph 3.A.(2) and make sure it is 200 megohms minimum.

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B. Resistance Check

- (1) Check resistance from PP1-G to un-anodized brew shelf and make sure that the resistance is 2.5 Milliohms maximum.
- (2) If applicable, check resistance from PP1-G to metal disc located in the center of the platen heater and make sure that the resistance is 3.2 milliohms maximum.
- (3) Connect a resistance bridge between PP1-F and faucet. The measured resistance shall not be greater than 3.0 milliohms.

C. Water Supply Temperature Test

- (1) If water supply is connected to the Beverage Maker, disconnect water supply from the Beverage Maker.
- (2) Measure the temperature of the water from the water supply using a thermometer. The temperature of the water should be 62° F (17° C) to 72° F (22° C).

NOTE: Temperatures outside this range will affect correct Beverage Maker operation. Results of other tests may not meet specifications.

D. Ambient Temperature Test

- (1) Using a thermometer, measure the temperature in the immediately adjacent area of the test environment. The temperature should be between 70° F (21° C) and 85° F (29° C).

NOTE: Temperatures outside this range will affect correct Beverage Maker operation. Results of other tests may not meet specifications.

E. Tank Pressure Test

NOTE: The following test is for some safety features, correct operation of plumbing system, and for detection of water leaks. If a Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

- (1) Replace the pressure relief valve with a 1/8 inch NPT plug prior to testing.
- (2) Connect the Beverage Maker to water supply.
- (3) Let tank fill with water. Open faucet to verify the tank is full.
- (4) Adjust water pressure to 130 psig (8.96 barg). Hold for a minimum of 5 minutes. Inspect tank for leaks. No leaks are allowed.
- (5) Release pressure then remove plug.
- (6) Put pressure relief valve back into the tank using thread seal tape.
- (7) Gradually increase the water pressure to the tank. The valve should remain closed at pressures below 75 psig (5.17 barg). Continue to increase pressure ensuring the valve is fully open prior to or at 100 psig (6.89 barg).

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NOTE: The pressure relief valve should be examined and have a pressure test for correct operation every six months or 1500 flight hours, whichever comes first.

- (8) Disconnect water supply and drain the Beverage Maker. Make sure the vent valve operates correctly.

NOTE: The vent valve lets the tank fill and drain, when required. If the tank fills (without continuous overflow) and drains, the vent valve is operating. A small amount of water may come out of the vent valve drain line as the tank completes filling. This is normal operation.

F. POWER Button and LOW WATER Indicator Light Tests

NOTE: The following test is for some safety features and correct operation of water level systems. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

- (1) Connect the power supply to the Beverage Maker.

NOTE: The small amount of current required for operation of the control circuits will not usually show on ammeters. Zero on the ammeters, described in these tests, includes the small amount of current required by the control circuits after heaters go off.

- (2) If connected, disconnect the water supply from the Beverage Maker.

NOTE: The Beverage Maker tank must be empty to continue this test. If the tank is not empty, drain the tank before continuing.

- (3) Press the POWER button.

- (4) The POWER button light should come on, the LOW WATER indicator light should come on, and the heaters should not come on (as indicated by the ammeters).

- (5) Press the POWER button. Make sure the POWER button light goes off.

- (6) Connect a water supply to the Beverage Maker and adjust water pressure to 24 to 29 psig (1.66 to 2.0 barg).

- (7) Make sure the Beverage Maker tank fills (as shown by the flow meter) and does not leak.

NOTE: A small amount of water may come out of the vent valve drain line as the tank completes filling. This is normal operation.

- (8) At the end of fill, press the POWER button. Make sure the POWER button light comes on. Make sure the LOW WATER indicator light is off, and then press the POWER button so the POWER button light goes off.

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G. RTD Circuit Test

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY FOR TESTING PURPOSES.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY FREE ENVIRONMENT.

NOTE: The following test is for correct operation of brew temperature control systems. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

- (1) With the Beverage Maker disconnected from the power supply and water supply, remove the cover from the power module assy and disconnect the RTD assy plug from the J4 connector on the controller board assy.
- (2) Toggle the switch on the RTD simulator harness to the higher temperature side and then connect the harness to the controller board assembly.
- (3) Connect the Beverage Maker to the power supply.
- (4) Press the POWER button. This should trip the safety latch. If the safety latch is tripped, there will be no current draw to the heaters and the POWER button light should go into the double-blink mode.
- (5) Disconnect the power supply from the Beverage Maker.
- (6) Toggle the switch on the High Low Temperature RTD wiring harness (refer to Table 1001) to the low side RTD harness and connect the High Low Temperature RTD wiring harness to the controller board assy.
- (7) Connect the power supply to the Beverage Maker.
- (8) Reset the safety latch by pressing the manual switch on the back of the power module assy under the grommet (refer to Figure 1002).

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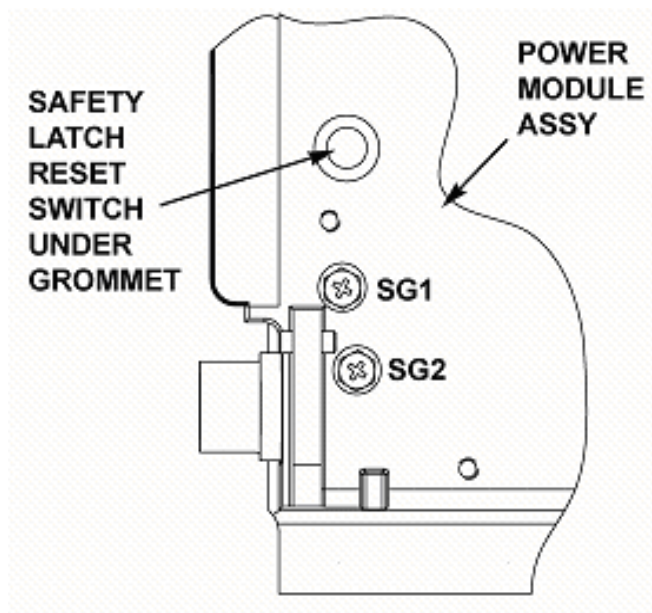


Figure 1002
Safety Latch Location

- (9) Press the POWER button. When there is no current drawn by the heaters, the safety latch should not trip and the POWER button light should not go into double-blink mode.
- (10) Disconnect the power supply from the Beverage Maker.
- (11) Disconnect the High Low Temperature RTD wiring harness from the controller board assy.
- (12) Connect the RTD assy to the controller board assy.
- (13) Put the cover on the power module assy.
- (14) Connect the power supply to the Beverage Maker.
- (15) Press the POWER button. This should not trip the safety latch. Make sure the POWER button light does not go into double-blink mode, and then press the POWER button again to turn off the Beverage Maker.

H. Tank Heater and Preheat Test

NOTE: The following test does a check of safety features and correct operation of heater control systems. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

NOTE: Preheat time is the elapsed time from heaters on to heaters off after the tank is first filled with water.

- (1) With the Beverage Maker connected to the power supply and water supply and the tank filled with water, be prepared to time preheating before pressing the POWER button.

NOTE: The LOW WATER indicator light should not come on during this test.

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- (2) Press the POWER button and start the stopwatch at the same time. Turn ON WARMER (if applicable).
- (3) Make sure the amperes for the phases are as follows:

Phase A	8.1 +0.6/-0.9 amperes
Phase B	7.8 +0.4/-0.7 amperes
Phase C	7.8 +0.4/-0.7 amperes

- (4) Stop the stopwatch when the current on Phase B goes to zero. The elapsed time should be 3 minutes, 30 seconds maximum, when starting with a cold tank.

NOTE: The small amount of current required for operation of the control circuits will not usually show on ammeters. Zero on the ammeters, described in these tests, includes the small amount of current required by the control circuits after the heaters go off.

NOTE: A small amount of water may come out of the pressure relief valve drain line as the tank completes preheating. This is normal operation.

J. BREW Test

NOTE: The following test puts the Beverage Maker through several brew cycles to do a check of safety features and correct operation. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

- (1) Start this test by making sure the tank is filled with water, the water supply pressure is set to 24 to 29 psig (1.66 to 2.0 barg), the Beverage Maker is connected to the water supply and power supply, and the POWER button light is on. The WARMER (where option is installed) must be on before starting this test. Put an empty server in the Beverage Maker and lower the brew handle.
- (2) Press the BREW button, observe the flow meter and when flow starts, start the stopwatch.

NOTE: Brew does not start until the water in the tank is heated. After the start of flow, there will be an interruption of flow for approximately 10 seconds. This lets water make coffee pillow pack wet during a brew cycle. This is normal operation and the time is included in the brew cycle time.

- (3) Stop the stopwatch when the flow meter goes to zero flow a second time. Elapsed time between when flow starts and stops should be 2 minutes 30 sec to 3 minutes 35 sec.

NOTE: The time between when the flow meter shows zero flow until the BREW button light goes off may be 10 to 20 seconds. This is a time that is not adjustable by a repair facility. This time between when the flow goes to zero and when the BREW button light goes off is a safety feature that lets most hot liquid have time to drain from the brew cup before the server is removed from the Beverage Maker.

- (4) Remove, empty, and install server in the Beverage Maker.
- (5) With the BREW button light off, press the BREW button to start a second brew.
- (6) Lift brew handle after the normal 10 second flow interruption. Lifting the brew handle should interrupt the brew cycle, as shown by the flow going to zero.

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- (7) Lower the brew handle. This should let the brew cycle continue.
 - (8) When the BREW button light goes off, press the BREW button. Another brew cycle should not start.
 - (9) Lift the brew handle and measure the temperature of liquid in the server. It should be between 175° F (79° C) and 195° F (91° C). The server should be full.
 - (10) Put the full server in the Beverage Maker and lower the brew handle.
 - (11) Press the BREW button. The BREW button light should come on for only a short time then go off again. The full server should be detected by the backup liquid level sensor for the server.
 - (12) Lift the brew handle, remove, empty, and install server in the Beverage Maker. Lower the brew handle.
- K. Install a server and a brew cup, then lower the brew handle. Pull on the server to make sure the server retainer is working correctly.

L. Optional Feature Tests

NOTE: The following tests are for optional features and correct operation. If the Beverage Maker fails any of these tests, refer to the FAULT ISOLATION Section (Paragraph 5) for recommendations. Start these tests by making sure the tank is filled with water, the beverage maker is connected to the water supply and power supply, and the POWER button light is on.

- (1) TEA Test (if option is installed)
 - (a) The WARMER (where option is installed) must be on before starting this test. Put an empty server in the Beverage Maker and lower the brew handle. The water supply pressure is set to 24 to 29 psig (1.66 to 2.0 barg).
 - (b) Press the TEA button, observe the flow meter and when flow starts, start the stopwatch.

NOTE: TEA brew does not start until the water in the tank is heated.
 - (c) Stop the stopwatch when the flow meter goes to zero flow. Elapsed time between when flow starts and stops should be 2 minutes 30 sec to 3 minutes 20 sec.
 - (d) Lift the brew handle and measure the temperature of liquid in the server. It should be between 175° F (79° C) and 195° F (91° C). The server should be full.
 - (e) Remove, empty, and install server in the Beverage Maker. Lower the brew handle.

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- (2) Platen Heater (WARMER) Test (if option is installed)
 - (a) Press the WARMER button when POWER button light is on. Make sure the WARMER light comes on and make sure the platen heater becomes warm.
- NOTE: The amperes required for the platen heater is so small it may not show on the ammeter for Phase A.
- (3) COLD Water from Faucet Test (if option is installed)
 - (a) Press the COLD button for cold water. Make sure cold water comes out of the faucet.
- (4) HOT Water from Faucet Test (if option is installed)
 - (a) Press HOT button for hot water. Make sure hot water comes out of the faucet. Continue to press button so hot water comes out of the faucet until heaters come on (as shown by ammeters).

4. Bench Test

A. RTD Assy Test

NOTE: Do this check of the RTD only if necessary for isolation of a fault in temperature control of the water heating system.

- (1) Disconnect RTD assy from the controller board assy.
- (2) Check the resistance across the disconnected wires with a standard ohmmeter. Resistance should be 1.1 kOhms $\pm$ 5% at room temperature (75° F (23° C) nominal).
- (3) If RTD assy is out of tolerance, replace RTD assy.

B. Pressure Relief Valve Calibration (PN 62-0223-1 Only)

NOTE: Calibration of the pressure relief valve is necessary if it is disassembled.

- (1) Assemble the valve (refer to Figure 1003).
- (2) Attach the valve to an adjustable pressure-regulated water supply.
- (3) Loosen the lock nut (210).
- (4) Turn the adjusting cap (215) on the valve body (220) so that water begins to come out of the fitting (250) when water pressure is at 75 psig. Water must flow freely out of the fitting at 100 psig.
- (5) When the adjusting cap is set so that the valve opens at 75 PSIG and is fully open at 100 psig, turn the lock nut (210) on the valve body (220) so that it keeps the adjusting cap (215) from turning.
- (6) After the lock nut (210) is tight, do a check of the valve to make sure it begins to open at 75 psig and becomes fully open at 100 psig.

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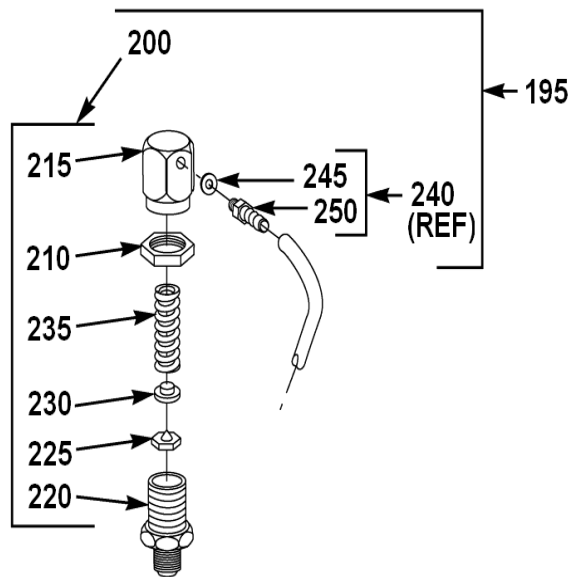


Figure 1003
Valve Assembly

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5. Fault Isolation

A. Table 1002 lists the probable causes for faults seen during use or testing.

Table 1002
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Brew handle hard to operate.	Chassis assy bent or broken.	Attempt to repair chassis assy before replacing.
	Hardware and/or fasteners bent or broken.	Replace hardware and/or fasteners.
	Brew handle binding (metal-to-metal contact).	
	Brew handle bent or broken.	Replace brew handle.
	Handle bushing(s) is deformed or broken.	Replace handle bushing(s).
	Brew head assy loose or binding.	
	Brew head casting bent or broken.	Attempt to repair brew head before replacing.
	Brew head guides worn or broken.	Replace brew head guides.
	Return spring weak or broken.	Replace spring. (Check bushing for abnormal wear).
	Server retainer assy binding (metal-to-metal contact).	
	Server retainer assy deformed or broken.	Attempt to repair server retainer assy before replacing.
	Server retainer spring(s) weak or broken.	Replace spring(s). (Check spring bushing(s) for abnormal wear).

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Table 1002 (Cont.)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
POWER button does not turn the Beverage Maker on or off. (Make sure unit is connected to main power source) or POWER button does not illuminate when required. or POWER button illuminates when not required. or WARMER button does not turn the platen heater on or off. (Make sure platen heater is good)	PC board switch assy failure. Faulty wiring and/or connector(s). controller board assy failure.	Replace PC board switch assy. Replace wiring and/or connector(s). Replace fuses. Replace controller board assy.
POWER button displays double-blink mode (overtemperature indication - safety latch is tripped).	Faulty wiring and/or connectors. RTD assy failure (fails OPEN simulating overtemperature condition in tank). Excessive steam head in tank assy. Low water sensor assy failure (fails SHORT simulating a full tank condition). Pressure relief valve failure (fails CLOSED allowing tank pressure to rise). Solid state relay failure (fails CLOSED).	Replace wiring and/or connector(s). Replace RTD assy. Replace low water sensor assy. Attempt to clean pressure relief valve before replacing. Replace controller board assy.

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Table 1002 (Cont.)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
POWER button displays triple-blink mode (phase loss indication).	Faulty wiring and/or connectors.	Replace wiring and/or connector(s).
	Thermostat buss bar failure (fails OPEN).	Repair/replace thermostat buss bar.
	Overtemperature thermostat(s) failure (tripped or failed OPEN).	Reset/replace over-temperature thermostat(s).
	Tank heater assy (s) failure.	Replace tank heater assy (s).
Beverage Maker will not accomplish a BREW or TEA cycle.	LLS backup probes failure (SHORTED simulating full server condition).	Replace infrared sensor assy. (Check for "wicking" and towel dry as needed. Check for water intrusion into cable assy).
or		
Beverage Maker will not dispense HOT water.	Brew head assy interlock failure (fails OPEN simulating brew handle up condition).	Repair/replace brew head assy.
	Solenoid valve assy failure.	Attempt to clean manifold assy before replacing.
	Solenoid valve assy plunger assy stuck.	Replace plunger assy.
	Manifold assy clogged.	Attempt to clean manifold assy before replacing.
Beverage Maker will not accomplish a BREW or TEA cycle.	Tank assy sensors.	
or		
Beverage Maker will not dispense HOT water.	Low water sensor assy failure (fails OPEN simulating false tank empty condition).	Attempt to clean low water sensor assy before replacing.
or		
POWER button displays double-blink mode (overtemperature indication).	RTD assy failure (fails OPEN simulating overtemperature condition in tank or fails SHORT simulating frozen condition in tank).	Replace RTD assy.

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Table 1002 (Cont.)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Beverage Maker will not dispense HOT or COLD water.	Main utilities (water and power) are not connected. Water inlet clogged or damage. Faucet assy clogged or damaged. Solenoid valve assy failure (failed CLOSED). Solenoid valve assy plunger assy stuck. Manifold clogged.	Connect to main utilities. Attempt to clean before replacing. Attempt to clean before replacing. Repair/replace solenoid valve assy. Replace plunger assy. Attempt to clean manifold assy before replacing.
(DELETED)		
Tank heaters will not heat	Thermostat buss bar failure (fails OPEN). Overtemperature thermostat(s) failure (tripped or failed OPEN). Faulty wiring and/or connectors. Low water sensor assy failure. RTD assy failure (fails OPEN simulating overtemperature condition in tank or fails SHORT simulating frozen condition in tank). controller board assy failure.	Repair/replace thermostat buss bar. Reset/replace over-temperature thermostat(s). Replace wiring and/or connector(s). Attempt to clean low water sensor assy before replacing. Replace RTD assy. Replace controller board assy.
Tank heater(s) will not turn off.	Tank assy sensors. Low water sensor assy failure (failed SHORT through inner sleeve). RTD assy failure (reporting false low temperatures). Controller board assy failure.	Replace low water sensor assy. Replace RTD assy. Replace controller board assy.

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Table 1002 (Cont.)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Faucet leaks.	Solenoid valve assy failure (fails OPEN).	Repair/replace solenoid valve assy.
	Solenoid valve assy plunger assy does not seal against internal valve seat.	Replace plunger assy.
Server does not completely fill.	Supply water pressure too low.	Adjust supply water pressure to 24 to 29 psig.
	Plumbing (tubing) is clogged.	Attempt to clean before replacing.
	LLS backup probes failure (SHORTED simulating false full server condition).	Replace infrared sensor assy. (Check for "wicking" and towel dry as needed. Check for water intrusion into cable assy)
Server overflows.	Faulty wiring and/or connectors.	Replace wiring and/or connector(s).
	LLS malfunction/failure. (If TP6 on controller board assy is 5 VDC, then photo transistor SHORTED) If voltage across R43 on controller board assy does not drop after BREW cycle initiation plus 35 seconds, then infrared LED is OPEN)	Replace infrared sensor assy.
	LLS backup probes failure (failed OPEN - will not detect an overflowing server condition)	Replace infrared sensor assy.
	Controller board assy failure.	Replace controller board assy.
Water tank will not drain correctly.	Vent valve failure (failed CLOSED).	Repair/replace vent valve assy.
	Plumbing (tubing) is clogged.	Attempt to clean before replacing.
	Water inlet clogged or damage.	Attempt to clean before replacing.

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Table 1002 (Cont.)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
BREW or TEA cycle too short or too long.	Flow washer failure (causing excessive or restricted water flow). Supply water pressure too low. Plumbing (tubing) is clogged. (check manifold, brew head, tank, tubing, and water inlet).	Replace flow washer. Adjust supply water pressure to 24 to 29 psig. Attempt to clean before replacing.
Resistance value(s) between PP1-G and/or PP1-F to measurement points are above maximum value(s).	Poor surface contact between chassis and ground locations due to contaminated/Corroded Grounding surfaces or loose ground stud hardware.	Refer to SIL 4510-SIL-007.

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SCHEMATICS AND WIRING DIAGRAMS

1. General

- A. Schematic and wiring diagrams are shown with associated parts and assemblies within this manual.
- B. For general reference, see the Beverage Maker plumbing schematic (refer to Figure 2001) and the wiring diagram (refer to Figure 2002).

NOTE: Plumbing schematic (Figure 2001) is shown with all options. Three (3) solenoids (maximum) may be installed.

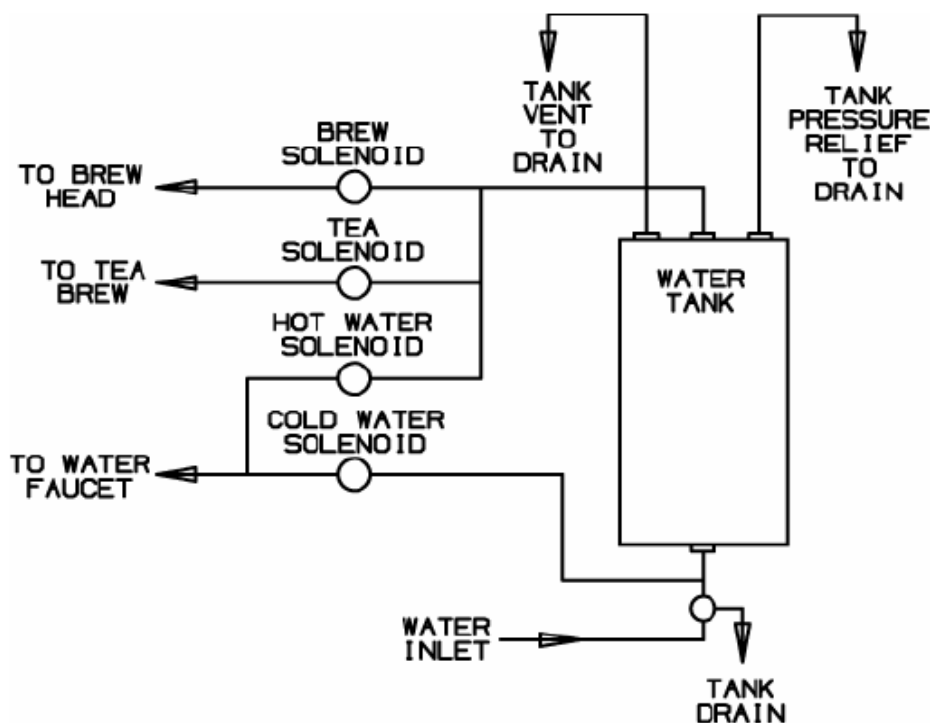


Figure 2001
Plumbing Schematic

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series

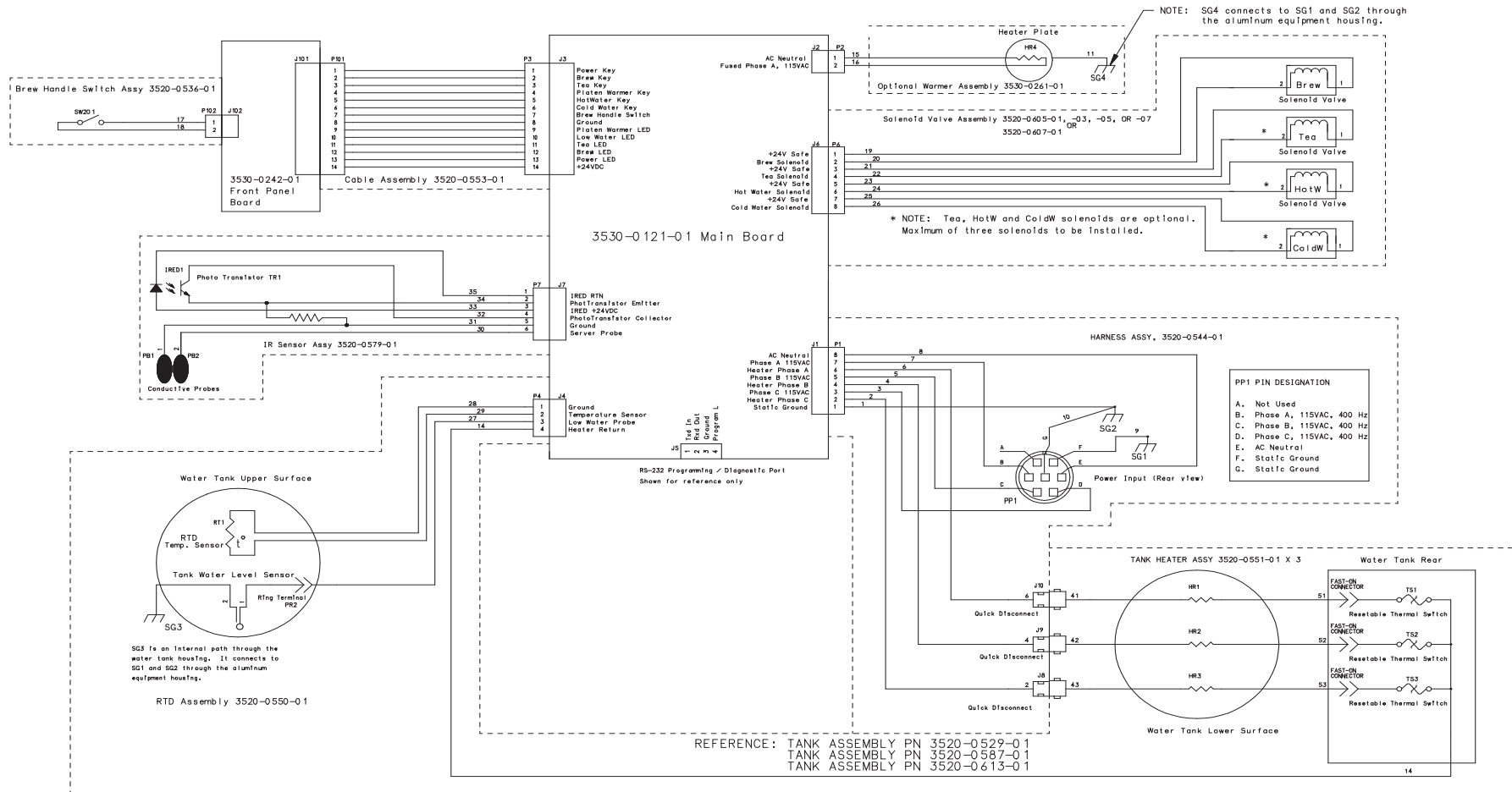


Figure 002
Wiring Diagram

DISASSEMBLY

1. General

WARNING: THE BEVERAGE MAKER AND WATER IN THE TANK MAY BE VERY HOT.

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY-FREE ENVIRONMENT.

CAUTION: MAINTENANCE ENGINEER MUST WEAR A CONDUCTIVE WRIST STRAP TO PROTECT THE CONTROLLER BOARD ASSEMBLY AGAINST ESD.

A. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK Sections prior to disassembly to establish the condition of the faulty component or most probable cause of its malfunction. This will determine the extent of disassembly required without completely tearing down and rebuilding the Beverage Maker or faulty component.

B. Refer to the ILLUSTRATED PARTS LIST (IPL) Section for disassembly and assembly relationships. The number parentheses after the noun name of a part refers to the item number in the IPL Figure.

C. Provide a clean space and working surface before beginning any disassembly operation.

NOTE: Removal of some screws may require use of an adjustable-torque screwdriver due to the use of thread sealant.

NOTE: Beverage residue and mineral deposits can be cooked into part connections making them hard to remove. The cleaning of some parts can help in their removal, especially parts exposed to coffee residue.

D. Before removing electronic components, note the polarity and position of the component.

2. Procedures

A. Beverage Maker (refer to IPL Figures 1, 1A, 1B, and 1C)

- (1) With the brew handle in the raised position, remove brew cup (20A or 20B) from the beverage maker subassembly (10).

Table 3001
Brew Cup Assembly PN Reference

UNIT PN	BREW CUP ASSEMBLY PN	DESCRIPTION
4510-XXUX-XX*	3520-0540-01	Brew cup assembly (upper)
4510-XXLX-XX*	3520-0540-03	Brew cup assembly (lower)

*X represents other numbers or letters in the Unit Part Number

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- (a) Remove two nuts (35), two screws (40), and spacer plate (30) from brew cup (25A, 25B) (refer to Figure 3001).

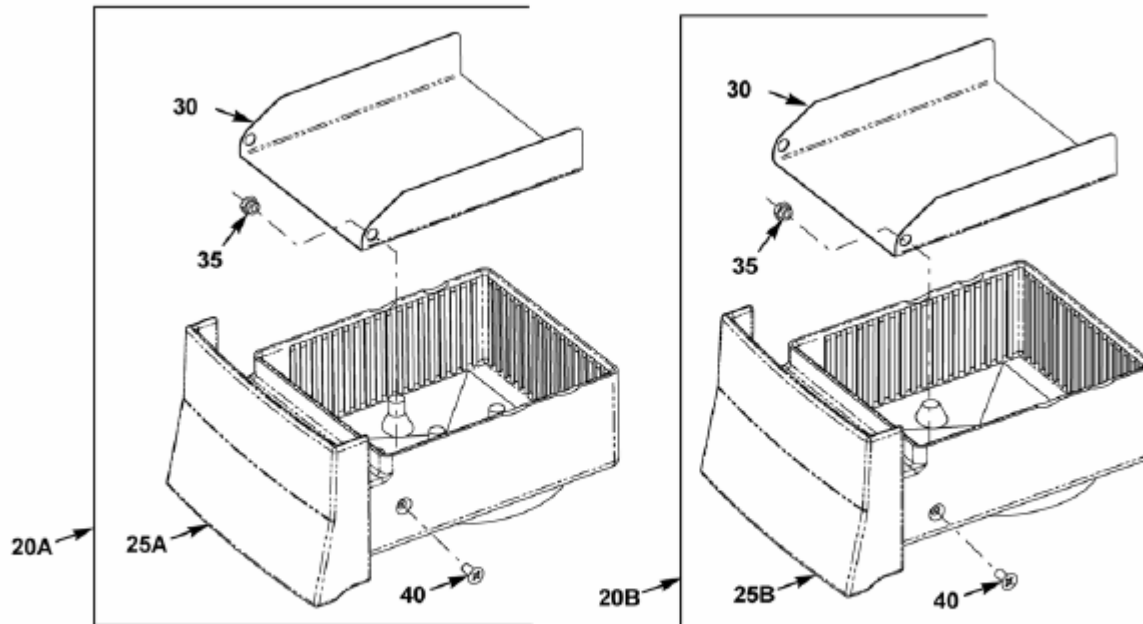


Figure 3001
Brew Cup Disassembly

- (2) Refer to DISASSEMBLY, Paragraph 2.B for disassembly of the beverage maker subassembly (1).

B. Beverage Maker Subassembly (refer to IPL Figures 2, 2A, 2B, 2C, and Figures 3002 through 3014)

CAUTION: DRAIN WATER FROM TANK ASSY (15) BEFORE DISASSEMBLY.

NOTE: Remove nylon wire tie downs as required to release wiring during the disassembly process.

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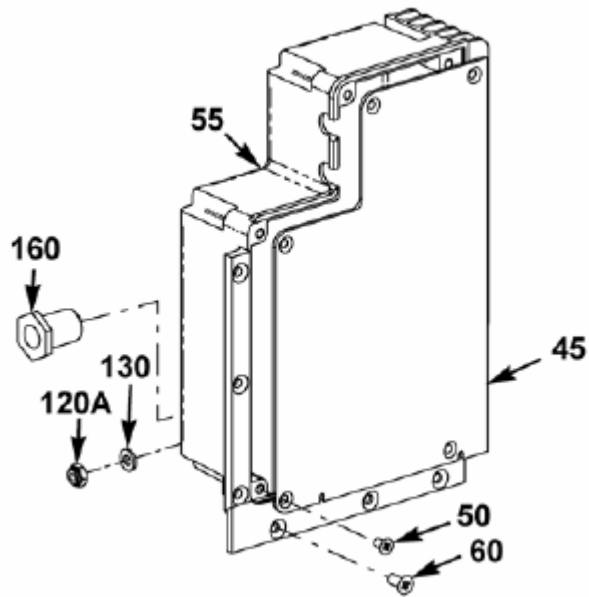


Figure 3002
Removal of Power Module Cover

- (1) Remove five screws (50) and power module assy cover (45) from the power module assy (55). This gives access to wiring harness (65) connections to the power controller board (75).

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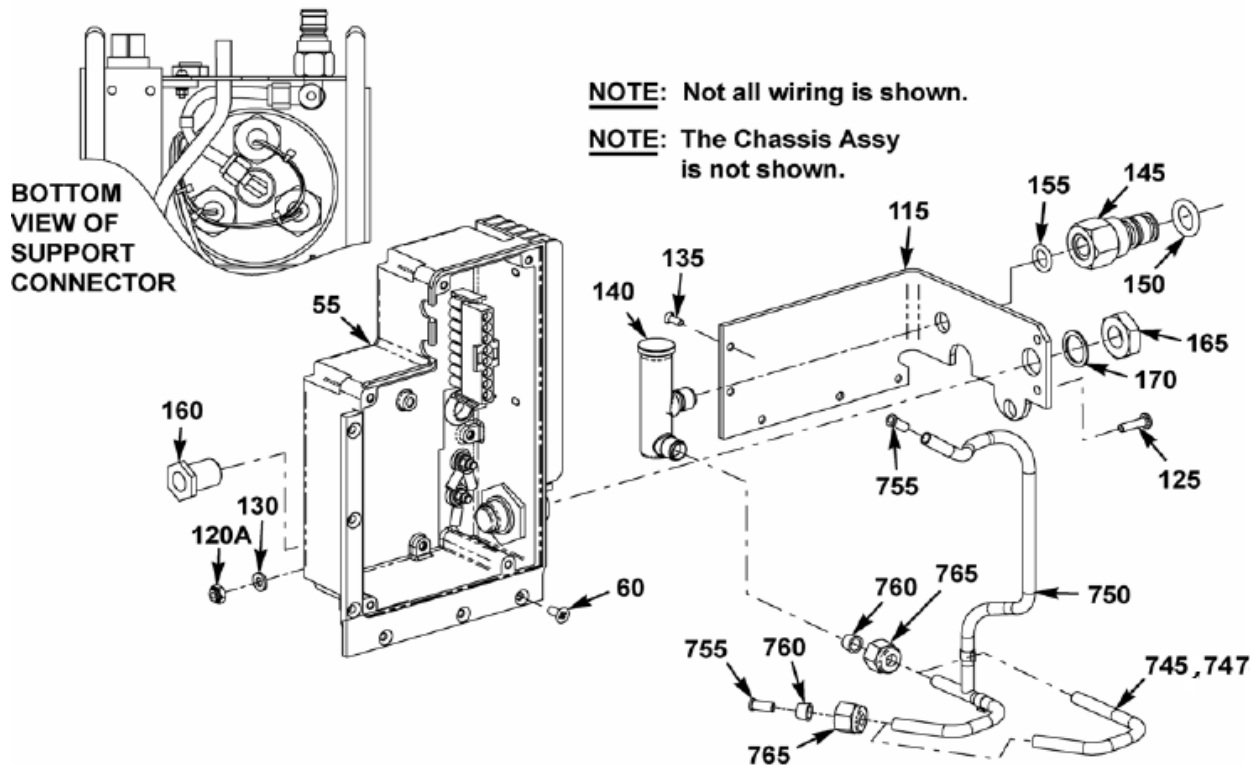


Figure 3003
Removal of Support Connector

- (2) Disconnect support connector (115) from the chassis assy (10) (refer to Figure 3003).
 - (a) Disconnect water supply tube assy (750, 747 or 745A) from drain valve assy (140) and tank assy (15).
 - 1 Remove ferrule (760), fitting (765), and insert (755) from water supply tube (745) if required for replacement or cleaning.
 - (b) Remove screw (125), flat washer (130), and lock nut (120A) from support connector (115) and power module assy (55).
 - (c) Remove hexagon nut (165), lock washer (170) and alignment pin guide (160) from support connector (115) and power module assy (55).
 - (d) Remove two screws (135) from the support connector (115) and the chassis assy (10 or 10A).
 - (e) Remove three screws (135) from the support connector (115) and the left side rail guide (175).

CAUTION: AFTER REMOVING ALL FASTENERS FROM THE CONNECTOR SUPPORT, CAREFULLY SLIDE THE SUPPORT CONNECTOR OFF THE REAR DRAIN TUBE (IF OPTION INSTALLED) TO AVOID DAMAGE TO THE DRAIN TUBE.

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- (f) Remove water shutoff plug (145), o-ring (155) and drain valve assy (140) from support connector(115). Refer to DISASSEMBLY, Paragraph 2.F for disassembly of drain valve assy (140).
- (g) Remove o-ring (150) from water shutoff plug assembly (145).

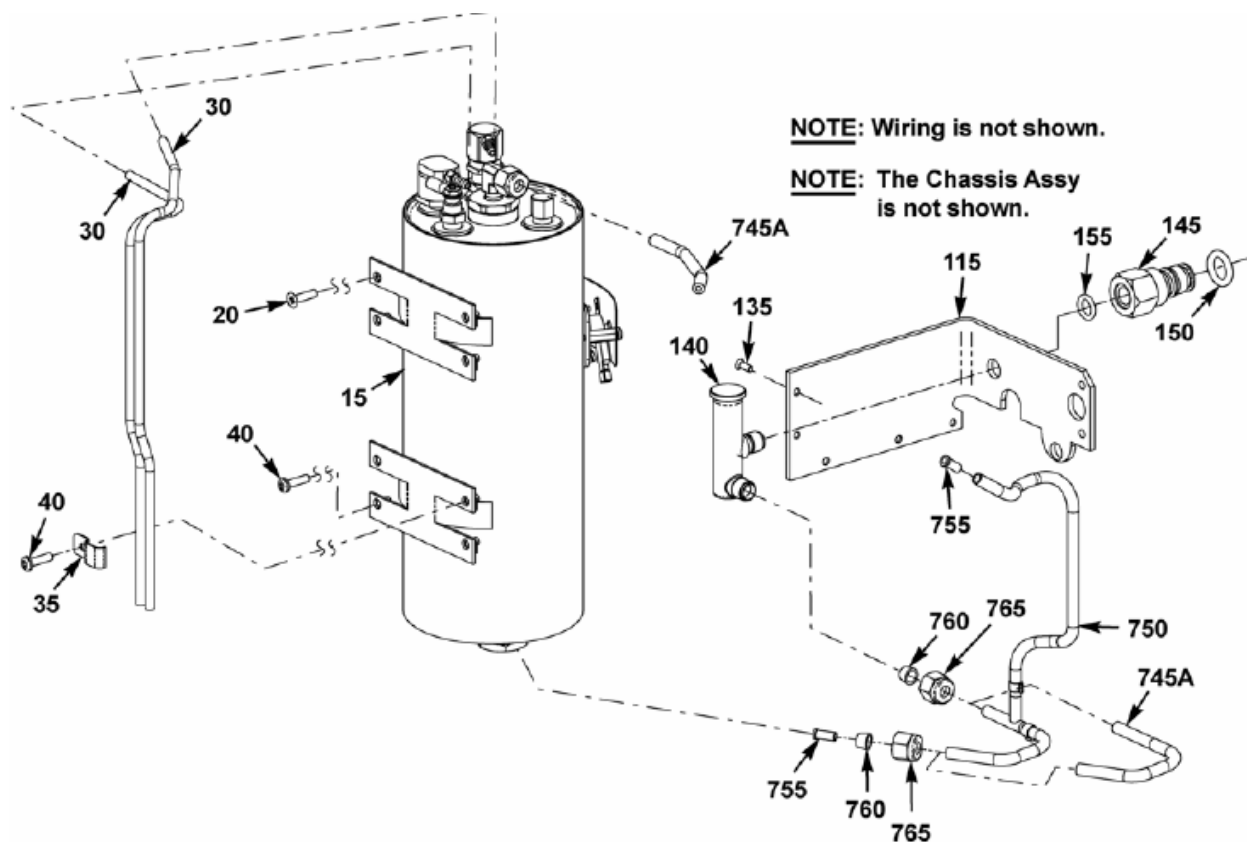


Figure 3004
Removal of Tank Assy

- (3) Remove the tank assy (15) from chassis assy (10 or 10A) (refer to Figure 3004).

NOTE: The tank assy (15) includes the low water sensor, overtemperature thermostats, vent valve, pressure relief valve, RTD assy, water inlet tube, heaters, water filter, and water tank.

CAUTION: IDENTIFY HEATER TERMINALS AND WIRE HARNESS ASSY TERMINALS THAT GO TOGETHER. THE WIRING TERMINALS MUST BE TURNED AND PULLED TO BE REMOVED FROM EACH OTHER. THEY ARE NOT EASY TO DISCONNECT BY HAND, SO TOOLS MAY BE REQUIRED. DO NOT USE TOO MUCH FORCE TO HOLD OR REMOVE THE PLASTIC TERMINALS FROM EACH OTHER.

- (a) Disconnect terminals of heaters (15, IPL Figure 3) from wire harness terminals.

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- (b) Disconnect tube assy (745A) from fitting (90, IPL Figure 3) of water filter on top of tank assy (15).
- (c) Inside the server cavity, remove screw (40) from tube clamp (35) on the drain tubes (30) that come from the top of the tank assy (15).
- (d) Remove the tubes (30) from barbed fittings (170 and 240, IPL Figure 3) on the top of the tank assy (15) if replacement or cleaning is required.
- (e) If this has not been done, remove five screws (50) and power module assy cover (45) from the power module assy (55). This gives access to RTD assy wiring plug connection to the power controller board (75).
- (f) Disconnect plug of RTD assy (190, IPL Figure 3) from power controller board (75) in the power module assy (55).

CAUTION: THE IDENTIFICATION PLACARD MUST BE KEPT WITH THE BEVERAGE MAKER WHEN THE SCREWS HOLDING THE PLACARD AND TANK ASSY ON THE CHASSIS ASSY (10) ARE REMOVED. THIS ID PLACARD IS NOT AVAILABLE AS A SPARE PART.

- (g) Remove four flat head screws (20) and four pan head screws (45, IPL Figures 1, 1A, 1B, 1C, 40, 25, IPL Figures 2, 2A, 2B, 2C) from the chassis assy (10 or 10A) and tank assy (15). Removal of all eight screws frees the tank assy (15) from the chassis assy (10 or 10A). Refer to DISASSEMBLY, Paragraph 2.C for disassembly of the tank assy (15).

NOTE: Removal of two screws (45, IPL Figures 1, 1A, 1B, 1C) low inside the chassis assy (10 or 10A) server cavity frees the ID placard from the Beverage Maker (1, IPL Figures 1, 1A, 1B, 1C).

NOTE: The brewhead assy (300) may need to be removed to get access to screws (20) on the upper part of the chassis assy (10 or 10A) and the tank assy (15) (refer to DISASSEMBLY, Paragraph 2.B.(12) to remove the brewhead assy (300)).

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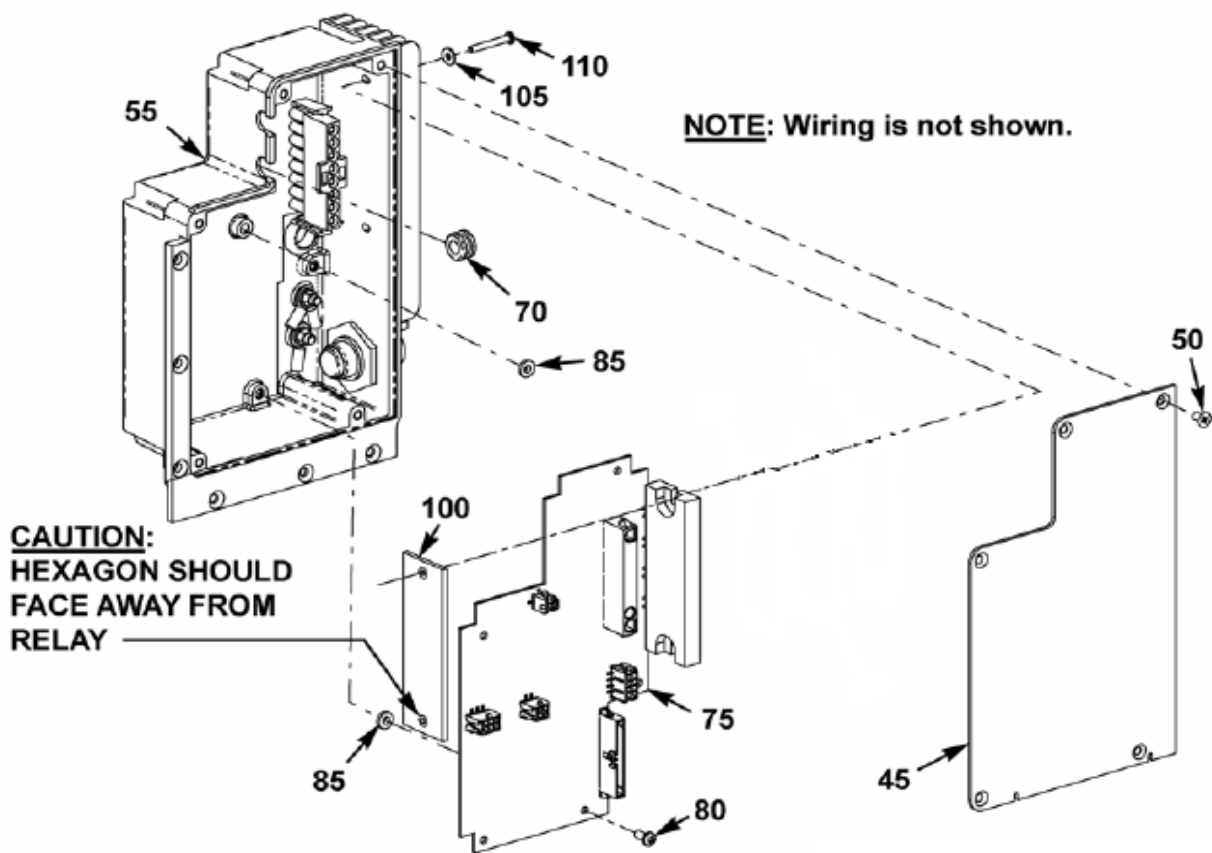


Figure 3005
 Removal of Controller Board Assy

- (4) Remove the power module assy (55) from the chassis assy (10 or 10A) (refer to Figures 3003 and 3005).

NOTE: Remove the support connector(115) from the power module assy (55) and chassis assy (10 or 10A) before removing the power module assy from the chassis assy. Refer to DISASSEMBLY, Paragraph 2.B.(2) to remove the support connector(115).

- (a) Remove three screws (60) from the power module assy (55) and guide rail (175).
- (b) Remove three screws (60) from the power module assy (55) and chassis assy (10 or 10A).

CAUTION: WHEN TOUCHING THE POWER CONTROLLER BOARD (75) OR ANY OF ITS INDIVIDUAL PARTS, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

- (c) Disconnect all wiring plugs from sockets in the power module assy (55).
- (d) Refer to DISASSEMBLY, Paragraph 2.D for disassembly of the power module assy (55).
- (e) Remove all wiring plugs from the controller board (75).

NOTE: Identify all wires and points of connection before disassembly to help in ASSEMBLY.

- (j) Carefully remove the relay and controller board assy (75), and relay retainer (100) from the power module assy (55). Isolate the controller board assy (75) from electrostatic discharge and exposure to water. Refer to DISASSEMBLY, Paragraph 2.E for disassembly of the controller board assy (75).

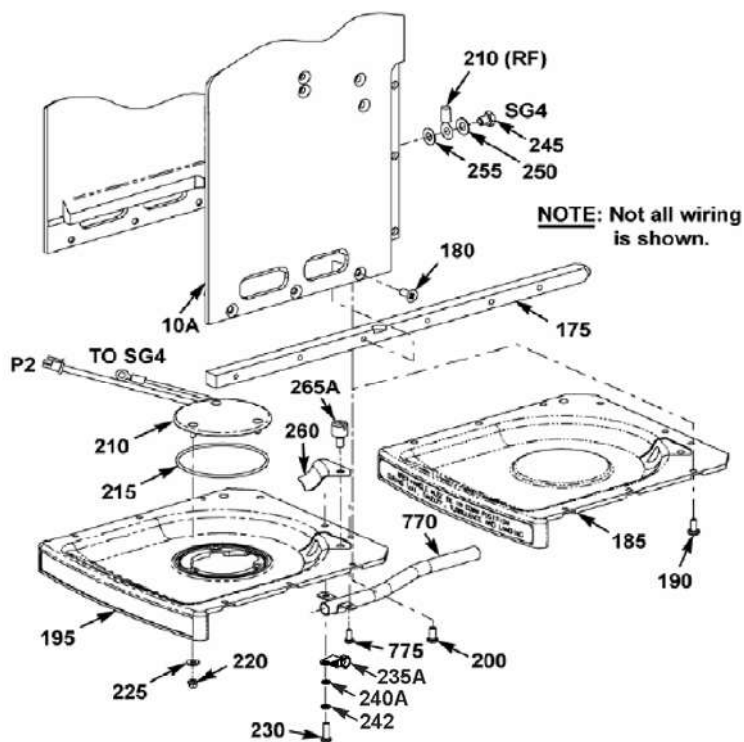


Figure 3006
Removal of Platen Assy

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- (f) Remove wiring and grommets (70) from holes in power module assy (55).
- (g) Remove five screws (80) and six shoulder washers (85) from the controller board assy (75) and the power module assy (55) (refer to Figure 3004).

NOTE: The six shoulder washers (85) are on six bosses in the power module assy (55).

- (h) Remove two screws (110) and two flat washers (105) from the power module assy (55) and the relay retainer (100).

CAUTION: DO NOT TOUCH THE RELAY OR PRY THE CONTROLLER BOARD (75) FROM THE HOUSING OF THE POWER MODULE ASSY (10, IPL FIGURE 4).

- (j) Carefully remove the relay and controller board assy (75), and relay retainer (100) from the power module assy (55). Isolate the controller board assy (75) from electrostatic discharge and exposure to water. Refer to DISASSEMBLY, Paragraph 2.E for disassembly of the controller board assy (75).

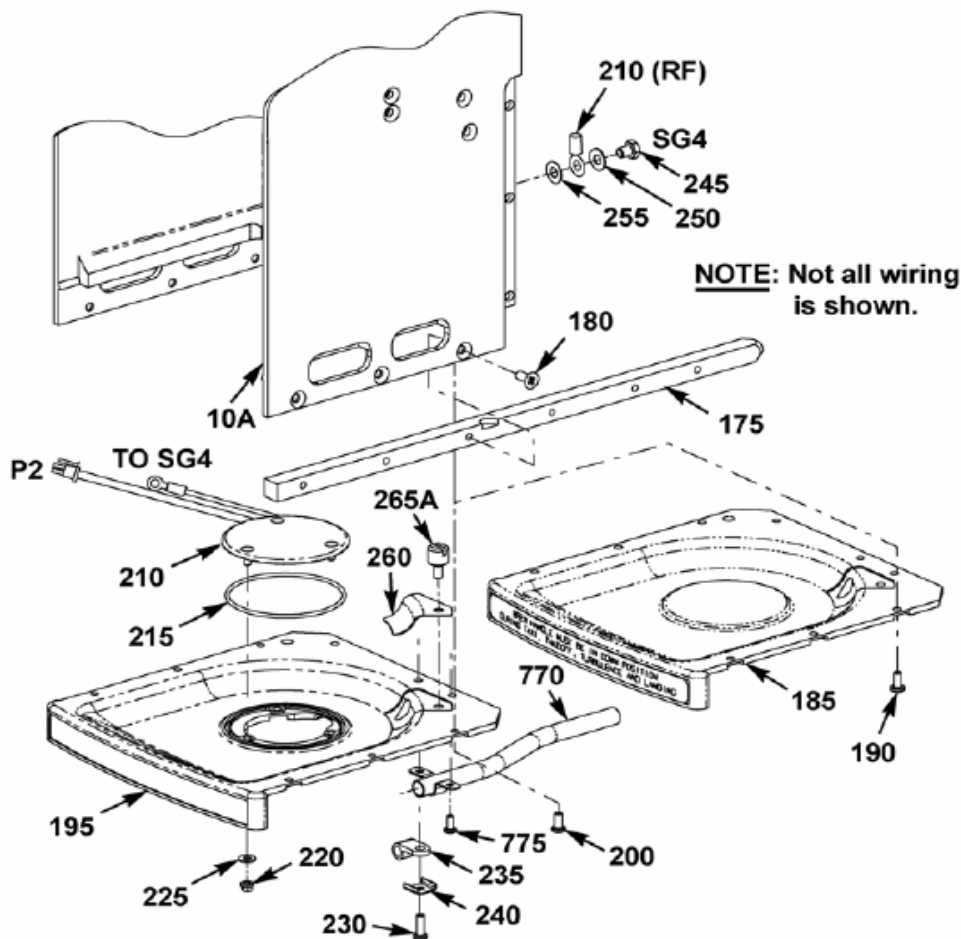


Figure 3006
Removal of Platen Assy

- (d) Remove screws (190) and platen assy (185, 195) from the chassis assy (10 or 10A).
- (e) Remove screw (230), lock washer (242), washer (240), and wire clamp (235) from wires for the platen heater (210) and the platen assy (195), where this option is installed.
- (f) Remove the drain screen fastener (265A) and drain screen (260) from the platen assy (185 or 195).

NOTE: If necessary, remove the platen inlay (205) from the platen assembly (185 or 195). To ensure a good bond of the platen inlay (205), make sure the platen assembly (185 or 195) is clean, free of debris and old adhesive.

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- (5) Remove the platen assy (185 or 195) from the chassis assy (10 or 10A) (refer to Figure 3006).

Table 3002
Platen Assembly Part Number Reference

UNIT PN	PLATEN ASSEMBLY PN
4510-X1XX-XX*	3520-0561-01 (no warmer)
4510-X2XX-XX*	3520-0561-01 (no warmer)
4510-X3XX-XX*	3520-0561-01 (no warmer)
4510-X4XX-XX*	3520-0561-03 (used with warmer)
4510-X5XX-XX*	3520-0561-03 (used with warmer)
4510-X6XX-XX*	3520-0561-03 (used with warmer)
4510-X7XX-XX*	3520-0561-01 (no warmer)
4510-X8XX-XX*	3520-0561-03 (used with warmer)

*X represents other numbers or letters in the Unit Part Number

- (a) Remove lock nut (220), flat washer (225), and o-ring (215) from platen heater (210) and platen assy (195), where this option is installed.
- (b) Remove the platen heater (210) plug from its connection on the controller board (75). The power module assy (55) may need to be opened to get access to the controller board (75).

NOTE: Remove the guide rails (175) and screws (180) from the chassis assy (10 or 10A) and the support connector(115) to get access to the screws that hold the platen assy (185 or 195) on the chassis assy (10 or 10A).

- (c) Remove two screws (230 and 775, or just 775) and the drain tube (770) from the platen assy (185, 195), where this option is installed.

Table 3003
Platen Drain Part Number Reference

UNIT PN	PLATEN DRAIN
4510-1XXX-XX*	Sump drain, no drain tube required
4510-2XXX-XX*	Rear drain, drain tube (770) required

*X represents other numbers or letters in the Unit Part Number

- (d) Remove screws (190) and platen assy (185, 195) from the chassis assy (10 or 10A).
- (e) Remove screw (230), washer (240), and wire clamp (235) from wires for the platen heater (210) and the platen assy (195), where this option is installed.
- (f) Remove the drain screen fastener (265A) and drain screen (260) from the platen assy (185 or 195).

NOTE: If necessary, remove the platen inlay (205) from the platen assembly (185 or 195). To ensure a good bond of the platen inlay (205), make sure the platen assembly (185 or 195) is clean, free of debris and old adhesive.

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- C. Remove screw (775) and drain tube (770) from platen assy (185 or 195) and chassis assy (10 or 10A). There may be adhesive sealant on the drain tube where it meets the platen assy (185 or 195).

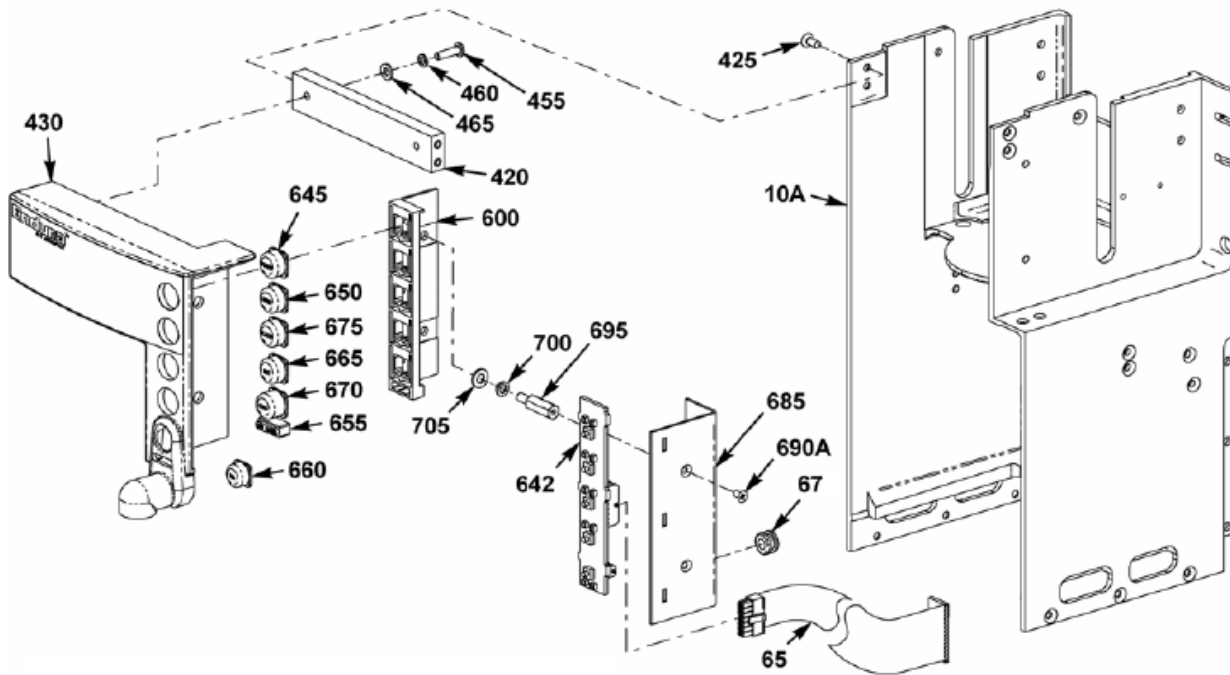


Figure 3007
Removal of Front Panel Assy

- (1) Remove the front panel assy (430, 435, 440, 445, or 450), brewhead assy (300), and brew handle (325) from the chassis assy (10 or 10A) (refer to Figure 3007).
 - (a) Remove two screws (690 or 690A) from the switch cover (685).
- CAUTION:** WHEN TOUCHING THE SWITCH ASSY, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.
- (b) Carefully remove switch assy (642) tabs from slots in the button frame (600) and the switch cover (685).
 - (c) Remove the wire harness assy (65) plug and switch assy (585) plug from the switch assy (642). Isolate the switch assy (642) from electrostatic discharge and exposure to water.

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- (d) Remove two standoffs (695), two lock washers (700), and two flat washers (705) from the button frame (600), front panel assy (430, 435, 440, 445, or 450), and chassis assy (10 or 10A).
- (e) Remove the harness assy (65), wires from switch assy (585), and grommet (680) from the switch cover (685).
- (f) Remove the grommet (680) from the harness assy (65) and wires from the switch assy (585).

NOTE: Before 2006/04/25, the switch cover (685) had only one wiring grommet slot.

- (g) Remove harness assy (65) and grommet (680) from slot in chassis assy (10 or 10A) near the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A).
- (h) Remove two screws (455), two lock washers (460), and two flat washers (465) from the front panel assy (430, 435, 440, 445, or 450) and chassis brace assy (420).
- (j) Remove the front panel assy (430, 435, 440, 445, or 450) from the brew handle (325), and faucet assy (270) or swivel faucet assy (275).
- (k) Remove the buttons (645, 650, 660, 665, 670, or 675) and indicator (655), where options are installed, from the button frame (600), only if necessary for cleaning or replacement.

NOTE: The buttons are press fit into the button frame and should only be removed if they need to be replaced or cleaned.

NOTE: The switch cover (685) must be removed to get access to screws (425) that attach the chassis brace assy (420) and screws (280A) that attach the faucet assy (270) or the swivel faucet assy (275) to the chassis assy (10 or 10A).

- (l) Remove four screws (425) from the chassis assy (10 or 10A) and chassis brace assy (420).

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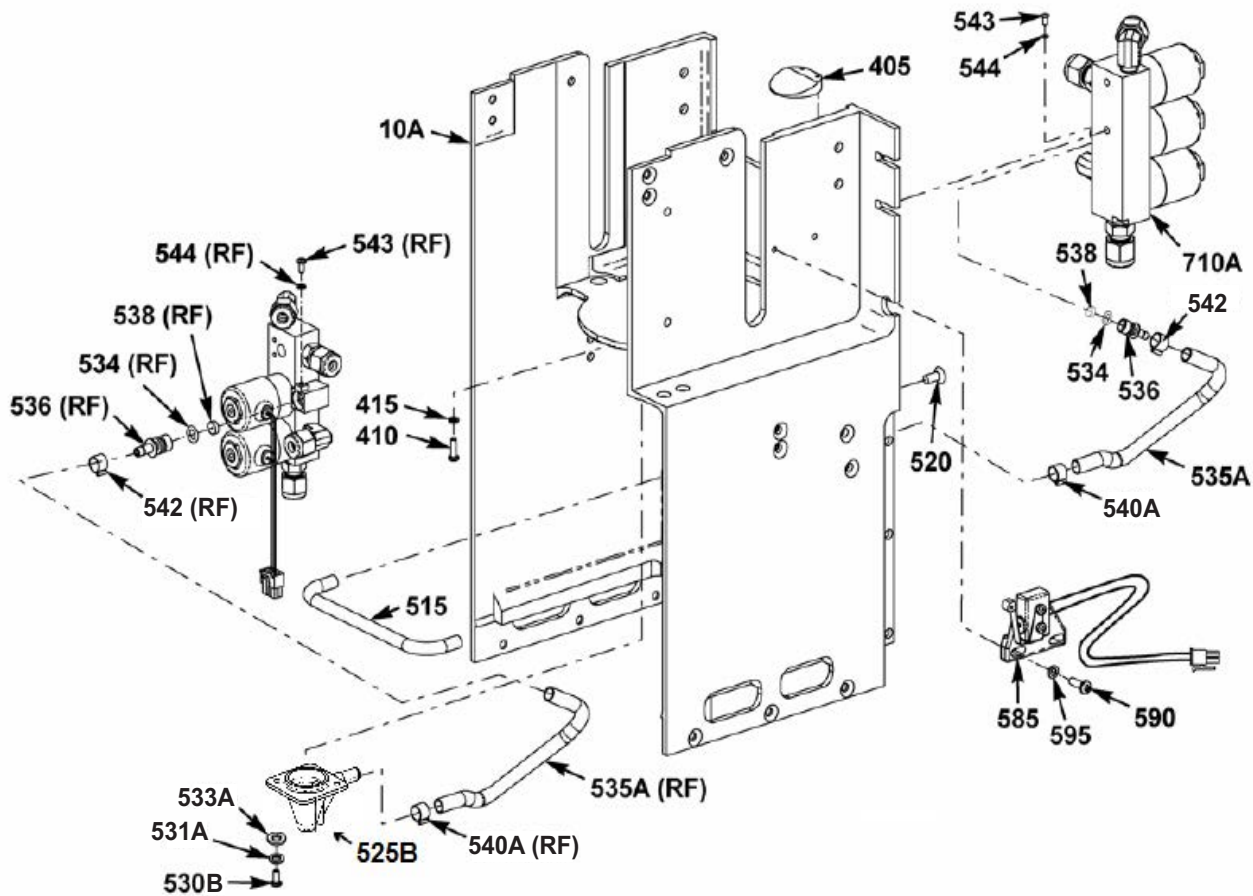


Figure 3008
 Removal of Switch Assy and Other Assy

- (2) Remove the switch assy (585) from the chassis assy (10 or 10A).
 - (a) The switch cover (685) may need to be removed to get access to the switch assy (642) connector for the switch assy (585) plug.

CAUTION: WHEN TOUCHING THE SWITCH ASSY (585), USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ESD.

 - (b) Disconnect the switch assy (585) plug from the switch assy (642).
 - (c) Remove two screws (590), two lock washers (595), and the switch assy (585) from the side of the chassis assy (10 or 10A).
- (3) Remove two screws (520) and handle (515) from the chassis assy (10 or 10A) (refer to Figure 3008).
- (4) Remove the tea funnel (525B) from the chassis assembly (10 or 10A) (refer to Figure 3008).

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- (a) Remove two screws (530B), two lock washers (531A), and two flat washers (533A) from the tea funnel (525B) and chassis assembly (10 or 10A).
- (b) Remove tube clamp (540A) and tea tube (535A) from the tea funnel (525B).
- (c) Remove tube clamp (542) and tube (535) or tea tube (535A) from the manifold assy (710, 710A, 725, or 725A).

NOTE: Removal of tube clamp (540A or 542) requires destruction of clamp that is usually installed with a crimping tool.

- (5) Remove two screws (410), two lock washers (415), and cup retainer (405) from the chassis assy (10 or 10A) (refer to Figure 3008).

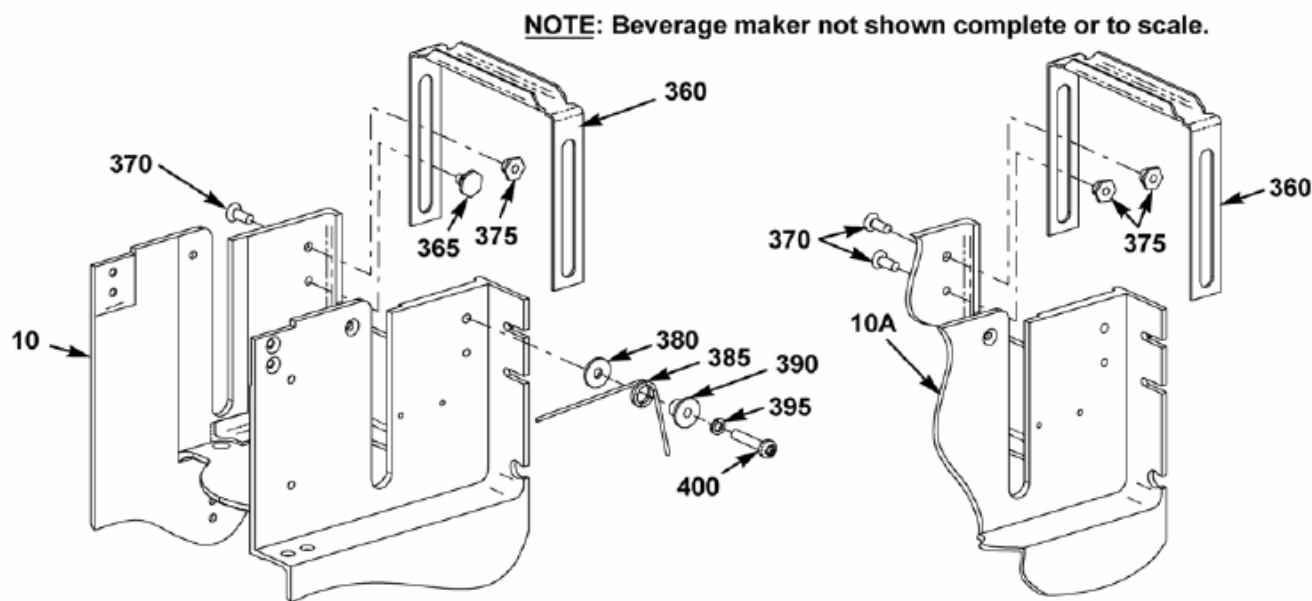


Figure 3009
Removal of Carrying Handle

- (6) Remove the carrying handle (360) from the chassis assy (10 or 10A) (refer to Figure 3009).
 - (a) Remove the tension spring (385) from slot in right side guide of brewhead assy (300).
 - (b) Remove screw (400), lock washer (395), spring retainer (390), tension spring (385), metallic washer (380), and handle bushing (375) from right side of the chassis assy (10 or 10A).
 - (c) Remove screw (370) and handle bushing (375) from left side of chassis assy (10 or 10A).
 - (d) Remove two guide screws (365) and carrying handle (360) from the chassis assy (10 or 10A).

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- (e) On units manufactured after 2006/01/20, remove two screws (370) and two handle bushings (375) from the left side of the chassis assy (10A) holding the carrying handle (360).
- (f) On units manufactured after 2006/01/20, remove a screw (370) and a handle bushing (375) from the right side of the chassis assy (10A) holding the carrying handle (360).

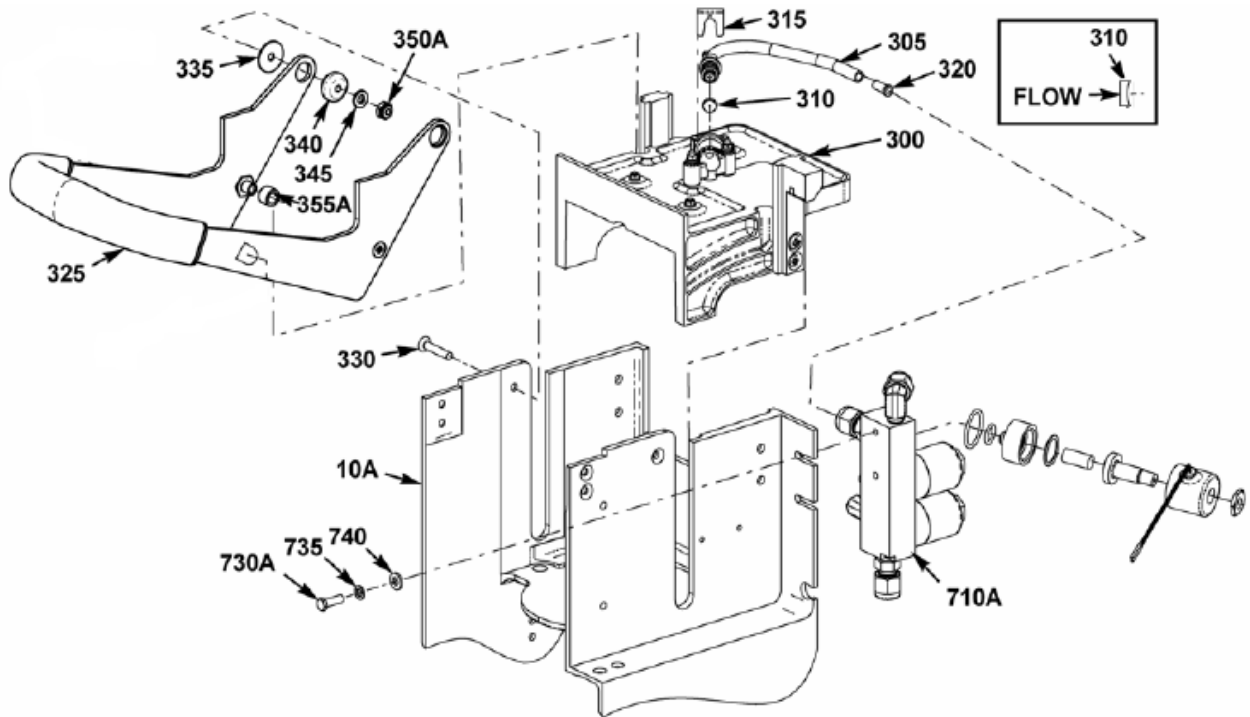


Figure 3010
Removal of Brewhead Assy

- (7) Remove brew handle (325) and brewhead assy (300) from the chassis assy (10 or 10A) and manifold assy (710, 715, 720, or 725) (refer to Figure 3010).

NOTE: Remove the front panel assy (430, 435, 440, 445, or 450) and chassis brace assy (425) from the chassis assy (10 or 10A) (refer to DISASSEMBLY, Paragraph 2.B.(6)) before removing the whole brewhead assy (300) and brew handle (325) from the chassis assy (10 or 10A).

- (a) Remove two screws (330), two handle washers (335), brew handle (325), two threaded bushings (340), two flat washers (345) and two lock nuts (350A) from the chassis assy (10 or 10A).
- (b) Remove the clip (315) and flow washer (310) from the brewhead assy (300) and brew tube assy (305).
- (c) Remove tube fittings from the brew tube assy (305) and the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A), then remove the insert (320) from the tube.

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- (d) Remove the brewhead assy (300) and brew handle (325) from the chassis assy (10 or 10A).
- (e) Remove the brew handle (325) from the brewhead assy (300) then remove two bushings (355 or 355A) from the brew handle.
- (f) Refer to DISASSEMBLY, Paragraph 2.J for disassembly of brewhead assy (300).
- (8) Remove the faucet assy (270) or swivel faucet assy (275) from the chassis assy (10 or 10A) (refer to Figure 3011).

Table 3004
 Faucet or Swivel Faucet Assy PN Reference

UNIT PN	FAUCET OR SWIVEL FAUCET ASSEMBLY PN	DESCRIPTION
4510-XXXF-XX*	3520-0525-01	Standard Faucet Assy
4510-XXXG-XX*	3520-0559-01	Swivel Faucet Assy

*X represents other numbers or letters in the Unit Part Number

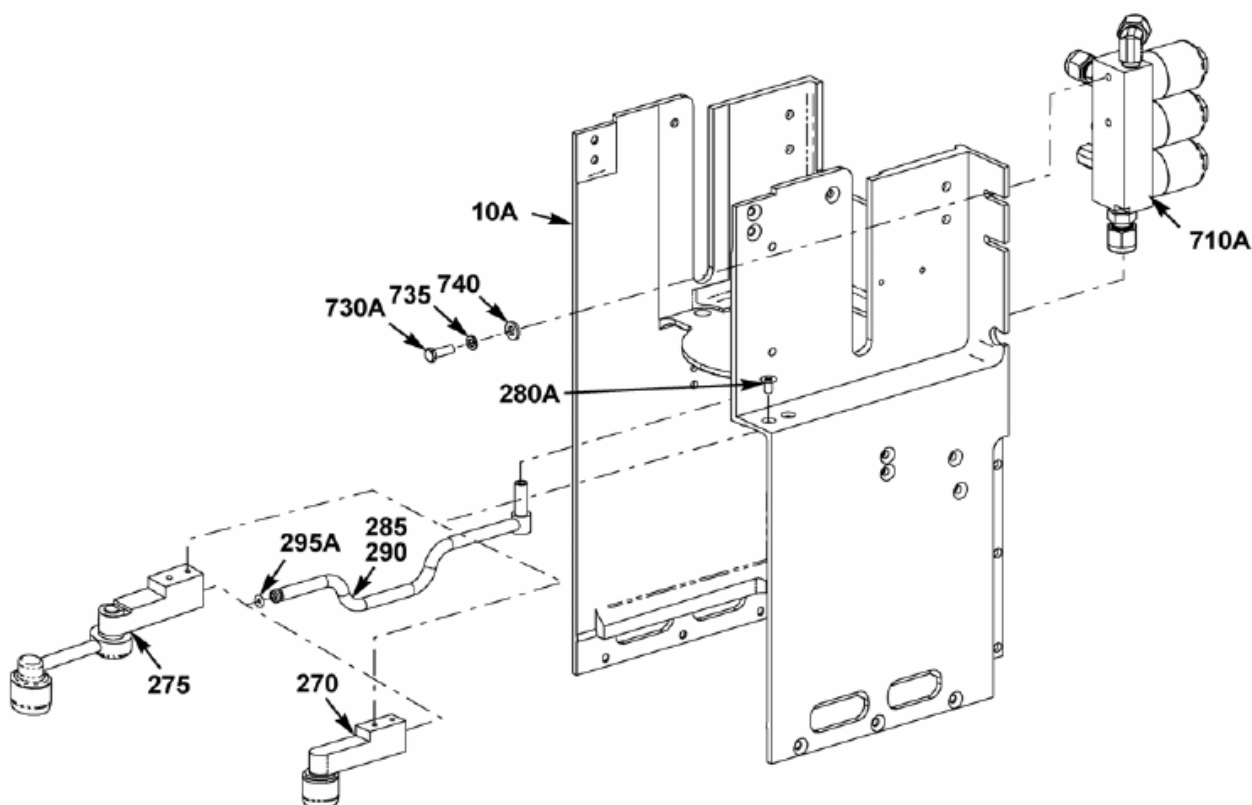


Figure 3011
 Removal of Faucet Assy or Swivel Faucet Assy

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NOTE: The switch cover (685) must be removed to get access to screws (280A) that attach faucet assy (270) or swivel faucet assy (275) to the chassis assy (10 or 10A).

- (a) Remove two screws (280 or 280A) from the faucet assy (270) or swivel faucet assy (275) and the chassis assy (10 or 10A).
- (b) Remove the faucet assy (270) or swivel faucet assy (275) from the faucet tube assy (285 or 290).
- (c) Remove o-ring (295 or 295A) from faucet tube assy (285 or 290).
- (d) Remove the faucet tube assy (285 or 290) and fittings from the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A). Refer to DISASSEMBLY, Paragraph 2.G for disassembly of the faucet assy (270) or DISASSEMBLY, Paragraph 2.H for disassembly of the swivel faucet assy (275).

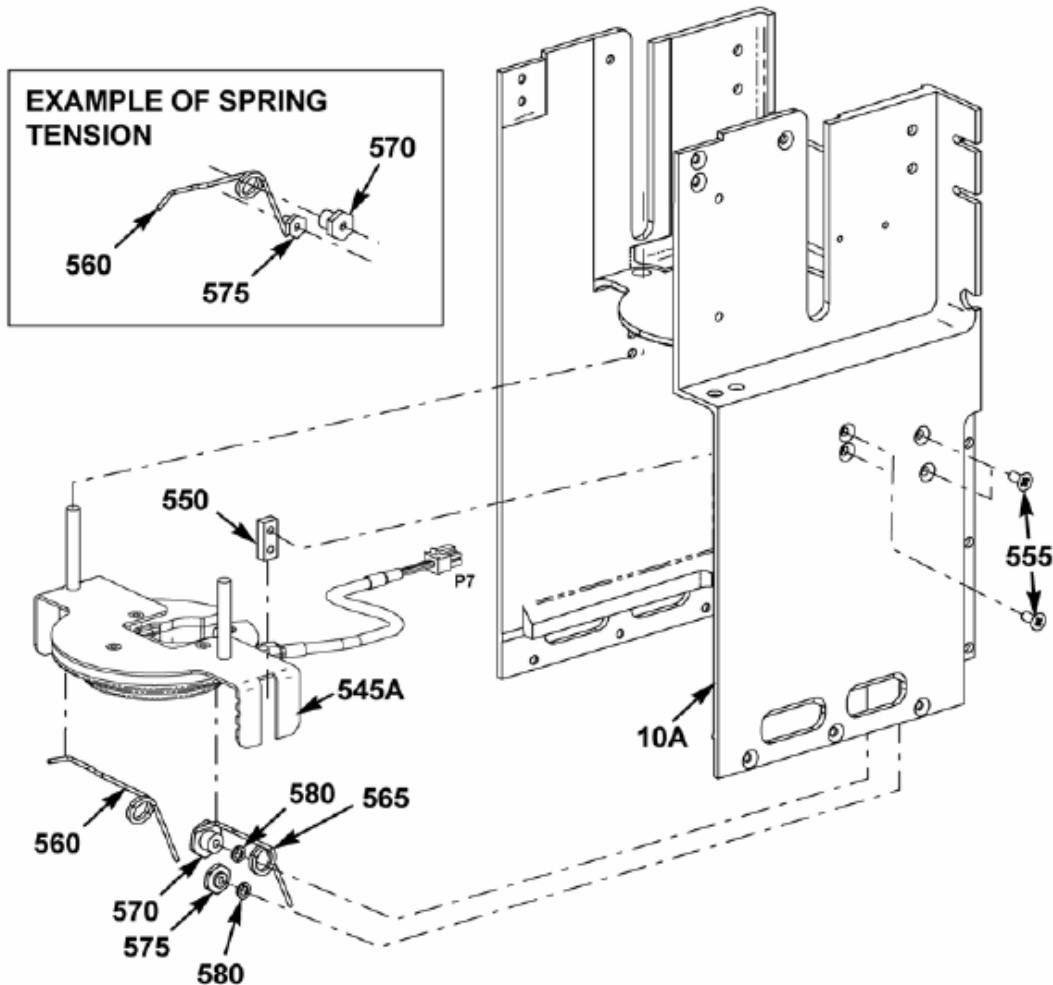


Figure 3012
Removal of Server Retainer Assy

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- (9) Remove the server retainer assy (545, 545A) from the chassis assy (10 or 10A) (refer to Figure 3012).

NOTE: The LLS potted assy (15, IPL Figure 10) can be removed without removal of the whole server retainer assy (545A) from the chassis assy (10 or 10A). Refer to DISASSEMBLY, Paragraph 2.K for disassembly of the server retainer assy (545A).

NOTE: The server retainer assy (545) is replaced with the new server retainer assy (545A) per Service Bulletin 4510-25-03.

- (a) Open the power module assy (55) to get access to the controller board assy (75) then remove the electrical connector of the server retainer assy (545 or 545A) from the controller board assy (75).

CAUTION: THE TWO SERVER RETAINER SPRINGS ARE UNDER TENSION SO THAT THEY HOLD UP THE SERVER RETAINER ASSY CORRECTLY. THEIR BOTTOM LEGS ARE HELD BY THE SPRING GUIDE BUSHINGS.

- (b) Remove two spring guide bushings (575) (holding the bottom leg of the server retainer springs (560, 565)), two lock washers (580), and two screws (555) from the chassis assy (10 or 10A).
- (c) Remove two screws (555), two lock washers (580), two server retainer springs (560, 565), and two spring bushings (570) from the chassis assy (10 or 10A).
- (d) Remove four screws (555) and two LLS guide assy (550) from the chassis assy (10 or 10A).
- (e) Remove the server retainer assy (545 or 545A) from the chassis assy (10 or 10A). Refer to DISASSEMBLY, Paragraph 2.K for disassembly of the server retainer assy (545 or 545A).

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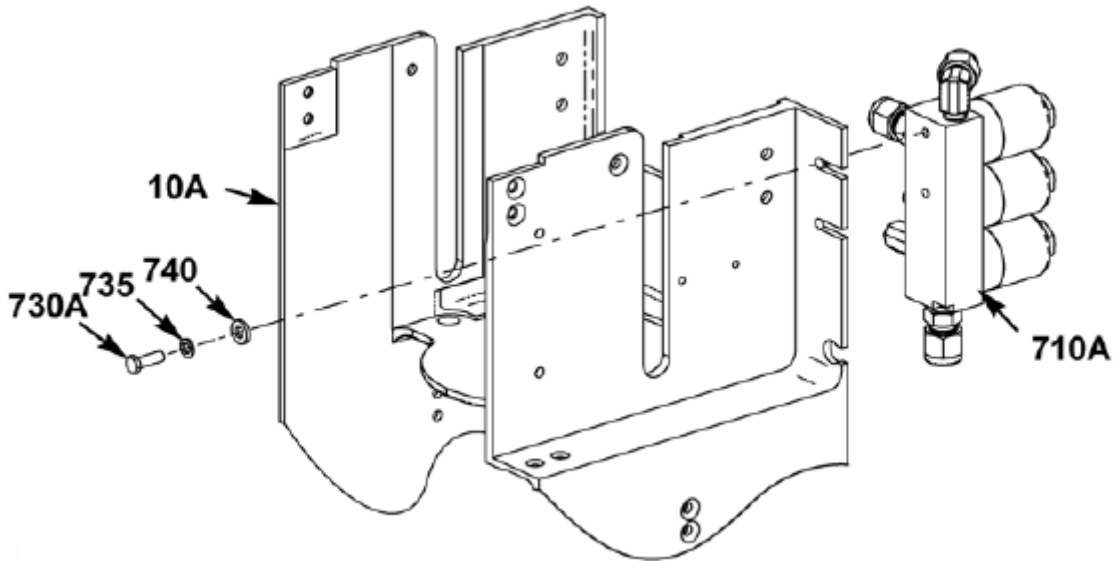


Figure 3013
Removal of Manifold Assy

- (10) Remove manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) from the chassis assy (10 or 10A) (refer to Figure 3013).
 - (a) Remove all plumbing connections from the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A).
 - (b) Remove wiring connections for the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) from the power module assy (55). The power module cover (45) may need to be removed to get access to the controller board (75).
 - (c) Remove two screws (730 or 730A), two lock washers (735), and two flat washers (740) from the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) and chassis assy (10 or 10A).
 - (d) Refer to DISASSEMBLY, Paragraph 2.L for disassembly of manifold assembly (710, 715, 720, or 725).
- (11) Remove the lockdown lever from the chassis assy (10 or 10A) (refer to Figure 3014).

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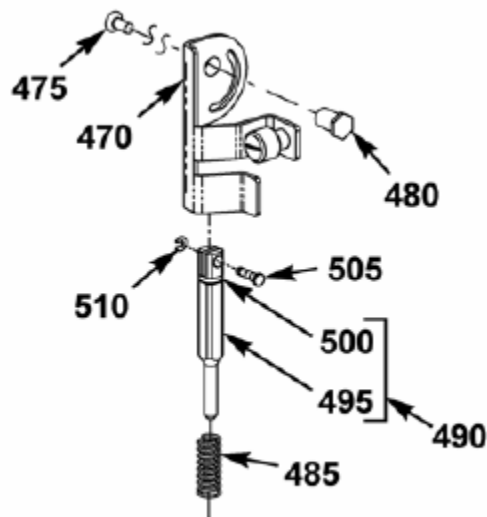


Figure 3014
Removal of the Lockdown Lever

- (a) Remove screw (475) and pin pivot assy (480) from the locating pin lever (470) and chassis assy (10 or 10A).
- (b) Remove pin locating lever (470), locating pin assy (490), and helical spring (485) from the chassis assy (10 or 10A).
 - 1 Remove o-ring (500) from machined pin (495).
- (c) Remove retaining ring (510) from pin (505) that holds locating pin assy (490) on locating pin lever (470).

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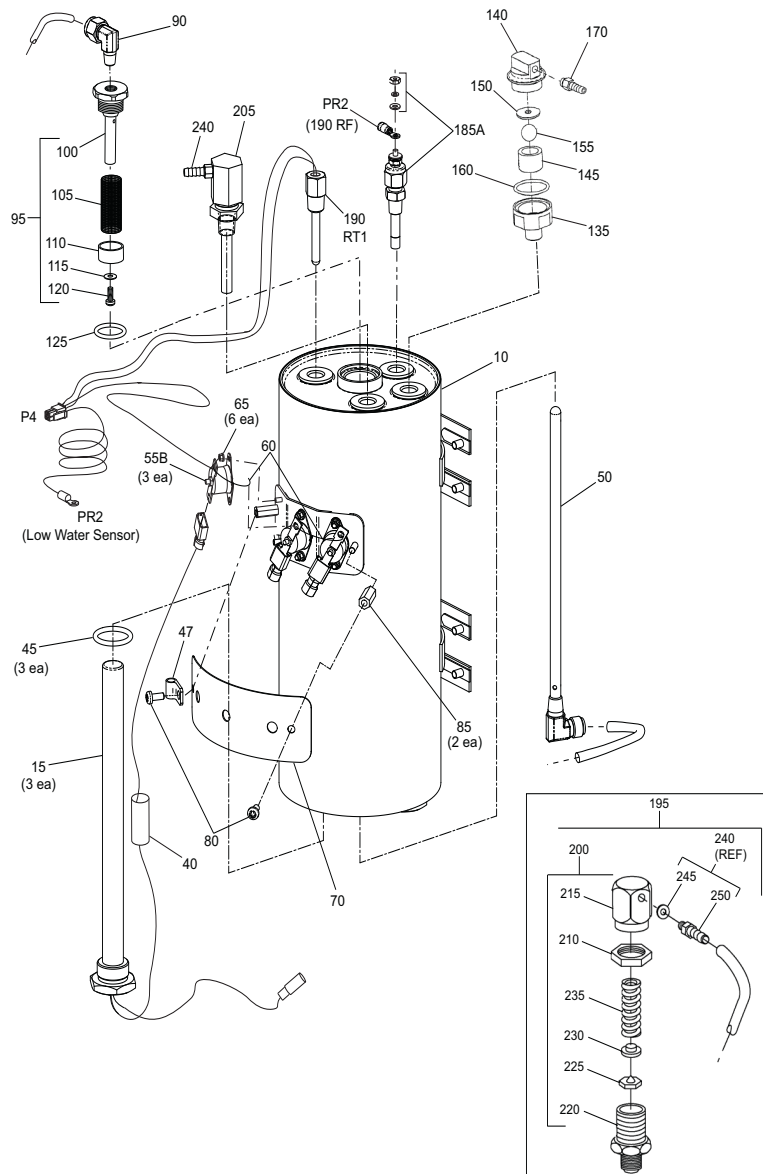


Figure 3015
Disassembly of Tank Assy

D. Tank Assy (refer to Figure 3015)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(3) for removal of tank assy (15, IPL Figure 2) from chassis assy (10 or 10A, IPL Figure 2).
- (2) Remove two screws (80), clamp (75) thermal switch cover (70), and two hex spacers (85) from welded tank assy (10).

CAUTION: WHEN HEATING SOLDER, TURN TANK SO HOT SOLDER FALLS AWAY FROM THE TANK AND THERMOSTATS.

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- (3) Heat solder and remove two wire busses (60) and an RTD assy (190) wire from thermostat (55) terminals.

CAUTION: CAREFULLY REMOVE HEATER ASSY WIRE TERMINAL FROM TERMINAL OF THE OVERTEMPERATURE THERMOSTAT SO THAT THE TERMINAL DOES NOT MOVE ON THE PLASTIC HOUSING OF THE THERMOSTAT. TERMINAL MOVEMENT CAN CRACK THE THERMOSTAT HOUSING AND RESULT IN FAILURE OF THE THERMOSTAT.

- (4) Remove three quick connect wiring terminals of heater assy (15) from terminals of overtemperature thermostats (55) on the welded tank assy (10).
- (5) Remove six lock nuts (65) and three thermostats (55A) from the welded tank assy (10).
- (6) Remove hex nut, lock washer, flat washer and RTD assy (190) wire terminal from low water sensor (185).
- (7) Remove low water sensor (185) from welded tank assy (10).
- (8) Remove tie downs from wiring for RTD assy (190).
- (9) Remove RTD assy (190) from welded tank assy (10).
- (10) Remove pressure relief valve assy (205) from welded tank assy (10).
- (11) Remove elbow fitting (90) from filter assy (95).
- (12) Remove filter assy (95) and o-ring (125) from welded tank assy (10).
- (a) Disassemble filter assy (95).
- 1 Remove screw (120), lock washer (115), retainer (110), filter screen (105) from the water jet (100).
- (13) Remove vent valve assy (130) from the welded tank assy (10) (refer to Figure 3016).
- (a) Unscrew vent valve body (140) from adapter (135).
- (b) Remove o-ring (160), ball (155), liner (145), and washer (150).
- (c) Remove barbed fitting (170) from the vent valve assy (130).
- 1 Disassemble the barbed fitting by removing the gasket (175) from the fitting (180).
- (14) Remove pressure relief valve assy (195) from welded tank assy (10).
- (15) Remove tube assy (50) from welded tank assy (10).
- NOTE:** Identify tank heater assy (15) and where they are installed in the bottom of the welded tank assy (10).
- (16) Remove three tank heater assy (15) and three o-rings (45) from welded tank assy (10).

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NOTE: Only if required for replacement, remove connector (25), connector pin (30) and terminal (20) from heater assy tank (35).

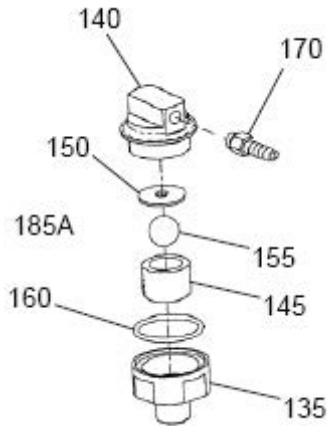


Figure 3016
Disassembly of Vent Valve Assembly

E. Power Module Assy (refer to Figure 3017)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(4) for removal of power module assy (55) from chassis assy (10 or 10A).
- (2) Remove SG1 and SG2 ground stack ups from the power module housing.
 - (a) Remove two lock nuts (45), two flat washers (55), wire terminals for harness assy (15), two flat washers (50), two flat washers (50), and two hex screws (60) from the power module housing assy (10).

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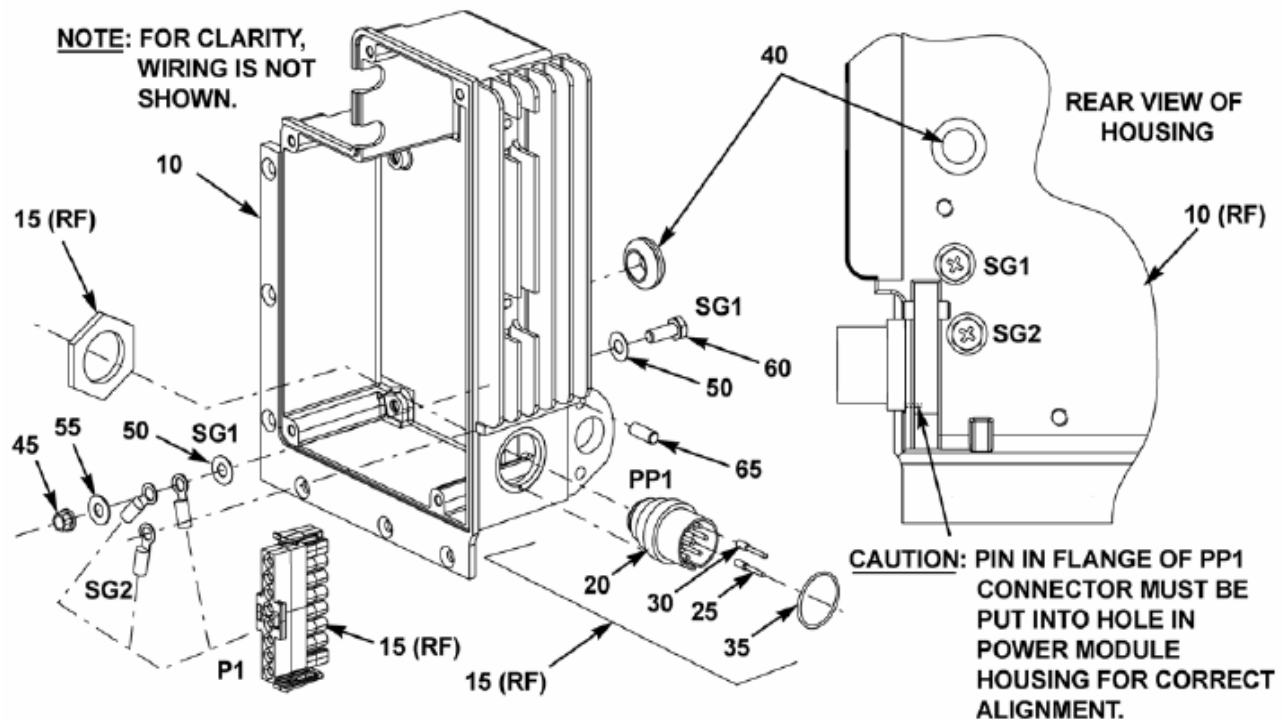


Figure 3017
 Disassembly of Power Module Assy

- (b) Remove pin (65) from power module housing assy (10).

NOTE: This pin helps align the support connector (115, IPL Figures 2, 2A, 2B, 2C) with the power module assy (55, IPL Figures 2, 2A, 2B, 2C).

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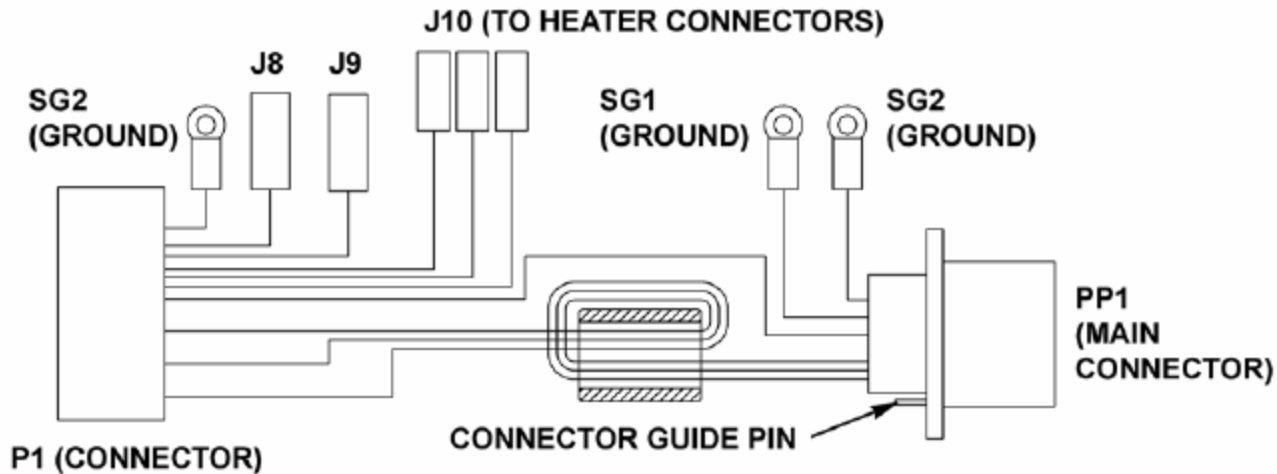


Figure 3018
Diagram of Harness Assy in Power Module Assy

CAUTION: THE P1 AND PP1 CONNECTORS MUST BE REMOVED FROM THE HARNESS ASSY TO REMOVE THE HARNESS ASSY THROUGH THE GROMMET AND HOLES IN THE POWER MODULE HOUSING ASSY (REFER TO FIGURE 3018).

(3) Remove the PP1 connector (20) from power module housing assy (10).

(a) Remove the hex nut (45) from the connector (20).

CAUTION: CAREFULLY REMOVE CONNECTOR FROM THE POWER MODULE HOUSING ASSY TO AVOID DAMAGE TO THE PP1 GUIDE PIN.

CAUTION: CAREFULLY REMOVE TERMINAL PINS (25 OR 30) FROM THE CONNECTOR TO FULLY REMOVE THE CONNECTOR FROM THE POWER MODULE HOUSING.

(b) Only if required for replacement, remove o-ring (35) and pins (25 or 30) from the connector (20).

(4) Remove harness assy (15) from the power module housing assy (10).

(5) Remove the grommet (40) from the back of the power module housing assy (10).

F. Controller Board Assy (refer to CMM 25-33-32)

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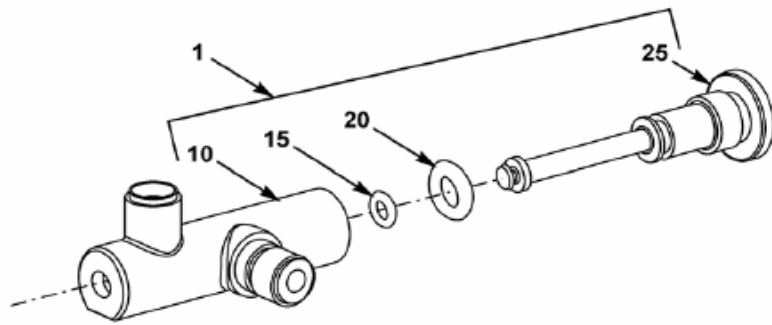


Figure 3019
Disassembly of Drain Valve Assy

G. Drain Valve Assy (refer to Figure 3019)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(2) (a) and (f) for removal of drain valve assy (1, and 140, IPL Figures 2, 2A, 2B, 2C) from support connector (115, IPL Figures 2, 2A, 2B, 2C).
- (2) Remove stem assy (25) from body (10).
 - (a) Remove upper o-ring (20) and lower o-ring (15) from stem assy (25).

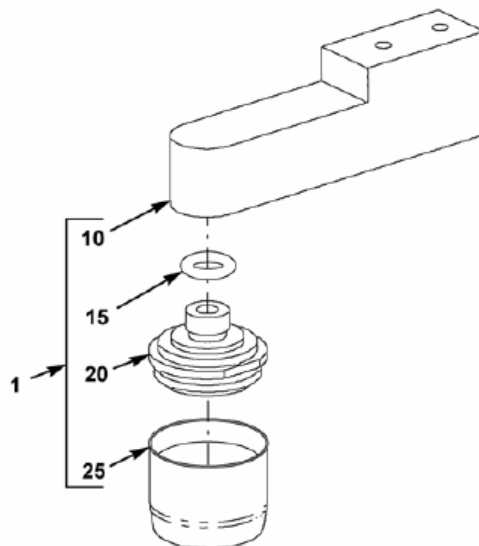


Figure 3020
Disassembly of Faucet Assy

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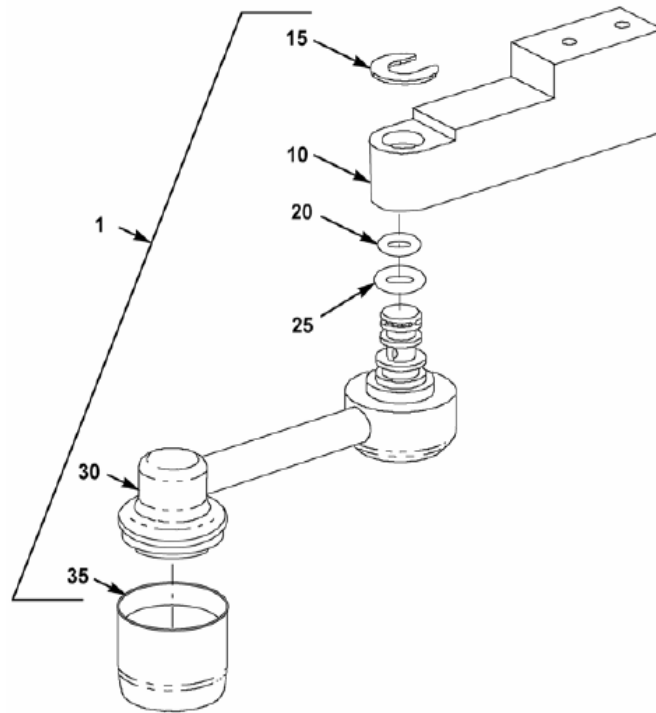


Figure 3021
Disassembly of Swivel Faucet Assy

H. Faucet Assy (refer to Figure 3020)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(13) for removal of faucet assy (270, IPL Figures 2, 2A, 2B, 2C) from chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C), if option is installed.
- (2) Remove aerator (25), aerator adapter (20) from faucet bar (10).
- (3) Remove o-ring (15) from aerator adapter (20).

I. Swivel Faucet Assy (refer to Figure 3021)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(13) for removal of swivel faucet assy (275, IPL Figures 2, 2A, 2B, 2C) from chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C), if option is installed.
- (2) Remove aerator (35) from welded faucet assy (30).
- (3) Remove retaining ring (15) from welded faucet assy (30).
- (4) Remove welded faucet assy (30) from swivel faucet bar (10).
- (5) Remove upper o-ring (20) and lower o-ring (25) from welded faucet assy (30).

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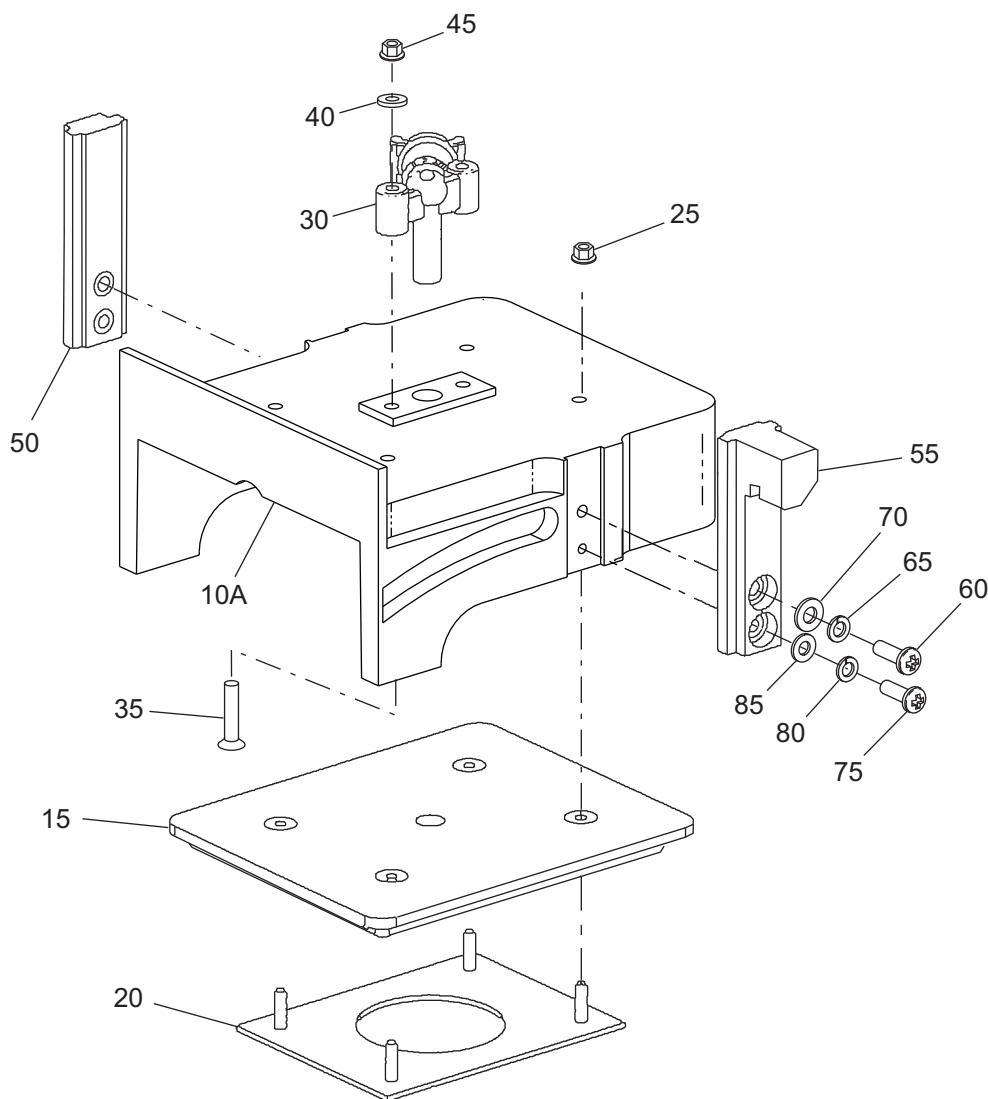


Figure 3022
 Disassembly of Brewhead Assy

J. Brewhead Assy (refer to Figure 3022)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(12) for removal of brewhead assy (300, IPL Figures 2, 2A, 2B, 2C) from chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C).

- (2) Remove four lock nuts (25), gasket retainer (20), and gasket (15) from the brew head (10).

NOTE: Coffee residue can cause parts to stick together in the brewhead assy.

- (3) Remove two screws (35), two lock nuts (45), two flat washers (40), and brew fitting (30) from the brew head (10).

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- (4) Remove four screws (60 and 75), four lock washers (65 and 80), four flat washers (70 and 85), and two guides (50 and 55) from the brew head.

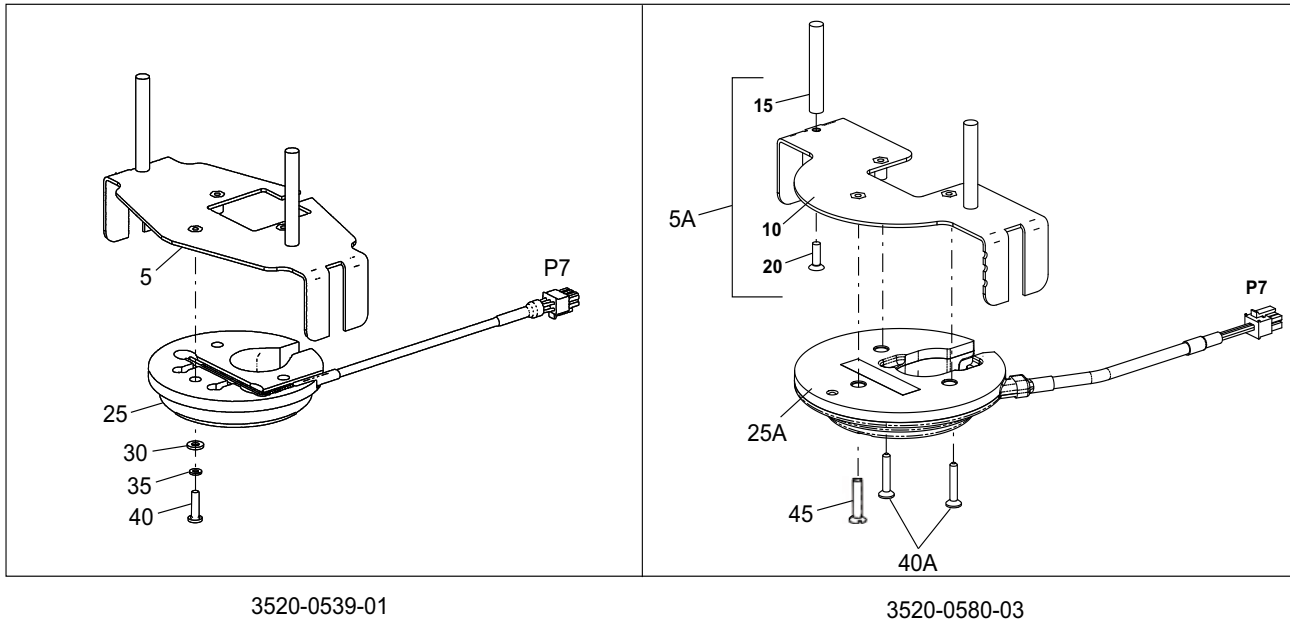


Figure 3023
Disassembly of Server Retainer Assemblies

K. Server Retainer Assy (refer to Figure 3023)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(14) for removal of server retainer assy (545 or 545A, IPL Figures 2, 2A, 2B, 2C) from chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C).

NOTE: The server retainer assy (545) is replaced with the new server retainer assy (545A) per Service Bulletin 4510-25-03.

- (2) Remove two screws (40B) and one screw (45) from the LLS retainer assy (25A) and the retainer assy (5A).
- (3) Only if required for replacement, remove two screws (20) from retainer rods (15) and cover (10).

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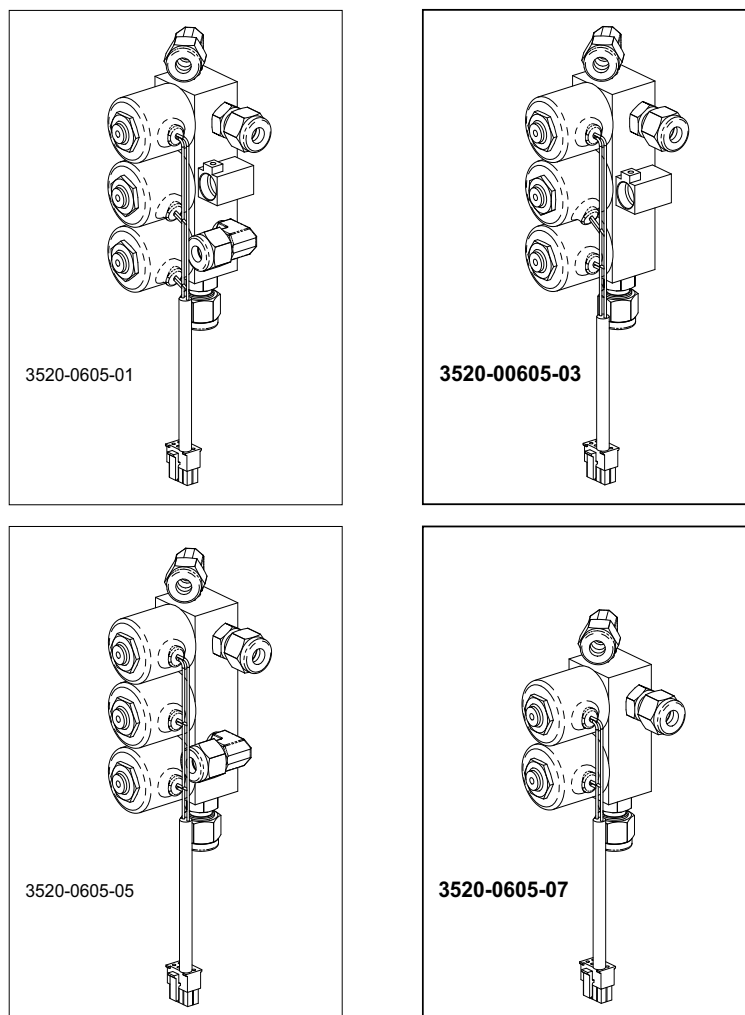


Figure 3024
Manifold Assy

L. Manifold Assy (refer to Figures 3024)

- (1) Refer to DISASSEMBLY, Paragraph 2.B.(15) for removal of manifold assy (710, 715, 720, 725, IPL Figures 2, 2A, 2B, 2C) from chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C).
- (2) Remove regulator fitting (15) from manifold body (5A, 5D, 5E, or 5H).
- (3) Remove straight fittings (20) from manifold body (5).
- (4) Remove elbow fittings (25A or 25B) from manifold body (5).

NOTE: After removing fittings from manifold body, clean thread seal tape from all threads.

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CAUTION: CAREFULLY REMOVE TERMINALS FROM THE P6 CONNECTOR, OR THE CONNECTOR MAY BE DAMAGED, RESULTING IN DAMAGE TO THE CONTROLLER BOARD.

- (5) Only if necessary for replacement, remove P6 connector (10) from wires for solenoid valve assy (30) using a Molex extractor tool part number 11-03-0043 or equivalent.

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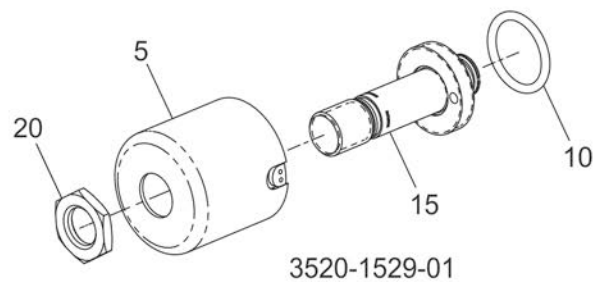
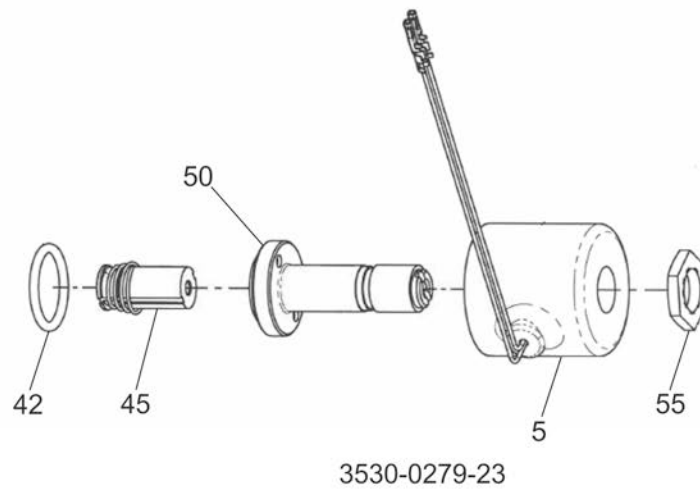
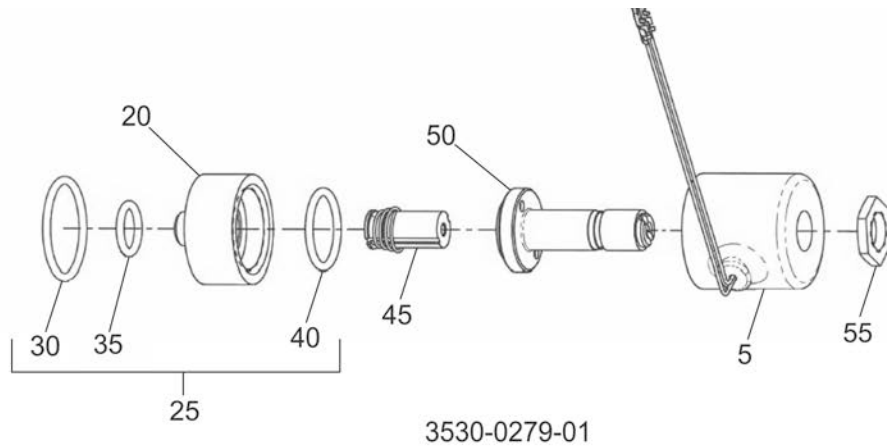


Figure 3025
Solenoid Valve Assembly

NOTE: Wiring for the coil assy needs to be removed from the connector (P6) before the solenoid assy can be fully disassembled.

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- (6) For PN 3530-0279-01, do the steps that follow to disassemble the solenoid valve assembly (refer to Figure 3025 and IPL Figure 12):
- (a) Remove two o-rings (30 and 35) from the valve body (20) or manifold body (refer to Figure 3024).
 - (b) Remove nut (55) from solenoid guide (50).
 - (c) Remove coil assembly (5) from solenoid guide (50).
 - (d) Remove solenoid guide (50) from valve body (20). Special tools are recommended for disassembly (refer to Table 9001 and Figure 9008).
 - (e) Remove plunger (45) from inside the solenoid guide (50).
 - (f) Remove o-ring (40) from the valve body (20).
 - (g) Only if necessary for repair, remove crimp terminals (15) from P6 connector using a Molex Extractor Tool part number 11-03-0043 or equivalent.
 - (h) Only if necessary for repair, remove crimp terminals (15) from the wiring to the solenoid coil (10).

NOTE: In the event that crimp terminals (15) are removed from wiring, the terminals cannot be reused.

- (7) For PN 3530-0279-23, do the steps that follow to disassemble the solenoid valve assembly (refer to Figure 3025 and IPL Figure 12):
- (a) Remove nut (55) from solenoid guide (50).
 - (b) Remove coil assembly (5) from solenoid guide (50).
 - (c) Remove solenoid guide (50) from solenoid manifold (5, IPL Figure 11). Special tools are recommended for disassembly (refer to Table 9001 and Figure 9009).
 - (d) Remove plunger (45) from inside the solenoid guide (50).
 - (e) Remove o-ring (42) from the solenoid guide (50).
 - (f) Only if necessary for repair, remove crimp terminals (15) from P6 connector using a Terminal Extractor Tool part number 11-03-0043 or equivalent.

NOTE: In the event that crimp terminals (15) are removed from P6 connector, the connector cannot be reused.

- (g) Only if necessary for repair, remove crimp terminals (15) from the wiring to the solenoid coil (10).

NOTE: In the event that crimp terminals (15) are removed from wiring, the terminals cannot be reused.

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- (8) For PN 3520-1529-01, do the steps that follow to disassemble the 2-way normally closed valve assembly (refer to Figure 3025, and IPL Figure 12A):
- (a) Remove nut (20) from plunger guide assembly (15).
 - (b) Remove coil assembly (5) from plunger guide assembly (15).
 - (c) Remove plunger guide assembly (15) from solenoid manifold (5, IPL Figure 11). Special tools are recommended for disassembly (refer to Table 9001 and Figure 9008).
 - (d) Remove O-ring (10) from the plunger guide assembly (15).
 - (e) Remove female crimp terminals (32, IPL Figure 11) from P6 connector (10, IPL Figure 11) only if replacement is necessary using a Terminal Extractor Tool part number 11-03-0043 or equivalent.

NOTE: If female crimp terminals (32, IPL Figure 11) are removed from P6 connector (10, IPL Figure 11), the connector cannot be reused and must be replaced.

- (f) Remove female crimp terminals (32, IPL Figure 11) from the wiring to the coil assembly (5) only if replacement is necessary.

NOTE: If female crimp terminals (32, IPL Figure 11) are removed from wiring, the terminals cannot be reused.

(9) Deleted.

(10) Deleted.

(11) Deleted.

(12) Deleted.

(13) Deleted.

NOTE: Deleted.

M. If CLEANING or CHECK will not be performed immediately, store all parts in a dry, dust-free container.

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CLEANING

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING CLEANING.

WARNING: WEAR EYE PROTECTION AND WATERPROOF PROTECTIVE GLOVES WHILE WORKING WITH CHEMICALS.

CAUTION: DO NOT LET MOISTURE TOUCH ELECTRICAL PARTS (FOR EXAMPLE, SOLENOID COILS OR PLATEN HEATER), WIRING, OR CONTROLLER BOARD ASSEMBLY.

CAUTION: DO NOT USE SOLUTIONS OR CONCENTRATIONS OTHER THAN THOSE RECOMMENDED BY B/E AEROSPACE.

CAUTION: USE ONLY THE RECOMMENDED CLEANER CONCENTRATION IN NORMAL ROOM TEMPERATURE SOLUTION WITH CLEAN WATER.

A. Most interior and exterior surfaces of the Beverage Maker can be cleaned with cloth towels that have been moistened with clean water.

NOTE: To avoid scratching surfaces, do not use paper towels.

B. Cleaning materials for the Beverage Maker are listed in Table 4001.

NOTE: Equivalent substitutes may be used for listed items.

Table 4001
Cleaning Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
A food-grade aqueous acidic cleaner that removes calcium, lime, and rust deposits (scale) without damage to unit form, fit, or function	None	Commercially Available
Brush, Wire, Soft-bristled	None	Commercially Available
Clean Water	None	Commercially Available
Cleaner, Mild Liquid Detergent (Dish Soap)	None	Commercially Available
Cloth, Cleaning, Low-Lint, Type 1	A-A-59323	Commercially Available
Cotton Swabs	None	Commercially Available
Isopropyl Alcohol	TT-I-735	Commercially Available
Nitrogen Gas	A-A-59503	Commercially Available
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2. Procedure

A. The Beverage Maker assembly is part of a galley configuration subject to heat and pressure variations. Therefore, the following cleanliness requirements must be complied with:

- (1) The work area for cleaning, assembling, and testing must be clean and free of hydrocarbon residue, oil, or other petroleum base lubricants.
- (2) The pressurized gas or air supply used to dry parts after cleaning must be dry, clean, and free of hydrocarbon contamination and excessive moisture.
- (3) New or used parts must be cleaned in accordance with Paragraphs 2.B. through 2.F. of this section before assembling into the Beverage Maker assemblies.

B. Clean the parts (except those parts mentioned in the preceding CAUTION) in a warm mild liquid detergent solution bath.

WARNING: WEAR EYE PROTECTION AND WATER PROOF PROTECTIVE GLOVES WHILE WORKING WITH CHEMICALS.

- (1) Clean electrical parts with isopropyl alcohol (refer to Table 4001).
- (2) Prepare a warm mild liquid detergent solution with clean water (refer to Table 4001) not hotter than 140° F (60° C). A mixture of approximately 13 ounces of cleaner concentrate in one gallon of clean water (refer to Table 4001), or 100 ml of cleaner concentrate in one liter of clean water (refer to Table 4001).
- (3) Fully submerge and agitate each part (except those parts mentioned in the preceding CAUTION) in the mild liquid detergent solution. Let the cleaner flush all internal and external surfaces. Scrub the chassis assembly and the top of the brew head assembly with a soft brush.
- (4) Mineral deposits may be removed by soaking affected parts in a solution of calcium, lime, and rust remover (refer the Table 4001). Follow manufacturer's precautions for any cleaner. Rinse with clean water (refer to Table 4001).
- (5) Thoroughly rinse the parts in de-ionized water at room temperature. Allow the parts to dry thoroughly at room temperature.

CAUTION: DO NOT ALLOW DAMPNES OR MOISTURE TO ENTER THE POWER MODULE ASSEMBLY, SOLENOID COILS, SWITCH ASSEMBLIES, OR ANY ELECTRICAL CONNECTIONS.

C. Clean the surfaces of the remaining parts that should not be submerged (refer to the preceding CAUTION and Table 4001).

- (1) Clean stained surfaces of the Beverage Maker with a soft, cotton cloth or swab dipped in the prepared mild liquid detergent solution.
- (2) Brew Cup
 - (a) Clean all surfaces inside brew cup with cloth moistened with water and the prepared mild liquid detergent solution.

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- (b) Wipe off any film with a clean, damp cloth.
 - (c) Lift flap inside brew cup and clean all surfaces under flap with cloth moistened with water and the prepared mild liquid detergent solution.
 - (3) Clean tube assemblies by running a flexible wire brush through the interior. Rinse with clear water.
 - (4) When the tank assembly is disassembled, clean thread seal tape from fittings.
 - (5) Clean server retainer assembly with a clean damp cloth.
 - (6) Clean the server and brew cup cavities of the chassis assembly with a soft cotton cloth or swab dipped in the prepared mild liquid detergent solution.
 - (7) Rinse the parts with a soft cotton cloth dipped in de-ionized water at room temperature.
- D. If coffee residue is on a part, remove the residue using a small cotton swab and isopropyl alcohol (refer to Table 4001). Rinse the parts with a soft cotton cloth dipped in room temperature de-ionized water.
- E. For flushing and de-calcification of the tank assembly, refer to UG-4XXX-25-002.
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 - (2) Deleted.
 - (3) Deleted.
 - (4) Deleted.
 - (5) Deleted.
 - (6) Deleted.
 - (7) Deleted.
- NOTE: Deleted.
- F. Allow the parts to dry thoroughly at room temperature, or dry each part using nitrogen gas (refer to Table 4001) or commercially bottled, dry, clean, compressed air.
- G. If INSPECTION/CHECK will not be performed immediately, store all parts in a dry, dust-free container.

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INSPECTION/CHECK

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING CHECK.

- A. Preliminary cleaning before initial inspection will help locate visible defects.
- B. Visually inspect working parts under strong light and magnification for cracks. Inspect structure for distortion, cracking, and excessive wear that could impair the function of equipment.

NOTE: The term, excessive wear, shall be defined as obvious deformation or deterioration of a part that might render the Beverage Maker inoperative. If doubt exists regarding the serviceability of a part, refer to the TESTING AND FAULT ISOLATION, REPAIR, and FITS AND CLEARANCES Sections of this book, as well as the guidelines listed below.

NOTE: Prepare a clean space and working surface before beginning any INSPECTION/CHECK Section operation.

NOTE: If repair of individual parts are necessary, refer to the REPAIR Section.

2. Procedures

- A. Inspect the Beverage Maker for signs of damage, breaks, overheating, cracks, or deformation. Use a strong light and some means of magnification.

- (1) Visually inspect all disassembled threaded parts for stripped or crossed threads.

- B. Electrical

- (1) Inspect electrical harnesses, wiring, and connection points for damage, deteriorated insulation, and loose wires.

- (2) Inspect wire routing and nylon wire tie downs to make sure no wiring is loose or exposed to chafing against other parts or heat sources.

- (3) Examine the Beverage Maker power connector PP1-3 for loose, bent, or broken pins, deformed shell, and evidence of arcing or flashover at pin insert insulation.

- (4) Examine wiring harnesses and terminals for cracked insulation and loose terminals.

- (5) Examine the controller board assembly for deformation, overheating, water intrusion, loose, bent or broken components and connections, and evidence of arcing or flashover.

- (6) Switches should not be broken or excessively worn. Make sure text on the switch buttons can be read and is complete. When pressed, switches should compress smoothly with a slight resistance until circuit connection is complete. The button light should come on when the button is pressed and go off when the button is pressed again.

- C. Plumbing

- (1) Examine tubing for leakage, flattening, or kinking, and cracked, split, loose, or worn tubing and fittings.

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- (2) Examine all plumbing for flattening, kinking, cracking, splitting, loose fittings, or ineffective sealing.
- (3) Examine tank assembly for leakage, overheating, deterioration, or cracks.
- (4) Examine solenoid valve assembly (Figure 5001) on manifold assembly for leakage, overheating, deterioration, flattening, cracking, splitting, loose fittings, or ineffective sealing. Special tools are recommended for disassembly (refer to Table 9001).

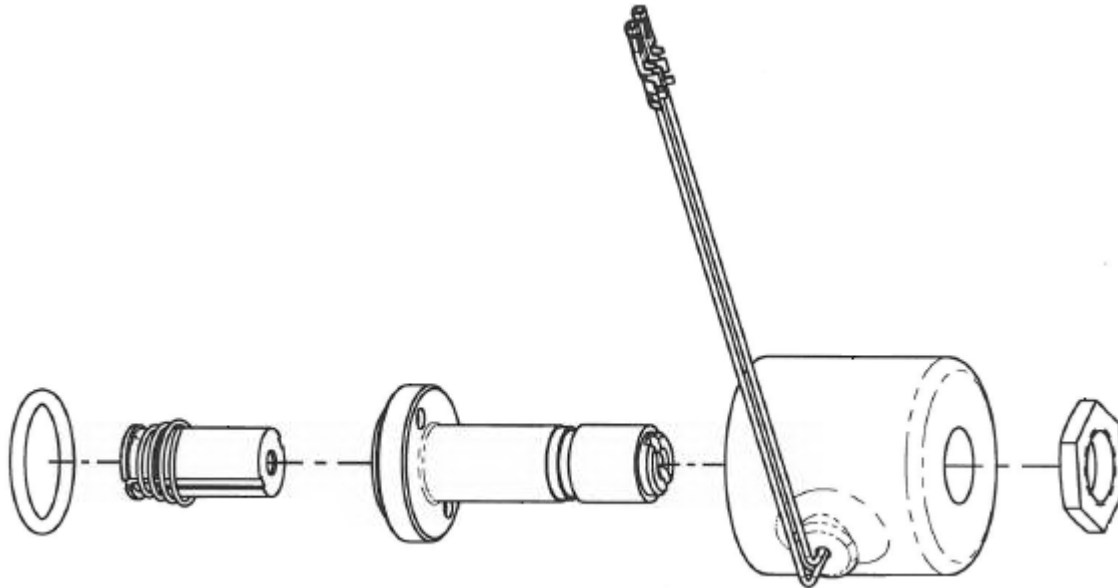


Figure 5001
Solenoid Valve Assembly

D. Vent Valve Assembly (refer to Figure 5002)

- (1) Disassemble vent valve assembly and inspect parts for wear/defects (refer to DISASSEMBLY section). Replace worn or defective parts. Clean body (140), adapter (135), and internal components (refer to CLEANING section). Reassemble vent valve assembly (refer to ASSEMBLY section).

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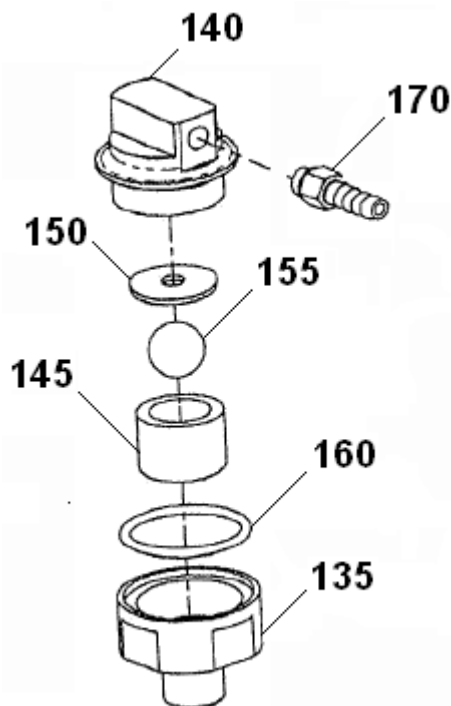


Figure 5002
Vent Valve Assembly

E. Mechanical

- (1) Examine the brew handle locking action for damage or excessive wear. Operation should be smooth and easy. Examine bushings on the brew handle pins. Repeated raising and lowering of the brew head assembly contributes to wear.
- (2) Examine brew head assembly action for damage or excessive wear. Lubricant is necessary on the sides of the upright channels in the chassis assembly for the brew head assembly guides. Operation should be smooth and easy.
- (3) Put a server in the Beverage Maker and fully lower the brew handle. The server retainer assembly should keep the server in place. Examine the server retainer assembly for damage or wear. Operation should be smooth and easy.
- (4) Installation and removal of server and brew cup in chassis assembly cavities should be smooth and easy.
- (5) The locking pin mechanism for the Beverage Maker on a rail assembly should operate smooth and easy without binding in either the raised or lowered position.
- (6) When lowered, the locking pin should keep the Beverage Maker on the rail assembly. When raised, the locking pin should let the Beverage Maker be removed from the rail assembly.
- (7) Installation and removal of the Beverage Maker on a rail assembly should be smooth and easy.

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- (8) Examine alignment of the platen assembly, chassis assembly, power module assembly, rail guides, electrical connector, and water supply connection. The bottom of the Beverage Maker should rest squarely on a flat surface enabling an easy fit into galley installation locations.
- (9) Examine the switch assembly for the brew head assembly that is attached to the right side of the chassis assembly. If parts have been bent or fasteners have become loose, the switch assembly may need to be aligned correctly (refer to the FITS AND CLEARANCES section for more information).
- (10) Instruction placards on the front of the Beverage Maker should be complete and easy to read.

3. Recommended Periodic Maintenance

NOTE: It is highly recommended that the relief valve be examined and tested for correct operation every six months or every 1500 flight hours, whichever comes first.

NOTE: The Beverage Maker should be removed from the aircraft for regular maintenance at a minimum of once a year. In areas where water has high mineral content, more frequent maintenance may be required. Water quality and airline experience dictate frequency of maintenance.

NOTE: Failure to perform regular maintenance as described may limit warranty claims.

NOTE: It is recommended to change the vent valve ball, PN MISC-50509, on every shop visit or at the annual cleaning and descale interval whichever comes first.

- A. Daily (on aircraft): Fully clean the Beverage Maker infrared sensor assembly, brew cup cavity, and switch panel.
- B. Weekly: Clean and inspect for obvious damage. Replace worn components.
- C. Every 1500 flight hours or six months (whichever comes first).
 - (1) Completely flush the entire Beverage Maker water system to remove mineral deposits (refer to the CLEANING section for cleaning procedures and materials).
 - (2) Clean or replace vent valve.
 - (3) Inspect and do a test of the pressure relief valve while the valve is installed on the unit. Replace valve if it does not operate correctly.
 - (4) Tighten all external screws.
 - (5) Check brew handle operation. Replace worn parts and lubricate new parts at installation.
 - (6) Test the Beverage Maker as described in the TESTING AND FAULT ISOLATION Section. Repair or replace parts so the Beverage Maker meets test requirements.
- D. Every 3000 flight hours or 12 months (whichever comes first).
 - (1) Perform 1500 hour maintenance procedures as outlined in Paragraph 3.C.
 - (2) Examine brew head flow washer, o-rings, fitting, and brew head gasket. Replace parts that are worn or damaged.

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- (3) Where TEA option is installed, examine flow washer for funnel tube. Replace o-ring and flow washer if worn.
- (4) Examine solenoid valve assembly. Replace o-ring if worn.
- E. Repeat cycle of maintenance procedures as outlined in Paragraphs 2.C. and 2.D.
- F. Create and maintain maintenance and repair records for the Beverage Maker by part number and serial number.

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REPAIR

1. General

CAUTION: B/E AEROSPACE RECOMMENDS THE FOLLOWING PROCEDURES BE UTILIZED BY A QUALIFIED, APPROVED, AND CERTIFIED REPAIR FACILITY. THE FOLLOWING INFORMATION IS SUPPLIED AT THE REQUEST OF OPERATING CUSTOMERS. IT IS RECOMMENDED THAT EQUIPMENT REQUIRING THIS LEVEL OF REPAIR BE RETURNED TO THE FACTORY. B/E AEROSPACE ACCEPTS NO LIABILITY FOR THE PERFORMANCE OF THE EQUIPMENT WHEN REPAIRS ARE DONE BY THE CUSTOMER USING THIS INFORMATION.

- A. Replace all parts worn or damaged beyond repair. Refer to IPL Figures 1 through 12 and corresponding Illustrated Parts Lists beginning on page 10001 of this manual, as a guide to identification and location of replaceable components.

NOTE: B/E Aerospace does not recommend the repair of broken metal parts. The integrity of the chassis assy, power module assy, rail guides and tank are not assured if these items are broken then welded. These items should be replaced in their entirety.

- B. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK sections to assist in identification of parts requiring replacement or adjustment.
- C. Anchor nuts are permanently attached to various assemblies in the Beverage Maker main subassembly. In the event that they become damaged, they may be replaced by drilling out attaching rivets and riveting on new anchor nuts.
- D. Some holes in castings use helical coil inserts. In the event that they become damaged, they may be removed and replaced.
- E. In the event that tubing is worn or damaged, the tubing should be replaced and not repaired.

2. Electrical

- A. Replace any worn or defective parts.
- B. Repair or replace any metal parts that have nicks or gouges.
- C. Do not repair wiring. Any damaged wiring should be removed and replaced.
- D. Refer to the DISASSEMBLY and ASSEMBLY Sections for removal and replacement of electrical components.
- E. Controller Board Assy

CAUTION: WHEN TOUCHING THE POWER CONTROLLER BOARD OR ANY OF ITS INDIVIDUAL PARTS, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ESD.

NOTE: B/E Aerospace does not recommend the part-level repair of the power controller board assy. This item should be replaced in its entirety.

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CAUTION: REMOVE THE PROGRAMMED MICROCONTROLLER ONLY WITH TOOLS
DESIGNED FOR SAFE REMOVAL OF THE PROGRAMMED MICROCONTROLLER
(REFER TO TABLE 9001 AND FIGURE 9003).

- (1) The controller board assy can have two fuses or the programmed micro controller removed and replaced. Refer to the DISASSEMBLY and ASSEMBLY Sections for removal and replacement of the controller board assy and these parts.

NOTE: The programmed micro controller and fuses can be removed from the controller board assy while the board is in the power module assy.

- (2) For controller board assy maintenance and repair, refer to CMM 25-33-32.

ASSEMBLY

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING ASSEMBLY.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE ASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORK STATION IN A STATIC ELECTRICITY FREE ENVIRONMENT.

CAUTION: MAINTENANCE ENGINEER MUST WEAR A CONDUCTIVE WRIST STRAP TO PROTECT THE CONTROLLER BOARD ASSEMBLY AGAINST ESD.

A. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK Sections prior to assembly. This will determine the extent of assembly required for testing procedures after rebuilding the Beverage Maker or removing a faulty component.

B. Refer to the ILLUSTRATED PARTS LIST (IPL) Section for disassembly and assembly relationships. The number in parentheses after the noun name of a part refers to the item number in the IPL Figure.

C. Provide a clean space and working surface before beginning any assembly operation.

NOTE: Installation of some parts may require use of an adjustable-torque wrench due to torque requirements.

NOTE: Beverage residue and mineral deposits can be cooked into part connections making them hard to remove. The cleaning of some parts can help in their installation, especially parts exposed to coffee residue.

D. Before installing electronic components, make sure polarity and position of the component will let the circuit operate correctly.

E. Refer to the figures in the ILLUSTRATED PARTS LIST Section and assemble the Beverage Maker in accordance with the following procedures. Refer to Table 7001 for required consumable items.

CAUTION: DO NOT USE THREAD LOCK ADHESIVE (REFER TO TABLE 7001) ON OR NEAR PLASTIC PARTS. LET PARTS INSTALLED INTO AN ASSEMBLY USING THREAD LOCK ADHESIVE CURE FOR 24 HOURS BEFORE ATTACHING THE ASSEMBLY TO ANY PARTS MADE OF PLASTIC. MAKE SURE TO FULLY CLEAN HANDS AFTER USING THREAD LOCK ADHESIVE SO NO ADHESIVE IS PUT ON OTHER PARTS OF THE UNIT. SOLVENT IN THE ADHESIVE CAN DAMAGE PLASTIC PARTS.

NOTE: Refer to the FITS AND CLEARANCES Section for adjustment of essential assemblies during assembly.

NOTE: Equivalent substitutes may be used for listed items.

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Table 7001
Assembly Materials and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
1 in X 3 in Deep Socket, 1/2 in Drive for heater and water filter removal (refer to Figure 9001)	3575-1318-01	B/E Aerospace, Inc. (V63367)
Alodine Chemical Coating	Alodine 1201 CM-50017	Henkel Corp. (V71410) B/E Aerospace, Inc. (V63367)
Black Adhesive Sealant	732 Black RTV-103 AD-052	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)
Clamp (Qty: 6)	None	Commercially Available
Epoxy Adhesive	E-Cast F-28 Resin F-14 Hardener AD-50029	United Resin Corp. (V08LA7) B/E Aerospace, Inc. (V63367)
Heat Sink Compound	10-8109 CM-50019	Waldom Inc. (V72653) B/E Aerospace, Inc. (V63367)
Loctite-5900	4481015-001	B/E Aerospace, Inc. (V16827)
Lubricant and Sealant, FDA	CM-50018	B/E Aerospace, Inc. (V63367)
Petroleum Jelly	None CM-50015	Commercially Available B/E Aerospace, Inc. (V63367)
Silicone Rubber Adhesive	736 Red RTV AD-50005	Dow Corning Corp. (V71984) B/E Aerospace, Inc. (V63367)
Soft-Cloth	None	Commercially Available
Solder	Kester No. 245, No Clean, Low Residue SD-50003	Kester Solder (Div of Litton Systems, Inc.) (V75297) B/E Aerospace, Inc. (V63367)
Syringe	None	Commercially Available
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Thread Lock Adhesive	242 Blue AD-300	Loctite Corp. (V05972) B/E Aerospace, Inc. (V63367)
Thread Lubricant	Never Seez CM-50016	Bostik Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)

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2. Procedure

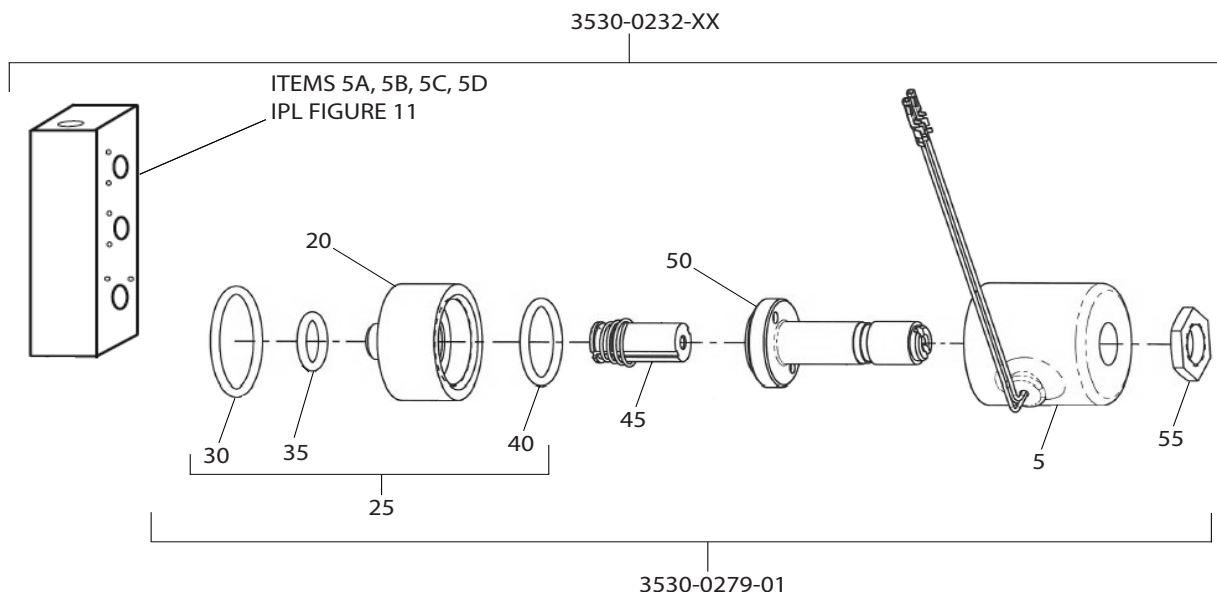


Figure 7001

Solenoid Valve Assembly (PN 3530-0279-01)/Manifold Assembly (PN 3530-0232-XX)

A. Solenoid Valve Assembly (PN 3530-0279-01)/Manifold Assembly (PN 3530-0232-XX) (refer to Figure 7001)

- (1) Only if necessary for repair, put crimp terminals (15) on the wiring to the solenoid coil (5), using a Molex terminal crimp tool part number 63819-0000 or equivalent.
- (2) Put plunger (45) in the solenoid guide (50), so that the seal end on the base of the plunger faces the open end of the solenoid guide.
- (3) Install o-ring (40) in valve body (20). Put solenoid guide (50) on valve body (20). Special tools are recommended for assembly (refer to Table 9001 and Figure 9008).
- (4) Install o-rings (30, 35) into the valve body (20). Install valve body (20) into manifold block (5, IPL Figure 11). Torque the solenoid guide (50) to 20 in-lbs to 30 in-lbs. When the torque on solenoid guide (50) is obtained, the valve body (20) will rotate until, torque listed above is obtained.
- (5) Put coil assy (5) on solenoid guide (50).
- (6) Put nut (55) on solenoid guide (50). Apply a torque of 18 to 22 in-lbs to the nut (55).

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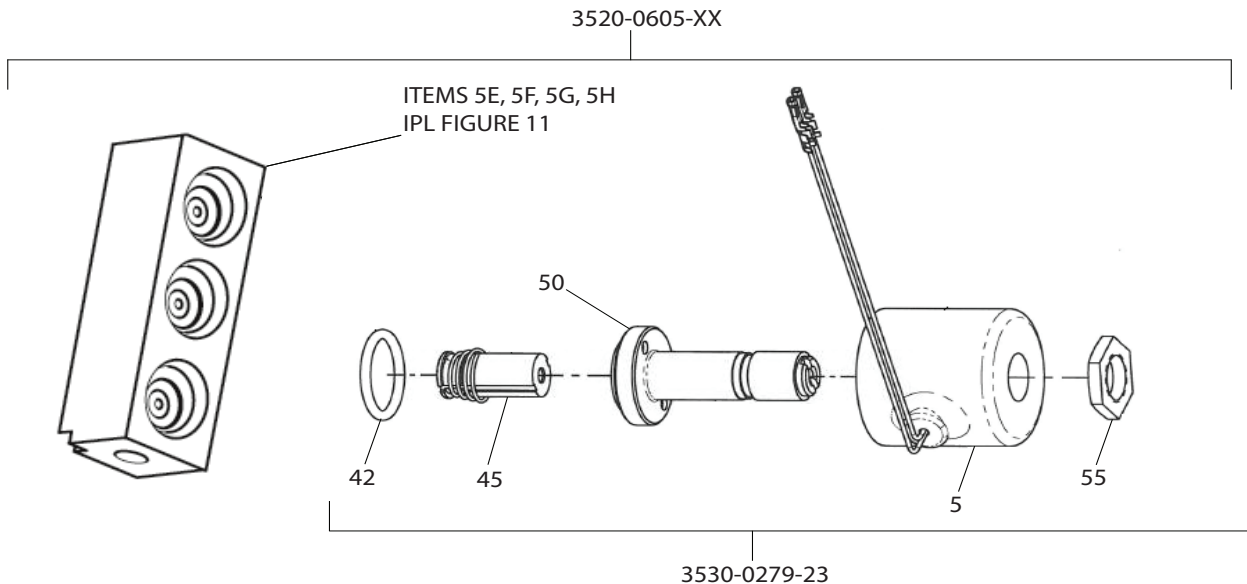


Figure 7002

Solenoid Valve Assembly (PN 3530-0279-23)/Manifold Assembly (PN 3520-0605-XX)

B. Solenoid Valve Assembly (PN 3530-0279-23) / Manifold Assembly (PN 3520-0605-XX) (refer to Figure 7002)

- (1) Only if necessary for repair, put crimp terminals (15) on the wiring to the solenoid coil (5), using a Molex terminal crimp tool part number 63819-0000 or equivalent.
- (2) Put plunger (45) in the solenoid guide (50), so that the seal end on the base of the plunger faces the open end of the solenoid guide.
- (3) Put the o-ring (42) in the solenoid block (5, IPL Figure 11).
- (4) Put solenoid guide (50) on solenoid block (5, IPL Figure 11). Special tools are recommended for assembly (refer to Table 9001 and Figure 9003). Torque the solenoid guide (50) into solenoid block (5, IPL Figure 11) to 20 in-lbs to 30 in-lbs.
- (5) Put coil assy (5) on solenoid guide (50).
- (6) Put nut (55) on solenoid guide (50). Apply a torque of 18 to 22 in-lbs to the nut (55).

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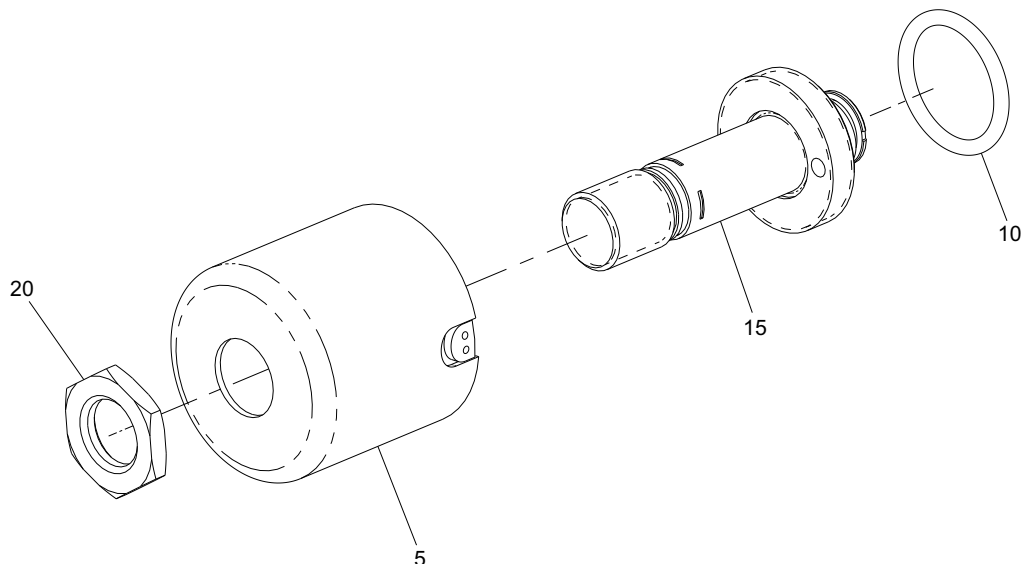


Figure 7003
2-Way Normally Closed Valve Assembly (PN 3520-1529-01)

C. 2-Way Normally Closed Valve Assembly (PN 3520-1529-01) (refer to Figure 7003 and IPL Figure 12A)

- (1) Only if female crimp terminals (32, IPL Figure 11) have been removed, put new female crimp terminals (32, IPL Figure 11) on the wiring to the coil assembly (5) using a Molex Terminal Crimp Tool part number 63819-0000 or equivalent.
- (2) Apply FDA Lubricant and Sealant to the O-ring (10).
- (3) Put the O-ring (10) in the solenoid manifold (5, IPL Figure 11).
- (4) Put plunger guide assembly (15) on solenoid manifold (5, IPL Figure 11). Apply a torque of 23 to 27 in-lbs (2.60 to 3.05 N-m) to the plunger guide assembly (15). Special tools are recommended for assembly. A minimum amount of Thread Lubricant should be applied to fill the threads. Any excess lubricant should be wiped clean.
- (5) Put coil assembly (5) on plunger guide assembly (15).
- (6) Put the nut (20) on plunger guide assembly (15). Apply a torque of 18 to 22 in-lbs (2.03 to 2.49 N-m) to the nut (20).

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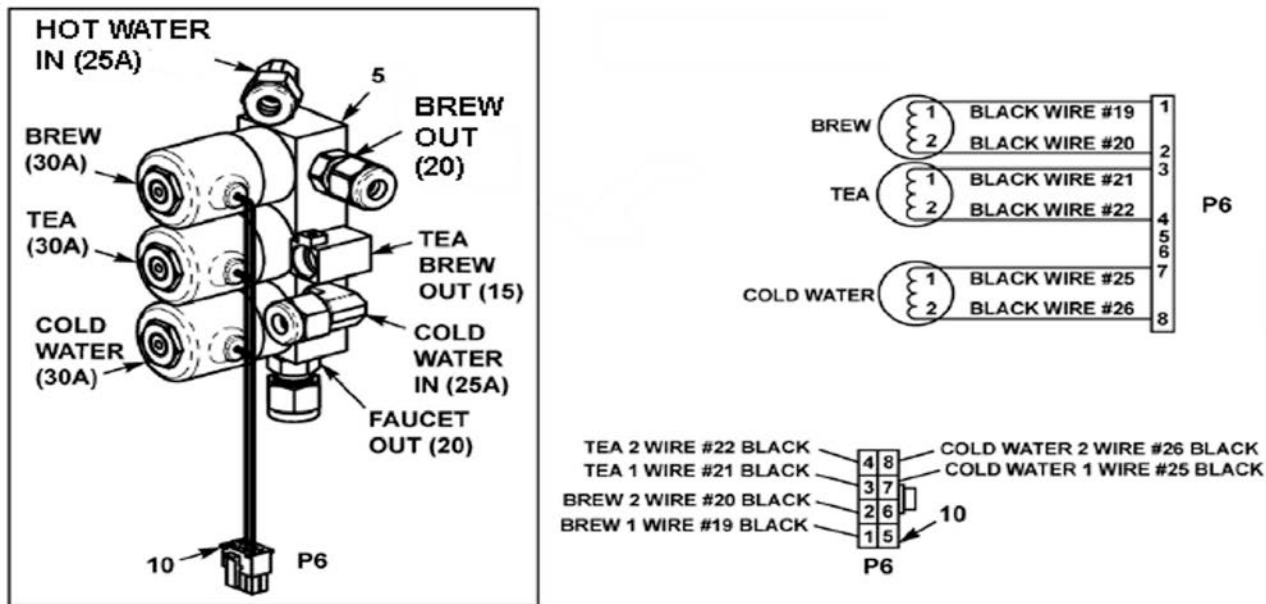


Figure 7004
 Manifold Assy (PN 3520-0605-01)

D. Manifold Assembly (refer to Figures 7004 thru 7008)

NOTE: Fittings (15, 20, 25A or 25B) are to be hand tightened plus 1 turn $\pm 1/2$ turn after installing onto manifold body (5).

- (1) Put thread seal tape on regulator fitting (15) and then put regulator fitting on manifold body (5A, 5D, 5E, or 5H).
- (2) Put thread seal tape on straight fittings (20) then put straight fittings on manifold body (5).
- (3) Put thread seal tape on elbow fittings (25A or 25B) then put elbow fittings on manifold body (5).

CAUTION: CORRECTLY CRIMP TERMINALS ON WIRE FOR SOLENOID COIL USING A MOLEX TERMINAL CRIMP TOOL PART NUMBER 63819-0000 OR EQUIVALENT.

- (4) Where necessary, install crimp terminals (15) on solenoid coil (10) wires.

CAUTION: CAREFULLY AND CORRECTLY INSTALL CRIMP TERMINALS IN THE P6 CONNECTOR, OR THE CONNECTOR MAY BE DAMAGED, RESULTING IN DAMAGE TO THE CONTROLLER BOARD.

- (5) Put solenoid valve assy (30) wiring crimp terminals in P6 connector (refer to Figure 7004).

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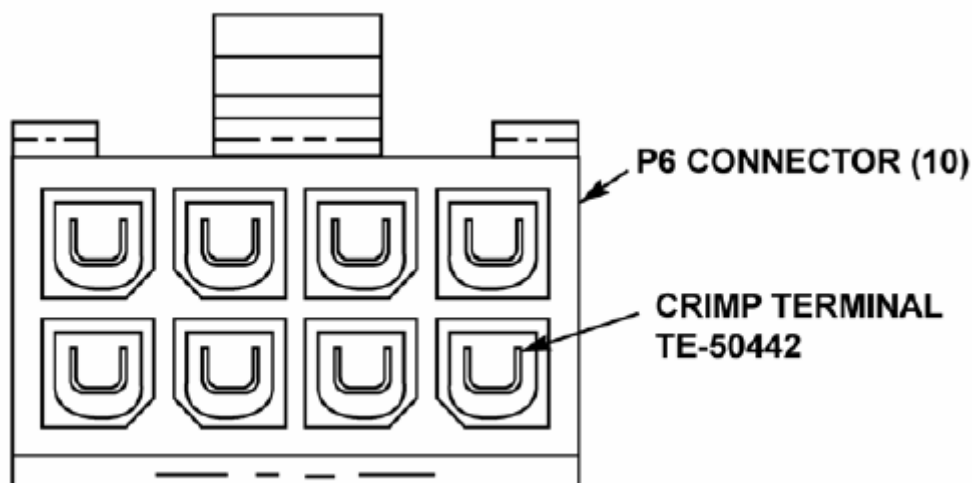


Figure 7005
 Crimp Terminal Alignment in P6 Connector
 (Terminal Side View of Connector)

- (6) Refer to ASSEMBLY, Paragraph 2.N.(2) for installation of manifold assy (710A, IPL Figures 2, 2A, 2B, 2C) on chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C).

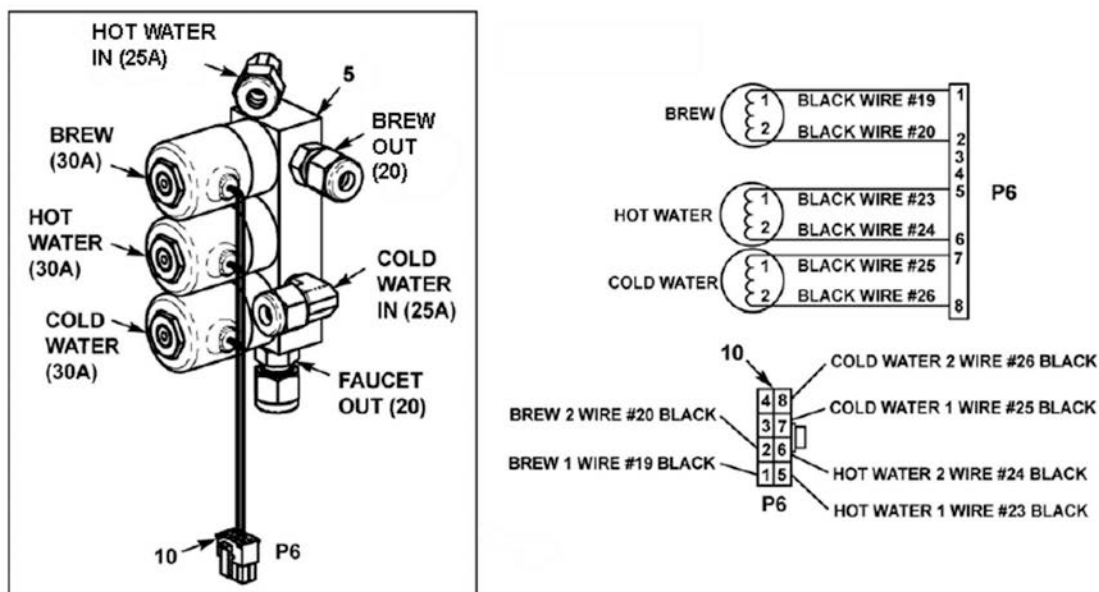


Figure 7006
 Manifold Assy (PN 3520-0605-05)

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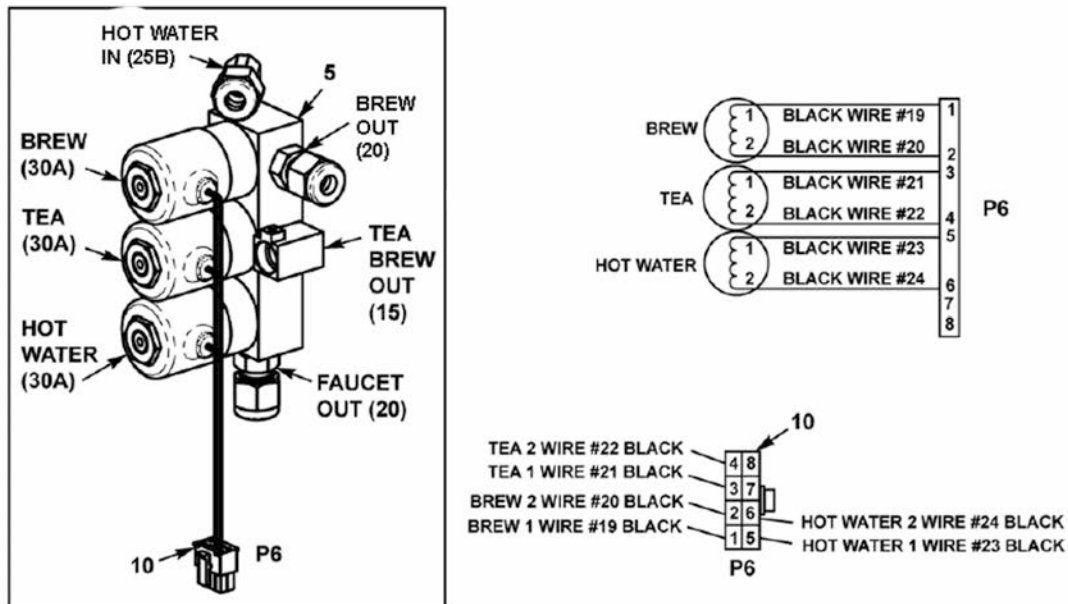


Figure 7007
 Manifold Assy (PN 3520-0605-03)

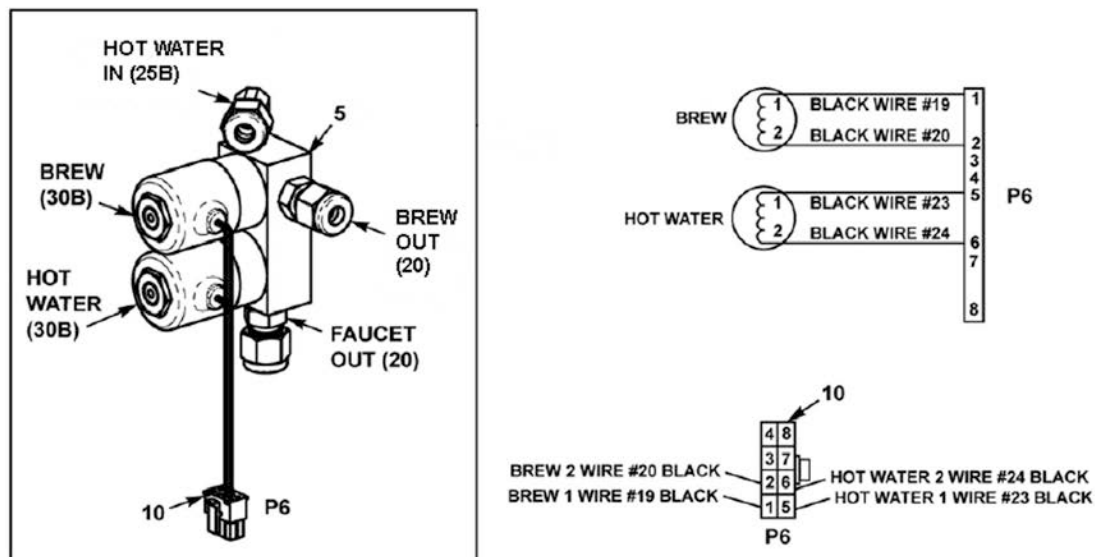


Figure 7008
 Manifold Assy (PN 3520-0605-07)

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E. Retainer Assy (refer to Figure 7009)

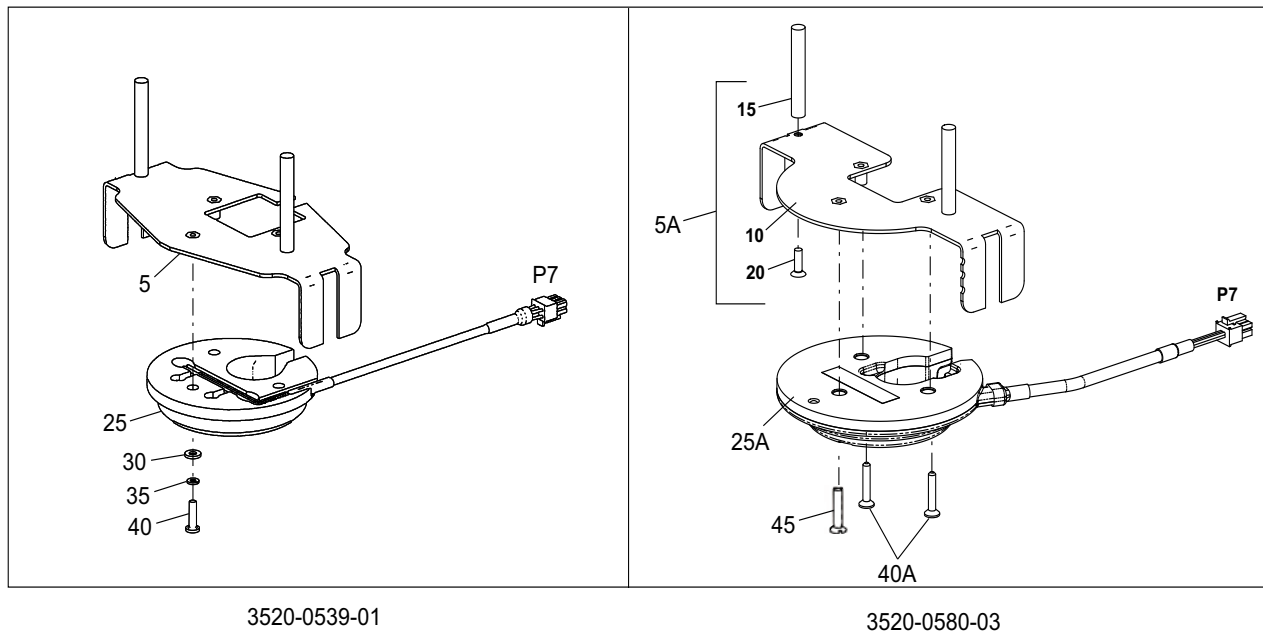


Figure 7009
Assembly of Server Retainer Assy

- (1) Put two LLS retainer rods (15) on the retainer plate assy (10) using two screws (20). Secure screws (20) with epoxy adhesive AD-50029 and tighten to a minimum 15 in-lbs (1.70 newton-meters).

NOTE: Epoxy Adhesive AD-50029 on the retainer assembly requires 24 hours of curing time before the Liquid Level Sensor (LLS) assembly may be installed on the retainer assembly.

CAUTION: A NYLON PATCH ON THE SCREW (40B) WILL CAUSE RESISTANCE TO TURNING OF THE SCREW (40B). DO NOT OVERTIGHTEN THE SCREW (40B) ON THE RETAINER PLATE AND ROD ASSY (5A) AND INFRARED SENSOR ASSY (25A) OR DAMAGE TO THE INFRARED SENSOR ASSY (25A) CAN RESULT. IN THE EVENT THAT THE SCREW (40B) IS TIGHTENED TOO MUCH, STRESS FRACTURES MAY BE STARTED IN THE INFRARED SENSOR ASSY (25A).

- (2) Put infrared sensor assy (25A) on the retainer plate and rod assy (5A) using two screws (40B) and one screw (45).
- (3) Tighten screws (40B) to 5 in-lbs (0.57 newton-meters) then fill screw cross recess with translucent silicone rubber adhesive/sealant (refer to Table 7001).

NOTE: Apply translucent silicone rubber adhesive sealant AD-50002 to threads of screw (45) before installing and tighten to 1 in-lb (0.11 newton-meters) then fill screw cross recess with translucent silicone rubber adhesive/sealant AD-50002.

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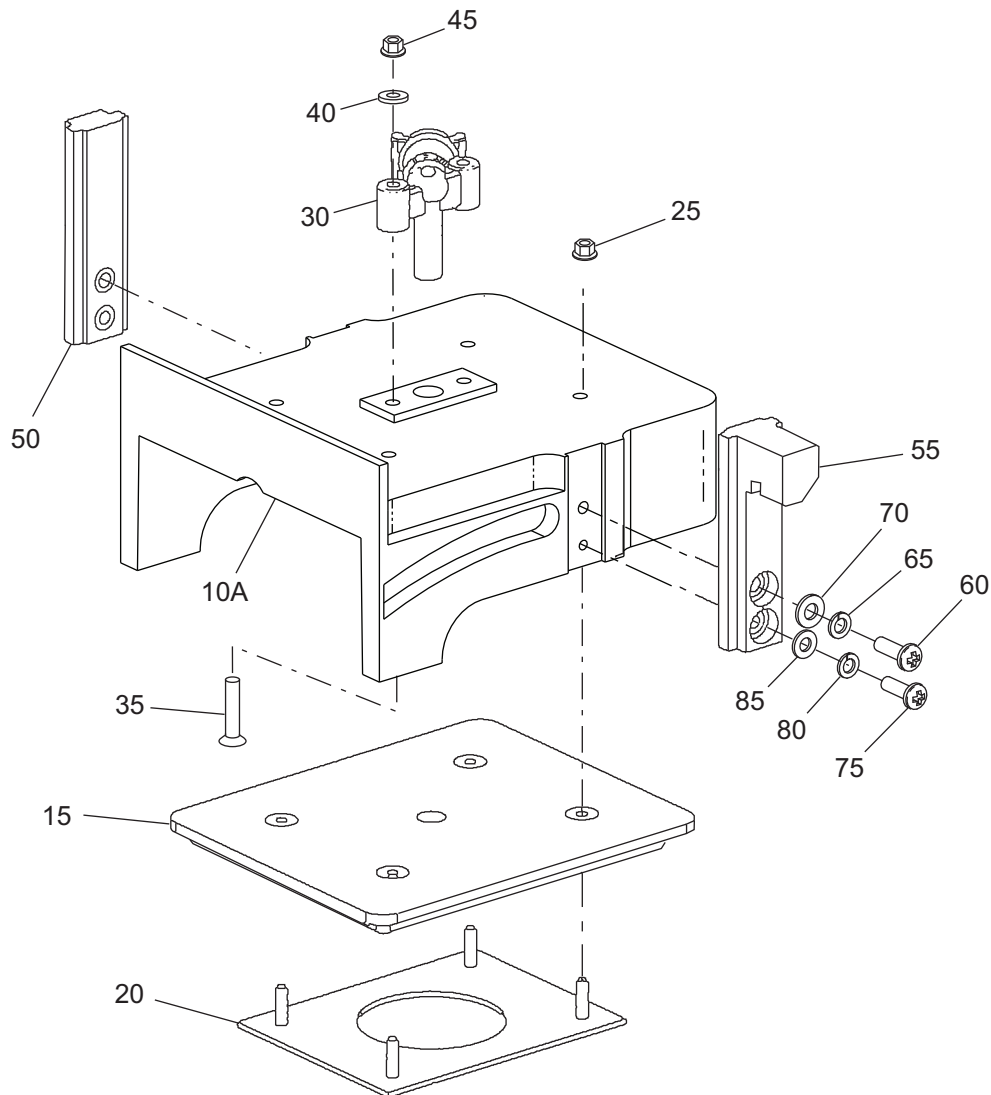


Figure 7010
 Assembly of Brewhead Assy

F. Brewhead Assy (refer to Figure 7009)

- (1) Install the brew fitting (30) in the brew head using two screws (35), two lock nuts (45), and two flat washers (40).
- (2) Install the gasket (15) on the brew head (10) using the gasket retainer (20) and four lock nuts (25).
- (3) Install the right side guide (55) on the brew head (10) using two screws (60 and 75), two lock washers (65 and 80), and two flat washers (70 and 85). Torque screws (60 and 75) to 10 to 12 in-lbs.

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- (4) Install the left side guide (50) on the brew head using two screws (60 and 75), two lock washers (65 and 80), and two flat washers (70 and 85). Torque screws (60 and 75) to 10 to 12 in-lbs.
- (5) Refer to ASSEMBLY, Paragraph 2.N.(9) for installation of brewhead assy (300, IPL Figures 2, 2A, 2B, 2C) on chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C).

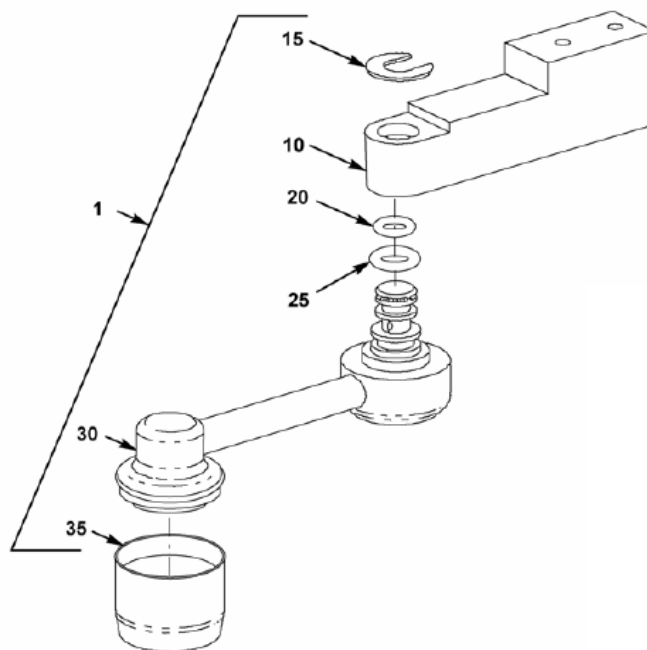


Figure 7011
Assembly of Swivel Faucet Assy

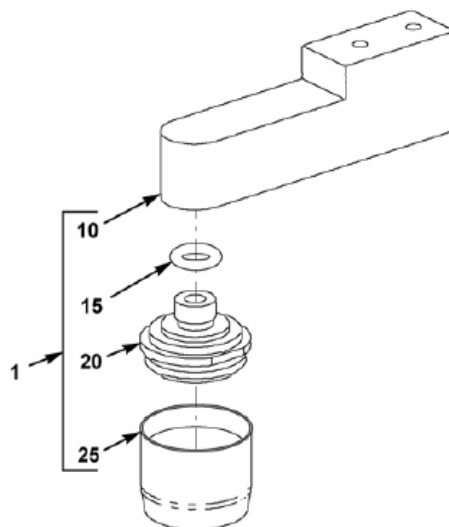


Figure 7012
Assembly of Faucet Assy

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G. Swivel Faucet Assy (refer to Figure 7011)

NOTE: Apply thread lubricant (refer to Table 7001) to o-rings (20 and 25).

- (1) Put upper o-ring (20) and lower o-ring (25) on welded faucet assy (30).
- (2) Put welded faucet assy (30) in swivel faucet bar (10).
- (3) Put retaining ring (15) on welded faucet assy (30).
- (4) Put aerator (35) on welded faucet assy (30).
- (5) Refer to ASSEMBLY, Paragraph 2.N.(8), for installation of swivel faucet assy (275, IPL Figures 2, 2A, 2B, 2C) on chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C), if option is installed.

H. Faucet Assy (refer to Figure 7012)

- (1) Put o-ring (15) on aerator adapter (20).
- (2) Put aerator (25) and aerator adapter (20) in faucet bar (10).
- (3) Refer to ASSEMBLY, Paragraph 2.N.(8) for installation of faucet assy (270, IPL Figures 2, 2A, 2B, 2C) on chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C), if option is installed.

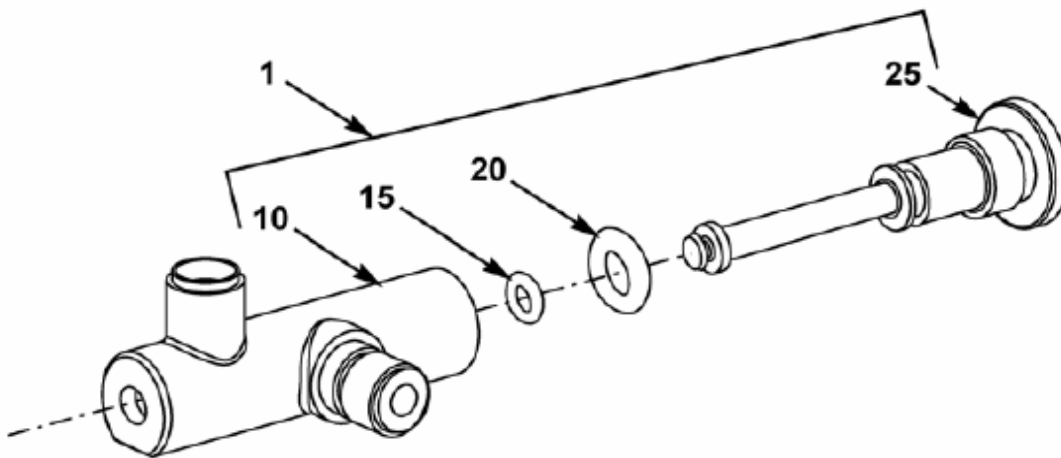


Figure 7013
Assembly of Drain Valve Assy

J. Drain Valve Assy (refer to Figure 7013)

- (1) Put upper o-ring (20) and lower o-ring (15) on stem assy (25).
- (2) Put stem assy (25) in body (10).
- (3) Refer to ASSEMBLY, Paragraph 2.N.(15) for installation of drain valve assy (1 and 140, IPL Figures 2, 2A, 2B, 2C) on support connector (115, IPL Figures 2, 2A, 2B, 2C).

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K. Controller Board (refer to CMM 25-33-32)

L. Power Module Assy (refer to Figures 7014 and 7015)

(1) Put the grommet (40) in hole in back of the power module housing assy (10).

NOTE: This grommet (40) covers the safety latch switch on the back of the controller board assy.

CAUTION: THE PP1 AND P1 CONNECTORS MUST BE INSTALLED ON THE HARNESS ASSY AFTER THE HARNESS ASSY IS PUT THROUGH THE GROMMET AND HOLE IN THE POWER MODULE HOUSING ASSY (REFER TO FIGURE 7013).

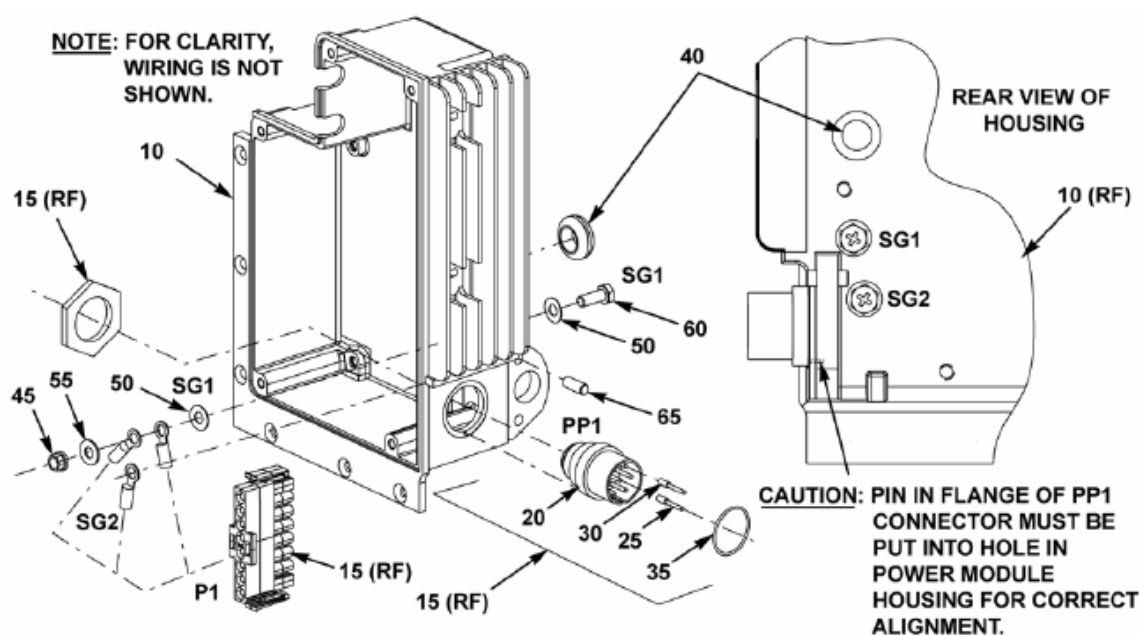


Figure 7014
Assembly of Power Module Assy

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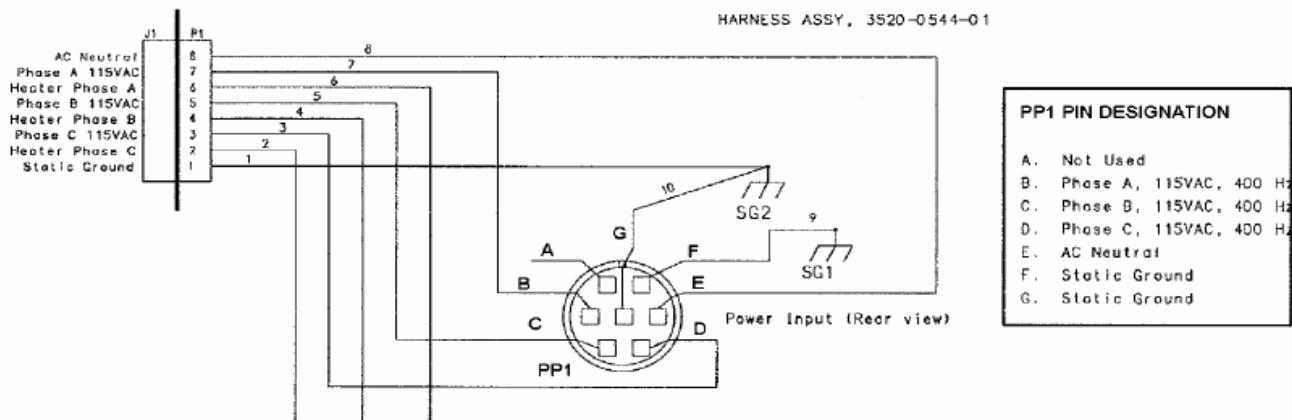


Figure 7015
 Wiring Diagram for PP1 Connector

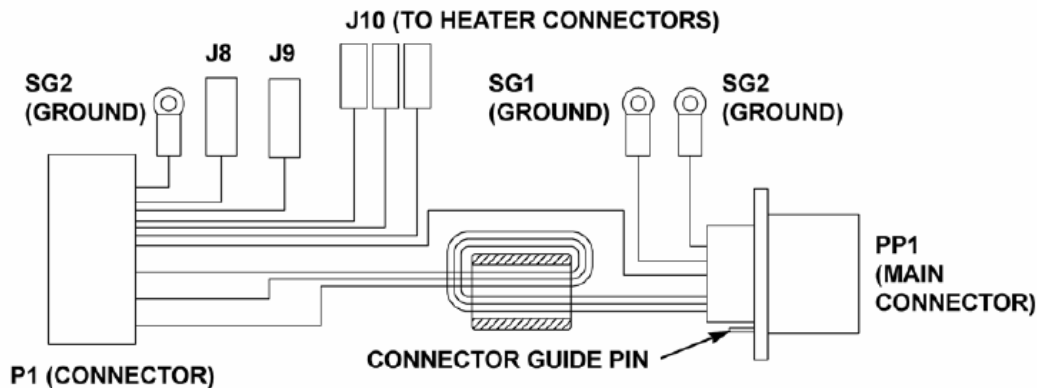


Figure 7016
 Harness Assy for Power Module Assy

- (2) Install pin (65) in power module housing assy (10).

NOTE: This pin helps align the support connector (115, IPL Figures 2, 2A, 2B, 2C) with the power module assy (55, IPL Figures 2, 2A, 2B, 2C).

- (3) Install the SG1 and SG2 ground stack ups for the harness assy (15) (refer to Figure 7014) in the power module housing assy (10).
 - (a) Install two lock nuts (45), two flat washers (55), wire terminals for harness assy (15), four flat washers (50), and two hex screws (60) on the power module housing assy (10).

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NOTE: The hex nut for the PP1 connector must be put on the harness assy and the PP1 connector must be put into the power module housing assy before the PP1 connector is attached to the harness assy (refer to Figure 7014).

CAUTION: DO NOT DAMAGE SMALL GUIDE PIN ON FLANGE OF PP1 CONNECTOR WHEN INSTALLING PP1 CONNECTOR ON POWER MODULE HOUSING.

- (4) Put PP1 connector (20) into power module housing assy (10).
 - (a) Make sure guide pin on flange of connector (20) goes into the hole in the power module housing assy (10).
 - (b) Only if removed for replacement put pins (25 and 30) into connector (20).
 - (c) Only if removed for replacement put o-ring (35) on connector.
 - (d) Attach connector (20) to power module housing assy (10) using hex nut.
- (5) Refer to ASSEMBLY, Paragraph 2.L. for installation of power module assy (55, IPL Figures 2, 2A, 2B, 2C) on chassis assy (10 or 10A, IPL Figures 2, 2A, 2B, 2C),.

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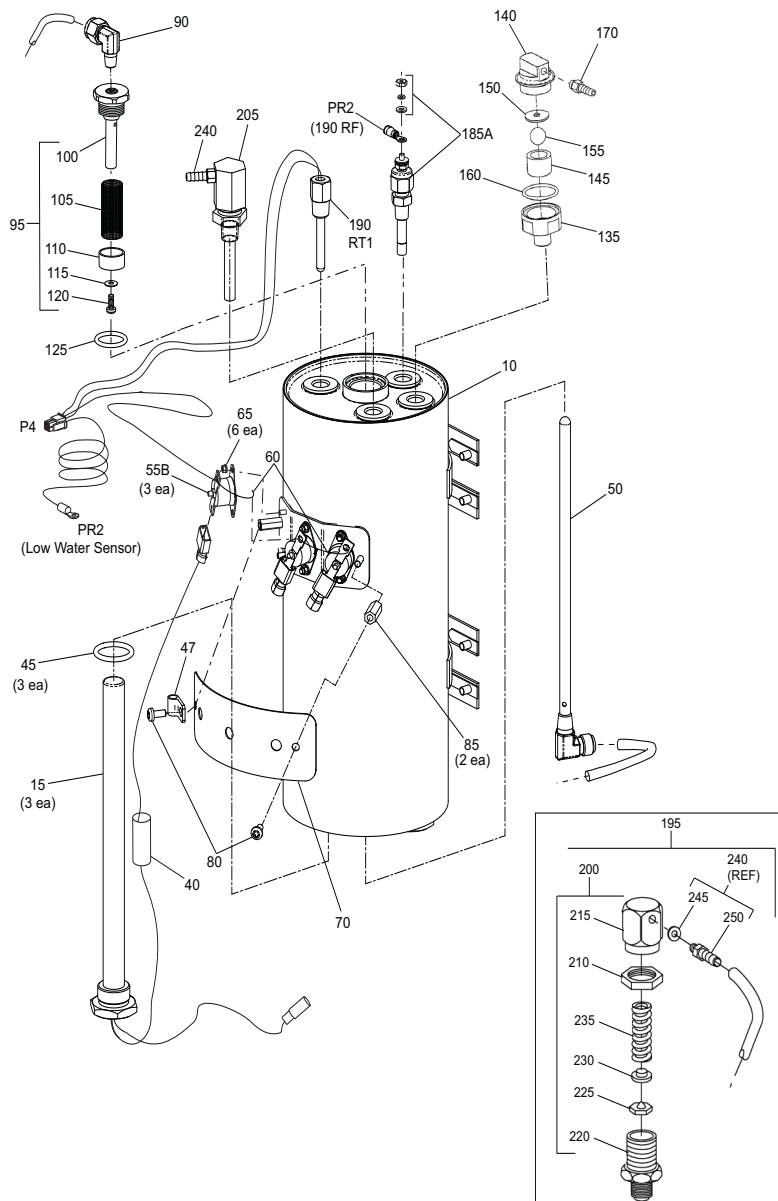


Figure 7017
Assembly of Tank Assy

M. Tank Assy (refer to Figures 7017, 7018, 7019, and 7020)

- (1) Install three tank heater assy (15) and three o-rings (45) in welded tank assy (10).

NOTE: Apply Thread Lubricant (refer to Table 7001) to the heater threads.

NOTE: Only if removed for replacement, put connector (25), connector pin (30) and terminal (20) on heater assy tank (35).

- (a) Put o-ring (45) on each tank heater assy (15).

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CAUTION: DO NOT TURN HEATER ASSY IN THE TANK TOO TIGHT.

- (b) Turn tank heater assy (15) into threaded hole in welded tank assy (10) to finger tightness, then use socket (refer to Figure 9001) and torque wrench to turn heater assembly to not exceed 10 ft/lbs torque.

NOTE: Install tank heater assy (15) correctly in the bottom of the welded tank assy (10) as shown in Figure 7018.

NOTE: Move wiring for tank heater assy (15) and connect it with wire tie downs to keep it from touching the welded tank assy (10).

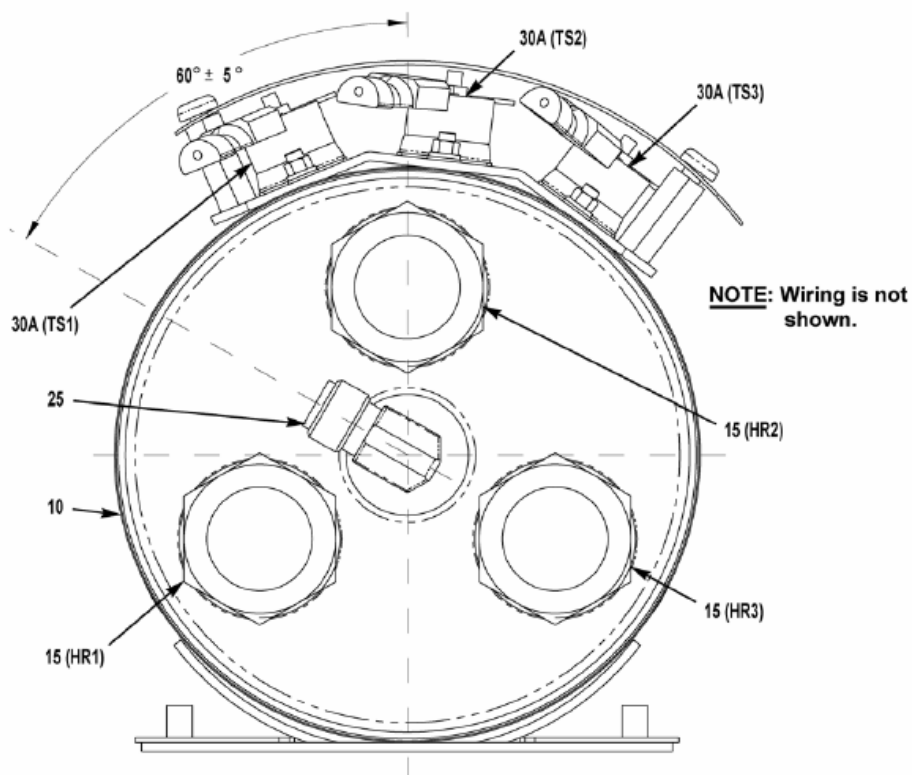


Figure 7018
Bottom View of Tank Assy

- (2) Install tube assy (50) in welded tank assy (10) using thread seal tape. Align elbow fitting as shown in Figure 7017.
- (3) Install pressure relief valve assy (205) in top of welded tank assy (10) using thread seal tape (refer to Figure 7018).

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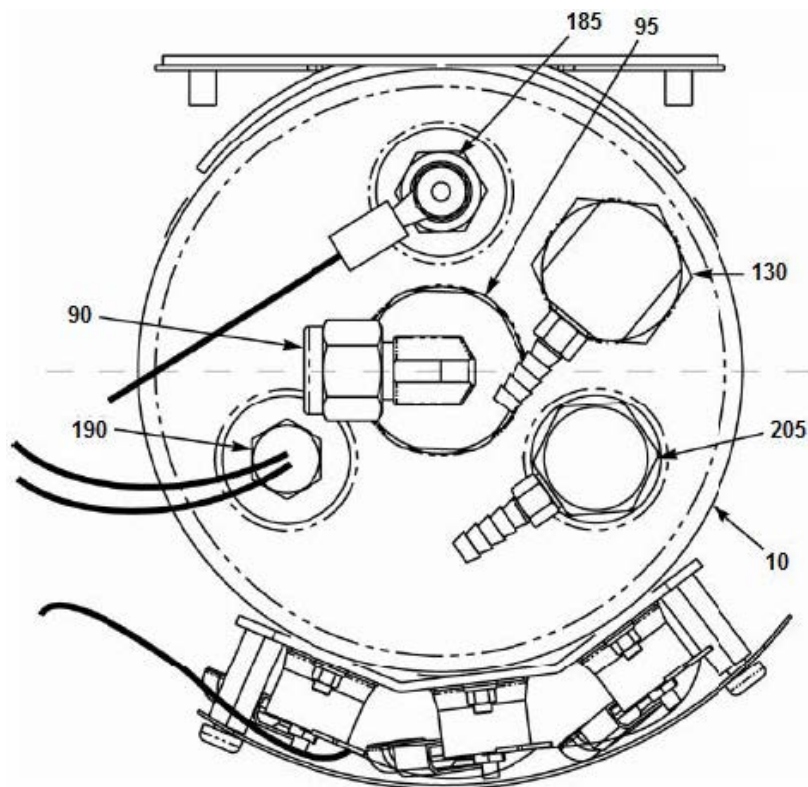


Figure 7019
Top View of Tank Assy

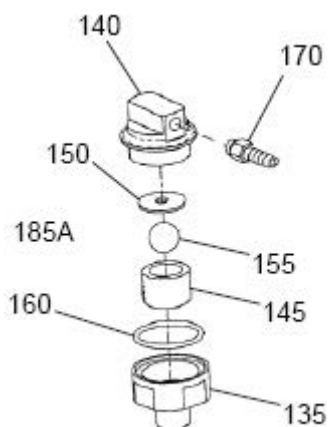


Figure 7020
Assembly of Vent Valve Assembly

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- (4) Install vent valve assy (130) in top of welded tank assy (10) using thread seal tape (refer to Figures 7019 and 7020).
 - (a) Install the gasket (175) on the fitting (180) to assemble the barbed fitting (170).
 - (b) Install the barbed fitting (170) on the vent valve assy (130).
 - (c) Fully insert washer (150) into body (140).
 - (d) Insert liner (145) into body (140).
 - (e) Insert ball (155) into liner (145).
 - (f) Install o-ring (160) onto adapter (135).
 - (g) Tighten body (140) onto adapter (135).
- (5) Assemble the filter assy (95) before installing it in the welded tank assy (10).
 - (a) Install the filter screen (105) and retainer (110) on the water jet (110) using a screw (120) and lock washer (115).
- (6) Install an elbow fitting (90) on the filter assy (95) using thread seal tape.
- (7) Install the filter assy (95) and o-ring (125) in the top of the welded tank assy (10).
 - (a) Put o-ring (125) on filter assy (95).
 - (b) Turn the filter assy (95) to finger tightness in the threaded hole in center of the top of the welded tank assy (10) then use socket (refer to Figure 9001) with torque wrench to turn filter assy to 10 ft-lbs torque.
- (8) Install the RTD assy (190) in the top of the welded tank assy (10) using thread seal tape.
- (9) Install the low water sensor (185) in the welded tank assy (10) using thread seal tape.
- (10) Install three thermostats (55A) on the welded tank assy (10) using six lock nuts (65).

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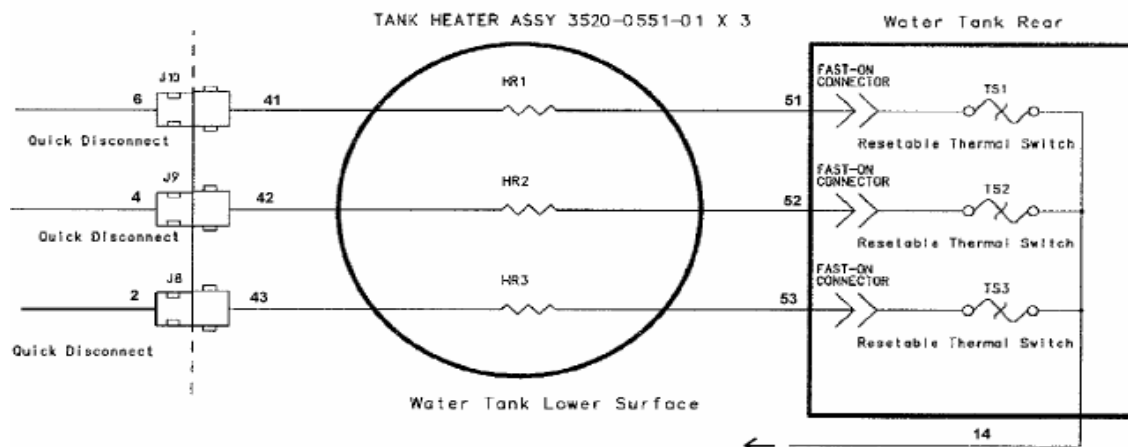


Figure 7021
 Schematic of Tank Heater Assembly Wiring

CAUTION: WHEN HEATING SOLDER, TURN TANK SO HOT SOLDER FALLS AWAY FROM THE TANK AND THERMOSTATS.

- (11) Install two wire busses (60) and one wire from RTD assy (190) on the upper terminals of the overtemperature thermostats (55) using solder.
 - (12) Attach RTD assy (190) wire that has ring terminal on it to the low water sensor (185) using hex nut.
 - (13) Attach plug of RTD assy (190) to controller board (75, IPL Figure 2, 2A, 2B, 2C). Power module assy (55, IPL Figure 2, 2A, 2B, 2C) and controller board (75, IPL Figure 2, 2A, 2B, 2C) may need to be installed before attaching wiring plug from RTD assy (190).
 - (14) Install two hex spacers (85) on the welded tank assy (10).
 - (15) Install a thermal switch cover (70) on the two hex spacers (85) using two screws (80) and clamp (75).
 - (16) Refer to ASSEMBLY, Paragraph 2.N.(14) for installation of tank assy (15) (refer to Figure 7036) on chassis assy (10 or 10A).
- N. Beverage Maker Subassembly (refer to IPL Figures 2, 2A, 2B, 2C, and Figures 7022 thru 7041).
- (1) Install the lockdown lever on the chassis assy (10 or 10A) (refer to Figure 7022).

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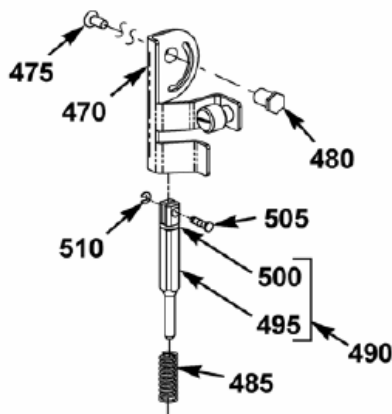


Figure 7022
Assembly of the Lockdown Lever

- (a) Install locating pin assy (490) on locating pin lever (470) using retaining ring (510) and pin (505).
 - 1 Install the o-ring (500) on the machined pin (495) that make up the locating pin assy (490).
- (b) Coat the locating pin assy (490) and helical spring (485) with petroleum jelly and put them into the hole in the boss in the left corner inside the chassis assy (10 or 10A) server cavity.

NOTE: The platen assy (185 or 195) will have to be installed on the chassis assy (10 or 10A) to hold the helical spring (485) in the chassis assy.
- (c) Install the locating pin lever (470) on the chassis assy (10 or 10A) using a screw (475) and pin pivot assy (480).

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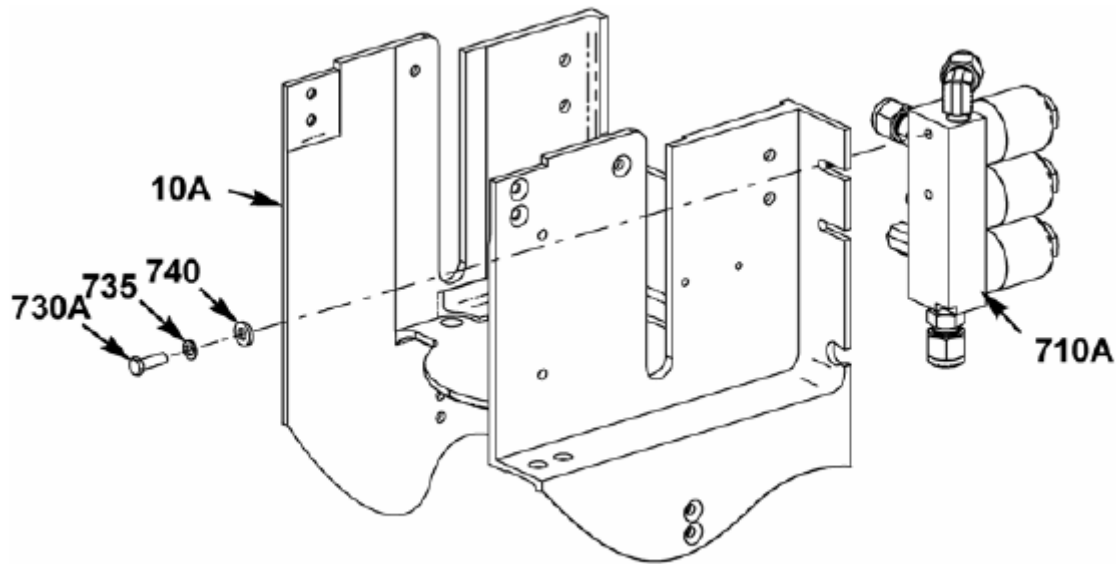


Figure 7023
Installation of Manifold Assy

- (2) Install manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) on the chassis assy (10 or 10A) (refer to Figure 7023).
 - (a) Install the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) on chassis assy (10 or 10A) using two screws (730A), two lock washers (735), and two flat washers (740).
 - (b) Attach wiring connections for the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) in the power module assy (55). Install the power module assy (55) on the chassis assy (10 or 10A) before attaching wiring.

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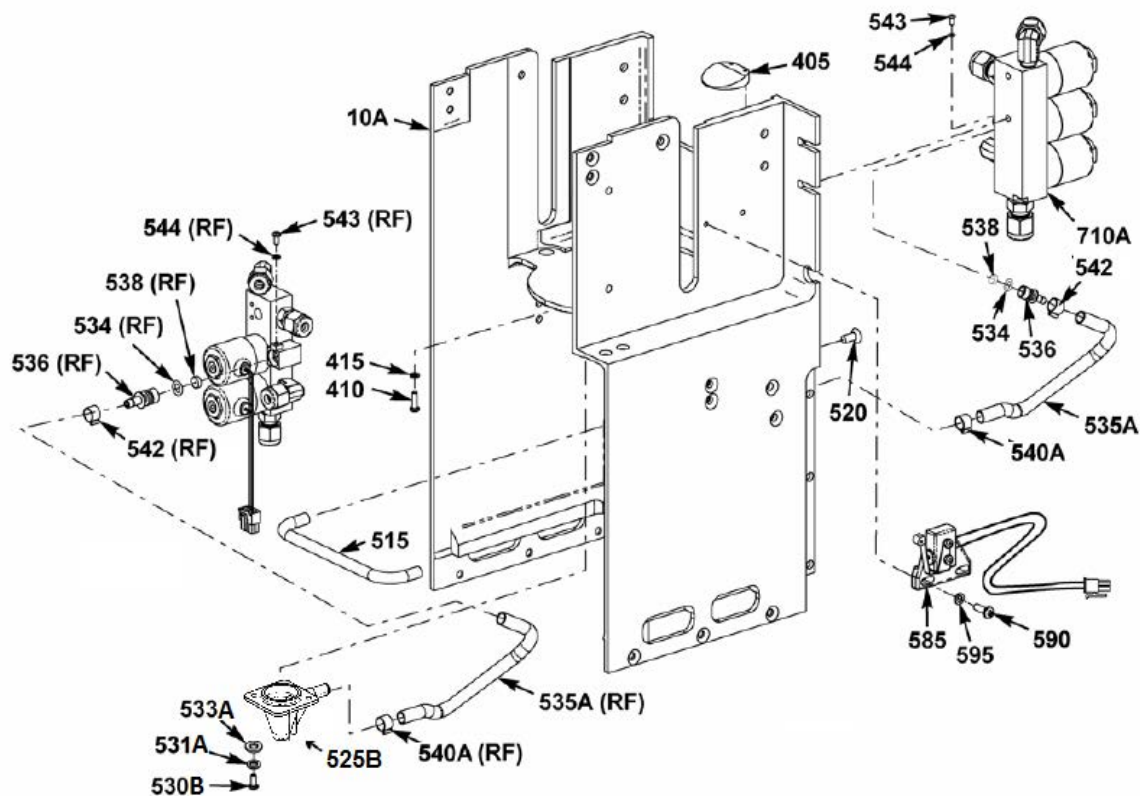


Figure 7024
 Installation of Funnel Tube

- (3) Install the cup retainer (405) on the chassis assy (10 or 10A) using two lock washers (415) and two screws (410) (refer to Figure 7024).
- (4) Install handle (515) on chassis assy (10 or 10A) using two screws (520) and thread lock adhesive (refer to Figure 7024).
- (5) Install the tea funnel (525B) on the chassis assembly (10 or 10A) (refer to Figure 7024).
 - (a) Install a tube (535A) on the tea funnel (525B) using a tube clamp (540A).

NOTE: The tube clamp requires the equivalent to an Oetiker tube clamp crimping tool for correct installation.

NOTE: The tube (535) is replaced by tea tube (535A) per Service Bulletin 4510-25-02.

- (b) Install tea funnel (525B) on chassis assembly (10 or 10A) using two screws (530B), two lock washers (531A), and two flat washers (533A).

NOTE: Apply black adhesive sealant (Refer to Table 7001) around the edge of the tea funnel (525B) for sealing purposes when mated to the housing brew shelf.

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- (c) Attach tube (535A) to correct fitting on the manifold assy (710 or 725) using a tube clamp (542) or attach tea tube (535A) to regulator cartridge (536) using a tube clamp (542).
- (d) Put o-ring (534) on regulator cartridge (536) after lubricating the o-ring with a small amount of petroleum jelly.

CAUTION: MAKE SURE THE CONCAVE SIDE OF THE FLOW WASHER IS FACING OUT OF THE REGULATOR CARTRIDGE. IN THE EVENT THAT THE FLOW WASHER IS NOT INSTALLED CORRECTLY, WATER WILL NOT FLOW CORRECTLY TO THE FUNNEL TUBE.

- (e) Put flow washer (538) in regulator (536). Put the concave side of flow washer (538) so it is facing out of the regulator (536).
- (f) Attach regulator cartridge (536) to the correct elbow fitting of the manifold assy (710A or 725A) using a screw (543) and lock washer (544).

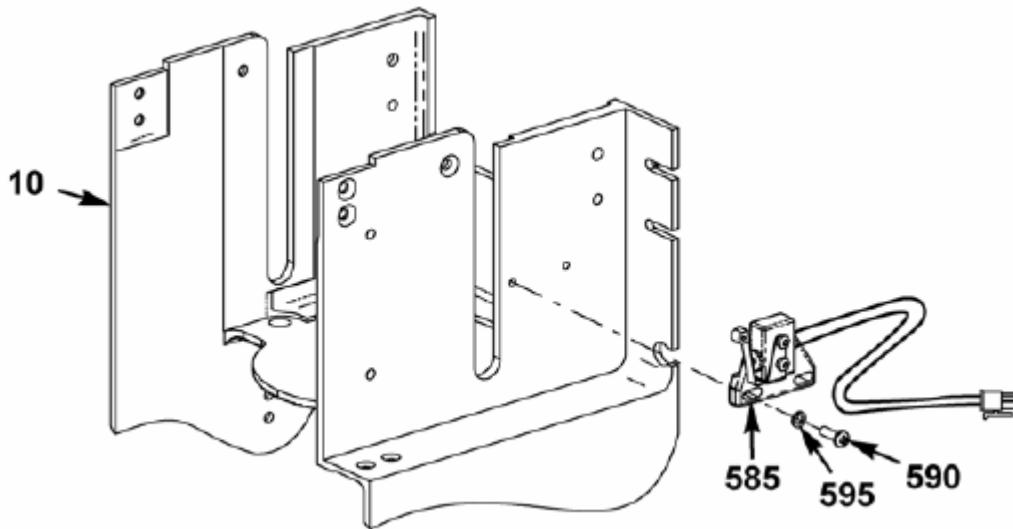


Figure 7025
Installation of Switch Assy

- (6) Attach the switch assy to the chassis assy (10 or 10A) (refer to Figure 7025).
 - (a) Put the harness assy (65) and wire for the switch assy (585) into the grommet (67), and then put the grommet into the slot in the switch cover (685).
 - (b) Attach the switch assy (585) to the right side of the chassis assy (10 or 10A) using two lock washers (595) and two screws (590).

NOTE: The screws holding the switch assy on the chassis assy should be tightened only after the switch actuator is correctly aligned with the guide on the brewhead assy (refer to FITS AND CLEARANCES).

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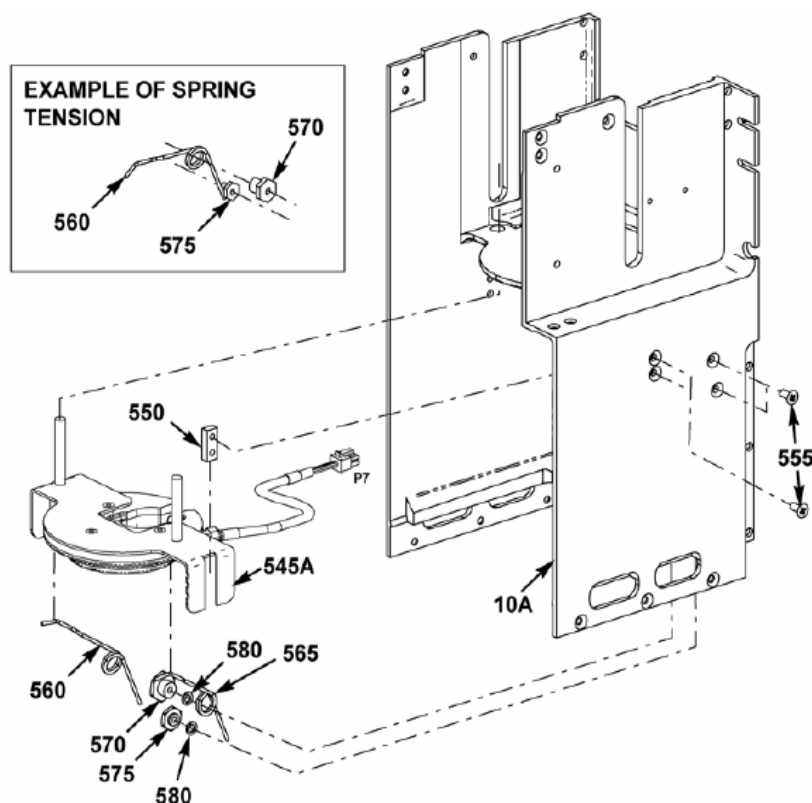


Figure 7026
Installation of Server Retainer Assy

- (7) Install the server retainer assy (545A) on the chassis assy (10 or 10A) (refer to Figure 7026).

NOTE: Server retainer assy (545) is replaced by new server retainer assy (545A) per Service Bulletin 4510-25-03.

- (a) Put the posts for the server retainer assy (545A) into holes in the chassis assy (10 or 10A).
- (b) Install two LLS guide assy (550) on the chassis assy (10 or 10A) using four screws (555).
- (c) Install two spring guide bushings (575) on the chassis assy (10 or 10A) using two lock washers (580) and two screws (555).

CAUTION: THE TWO SERVER RETAINER SPRINGS MUST BE PUT UNDER TENSION DURING INSTALLATION, SO THAT THEY HOLD UP THE SERVER RETAINER ASSY CORRECTLY.

- (d) Install two server retainer springs (560, 565), on the chassis assy (10 or 10A) using two spring bushings (570), two lock washers (580) and two screws (555). Put the bottom legs of the server retainer springs (560, 565) in front of the spring guide bushings (575) (refer to Figure 7026).

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- (e) Install two spring guide bushings (575) on the chassis assy (10 or 10A) using two lock washers (580), and two screws (555).
 - (f) Connect the plug for the server retainer assy (545A, per Service Bulletin 4510-25-03) to the controller board (75). The cover (45) of the power module assy (55) may need to be removed to get access to the controller board assy (75).
- (8) Install the faucet assy (270) or swivel faucet assy (275) on the chassis assy (10 or 10A) (refer to Figure 7027).

Table 7002
 Faucet or Swivel Faucet Assy PN Reference

UNIT PN	FAUCET OR SWIVEL FAUCET ASSEMBLY PN	DESCRIPTION
4510-XXXF-XX*	3520-0525-01	Standard Faucet Assy
4510-XXXG-XX*	3520-0559-01	Swivel Faucet Assy

*X represents other numbers or letters in the Unit Part Number

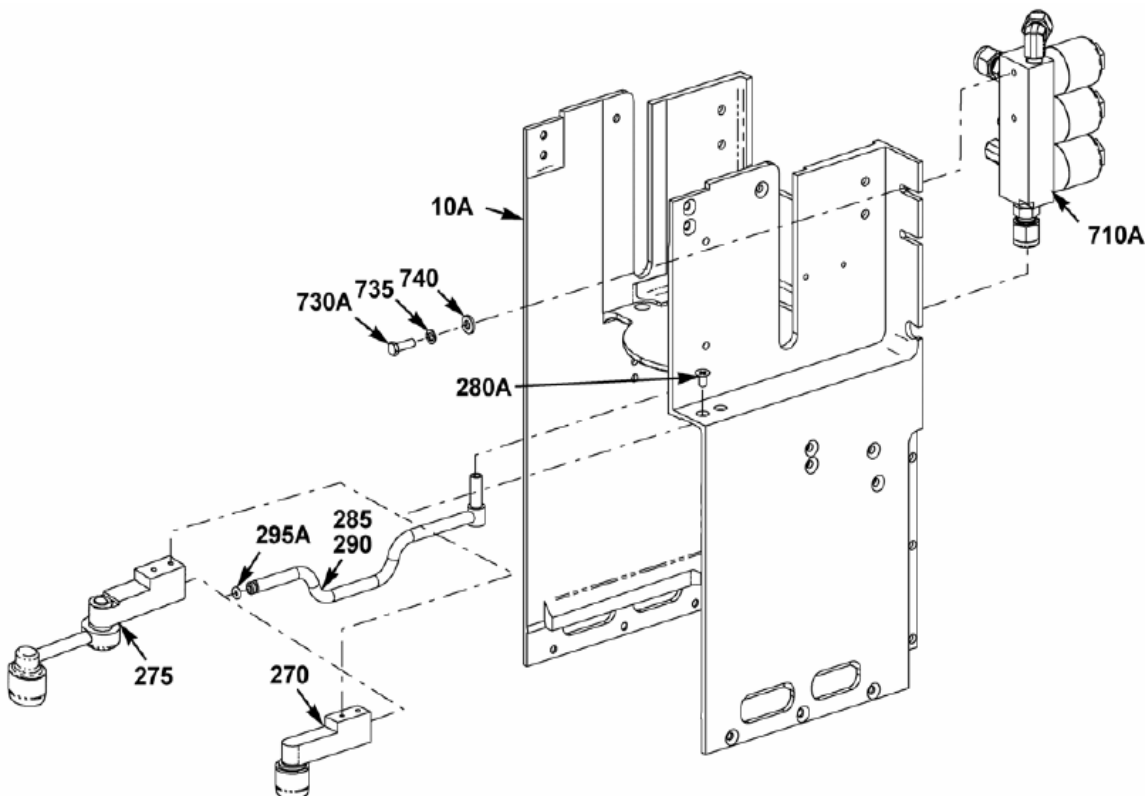


Figure 7027
 Installation of Faucet Assy or Swivel Faucet Assy

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- (a) Refer to ASSEMBLY, Paragraph 2.G. for assembly of swivel faucet assy (275) or, Paragraph 2.H. for assembly of faucet assy (270).
- (b) Install o-ring (295A) on faucet tube assy (285 or 290).
- (c) Install the faucet tube assy (285 or 290) on swivel faucet assy (275) or faucet assy (270).
- (d) Install faucet tube assy (285 or 290) on manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A).
- (e) Attach the swivel faucet assy (275) or faucet assy (270) on chassis assy (10 or 10A) using two screws (280).

NOTE: The switch cover (685) must be removed to get access to screws (280A) that attach faucet assy (270) or swivel faucet assy (275) to the chassis assy (10 or 10A).

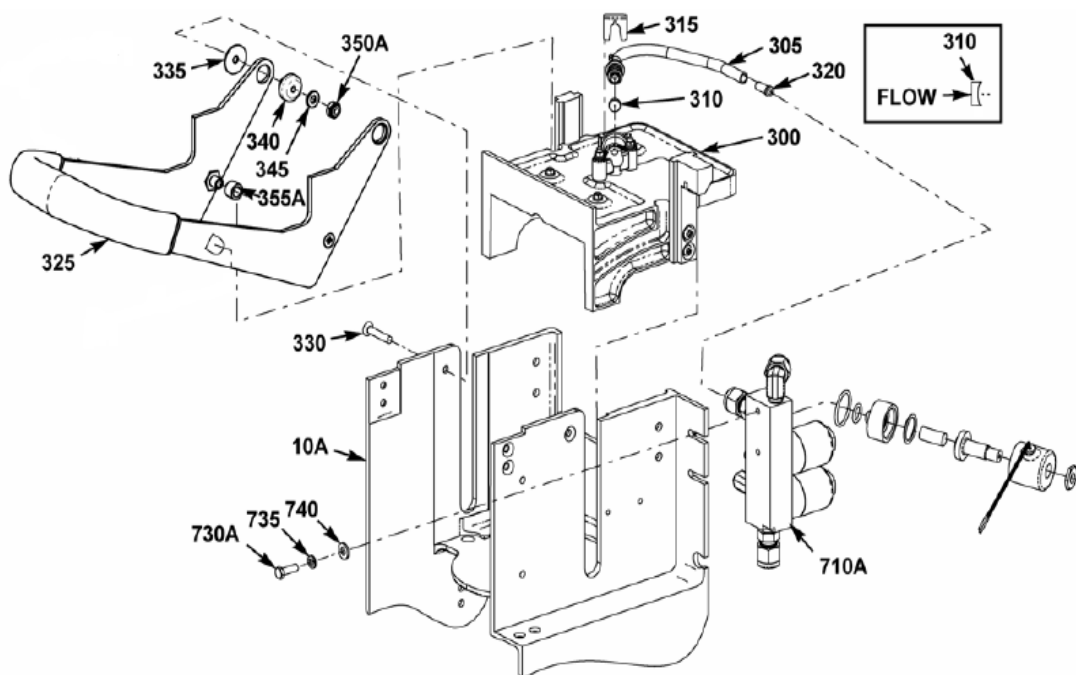


Figure 7028
Installation of Brewhead Assy

- (9) Attach the brewhead assy (300) to the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A) and chassis assy (10 or 10A) and brew handle (325) to the chassis assy (10 or 10A) (refer to Figure 7028).
- (a) Refer to ASSEMBLY, Paragraph 2.F. for assembly of brewhead assy (300).

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CAUTION: CORRECTLY INSTALL FLOW WASHER INTO BREWHEAD FITTING WITH FLAT SIDE OF THE FLOW WASHER FACING THE DIRECTION FROM WHICH THE WATER FLOWS (I.E, CONCAVE SIDE SHOULD FACE INWARDS INTO BREW HEAD FITTING). IN THE EVENT THAT THE FLOW WASHER IS NOT INSTALLED CORRECTLY, FLOW RATE AND BEVERAGE QUALITY WILL BE AFFECTED.

- (b) Install flow washer (310) into brewhead fitting, which is located on top of brewhead assembly (300), with concave side facing inward. Install brew tube assembly (305) into brewhead fitting and secure with clip (315).
- (c) Remove nut and ferrule from compression fitting on manifold assembly (710) and slide onto brew tube assembly. Install tube insert (320) into tube assembly (305) and secure brew tube assembly (305) to manifold assembly fitting by tightening compression nut.
- (d) Put two bushings (355A) on brew handle (325).
- (e) Put a small amount of petroleum jelly on the sides of guides on brewhead assy (300), and in the channel in brewhead assy for the two brew handle bushings (355A).
- (f) Put brew handle (325) on brewhead assy (300).
- (g) Put brew handle (325) and brewhead assy (300) on chassis assy (10 or 10A).

NOTE: The guides on the brewhead assy (300) are put into the channels in the chassis assy (10 or 10A) that are coated with petroleum jelly.

NOTE: Screws (20) holding the tank assy (15) on the chassis assy (10 or 10A) will be covered when the brewhead assy (300) is put on the chassis assy. The tank assy (15) should be attached to the chassis assy (10 or 10A) before the brewhead assy (300) is attached.

- (h) Attach the brew handle (325) to the chassis assy (10 or 10A) using two screws (330), two handle washers (335), brew handle (325), two threaded bushings (340), two flat washers (345) and two lock nuts (350A).
 - (i) Put insert (320) into brew tube assy (305).
 - (j) Attach brew tube assy (305) to the manifold assy (710, 710A, 715, 715A, 720, 720A, 725, or 725A).
- (10) Attach the carrying handle (360) to the chassis assy (10 or 10A) (refer to Figure 7029).

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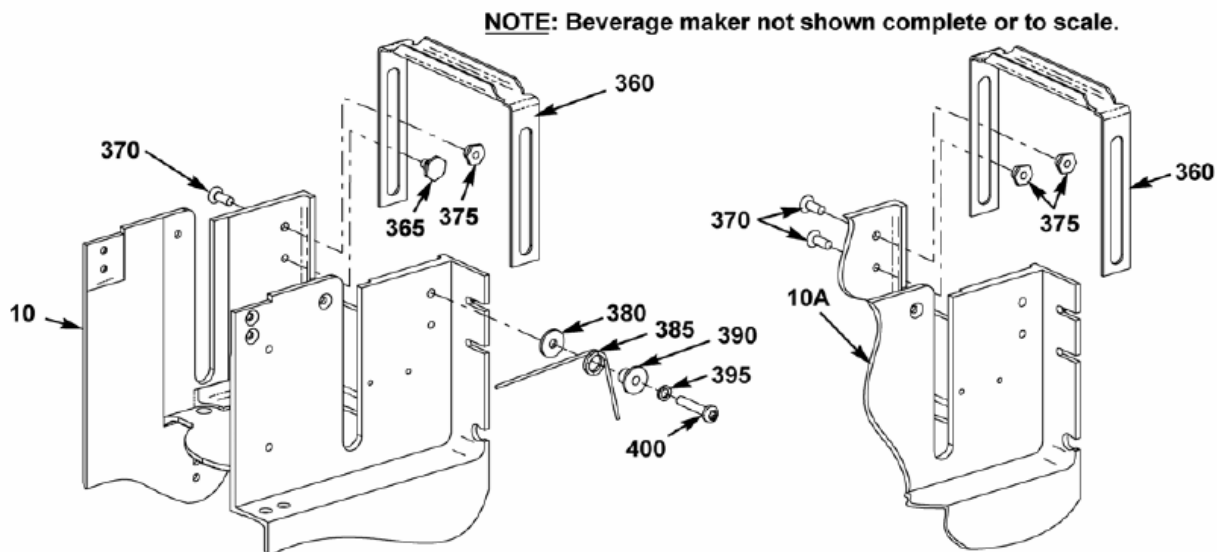


Figure 7029
 Installation of Carrying Handle

- (a) Attach the carrying handle (360) to the chassis assy (10) using two guide screws (365) or attach the carrying handle to the chassis assy (10A) (units manufactured after 2006/01/20) using three screws (370) and three handle bushings (375).
- (b) Attach the carrying handle (360) to the left side of the chassis assy (10) using a screw (370) and handle bushing (375) (on units manufactured before 2006/01/20).
- (c) Attach the carrying handle (360) to the right side of the chassis assy (10 or 10A) using a screw (400), lock washer (395), spring retainer (390), tension spring (385), metallic washer (380), and handle bushing (375).

NOTE: The end of the tension spring (385) must be placed in the channel in the guide on the right side of the brewhead assy.

Table 7003
 Front Panel Assembly Part Numbers

UNIT PN	FRONT PANEL ASSEMBLY PN
4510-X1XX-XX*	3520-0560-05
4510-X2XX-XX*	3520-0560-03
4510-X3XX-XX*	3520-0560-09
4510-X4XX-XX*	3520-0560-03
4510-X5XX-XX*	3520-0560-07

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Table 7003 (Cont.)
 Front Panel Assembly Part Numbers

UNIT PN	FRONT PANEL ASSEMBLY PN
4510-X6XX-XX*	3520-0560-01
4510-X7XX-XX*	3520-0560-03
4510-X8XX-XX*	3520-0560-01

*X represents other numbers or letters in the Unit Part Number

- (11) Attach the front panel assy (430, 435, 440, 445, or 450) and switches to the chassis assy (10 or 10A) (refer to Figure 7030).
- (a) Attach the chassis brace assy (420) to the chassis assy (10 or 10A) using four screws (425).

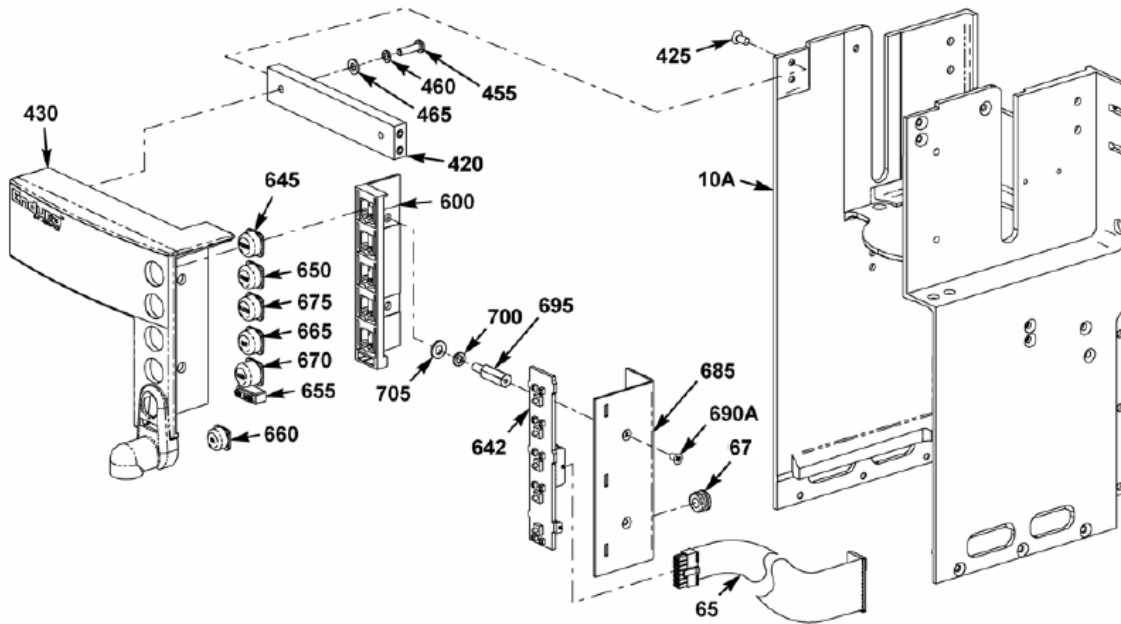


Figure 7030
 Installation of Front Panel Assy

- (b) Put the buttons (645, 650, 660, 665, 670, or 675), where options are installed, in the button frame (600) (refer to Figures 7030 and 7031).

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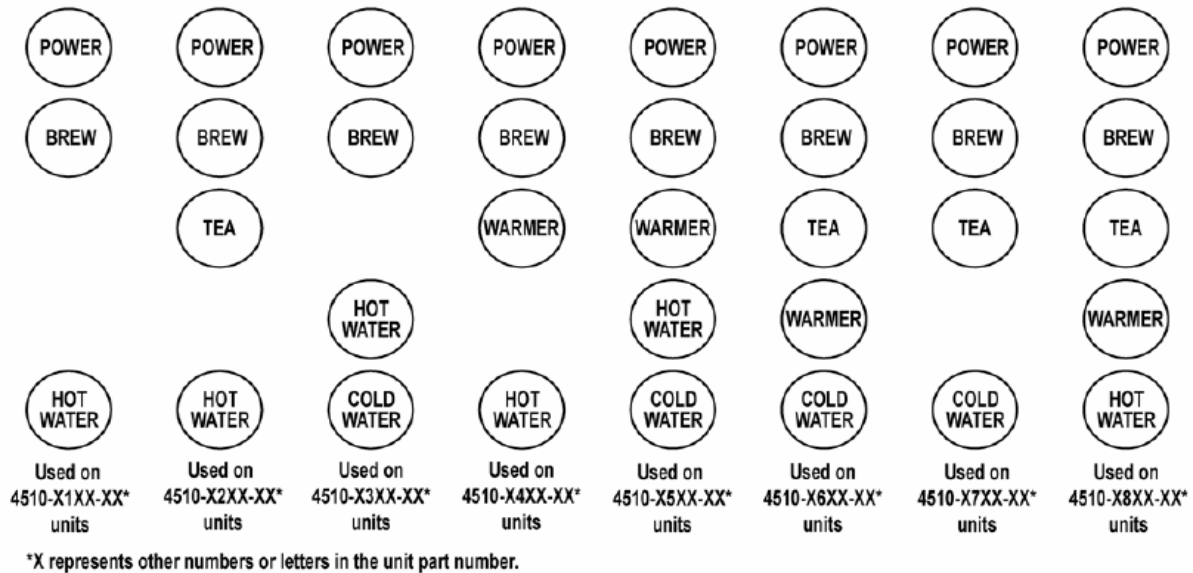


Figure 7031
 Installation of Buttons in Button Frame

- (c) Put the indicator (655) in the front panel assy (430, 435, 440, 445, or 450) (refer to Figures 7030 and 7032).

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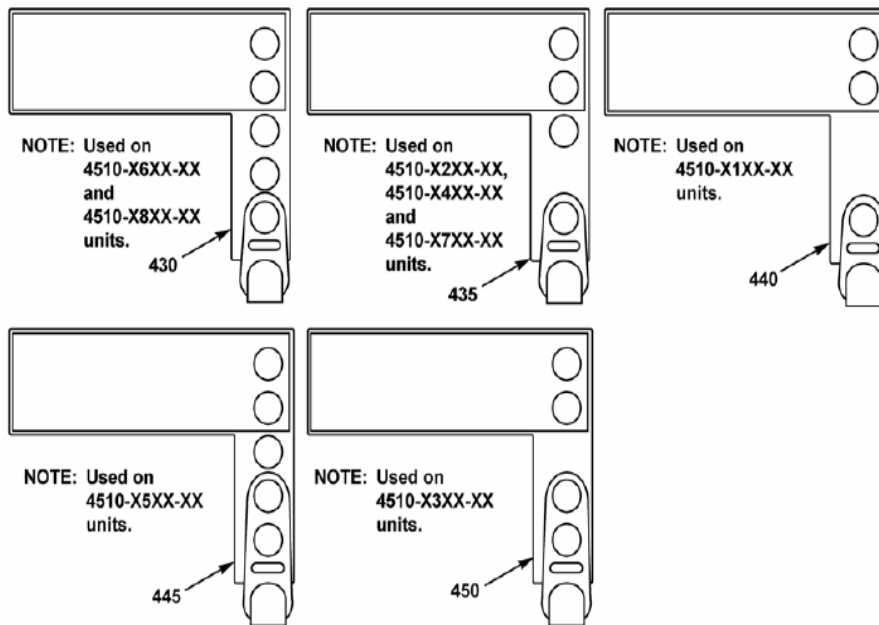
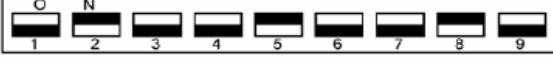
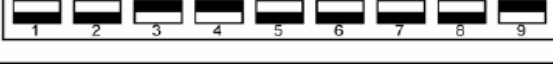
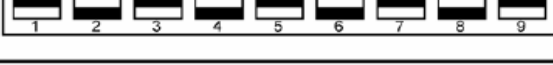


Figure 7032
Identification of Front Panel Assy

- (d) Put the front panel assy (430, 435, 440, 445, or 450) over the brew handle (325), and faucet assy (270) or swivel faucet assy (275).
- (e) Attach the button frame (600) and front panel assy (430, 435, 440, 445, or 450) to the chassis assy (10 or 10A) using two flat washers (705), two lock washers (700), and two standoffs (695).
- (f) Attach the front panel assy (430, 435, 440, 445, or 450) to the chassis brace assy (420) using two flat washers (465), two lock washers (460), and two screws (455).
- (g) Put harness assy (65) into grommet (67).

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SWITCH ASSEMBLY PN	BEVERAGE MAKER OPTION (FOR REFERENCE ONLY)	DIP SWITCH CONFIGURATION
3520-0556-01	POWER, BREW, HOT WATER	
3520-0556-02	POWER, BREW, TEA, HOT WATER	
3520-0556-03	POWER, BREW, COLD WATER, HOT WATER	
3520-0556-04	POWER, BREW, WARMER, HOT WATER	
3520-0556-05	POWER, BREW, WARMER, COLD WATER, HOT WATER	
3520-0556-06	POWER, BREW, TEA, WARMER, COLD WATER	
3520-0556-07	POWER, BREW, TEA, COLD WATER	
3520-0556-08	POWER, BREW, TEA, WARMER, HOT WATER	

NOTE: Black = Switch position. For example, if bottom half of box is filled, then switch is in down or "off" position.

Figure 7033
Position of DIP Switches

CAUTION: WHEN TOUCHING THE SWITCH CONTROLLER BOARD USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE. USE ONLY A NON-CONDUCTIVE TOOL TO MOVE THE DIP SWITCHES.

- (h) Use a non-conductive tool to put DIP switches on switch assembly (pc board) (642) in correct positions for options on beverage maker (refer to Figure 7035).
- (j) Attach plugs on harness assy (65) to configured switch assembly (605A, 610A, 615A, 620A, 625A, 630A, 635A, or 640A).
- (k) Attach configured switch assembly (605A, 610A, 615A, 620A, 625A, 630A, 635A, or 640A) to button frame (600). Tabs on the switch assembly (642) go into slots in the button frame (600).

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- (l) Put harness assy (65) and grommet (67) into slot on switch cover (685).
 - (m) Push wires under the switch cover (685) then put tabs on the switch assembly (642) in slots in the switch cover.
 - (n) Attach switch cover (685) to two standoffs (695) using two screws (690A).
- (12) Attach platen assy (185, 195) to chassis assy (10 or 10A) (refer to Figure 7036).

Table 7004
Platen Assembly Part Number Reference

UNIT PN	PLATEN ASSEMBLY PN
4510-X1XX-XX*	3520-0561-01 (no warmer)
4510-X2XX-XX*	3520-0561-01 (no warmer)
4510-X3XX-XX*	3520-0561-01 (no warmer)
4510-X4XX-XX*	3520-0561-03 (used with warmer)
4510-X5XX-XX*	3520-0561-03 (used with warmer)
4510-X6XX-XX*	3520-0561-03 (used with warmer)
4510-X7XX-XX*	3520-0561-01 (no warmer)
4510-X8XX-XX*	3520-0561-03 (used with warmer)

*X represents other numbers or letters in the Unit Part Number

- (a) If removed, install platen inlay (205) on the platen assembly (185, 195) as follows:

WARNING: USE THE APPROVED HAND AND EYE PROTECTION WHEN YOU APPLY THE LOCTITE-5900.

- 1 Apply loctite-5900 (refer to Table 7001) on the platen assembly (185, 195) using a syringe (refer to Table 7001), (refer to Figure 7034).

- 2 Place platen inlay (205) to the platen assembly (185, 195).

NOTE: Make sure to install the platen inlay (205) within 15 minutes of loctite-5900 application.

- 3 Keep platen inlay (205) in position using clamps (refer to Table 7001) and let it cure for 48 hours (refer to Figure 7035).

- 4 After curing, remove the excess loctite-5900 from the platen assembly (185, 195) using soft-cloth (refer to Table 7001).

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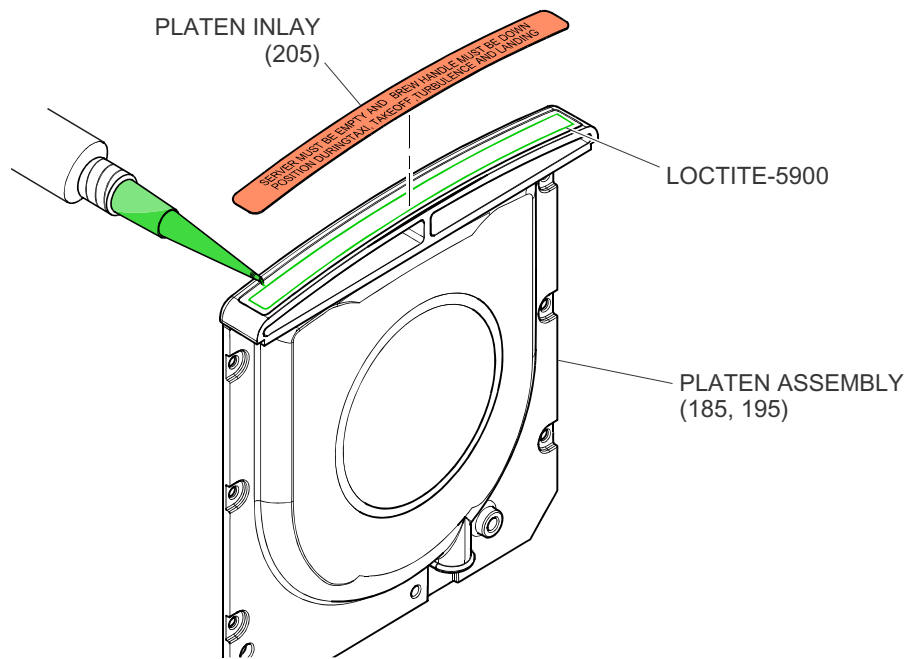


Figure 7034
Application of Loctite-5900

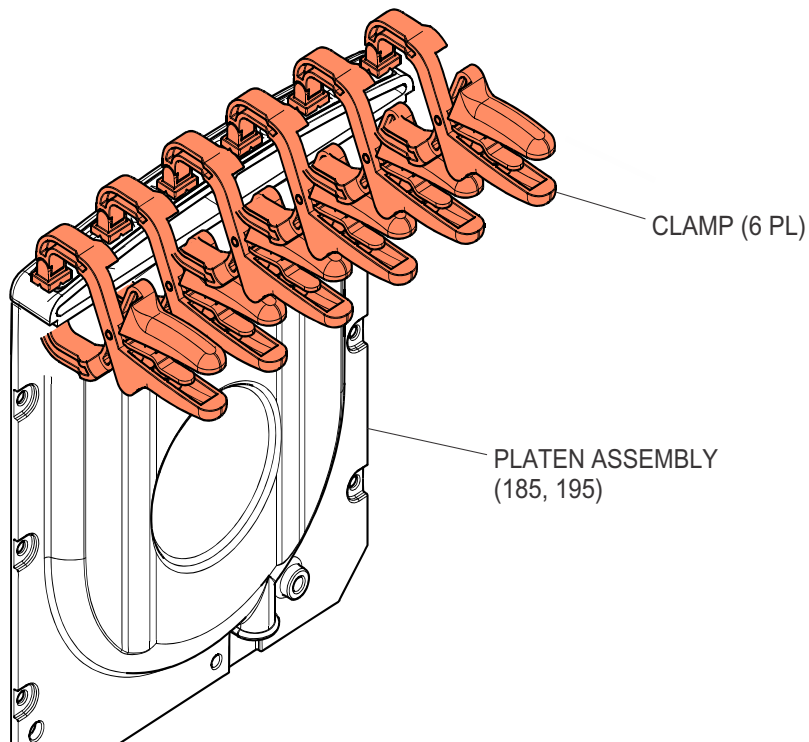


Figure 7035
Installation of Platen Inlay

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- (b) Where this option is installed, attach drain tube (770) to platen assy (185 or 195) using black adhesive sealant and a screw (775).

NOTE: The platen assy (185 or 195) may need to be installed on the chassis assy (10 or 10A) to attach the screw (775) for the drain tube (770).

Table 7005
Platen Drain Part Number Reference

UNIT PN	PLATEN DRAIN
4510-1XXX-XX*	Sump drain, no drain tube required
4510-2XXX-XX*	Rear drain, drain tube (770) required

*X represents other numbers or letters in the Unit Part Number

- (c) Attach drain screen (260) to the platen assy (185 or 195) using drain screen fastener (265A).
- (d) Put o-ring (215) in slot on platen assy (195), where platen heater (210) option is installed (refer to Table 7005).

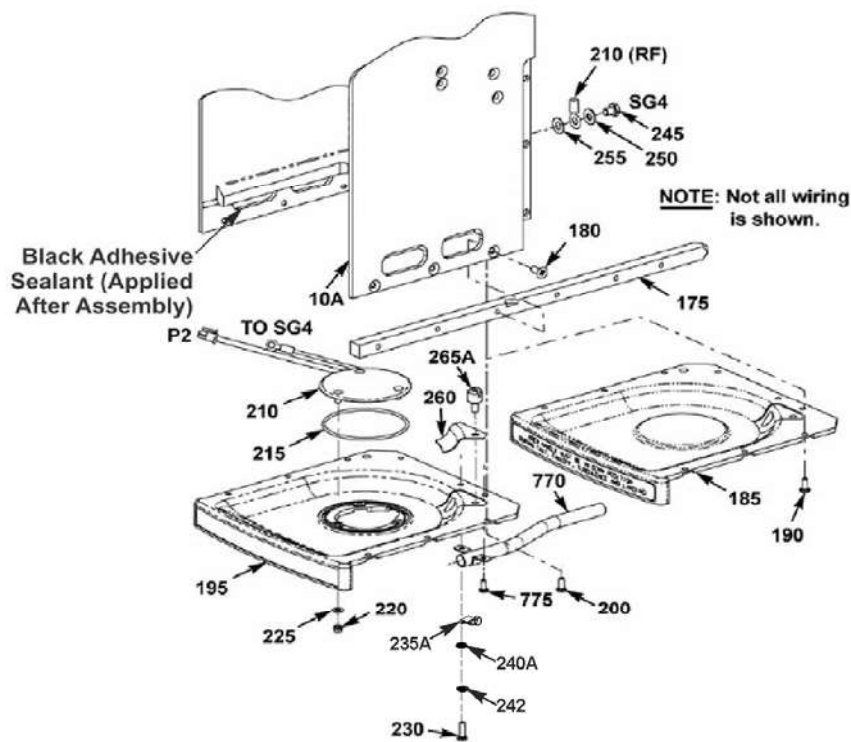


Figure 7036
Installation of Platen Assembly

- (e) Where this option is installed, attach platen heater (210) to platen assy (195) using three flat washers (225) and three lock nuts (220).

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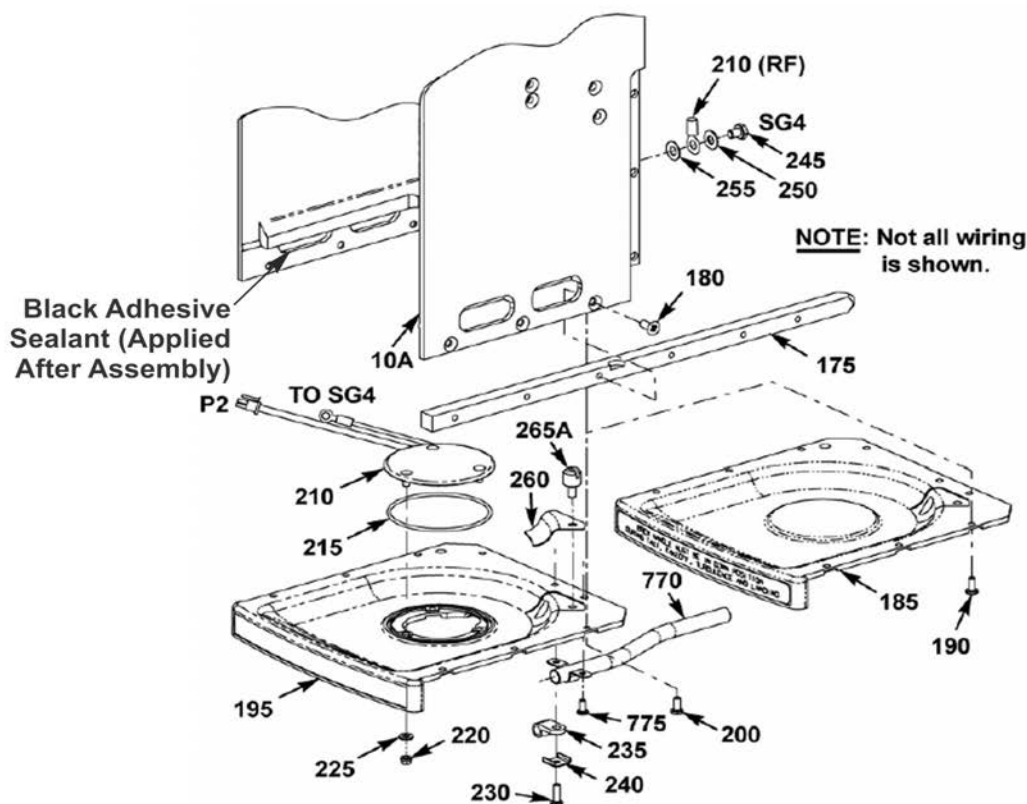


Figure 7036
 Installation of Platen Assembly

- (e) Where this option is installed, attach platen heater (210) to platen assy (195) using three flat washers (225) and three lock nuts (220).
- (f) Attach platen assy (185, 195) to the chassis assy (10 or 10A) using screws (190).

Figure 7037
Installation of Wire Clamp

- (g) Where this option is installed, attach platen heater (210) wiring to chassis assy (10 or 10A) and platen assy (195) using wire clamp (235), washer (240), lock washer (242), and screw (230) (refer to Figures 7033 and 7036).
- (h) Where this option is installed, attach platen heater (210) ground wire to chassis assy (10 or 10A) using a flat washer (255), wire terminal, flat washer (250), and hex screw (245).

NOTE: Apply alodine (refer to Table 7001) on flat washer (255) prior to installation.

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CAUTION: MAKE SURE WIRES IN WIRE CLAMP ARE SIDE BY SIDE AND NOT ON TOP OF EACH OTHER (REFER TO FIGURE 7037).

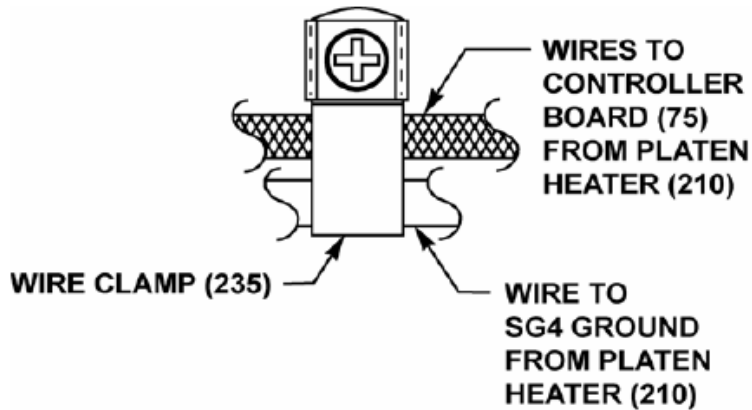


Figure 7037
Installation of Wire Clamp

- (g) Where this option is installed, attach platen heater (210) wiring to chassis assy (10 or 10A) and platen assy (195) using wire clamp (235), washer (240), and screw (230) (refer to Figures 7033 and 7036).
- (h) Where this option is installed, attach platen heater (210) ground wire to chassis assy (10 or 10A) using a flat washer (255), wire terminal, flat washer (250), and hex screw (245).

NOTE: Apply alodine (refer to Table 7001) on flat washer (255) prior to installation.

- (j) Attach the platen heater (210) plug to its connection on the controller board (75). The power module assy (55) may need to be installed for correct routing of the wiring to the controller board (75).

NOTE: The guide rails (175) and screws (180) to the chassis assy (10 or 10A) and the support connector(115) will have to be installed after the platen assy (185 or 195) is installed on the chassis assy (10 or 10A).

- (k) Apply a bead of black adhesive sealant (refer to Table 7001) to the edge of the chassis assembly (10 or 10A) where it touches the platen assembly (185, 195).

- (13) Install the power module assy (55) on the chassis assy (10 or 10A, IPL Figure 2).

CAUTION: WHEN TOUCHING THE POWER CONTROLLER BOARD (75) OR ANY OF ITS INDIVIDUAL PARTS, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

- (a) Install controller board assy (75) in power module assy (55) using five screws (80) and six shoulder washers (85). One shoulder washer (85) needs to be centered on the back of the controller board, where there will be no screw hole, and no interference with board circuitry (refer to ASSEMBLY, Paragraph 2.K. for assembly of controller board assy (75)).

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- (b) Put two screws (110) and two flat washers (105) into the power module assy (55).

CAUTION: PUT THE SIDE OF THE RELAY RETAINER (100) THAT HAS VISIBLE THREADS IN THE HOLES NEAR THE RELAY, AND THE SIDE THAT SHOWS THE HEXAGON OF THE THREADED INSERT SHOULD FACE AWAY FROM THE RELAY. DAMAGE TO THE RELAY CAN RESULT FROM INSTALLATION WITH THE RELAY RETAINER (100) HEXAGONS OF THE FASTENERS FACING TO THE RELAY.

- (c) Before inserting the screws (110) place a small amount of thread lock AD-300 on the threaded holes in the relay retainer (100).

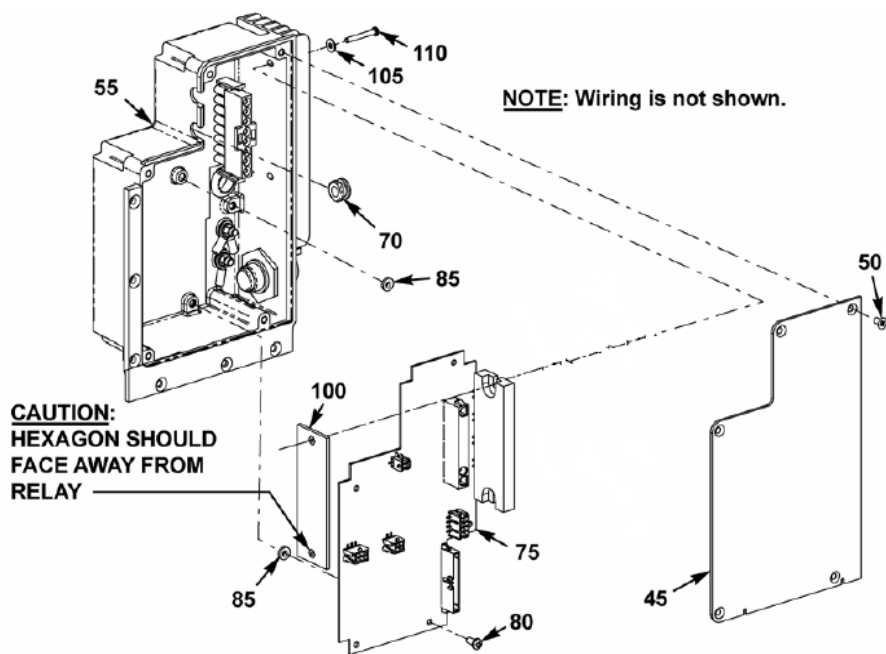


Figure 7038
Installation of Controller Board Assy

- (d) Evenly pull the relay to the side of the power module assy (55) by tightening the screws (110).

CAUTION: DO NOT TIGHTEN SCREWS TOO MUCH. THE RELAY CAN BE DESTROYED IF SCREWS ARE PUT ON TOO TIGHT.

- (e) Use a torque wrench to tighten each screw (110) to one inch-pound force (0.11 newton-meter) of torque. Then tighten each screw 110) to two inch-pounds force (0.22 newton-meter) of torque.
- (f) Attach the power module assy (55) to the chassis assy (10 or 10A) using three screws (60) and thread seal adhesive.
- (g) Attach the power module assy (55) to the guide rail (175) using three screws (60) and thread seal adhesive.

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NOTE: The guide rail (175) may need to be attached to the chassis assy (10 or 10A) after the platen assy (185 or 195) has been installed.

- (h) Connect all wire harness (40 and 65, IPL Figure 2 and 15, IPL Figure 4) plugs to sockets on the controller board assy (75).
- (j) Put grommets (70) on the wire harnesses (40 and 65, IPL Figure 2 and 15, IPL Figure 4) and wires from equipment options and put grommets in slots in power module assy (55) housing. Put open side of grommet near the side of the power module assy (55).

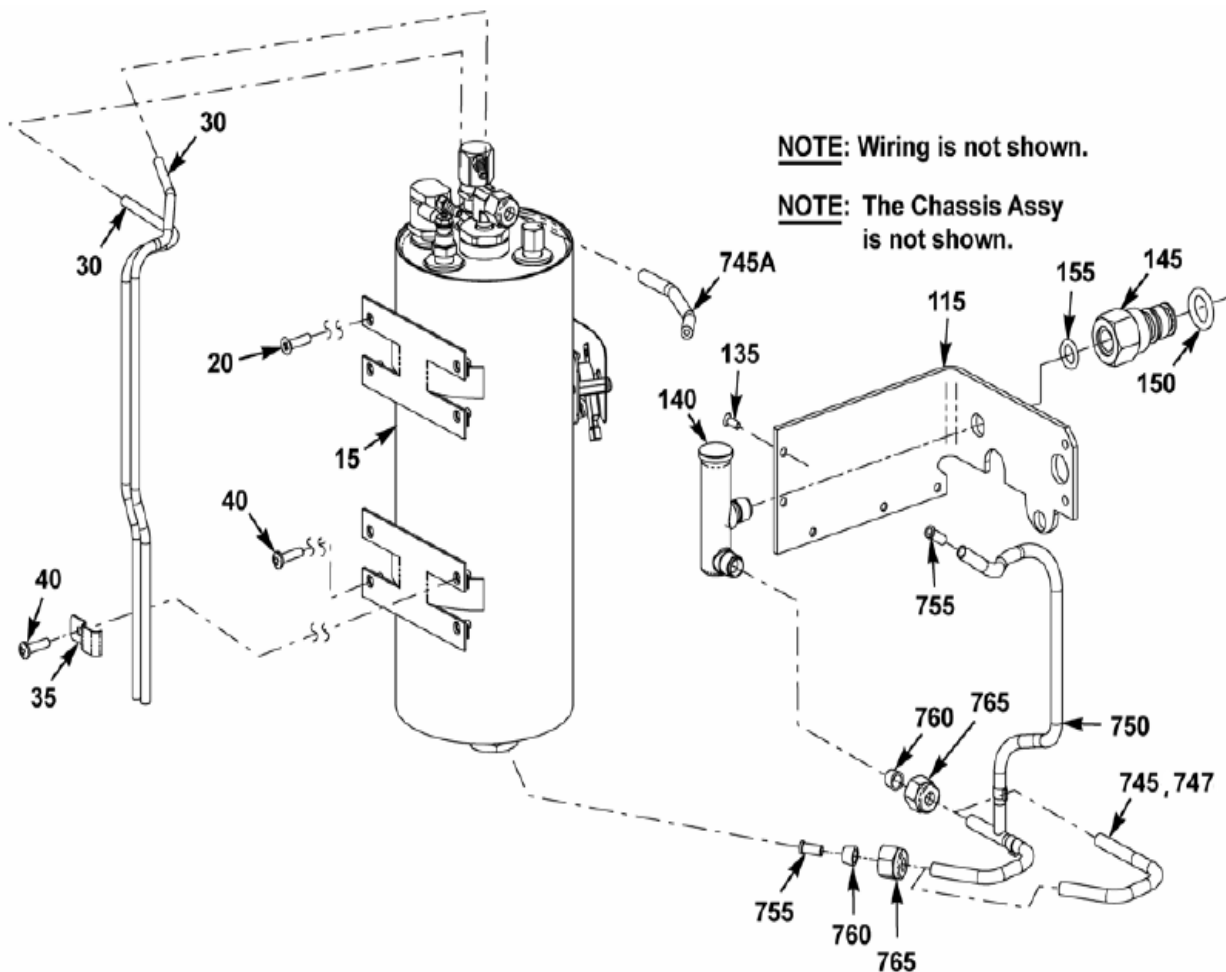


Figure 7039
 Installation of Tank Assy

- (14) Attach the tank assy (15) to the chassis assy (10 or 10A) (refer to Figure 7039).

NOTE: The brewhead assy (300) needs to be attached to the chassis assy (10 or 10A) after the tank assy (15) is installed to get access to screws (20) on the upper part of the chassis assy (10 or 10A) for the tank assy (15). Refer to ASSEMBLY, Paragraph 2.F. to install the brewhead assy (300).

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CAUTION: THE IDENTIFICATION PLACARD MUST BE KEPT WITH THE BEVERAGE MAKER WHEN THE SCREWS HOLDING THE PLACARD AND TANK ASSY ON THE CHASSIS ASSY (10) ARE REMOVED. THIS ID PLACARD IS NOT AVAILABLE AS A SPARE PART.

- (a) Attach the tank assy (15) to the chassis assy (10 or 10A) using four pan head screws (45, IPL Figure 1 and 40, 25 IPL Figure 2 and Figure 7039) and four flat head screws (20). Refer to ASSEMBLY, Paragraph 2.M. for assembly of the tank assy (15).

NOTE: Installation of two screws (45, IPL Figures 1, 1A, 1B, 1C) low inside the chassis assy (10 or 10A) server cavity attaches the ID placard to the chassis assy.

- (b) Put wires for RTD (190, IPL Figure 3) through grommet in side of power module assy (55) and connect plug of RTD (190, IPL Figure 3) wire to controller board (75) in the power module assy.
- (c) Attach tubes (30) to barbed fittings (170 and 240, IPL Figure 3) on the top of the tank assy (15) if replacement or cleaning was required.
- (d) Put tubes (30) through holes in chassis assy (55) and attach the tubes inside the server cavity using a tube clamp (35) and screw (40). The tubes should be no less than one inch above the drain in the platen assy (185 or 195).
- (e) Connect tube assy (745A) to fitting (90, IPL Figure 3) of water filter on top of tank assy (15).
- (f) Use solder to attach wire to left side terminal of overtemperature thermostat on tank assy (15).

CAUTION: IDENTIFY HEATER TERMINALS AND WIRE HARNESS TERMINALS THAT GO TOGETHER. THE WIRING TERMINALS MUST BE PUSHED AND TURNED TO BE ATTACHED TO EACH OTHER. THEY ARE NOT EASY TO CONNECT BY HAND, SO TOOLS MAY BE REQUIRED. DO NOT USE TOO MUCH FORCE TO HOLD OR ATTACH THE PLASTIC TERMINALS TO EACH OTHER.

- (g) Connect wire harness terminals to terminals of heater assy (15, IPL Figure 3).

CAUTION: CAREFULLY ATTACH WIRE TERMINALS TO TERMINALS OF THE OVERTEMPERATURE THERMOSTATS SO THAT THE TERMINALS DO NOT MOVE ON THE PLASTIC HOUSING OF THE THERMOSTATS. MOVEMENT OF A TERMINAL CAN CRACK THE THERMOSTAT HOUSING THAT CAN RESULT IN FAILURE OF THE THERMOSTAT.

- (h) Attach three quick connect wiring terminals to terminals of overtemperature thermostats (55A, IPL Figure 3) on the tank assy (15).
- (i) Attach ring terminal to hex nut on low water sensor (185, IPL Figure 3) on the top of the tank assy (15) using a hex nut on low water sensor.

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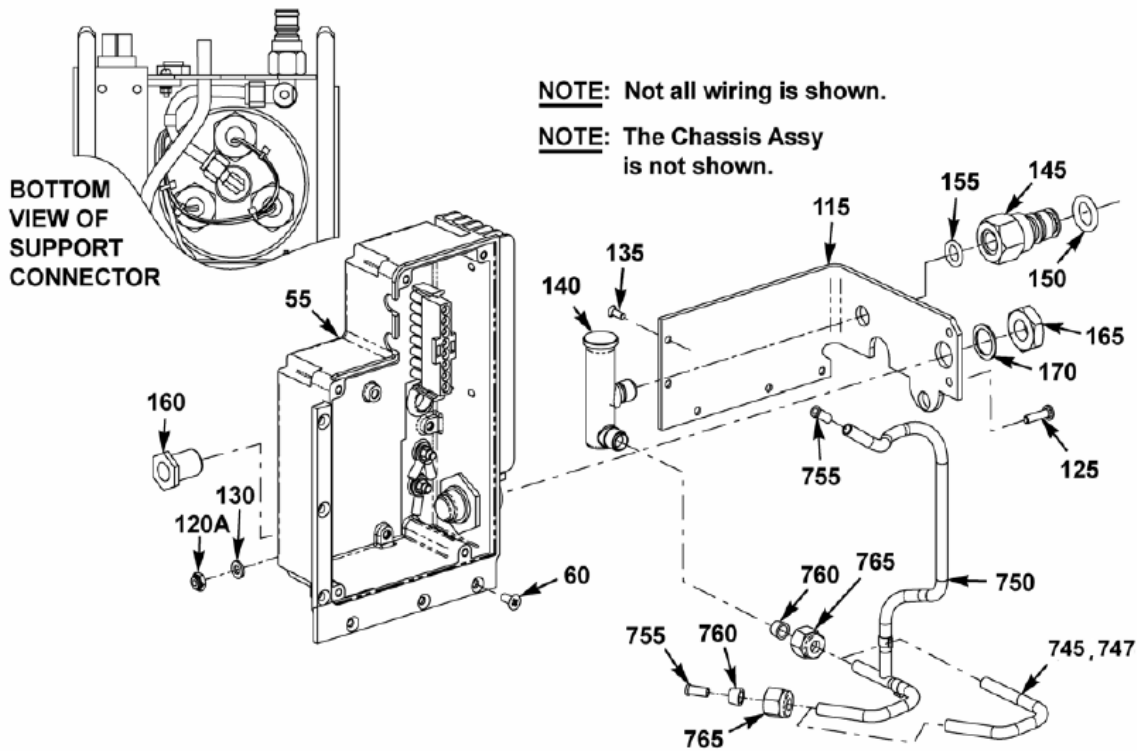


Figure 7040
 Installation of Support Connector

- (15) Attach support connector(115) to chassis assy (10 or 10A).
 - (a) Refer to ASSEMBLY, Paragraph 2.J. for assembly of the drain valve assy (140).
 - (b) If removed, install o-ring (150) on water shutoff plug (145).
 - (c) Attach the drain valve assy (140) to the support connector (115) using an o-ring (155) and water shutoff plug (145).
- CAUTION:** CAREFULLY PUT THE SUPPORT CONNECTOR ON THE REAR DRAIN TUBE (IF OPTION INSTALLED) TO AVOID DAMAGE TO THE DRAIN TUBE.
- (d) Attach the support connector (115) to the left side rail guide (175) using three screws (135) and thread seal adhesive.
 - (e) Attach the support connector (115) to the left side of the chassis assy (10 or 10A) using two screws (135) and thread seal adhesive.
 - (f) Attach the support connector (115) to the power module assy (55) using screw (125), flat washer (130), and lock nut (120A).
 - (g) Attach the alignment pin guide (160) to the support connector (115) and power module assy (55) using a lock washer (170) and hexagon nut (165).

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- (h) Put the ferrule (760), fitting (765), and insert (755) on the water supply tube assy (745A or 750).
- (j) Attach the water supply tube assy (750, 747 or 745A) to the drain valve assy (140) and tank assy (15).
- (16) After necessary wire connections (refer to Figure 7014) are made to the controller board (75) in the power module assy (55), attach the power module assy cover (45) to the power module assy (55) using five screws (50) (refer to Figure 7041).

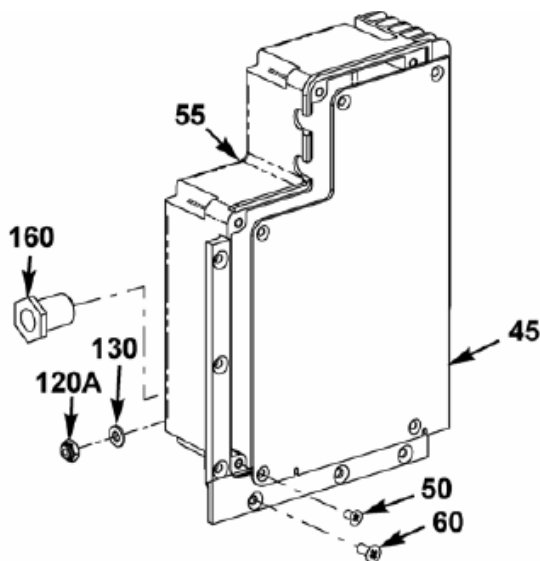


Figure 7041
Installation of Power Module Cover

- (17) Attach wire tie downs to wiring, spacing the tie downs approximately one and a half inches apart, to keep wiring from rubbing or touching heat sources (refer to FITS AND CLEARANCES, Figures 8004 and 8005 for examples of wire routing).

P. Assembly of Beverage Maker (refer to IPL Figures 1, 1A, 1B, 1C and Figure 7042).

Table 7006
Brew Cup Assembly PN Reference

UNIT PN	BREW CUP ASSEMBLY PN	DESCRIPTION
4510-XXUX-XX*	3520-0540-01	Brew cup assembly (upper)
4510-XXLX-XX*	3520-0540-03	Brew cup assembly (lower)

*X represents other numbers or letters in the Unit Part Number

- (1) Attach the spacer plate (30) to the brew cup (25A, 25B) using two screws (40) and two nuts (35).

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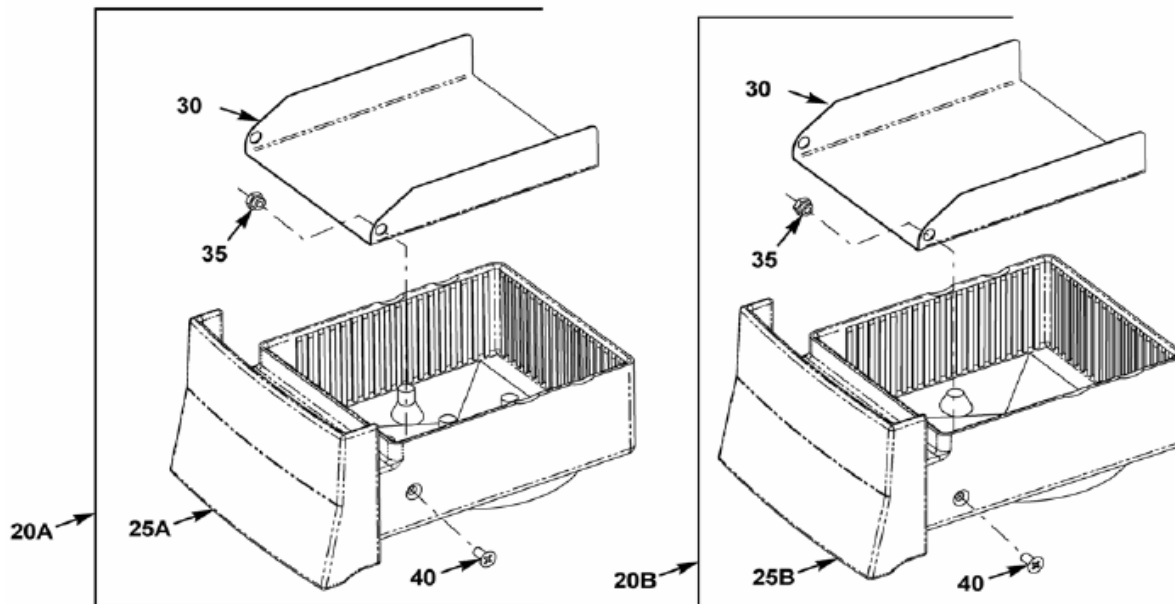


Figure 7042
Brew Cup Assembly

- (2) With the brew handle in the raised position, put the brew cup (20A or 20B) (refer to Table 7006) in the cavity of the beverage maker subassembly (10).
- (3) Refer to FITS AND CLEARANCES for adjustment of essential assemblies.
- (4) Refer to TESTING AND FAULT ISOLATION for procedures to test the beverage maker following ASSEMBLY.

FITS AND CLEARANCES

1. General

- A. Refer to the TESTING AND FAULT ISOLATION and CHECK Sections for assistance in identifying parameters and adjustments required for full function of the Beverage Maker.

2. Procedures

A. Plumbing

- (1) Do not tighten nut (holding solenoid coil on the guide assy) too much. Putting this nut on too tight can damage or hinder function of the solenoid and solenoid coil.
- (2) When putting Teflon tape on plumbing fittings:
 - (a) Make sure all threads are clean.
 - (b) Start tape below the first thread and wrap in the direction of the spiral of the threads.
 - (c) Wrap the threads with no more than two layers of tape, pull tape tight so it conforms to threads, and then cut off the tape.
- (3) Trim tubes to a flat face perpendicular to the wall of the tube. An irregular face on the tube end contributes to a seal that can fail under pressure. Align tube, insert, and ferrule carefully when tightening nut and tube on fittings to make sure there is an effective seal.
- (4) When tightening plumbing fittings that use a flow washer, o-ring, or gasket (on vent valve and pressure relief valve fittings), make sure these parts are not crushed or pinched, or an ineffective seal may result.
- (5) The flow washer (310 and 538) should be installed so the concave side faces away from water flows (refer to Figure 8001).

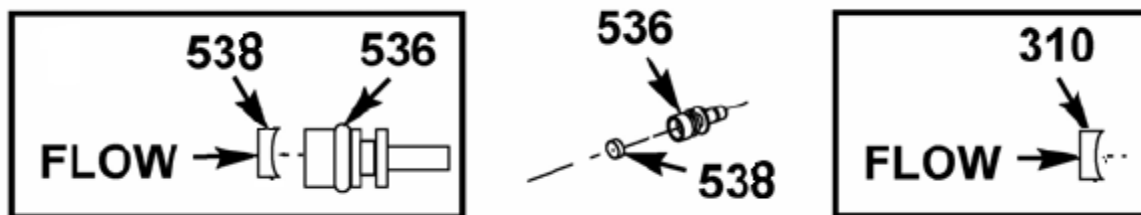


Figure 8001
Flow Washer Detail

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- (6) The drain tubes for the relief valve and the vent valve should be retained by the fastener and clamp on the back wall of the server cavity of the chassis assy. The tubing should follow a smooth curve from the tank assy valves to the back wall of the server cavity of the chassis assy, without otherwise touching the tank assy. The tubes should end no less than one inch (2.5 cm) above the platen assy to avoid backup from the drain (this is a requirement that helps avoid contamination of potable water).
- (7) The brew head tubing should not touch the chassis assy but fit a smooth curve from the manifold assy to the brew head assy.
- (8) In the event that the tubing must be replaced, it must be heat formed to fit so the tubing does not flatten or become kinked at installation.
- (9) All plumbing and tubing must fit within the envelope of the chassis assy, power module assy and tank assy, to allow the Beverage Maker to smoothly and easily fit into galley installations.

B. Mechanical and Structural

- (1) Align the switch assy (585, IPL Figures 2) with the brew head guide on the brew head assy (300). Tighten screws (590) so that the actuator wheel touches the brew head guide when the brew head assy is lowered. When the brew head assy is lowered, the guide should cause the actuator to push the button on the switch (refer to Figure 8002).

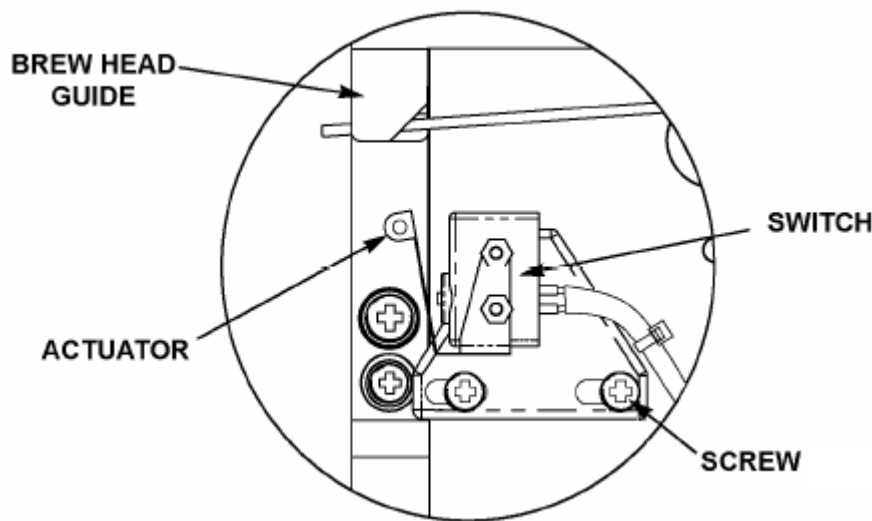


Figure 8002
Switch Assy Detail

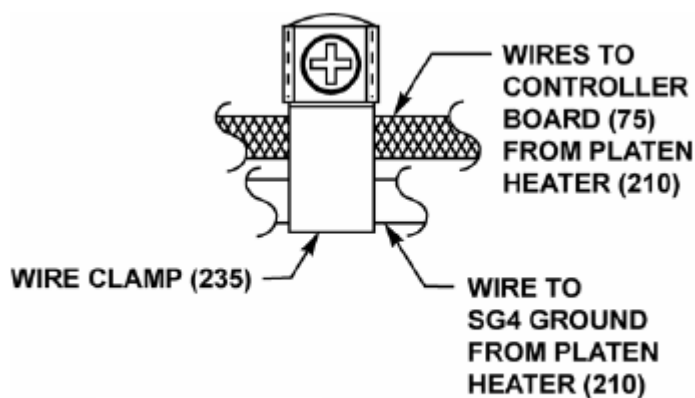
- (2) Seal all external screw heads inside the server cavity with white RTV seal.
- (3) The lockdown lever mechanisms (470 and 490, IPL Figure 2) should work smoothly and easily, without binding when being moved from either the lifted or lowered positions.
- (4) The pin on the power module assy (55, IPL Figure 2) should be at or below the surface of the outside of the connector support (115).

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- (5) All fasteners on the outer walls of the Beverage Maker should be at or below the surface of the wall of the Beverage Maker.

C. Electrical

- (1) When installing the power connector (PP1) for the wire harness on the back of the power module assy (IPL Figure 4), adequate wire length must be provided so no tension is put on the connector pins in the connector. Strained wiring can result in bent pins, hindering secure and complete electrical connection between the Beverage Maker and the rail assembly.
- (2) All wiring must fit within the external envelope of the chassis assy, power module assy, and tank assy, to allow the Beverage Maker to fit smoothly and easily into galley installations.
- (3) The screws holding the relay (on the controller board assy) to the side of the power module housing assy should not be tightened to more than 2 in-lbs (0.22 newton-meter) of torque.



- (4) The platen heater wires should point toward the rear of the Beverage Maker to allow the wiring to be fastened under the clamp on the bottom of the platen assy.

CAUTION: MAKE SURE WIRES IN WIRE CLAMP ARE SIDE BY SIDE AND NOT ON TOP OF EACH OTHER (REFER TO FIGURE 8003).

- (5) Attach nylon wire tie downs to wiring, spacing the tie downs approximately one and a half inches apart, to keep wiring from rubbing or touching heat sources (refer to Figures 8004 and 8005 for examples of wire routing). Spacing between the nylon wire tie downs is recommended to be between 0.5 and 1.5 in (1.27 and 3.81 cm).
- (6) Put wiring in grommet, install it in the slot, then turn grommet so the side opening is near the closed end of the slot, so the grommet holds wiring correctly (refer to Figure 8006).

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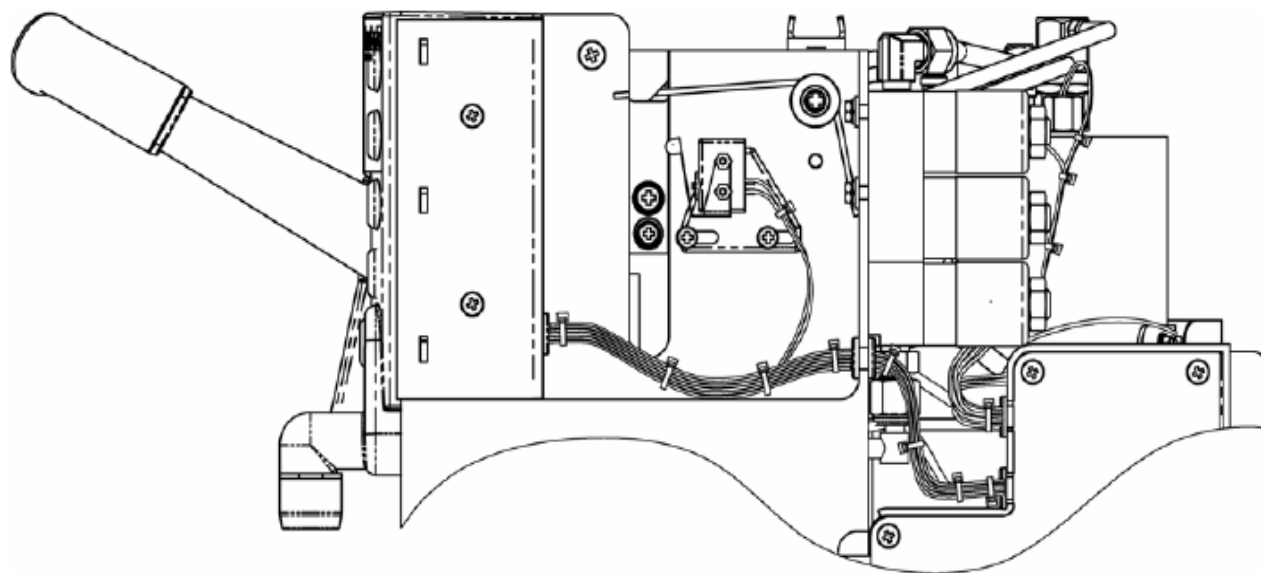


Figure 8004
Example of Wire Routing

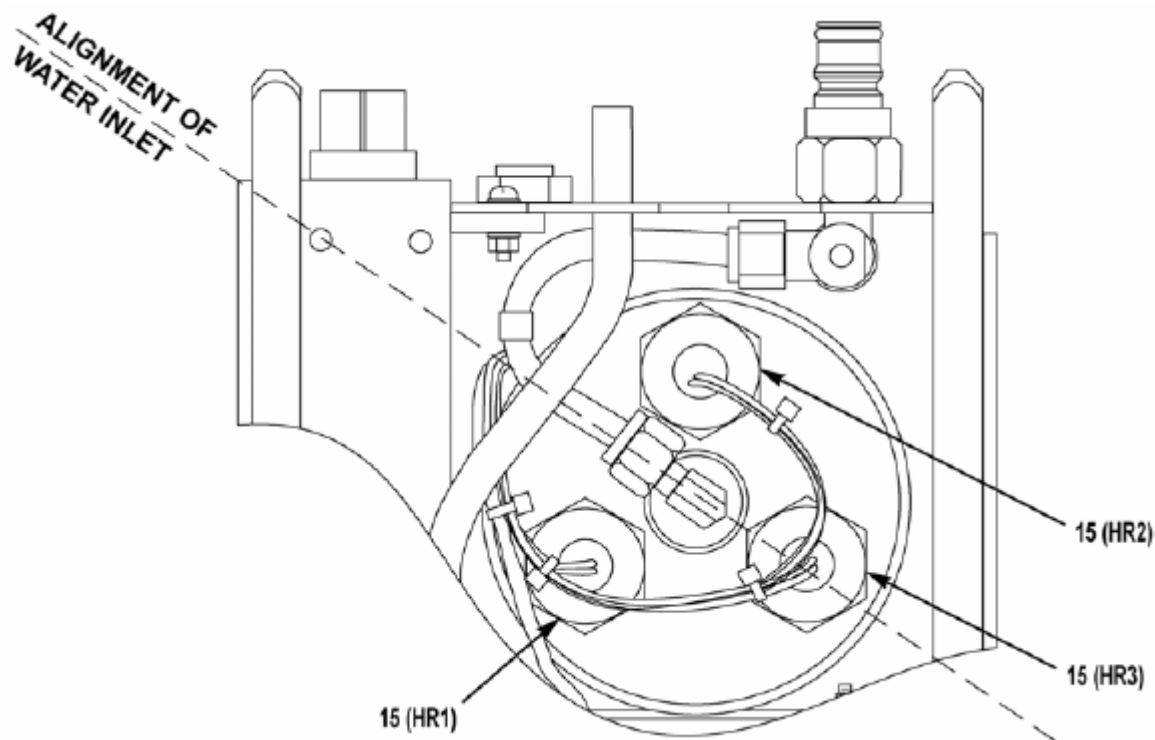


Figure 8005
Example of Heater Assy Wire Routing

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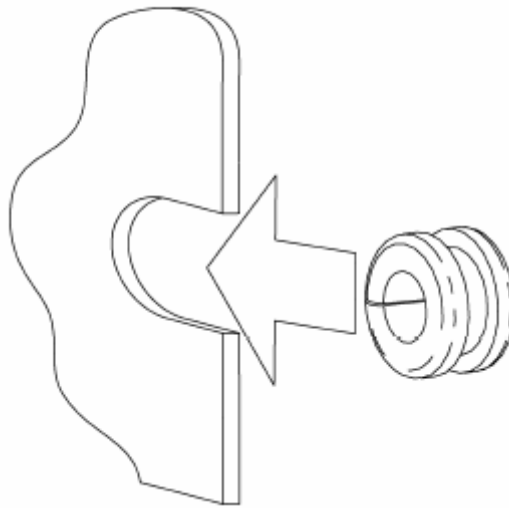


Figure 8006
Grommet Installation

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SPECIAL TOOLS, FIXTURES AND EQUIPMENT

1. General

- A. All work described in this manual may be conducted in a normal work area with ambient environmental and physical conditions and whose environment has been cleaned for Beverage Maker equipment repair.
- B. Special tools, fixtures and equipment required to perform the work function listed in all paragraphs of this manual are listed in Table 9001.

NOTE: Equivalent substitutes may be used for listed items.

Table 9001
Special Tools, Fixtures and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
1 in X 3 in Deep Socket, 1/2 in Drive for heater and water filter removal (refer to Figure 9001)	3575-1318-01	B/E Aerospace, Inc. (V63367)
1/8 inch NPT plug	AN933-1D or MIT-022P	Commercially Available or FasTest, Inc. 1646 Terrace Drive Roseville, MN 55113-1317
Alodine Chemical Coating	Alodine 1201 CM-50017	Henkel Corp, (V71410) B/E Aerospace, Inc. (V63367)
A food-grade aqueous acidic cleaner that removes calcium, lime, and rust deposits (scale) without damage to unit form, fit, or function	None	Commercially Available
Black Adhesive Sealant	732 Black RTV-103 AD-052	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)
Brush, Wire, Soft-bristled	None	Commercially Available
Clamp (Qty: 6)	None	Commercially Available
Clean Water	None	Commercially Available
Cleaner, Mild Liquid Detergent (Dish Soap)	None	Commercially Available
Cloth, Cleaning, Low-Lint, Type 1	A-A-59323	Commercially Available
Cotton Swabs	None	Commercially Available
Electrical connector cable between Beverage Maker and power supply	998552	B/E Aerospace, Inc. (V63367)

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Table 9001 (Cont.)
Special Tools, Fixtures and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Epoxy Adhesive	E-Cast F-28 Resin F-14 Hardener AD-50029	United Resin Corp. (V08LA7) B/E Aerospace, Inc. (V63367)
ESD Conductive Mat	None	Commercially Available
ESD Grounding Wrist Strap	None	Commercially Available
Flow Washer Tool (Use one flow washer tool to remove flow washer and use two tools to install flow washer) (refer to Figure 9002)	3575-1319-01	B/E Aerospace, Inc. (V63367)
Heat Sink Compound	10-8109 CM-50019	Waldom Inc. (V72653) B/E Aerospace, Inc. (V63367)
High Low Temperature RTD wiring harness with 4-position connector	997112	B/E Aerospace, Inc. (V63367)
IC (Microcontroller) Extraction Tool, 20-128 pins non-ESD pull tool (refer to Figure 9003)	EX-5	OK Industries (V9X418)
Insertion Tool (for PP1 connector) (refer to Figure 9004)	MS24256A16	Commercially Available
Isopropyl Alcohol	TT-I-735	Commercially Available
Loctite-5900	4481015-001	B/E Aerospace, Inc. (V16827)
Lubricant and Sealant, FDA	CM-50018	B/E Aerospace, Inc. (V63367)
Megohmmeter	None	Commercially Available
Milliohmmer	QuadTech Model 1880	QuadTech Inc (V0PK96)
Nylon Wire Tie Downs	MS3367-4-9	Commercially Available
Nitrogen Gas	A-A-59503	Commercially Available
Ohmmeter	None	Commercially Available
Petroleum Jelly	None CM-50015	Commercially Available B/E Aerospace, Inc. (V63367)
Pincers (for tea tube clamps) (refer to Figure 9005)	14100082 993973	Oetiker, Inc. (V25281) B/E Aerospace, Inc. (V63367)
Power Module Wrench (for nut on PP1 connector) (refer to Figure 9006)	3575-1323-03	B/E Aerospace, Inc. (V63367)

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Table 9001 (Cont.)
Special Tools, Fixtures and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Power Supply, including: 3-phase 115 ± 3 VAC, 400 Hz, 10 amps per phase; 3 ammeters (0-25 amp); and 3 voltmeters (0-250 volt) (refer to Figure 1001)	None	Commercially Available
Removal Tool (for PP1 connector) (refer to Figure 9007)	MS24256R16	Commercially Available
Server	MISC-3201	B/E Aerospace, Inc. (V63367)
Silicone Rubber Adhesive	736 Red RTV AD-50005	Dow Corning Corp. (V71984) B/E Aerospace, Inc. (V63367)
Sleeving, Non-Fray	Varglas Type HO, Size 3/8 in TU-50039	Varflex Corp. (V79074) B/E Aerospace, Inc. (V63367)
Socket Key Tool (remove plunger guide assy from solenoid valve assy body or from solenoid manifold) (refer to Figure 9008)	3575-1321-01	B/E Aerospace, Inc. (V63367)
Deleted		
Soft-Cloth	None	Commercially Available
Solder	Kester No. 245, No Clean, Low Residue SD-50003	Kester Solder (Div of Litton Systems, Inc.) (V75297) B/E Aerospace, Inc. (V63367)
Stopwatch	None	Commercially Available
Syringe	None	Commercially Available
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Terminal Extractor Tool (refer to Figure 9009)	0011030043	Molex (V27264)
Thermometer, 50-250° F (10-121° C)	None	Commercially Available
Thread Lock Adhesive	242 Blue AD-300	Loctite Corp. (V05972) B/E Aerospace, Inc. (V63367)
Thread Lubricant	Never Seez CM-50016	Bostik Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Torque Tool (handle for socket key tool) (refer to Figure 9010)	3575-1320-01	B/E Aerospace, Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)

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Table 9001 (Cont.)
Special Tools, Fixtures and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Water Supply, including: Pressure regulator (0-150 psig, 0-10.34 barg) and flow meter (0-0.5 GPM, 0-1.9 LPM) (refer to Figure 1001)	None	Commercially Available

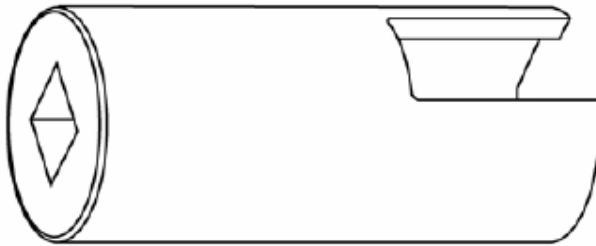


Figure 9001
1in X 3 in Deep Socket, 1/2 in Drive

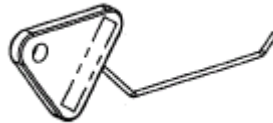


Figure 9002
Flow Washer Tool

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Figure 9003
Extraction Tool



Figure 9004
Insertion Tool

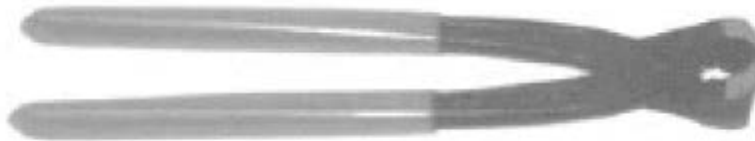


Figure 9005
Pincers

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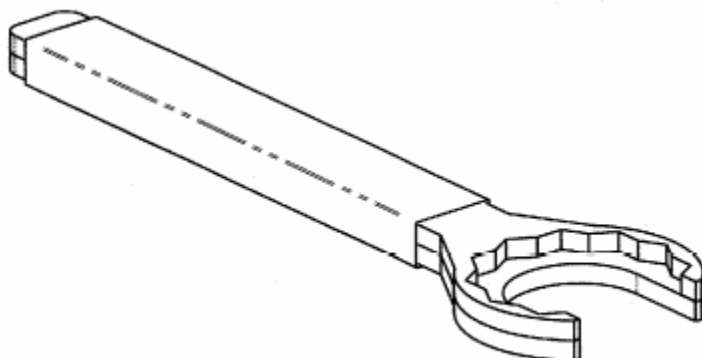


Figure 9006
Power Module Wrench



Figure 9007
Removal Tool

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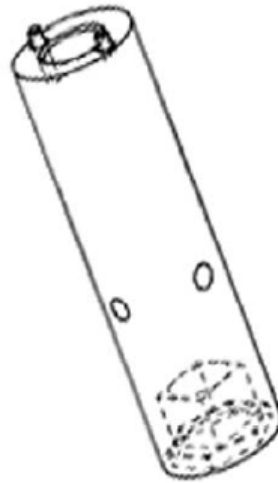


Figure 9008
Socket Key Tool



Figure 9009
Terminal Extractor Tool

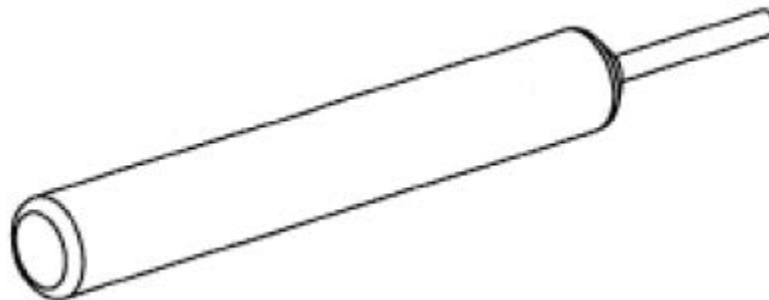


Figure 9010
Torque Tool Handle for Socket Key Tool

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STORAGE

1. **General**

A. This section contains the storage procedures for the Beverage Maker.

2. **Procedure**

A. Storage

- (1) After completing the Beverage Maker assembly, final adjustments, and final testing, dry out the Beverage Maker and place the Beverage Maker in a clean, dust free, moisture proof container. A sealed polyethylene bag is acceptable.
- (2) The Beverage Maker may be stored in any normal ambient temperature, pressure, and humidity environment.
- (3) Mark the Beverage Maker container with the correct part identification. Include the name of the Beverage Maker, part number, Beverage Maker condition, and date of repair.

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ILLUSTRATED PARTS LIST

1. Introduction

The ILLUSTRATED PARTS LIST is intended for use in provisioning, storing, and issuing replaceable parts for the component, and in identification of new and reclaimed parts.

2. Vendor's Parts

Vendor's parts used in the unit and which are not altered by B/E Aerospace, Inc. are listed in the Detailed Parts List by the vendor's part number, vendor's description of the part, and vendor's code listed in parentheses following the description. Vendors' code symbols used in this publication are taken from the CAGE Cataloging Handbook, and consist of the applicable code symbol preceded by the Letter "V." Table 10001 shows a numerically arranged list of vendor codes used in this publication.

Table 10001
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V00779	Tyco Electronics Corporation 2800 Fulling Mill Rd Bldg 38 Middletown, PA 17057
V01139	Momentive Performance Materials Inc. 260 Hudson River Road Saratoga, NY 12188
V08LA7	United Resin Corp 1305 E Street Gertrude Pl Suite B Orange, CA 92705
V02570	Swagelock Company 29500 Solon Road Cuyahoga, OH 44139-3449
V02697	Parker-hannifin Corporation 2360 Palumbo DR Lexington, KY 40509
V05972	Loctite Corporation 1001 Trout Brook Crossing Hartford, CT 06067-3582
V0PK96	Quadtech Inc 734 Forest St 500 Marlborough, MA 01752
V1Y0Z7	Bostik Inc 205 Oliver Dr Marshall, MI 49068
V25281	Oetiker Inc. 6317 Euclid St Marlette, MI 48453

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Table 10001 (Cont.)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V27264	Molex, Inc. DBA Molex Automotive Division 2222 Wellington Court Lisle, IL 60532-1682
V63367	B/E Aerospace, Inc. 15701 W. 95th Street Lenexa, KS 66219, USA
V71984	Dow Corning Corporation 2200 West Salzburg Road Midland, MI 48686-0001
V72653	GC/waldom Inc 1801 Morgan St Rockford, IL 61102
V75297	Kester, Inc. 800 W Thorndale Avenue Dupage, IL 60143
V79074	Varflex Corporation 512 W Court St Rome, NY 13440
V9X418	OK Industries Inc. 134 Marbledale Road Tuckahoe, NY 10707

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3. Use of the Illustrated Parts List

A. When the part number is known:

Turn to the Numerical Index and locate the part number. Figure and item numbers on the illustration where the part appears are listed in the column to the right of the part number. Turn to the illustration and locate the item number. The corresponding item number in the accompanying Detailed Parts List will give part number, description, assembly relationship, quantity required for that particular application, and whether or not the part appears in the illustrations.

B. When the part number is not known:

Look through the figures and identify the part by appearance. Note the figure number and the item number. The corresponding Fig-Item number in the Detailed Parts List shows the part number, description, assembly relationship, effectivity code, and the number of units per assembly.

4. Material Arrangement

A. Numerical Index

The Numerical Index is an alphanumeric listing of all the part numbers in the Detailed Parts List. Each part number is listed together with its corresponding item number, ILLUSTRATED PARTS LIST figure number, and the total required number of units per assembly. The figure-item number is composed of two parts. To the left of the dash is the figure number. It is the number of the figure in the ILLUSTRATED PARTS LIST that shows the part. To the right of the dash is the item number. It is the number that identifies a part in a figure.

B. Illustrations

The ILLUSTRATED PARTS LIST illustrations interrelate to the Numerical Index and the Detailed Parts List. The separate illustrations pictorially describe recommended disassembly/assembly sequence and support DISASSEMBLY and ASSEMBLY sections as well as the Numerical Index and the Detailed Parts List of this manual.

C. Detailed Parts List

(1) Fig-Item

The Detailed Parts List is arranged numerically by item numbers. These are the same item numbers listed in the Numerical Index. In the Detailed Parts List, the figure number portion of the figure-item number is listed only once per column. Immediately below a figure number are the item numbers that appear on that figure. A dash in front of an item number means the corresponding part is not shown in that or any other figure. Each item number is listed together with its corresponding part number, nomenclature (description, assembly relationship), effectivity code, and units per assembly.

(2) Part Number

The part number column contains the B/E Aerospace Part Number or if there is no B/E Aerospace Part Number the original manufacturer's part number will be shown. When standard parts are used, the standard part number is listed in this column.

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(3) Airline Part Number

This column is left blank for airline internal use.

(4) Nomenclature

(a) Indenture System

The indenture system shows the relationship of parts and assembly to next higher assembly or installations, as follows:

1 2 3 4 5 6 7

Unit (Product which this manual describes)

- Detail parts for assembly into the unit
- Subassembly
- Attaching parts for subassembly
- * * *

- • Detail parts for subassembly
- • Sub-subassembly
- • Attaching parts for sub-subassembly
- * * *
- • • Detailed parts for sub-subassembly

(b) Attaching Parts

Attaching parts are captioned ATTACHING PARTS and are listed immediately following the parts attached. The * * * symbol follows the last item of the attaching parts group.

(c) Vendor Code

Parts manufactured by companies other than B/E Aerospace, Inc. are identified by an appropriate vendor's CAGE code following the nomenclature, and are preceded by the letter "V." Standard parts such as AN, MS, etc., are not identified by a vendor code.

(5) Effectivity Code

When two or more units (main assembly) are listed in the same ILLUSTRATED PARTS LIST, a code ("A", "B", "C", etc.) is assigned to each main unit. All subcomponents that are peculiar to a particular unit are identified by the same effectivity code letter as the unit. If parts are common to all units, the effectivity code column is left blank. The coding used in the ILLUSTRATED PARTS LIST in this manual is shown in Table 10002.

Table 10002
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-11UF-00	1	A
4510-11LF-00	1	B
4510-12UF-00	1	C
4510-12LF-00	1	D

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Table 10002 (Cont.)
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-13UF-00	1	E
4510-13LF-00	1	F
4510-14UF-00	1	G
4510-14LF-00	1	H
4510-15UF-00	1	J
4510-15LF-00	1	K
4510-16UF-00	1	L
4510-16LF-00	1	M
4510-17UF-00	1	N
4510-17LF-00	1	P
4510-18UF-00	1	R
4510-18LF-00	1	T
4510-21UF-00	1A	A
4510-21LF-00	1A	B
4510-22UF-00	1A	C
4510-22LF-00	1A	D
4510-23UF-00	1A	E
4510-23LF-00	1A	F
4510-24UF-00	1A	G
4510-24LF-00	1A	H
4510-25UF-00	1A	J
4510-25LF-00	1A	K
4510-26UF-00	1A	L
4510-26LF-00	1A	M
4510-27UF-00	1A	N
4510-27LF-00	1A	P
4510-28UF-00	1A	R
4510-28LF-00	1A	T
4510-11UG-00	1B	A
4510-11LG-00	1B	B
4510-12UG-00	1B	C

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Table 10002 (Cont.)
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-12LG-00	1B	D
4510-13UG-00	1B	E
4510-13LG-00	1B	F
4510-14UG-00	1B	G
4510-14LG-00	1B	H
4510-15UG-00	1B	J
4510-15LG-00	1B	K
4510-16UG-00	1B	L
4510-16LG-00	1B	M
4510-17UG-00	1B	N
4510-17LG-00	1B	P
4510-18UG-00	1B	R
4510-18LG-00	1B	T
4510-21UG-00	1C	A
4510-21LG-00	1C	B
4510-22UG-00	1C	C
4510-22LG-00	1C	D
4510-23UG-00	1C	E
4510-23LG-00	1C	F
4510-24UG-00	1C	G
4510-24LG-00	1C	H
4510-25UG-00	1C	J
4510-25LG-00	1C	K
4510-26UG-00	1C	L
4510-26LG-00	1C	M
4510-27UG-00	1C	N
4510-27LG-00	1C	P
4510-28UG-00	1C	R
4510-28LG-00	1C	T

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(6) Units per assembly

Quantities specified in the "UNITS PER ASSY" column are the total number of each part required per assy or subassy and are not necessarily the total used per the complete unit.

- (a) "AR" indicates "as required" and refers to variable quantity items, such as shims, spacers, or similar items.
- (b) "RF" indicates "reference" and refers to items and their quantities which are listed elsewhere and are provided here as a reference only.
- (c) "NP" indicates "non procurable" and refers to items which are not procurable and may not be ordered.

5. Parts Replacement Data

Interchangeability relationship between parts is identified in the "NOMENCLATURE" column. A list of terms used and their definition is as follows.

TERM	ABBREVIATION	DEFINITION
Alternate	ALT	Part fully meets functional and structural specifications, but differs either in overall external dimensions, connection installation, and/or mounting provisions, and may require additional rework or modifications to install in a specific application.
Optional	OPT	The part is fully interchangeable in form, fit, and function with other item numbers shown.
Preferred	PRFD	Part is preferred over the other optional parts shown.
Replaced By	REPLD BY	Part is replaced by and is interchangeable with item number listed in notation.
Replaces	REPLS	Part replaces and is interchangeable with number listed in notation.
Superseded By	SUPSD BY	Part is replaced by and is not interchangeable with item number listed in notation.
Supersedes	SUPSDS	Part replaces and is not interchangeable with item number listed in notation.
Service Bulletins	PRE SB POST SB RWK SB	Identifies when a service bulletin (SB) has an effect on a part. PRE shows the condition of the part before the work in the service bulletin is done; POST shows the condition after the work is done; RWK shows that the part is reworked by the service bulletin.

6. Alpha/Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
AN316C9R		2	-165A	1
		2A	-165A	1
		2B	-165A	1
		2C	-165A	1
CL315		3	75	1
CL3201		2	-235	1
		2A	-235	1
		2B	-235	1
		2C	-235	1
CL50010		2	307	1
		2	542	1
		2A	307	1
		2A	542	1
		2B	307	1
		2B	542	1
		2C	307	1
		2C	542	1
		2	235A	1
		2A	235A	1
CL50032		2B	235A	1
		2C	235A	1
		2	-540	2
CL50034				

- ITEM NOT ILLUSTRATED

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6. Alpha/Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
CL315		3	75	1
CL3201		2	235	1
		2A	235	1
		2B	235	1
		2C	235	1
CL50010		2	307	1
		2	542	1
		2A	307	1
		2A	542	1
		2B	307	1
		2B	542	1
		2C	307	1
		2C	542	1
CL50034		2	-540	2
		2	540A	1
		2A	-540	2
		2A	540A	1
		2B	-540	2
		2B	540A	1
		2C	-540	2
		2C	540A	1
CO50311		4	-16	3
CO50312		4	-17	3
CO50313		3	-25	1
CO50314		3	-30	1
CO50409		11	10	1
CO6505		4	20	1
CO6505-01		4	25	1
CO6505-03		4	30	6
CO6505-05		4	35	1
GR098		2	67	2
		2	70	2
		2	587	1
		2	-680	1
		2	-680A	DEL
		2A	67	2
		2A	70	2
		2A	587	1
		2A	-680	1
		2A	-680A	DEL
		2B	67	2

- ITEM NOT ILLUSTRATED

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
GR098		2B	70	2
		2B	587	1
		2B	-680	1
		2B	-680A	DEL
		2C	67	2
		2C	70	2
		2C	587	1
		2C	-680	1
		2C	-680A	DEL
GR50053		4	40	1
LW163		2	170	1
		2A	170	1
		2B	170	1
		2C	170	1
MISC013		7	25	1
		8	35	1
MISC3201		1	60*	AR
		1A	60*	AR
		1B	60*	AR
		1C	60*	AR
MISC50389		3	-185	1
MISC50411		3	155	1
MISC50470		3	-165	1
MISC50509		3	-165A	1
MS15795-804		2	105	2
		2	533	2
		2	533A	2
		2A	105	2
		2A	533	2
		2A	533A	2
		2B	105	2
		2B	533	2
		2B	533A	2
		2C	105	2
		2C	533	2
		2C	533A	2
		9	40	2
		9	85	2
MS15795-805		10	30	3
		2	-225	3
		2A	-225	3
		2B	-225	3
		2C	-225	3

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS15795-807		9	70	2
		2	130	1
MS15795-807		2	345	2
		2	465	2
		2	740	2
		2A	130	1
		2A	345	2
		2A	465	2
		2A	740	2
		2B	130	1
		2B	345	2
		2B	465	2
		2B	740	2
		2C	130	1
		2C	345	2
		2C	465	2
MS15795-808		2C	740	2
		2	705	2
		2A	705	2
		2B	705	2
MS16633-4012		2C	705	2
		2	510	1
		2A	510	1
		2B	510	1
MS16633-4031		2C	510	1
MS21042L3		8	15	1
MS21043-04		4	45	2
MS21043-06		3	65	6
		9	25	4
		9	45	2
		2	220	3
		2A	220	3
		2B	220	3
		2C	220	3
MS21209C0810		2	12	1
		2A	12	1
		2B	12	1
		2C	12	1
MS24693C1		2	-530	2
		2A	-530	2
		2B	-530	2
		2C	-530	2
MS24693C24		2	50	5

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

Temporary
Revision No. 8A

COLLINS AEROSPACE PROPRIETARY U.S. Export Classification: EAR99 This watermark supersedes all previous export classifications or markings as of 12 Jul 2023.

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS24693C48		2	60	6
		2	135	5
		2	180	6
		2	425	4
		2	475	1
		2	520	2
		2	555	4
		2A	60	6
		2A	135	5
		2A	180	6
		2A	425	4
		2A	475	1
		2A	520	2
		2A	555	4
		2B	60	6
		2B	135	5
		2B	180	6
		2B	425	4

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS24693C24		2	280A	2
		2	690A	2
		2A	50	5
		2A	280A	2
		2A	690A	2
		2B	50	5
		2B	280A	2
		2B	690A	2
		2C	50	5
		2C	280A	2
		2C	690A	2
MS24693C26		2	-280	2
		2	-690	2
		2A	-280	2
		2A	-690	2
		2B	-280	2
		2B	-690	
		2C	-280	2
		2C	-690	2
MS24693C27		10	20	2
MS24693C272		2	370	3
		2A	370	3
		2B	370	3
		2C	370	3
MS24693C31		9	35	2
MS24693C48		2	60	6
		2	135	5
		2	180	6
		2	425	4
		2	475	1
		2	520	2
		2	555	8
		2A	60	6
		2A	135	5
		2A	180	6
		2A	425	4
		2A	475	1
		2A	520	2
		2A	555	8
		2B	60	6
		2B	135	5
		2B	180	6
		2B	425	4

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS24693C48		2B	475	1
		2B	520	2
		2B	555	4
		2C	60	6
		2C	135	5
		2C	180	6
		2C	425	4
		2C	475	1
		2C	520	2
		2C	555	4
		2	-555A	4
		2A	-555A	4
MS24693C49		2B	-555A	4
MS24693C51		2C	-555A	4
		2	20	4
MS35338-136		2	242	1
		2	595	2
		2A	242	1
		2A	595	2
		2B	242	1
		2B	595	2
		2C	242	1
		2C	595	2
		3	115	1
		9	80	2
		10	35	3
		9	65	2
MS35338-137				

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS24693C48		2B	475	1
		2B	520	2
		2B	555	8
		2C	60	6
		2C	135	5
		2C	180	6
		2C	425	4
		2C	475	1
		2C	520	2
		2C	555	8
MS24693C51		2	20	4
		2A	20	4
		2B	20	4
		2C	20	4
MS24693C53		2	330	2
		2A	330	2
		2B	330	2
		2C	330	2
MS35338-135		2	415	2
		2	531	2
		2	531A	2
		2	544	1
		2A	415	2
		2A	531	2
		2A	531A	2
		2A	544	1
		2B	415	2
		2B	531	2
		2B	531A	2
		2B	544	1
		2C	415	2
		2C	531	2
		2C	531A	2
		2C	544	1
MS35338-136		9	80	2
		10	35	3
		2	595	2
		2A	595	2
		2B	595	2
		2C	595	2
		3	115	1
MS35338-137		9	65	2
		2	460	2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS35338-137		2	580	4
		2	735	2
		2A	460	2
		2A	580	4
		2A	735	2
		2B	460	2
		2B	580	4
		2B	735	2
		2C	460	2
		2C	580	4
		2C	735	2
MS35338-138		2	395	1
		2	700	2
		2A	395	1
		2A	700	2
		2B	395	1
		2B	700	2
		2C	395	1
		2C	700	2
MS51849-53C		2	730A	2
		2A	730A	2
		2B	730A	2
		2C	730A	2
MS51849C53		2	-730	2
		2A	-730	2
		2B	-730	2
		2C	-730	2
MS51957-13		2	530A	2
		2	530B	2
		2	543	1
		2A	530A	2
		2A	530B	2
		2A	543	1
		2B	530A	2
		2B	530B	2
		2B	543	1
		2C	530A	2
		2C	530B	2
		2C	543	1
MS51957-15		2	410	2
		2A	410	2
		2B	410	2
		2C	410	2

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS51957-26		3	120	1
MS51957-27		2	80	5
		2	190	8
		2	200	8
		2A	80	5
		2A	190	8
		2A	200	8
		2A	-775	1
		2B	80	5
		2B	190	8
		2B	200	8
		2C	80	5
		2C	190	8
		2C	200	8
		2C	-775	1
		3	80	2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS51957-20		2	110	2
		2A	110	2
MS51957-20		2B	110	2
		2C	110	2
MS51957-26		3	120	1
MS51957-27		2	80	5
		2	190	9
		2	200	8
		2A	80	5
		2A	190	9
		2A	200	8
		2A	-775	1
		2B	80	5
		2B	190	9
		2B	200	8
		2C	80	5
		2C	190	9
		2C	200	8
		2C	-775	1
		3	80	2
MS51957-29		9	75	2
		2	230	1
		2	590	2
		2A	230	1
		2A	590	2
		2B	230	1
		2B	590	2
		2C	230	1
		2C	590	2
MS51957-30		10	40	3
MS51957-45		9	60	2
MS51957-46		1	45	2
		1A	45	2
		1B	45	2
		1C	45	2
		2	25	1
		2	40	1
		2	125	1
		2	455	2
		2A	25	1
		2A	40	1
		2A	125	1
		2A	455	2

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

Temporary
Revision No. 8A

COLLINS AEROSPACE PROPRIETARY U.S. Export Classification: EAR99 This watermark supersedes all previous export classifications or markings as of 12 Jul 2023.

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
NAS1801-3-8		4	60	2
NAS620C6		2	240A	1
		2A	240A	1
		2B	240A	1
		2C	240A	1
NS50113		3	85	2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS51957-46		2B	25	1
		2B	40	1
		2B	125	1
		2B	455	2
		2C	25	1
		2C	40	1
		2C	125	1
		2C	455	2
MS51958-68		2	400	1
		2A	400	1
		2B	400	1
		2C	400	1
NAS1149D0316H		2	255	1
		2A	255	1
		2B	255	1
		2C	255	1
NAS1149F0332P		4	50	4
		2	250	1
		2A	250	1
		2B	250	1
NAS1189E06P5L		2C	250	1
		4	55	2
		1	40	2
		1A	40	2
NAS1801-3-4		1B	40	2
		1C	40	2
		2	245	1
		2A	245	1
NAS1801-3-8		2B	245	1
		2C	245	1
		4	60	2
		3	85	2
NS50113		6	15	1
OR0341		2	-295	1
OR0361		2A	-295	1
OR0381		2B	-295	1
		2C	-295	1
		2	295A	1
		2A	295A	1
OR0571		2B	295A	1
		2C	295A	1
		6	20	1
		7	15	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
OR0571		8	25	1
		2	306	2
		2	500	1
		2	534	1
		2A	306	2
		2A	500	1
		2A	534	1
		2B	306	2
		2B	500	1
		2B	534	1
		2C	306	2
		2C	500	1
OR0662		2C	534	1
		2	155	1
		2A	155	1
		2B	155	1
OR067		2C	155	1
		2	150	1
		2A	150	1
		2B	150	1
OR0941		2C	150	1
		3	45	3
		3	125	1
		8	20	1
OR50001		3	160	1
OR50019		2	215	1
OR50039		2A	215	1
		2B	215	1
		2C	215	1
		11	25A	2
PF102		11	25B	1
		3	90	1
PF110		11	20	2
PF143		2	765	2
		2A	765	2
		2B	765	2
		2C	765	2
PF202		2	320	1
		2	755	2
		2A	320	1
		2A	755	2
		2B	320	1
		2B	755	2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
PF50002		2C	320	1
		2C	755	2
		2	760	2
PF50002		2A	760	2
		2B	760	2
		2C	760	2
PN5855		4	65	1
SC50268-30		10	40A	2
SC50270		10	45	1
SC50280-29		9	-75A	2
SC50280-45		9	-60A	2
SN431		2	-120	1
		2	350	2
		2A	-120	1
		2A	350	2
		2B	-120	1
		2B	350	2
		2C	-120	1
		2C	350	2
TE50432		3	-20	1
TE50442		11	-32	AR
TU379		12	-15	2
		2	30	2
		2A	30	2
		2B	30	2
		2C	30	2
TU4505		2	308	1
		2	747	1
		2A	308	1
		2A	747	1
		2B	308	1
		2B	747	1
		2C	308	1
		2C	747	1
		2	-745	2
		2	-745A	2
TU50020		2	745B	1
		2A	-745	2
		2A	-745A	2
		2A	745B	1
		2B	-745	2
		2B	-745A	2
		2B	745B	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
VA50107		3	205	1
WA120		2	-240	1
		2A	-240	1
		2B	-240	1
		2C	-240	1
WA50036		2	85	6

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2C	-745	2
		2C	-745A	2
		2C	745B	1
TU50039		3	40	1
TU50052		11	35	1
TU50242		2	-535	1
		2A	-535	1
TU50242		2B	-535	1
		2C	-535	1
VA1003		3	230	RF
		3	-230A	DEL
VA1004		3	235	RF
		3	-235A	DEL
VA1031		3	225	RF
		3	-225A	DEL
VA50107		3	205	1
WA120		2	240	1
		2A	240	1
		2B	240	1
		2C	240	1
WA50036		2	85	6
		2A	85	6
		2B	85	6
		2C	85	6
WI156		3	60	1
		2	-602A	DEL
		2A	-602A	DEL
		2B	-602A	DEL
		2C	-602A	DEL
0R50099		12A	10	1
352-0983-3		3	55B	3
3520-0236-01		2	-525	1
		2A	-525	1
		2B	-525	1
		2C	-525	1
3520-0242-01		2	-602	1
		2A	-602	1
		2B	-602	1
		2C	-602	1
3520-0509-01		3	50	1
3520-0523-01		3	-130	1
3520-0525-01		7	-1	RF
		2	270	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0526-11		2A	270	1
		1	10A	1
3520-0526-12		2	-1A	RF
		1	10B	1
3520-0526-13		2	-1B	RF
		1	10C	1
3520-0526-14		2	-1C	RF
		1	10D	1
3520-0526-15		2	-1D	RF
		1	10E	1
3520-0526-16		2	-1E	RF
		1	10F	1
3520-0526-17		2	-1F	RF
		1	10G	1
3520-0526-18		2	-1G	RF
		1	10H	1
3520-0526-21		2	-1H	RF
		1A	10A	1
3520-0526-22		2A	-1A	RF
		1A	10B	1
3520-0526-23		2A	-1B	RF
		1A	10C	1
3520-0526-24		2A	-1C	RF
		1A	10D	1
3520-0526-25		2A	-1D	RF
		1A	10E	1
3520-0526-26		2A	-1E	RF
		1A	10F	1
3520-0526-27		2A	-1F	RF
		1A	10G	1
3520-0526-28		2A	-1G	RF
		1A	10H	1
3520-0526-31		2A	-1H	RF
		1B	10A	1
3520-0526-32		2B	-1A	RF
		1B	10B	1
3520-0526-33		2B	-1B	RF
		1B	10C	1
3520-0526-34		2B	-1C	RF
		1B	10D	1
3520-0526-35		2B	-1D	RF
		1B	10E	1
		2B	-1E	RF

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0526-36		1B	10F	1
		2B	-1F	RF
3520-0526-37		1B	10G	1
		2B	-1G	RF
3520-0526-38		1B	10H	1
		2B	-1H	RF
3520-0526-41		1C	10A	1
		2C	-1A	RF
3520-0526-42		1C	10B	1
		2C	-1B	RF
3520-0526-43		1C	10C	1
		2C	-1C	RF
3520-0526-44		1C	10D	1
		2C	-1D	RF
3520-0526-45		1C	10E	1
		2C	-1E	RF
3520-0526-46		1C	10F	1
		2C	-1F	RF
3520-0526-47		1C	10G	1
		2C	-1G	RF
3520-0526-48		1C	10H	1
		2C	-1H	RF
3520-0528-01		2A	770	1
		2C	770	1
3520-0529-01		2	15	1
		2A	15	1
		2B	15	1
		2C	15	1
		3	-1	RF
3520-0530-01		2	55	1
		2A	55	1
		2B	55	1
		2C	55	1
		4	-1	RF
3520-0531-01		4	10	1
3520-0533-01		2	325	1
		2A	325	1
		2B	325	1
		2C	325	1
3520-0534-01		9	-1	RF
		2	300	1
		2A	300	1
		2B	300	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0535-01		2C	300	1
		2	285	1
		2A	285	1
		2B	285	1
		2C	285	1
3520-0535-03		2	-290	1
		2A	-290	1
		2B	-290	1
		2C	-290	1
3520-0536-01		2	585	1
		2A	585	1
3520-0536-01		2B	585	1
		2C	585	1
3520-0537-01		2	550	2
		2A	550	2
		2B	550	2
		2C	550	2
3520-0539-01		10	-1	RF
		2	-545	1
		2A	-545	1
		2B	-545	1
		2C	-545	1
3520-0540-01		1	20A	1
		1A	20A	1
		1B	20A	1
		1C	20A	1
3520-0540-03		1	20B	1
		1A	20B	1
		1B	20B	1
		1C	20B	1
3520-0542-01		2	420	1
		2A	420	1
		2B	420	1
		2C	420	1
3520-0543-01		10	5	1
3520-0544-01		4	15	1
3520-0548-01		2	-10	1
		2	10A	1
		2A	-10	1
		2A	10A	1
		2B	-10	1
		2B	10A	1
		2C	-10	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0550-01		2C	10A	1
		3	190	1
		3	15	3
3520-0551-01		2	305	1
		2A	305	1
		2B	305	1
3520-0552-01		2C	305	1
		2	65	1
		2A	65	1
3520-0553-01		2B	65	1
		2C	65	1
		2	750	1
3520-0554-01		2A	750	1
		2B	750	1
		2C	750	1
3520-0555-01		3	10	1
		2	-605	1
		2A	-605	1
3520-0556-01		2B	-605	1
		2C	-605	1
		2	-610	1
3520-0556-02		2A	-610	1
		2B	-610	1
		2C	-610	1
3520-0556-03		2	-615	1
		2A	-615	1
		2B	-615	1
3520-0556-04		2C	-615	1
		2	-620	1
		2A	-620	1
3520-0556-05		2B	-620	1
		2C	-620	1
		2	-625	1
3520-0556-06		2A	-625	1
		2B	-625	1
		2C	-625	1
3520-0556-07		2	-630	1
		2A	-630	1
		2B	-630	1
		2C	-630	1
		2	-635	1
		2A	-635	1
		2B	-635	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0556-08		2C	-635	1
		2	-640	1
		2A	-640	1
		2B	-640	1
		2C	-640	1
3520-0557-01		10	25	1
3520-0558-01		8	30	1
3520-0559-01		8	-1	RF
3520-0560-01		2B	275	1
		2C	275	1
		2	430	1
		2A	430	1
		2B	430	1
3520-0560-01		2C	430	1
3520-0560-03		2	-435	1
3520-0560-03		2A	-435	1
		2B	-435	1
		2C	-435	1
		2	-440	1
		2A	-440	1
3520-0560-05		2B	-440	1
		2C	-440	1
		2	-445	1
		2A	-445	1
		2B	-445	1
3520-0560-07		2C	-445	1
		2	-450	1
		2A	-450	1
		2B	-450	1
		2C	-450	1
3520-0560-09		2	-450	1
		2A	-450	1
		2B	-450	1
		2C	-450	1
		2	185	1
3520-0561-01		2A	185	1
		2B	185	1
		2C	185	1
		2	195	1
		2A	195	1
3520-0561-03		2B	195	1
		2C	195	1
		2	100	1
		2A	100	1
		2B	100	1
3520-0566-01		2C	100	1
		2	100	1
		2A	100	1
		2B	100	1
		2C	100	1
3520-0578-03		10	10	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0579-01		10	25A	1
3520-0580-03		10	-1A	RF
		2	545A	1
		2A	545A	1
		2B	545A	1
		2C	545A	1
3520-0581-01		3	185A	1
3520-0594-01		3	-130B	1
3520-0594-03		3	-130A	1
3520-0595-03		10	5A	1
3520-0603-01		1	-70A*	AR
		1A	-70A*	AR
		1B	-70A*	AR
		1C	-70A*	AR
3520-0603-03		1	-70B*	AR
3520-0603-03		1A	-70B*	AR
		1B	-70B*	AR
		1C	-70B*	AR
3520-0605-01		11	-1E	RF
		2	-710B	1
		2A	-710B	1
		2B	-710B	1
		2C	-710B	1
3520-0605-03		11	-1H	RF
		2	-725B	1
		2A	-725B	1
		2B	-725B	1
		2C	-725B	1
3520-0605-05		11	-1F	RF
		2	-715B	1
		2A	-715B	1
		2B	-715B	1
		2C	-715B	1
3520-0605-07		11	-1G	RF
		2	-720B	1
		2A	-720B	1
		2B	-720B	1
		2C	-720B	1
3520-1529-01		11	-30E	3
		11	-30F	2
		12A	-1	RF
3520-1616-01		12A	5	1
3520-1620-01		12A	15	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0121-01		5	-1	DEL
		2	75	1
		2A	75	1
		2B	75	1
		2C	75	1
3530-0232-01		11	-1A	RF
		2	-710	1
		2	710A	1
		2A	-710	1
		2A	710A	1
		2B	-710	1
		2B	710A	1
		2C	-710	1
		2C	710A	1
3530-0232-03		11	-1D	RF
		2	-725A	1
		2A	-725A	1
		2B	-725A	1
		2C	-725A	1
3530-0232-05		11	-1B	RF
3530-0232-05		2	-715A	1
		2A	-715A	1
		2B	-715A	1
		2C	-715A	1
3530-0232-07		11	-1C	RF
		2	-720A	1
		2A	-720A	1
		2B	-720A	1
		2C	-720A	1
3530-0233-01		2	-715	1
		2A	-715	1
		2B	-715	1
		2C	-715	1
3530-0236-01		2	-720	1
		2A	-720	1
		2B	-720	1
		2C	-720	1
3530-0241-01		2	600	1
		2A	600	1
		2B	600	1
		2C	600	1
3530-0242-01		2	642	1
		2A	642	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0244-01		2B	642	1
		2C	642	1
		2	645	1
		2A	645	1
		2B	645	1
3530-0245-01		2C	645	1
		2	650	1
		2A	650	1
		2B	650	1
		2C	650	1
3530-0246-01		2	660	1
		2A	660	1
		2B	660	1
		2C	660	1
		2	675	1
3530-0247-01		2A	675	1
		2B	675	1
		2C	675	1
		2	665	1
		2A	665	1
3530-0248-01		2B	665	1
		2C	665	1
		2	670	1
		2A	670	1
		2B	670	1
3530-0249-01		2C	670	1
		2	655	1
		2A	655	1
		2B	655	1
		2C	655	1
3530-0251-01		1	-25A	RF
		1A	-25A	RF
		1B	-25A	RF
		1C	-25A	RF
		1	-25B	RF
3530-0254-01		1A	-25B	RF
		1B	-25B	RF
		1C	-25B	RF
		2	210	1
		2A	210	1
3530-0254-03		2B	210	1
		2C	210	1
		2	-725	1
3530-0261-01				
3530-0265-01				

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0268-01 3530-0272-01 3530-0274-01		2A	-725	1
		2B	-725	1
		2C	-725	1
		3	-55	3
		9	10B	1
		2	-90	2
		2	-90A	DEL
		2A	-90	2
		2A	-90A	DEL
		2B	-90	2
		2B	-90A	DEL
		2C	-90	2
		2C	-90A	DEL
3530-0279-01		11	30A	3
		11	30C	2
		12	-1A	RF
		12	25	1
3530-0279-03		12	20	1
3530-0279-05		12	50	1
3530-0279-07		12	45	1
3530-0279-09		12	5	1
3530-0279-11		12	55	1
3530-0279-13		12	30	1
3530-0279-15		12	35	1
3530-0279-17		12	40	1
3530-0279-19		12	-10	1
3530-0279-21 3530-0279-23		11	30B	3
		11	30D	2
		12	-1B	RF
		12	42	1
3530-0279-25		3	-55A	3
3530-0280-01 3530-0289-01		2	-525A	1
		2A	-525A	1
		2B	-525A	1
		2C	-525A	1
3530-0305-01		1	-55*	AR
		1A	-55*	AR
		1B	-55*	AR
		1C	-55*	AR
3530-0310-01		1	-75*	AR
		1A	-75*	AR
		1B	-75*	AR
		1C	-75*	AR

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0311-01		1	-72*	AR
		1A	-72*	AR
		1B	-72*	AR
		1C	-72*	AR
3530-0879-01		2	525B	1
		2A	525B	1
		2B	525B	1
		2C	525B	1
3575-0010-01		1	30	1
		1A	30	1
		1B	30	1
		1C	30	1
3575-0050-01		2	560	1
		2A	560	1
		2B	560	1
		2C	560	1
3575-0050-02		2	565	1
		2A	565	1
		2B	565	1
		2C	565	1
3575-0189-01		9	30	1
3575-0388-01		1	35	2
3575-0388-01		1A	35	2
		1B	35	2
		1C	35	2
3575-0435-01		2	515	1
		2A	515	1
		2B	515	1
		2C	515	1
3575-0613-01		2	360	1
		2A	360	1
		2B	360	1
		2C	360	1
3575-0630-01		3	70	1
3575-0647-01		3	100	1
3575-0650-01		2	470	1
		2A	470	1
		2B	470	1
		2C	470	1
3575-1211-01		11	5A	1
3575-1211-03		11	5D	1
3575-1211-05		11	5B	1
3575-1211-07		11	5C	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3575-1214-01		2	260	1
		2A	260	1
		2B	260	1
		2C	260	1
3575-1215-01		2	-265	1
		2	265A	1
		2A	-265	1
		2A	265A	1
		2B	-265	1
		2B	265A	1
		2C	-265	
		2C	265A	1
3575-1220-01		2	695	2
		2A	695	2
		2B	695	2
		2C	695	2
3575-1223-01		2	45	1
		2A	45	1
		2B	45	1
		2C	45	1
3575-1225-01		7	20	1
3575-1227-01		7	10	1
3575-1230-01		9	-10	1
		9	-10A	1
3575-1236-05		10	15	2
3575-1244-01		2	685	1
		2A	685	1
		2B	685	1
		2C	685	1
3575-1246-01		8	10	1
3575-1250-01		4	12	1
3575-1253-01		2	-95	1
		2	-95A	DEL
		2A	-95	1
		2A	-95A	DEL
		2B	-95	1
		2B	-95A	DEL
		2C	-95	1
		2C	-95A	DEL
3575-1269-01		2	-205	1
		2A	-205	1
		2B	-205	1
		2C	-205	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3575-1283-01		2	536	1
		2A	536	1
		2B	536	1
		2C	536	1
3575-1284-01		11	15	1
3575-1312-01		2	535A	1
		2A	535A	1
		2B	535A	1
		2C	535A	1
3575-1324-01		3	135	1
3575-1325-01		3	140	1
3575-1327-01		3	145	1
3575-1329-01		3	150	1
3575-1351-01		11	5E	1
3575-1351-03		11	5H	1
3575-1351-05		11	5F	1
3575-1351-07		11	5G	1
3575-1356-01		2	290A	1
		2A	290A	1
		2B	290A	1
		2C	290A	1
3575-1362-01		2	225A	3
		2A	225A	3
		2B	225A	3
3575-1362-01		2C	225A	3
3575-1480-07		3	-35	1
3575-2785-01		3	-145A	1
3575-2791-01		3	-150A	1
3575-3037-01		12A	20	1
411-0676-3		2	385	1
		2A	385	1
		2B	385	1
		2C	385	1
411-0686-1		2	570	2
		2A	570	2
		2B	570	2
		2C	570	2
411-0689-1		2	575	2
		2A	575	2
		2B	575	2
		2C	575	2
411-0692-1		2	-355	2
		2	355A	2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2A	-355	2
		2A	355A	2
		2B	-355	2
		2B	355A	2
		2C	-355	2
		2C	355A	2
411-0693-1		9	15	1
411-0709-1		9	20	1
411-0724-1		2	335	2
		2A	335	2
		2B	335	2
		2C	335	2
		2	340	2
		2A	340	2
411-0725-1		2B	340	2
		2C	340	2
		9	50	1
		9	55	1
411-0748-1		9	55	1
411-0748-3		9	55	1
411-0926-1		2	-365	
		2A	-365	
		2B	-365	2
		2C	-365	
416-0463-1		2	405	1
		2A	405	1
		2B	405	1
		2C	405	1
416-0463-1		2	405	1
423-0078-11		2	175	2
423-0078-11		2A	175	2
		2B	175	2
		2C	175	2
		2	-356	2
425-0702-1		2A	-356	2
		2B	-356	2
		2C	-356	2
		2	-327	2
430531		2	-327A	DEL
		2A	-327	2
		2A	-327A	DEL
		2B	-327	2
		2B	-327A	DEL
		2C	-327	2
		2C	-327A	DEL
		1	-1B	RF
4510-11LF-00		1	-1B	RF

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
4510-11LG-00		1B	-1B	RF
4510-11UF-00		1	-1A	RF
4510-11UG-00		1B	-1A	RF
4510-12LF-00		1	-1D	RF
4510-12LG-00		1B	-1D	RF
4510-12UF-00		1	-1C	RF
4510-12UG-00		1B	-1C	RF
4510-13LF-00		1	-1F	RF
4510-13LG-00		1B	-1F	RF
4510-13UF-00		1	-1E	RF
4510-13UG-00		1B	-1E	RF
4510-14LF-00		1	-1H	RF
4510-14LG-00		1B	-1H	RF
4510-14UF-00		1	-1G	RF
4510-14UG-00		1B	-1G	RF
4510-15LF-00		1	-1K	RF
4510-15LG-00		1B	-1K	RF
4510-15UF-00		1	-1J	RF
4510-15UG-00		1B	-1J	RF
4510-16LF-00		1	-1M	RF
4510-16LG-00		1B	-1M	RF
4510-16UF-00		1	-1L	RF
4510-16UG-00		1B	-1L	RF
4510-17LF-00		1	-1P	RF
4510-17LG-00		1B	-1P	RF
4510-17UF-00		1	-1N	RF
4510-17UG-00		1B	-1N	RF
4510-18LF-00		1	-1T	RF
4510-18LG-00		1B	-1T	RF
4510-18UF-00		1	-1R	RF
4510-18UG-00		1B	-1R	RF
4510-21LF-00		1A	-1B	RF
4510-21LG-00		1C	-1B	RF
4510-21UF-00		1A	-1A	RF
4510-21UG-00		1C	-1A	RF
4510-22LF-00		1A	-1D	RF
4510-22LG-00		1C	-1D	RF
4510-22UF-00		1A	-1C	RF
4510-22UG-00		1C	-1C	RF
4510-23LF-00		1A	-1F	RF
4510-23LG-00		1C	-1F	RF
4510-23UF-00		1A	-1E	RF
4510-23UG-00		1C	-1E	RF

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
4510-24LF-00		1A	-1H	RF
4510-24LG-00		1C	-1H	RF
4510-24UF-00		1A	-1G	RF
4510-24UG-00		1C	-1G	RF
4510-25LF-00		1A	-1K	RF
4510-25LG-00		1C	-1K	RF
4510-25UF-00		1A	-1J	RF
4510-25UG-00		1C	-1J	RF
4510-26LF-00		1A	-1M	RF
4510-26LG-00		1C	-1M	RF
4510-26UF-00		1A	-1L	RF
4510-26UG-00		1C	-1L	RF
4510-27LF-00		1A	-1P	RF
4510-27LG-00		1C	-1P	RF
4510-27UF-00		1A	-1N	RF
4510-27UG-00		1C	-1N	RF
4510-28LF-00		1A	-1T	RF
4510-28LG-00		1C	-1T	RF
4510-28UF-00		1A	-1R	RF
4510-28UG-00		1C	-1R	RF
467-0865-1		2	309	1
		2A	309	1
		2B	309	1
		2C	309	1
467-0912-1		2	315	1
467-0912-1		2A	315	1
		2B	315	1
467-0912-1		2C	315	1
60-2001-1		2	380	1
		2A	380	1
		2B	380	1
		2C	380	1
		2	165	1
		2A	165	1
		2B	165	1
		2C	165	1
60-2013-1		2	485	1
		2A	485	1
		2B	485	1
		2C	485	1
62-0007-1		3	105	1
62-0008-1		6	10	1
62-0010-1		2	145	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
62-0144-1		2	35	1
		2A	35	1
62-0196-3		2	-747A	1
		2A	-747A	1
		2B	-747A	1
		2C	-747A	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
62-0011-1		2A	145	1
		2B	145	1
		2C	145	1
		3	110	1
		6	25	1
62-0013-1		6	-1	RF
62-0018-1		2	140	1
		2A	140	1
		2B	140	1
		2C	140	1
		2	115	1
		2A	115	1
		2B	115	1
		2C	115	1
		3	95	1
		2	390	1
62-0020-1		2A	390	1
		2B	390	1
		2C	390	1
		2	160	1
		2A	160	1
62-0025-1		2B	160	1
		2C	160	1
		2	375	4
		2A	375	4
		2B	375	4
62-0037-1		2C	375	4
		2	310	1
		2	538	1
		2A	310	1
		2A	538	1
62-0061-1		2B	310	1
		2B	538	1
		2C	310	1
		2C	538	1
		3	-180	1
62-0061-1		3	250	1
		3	170	1
		3	240	1
		3	-175	1
		3	245	1
62-0066-7		2	35	1
		2A	35	1
62-0136-11				
62-0136-7				
62-0136-9				
62-0144-1				

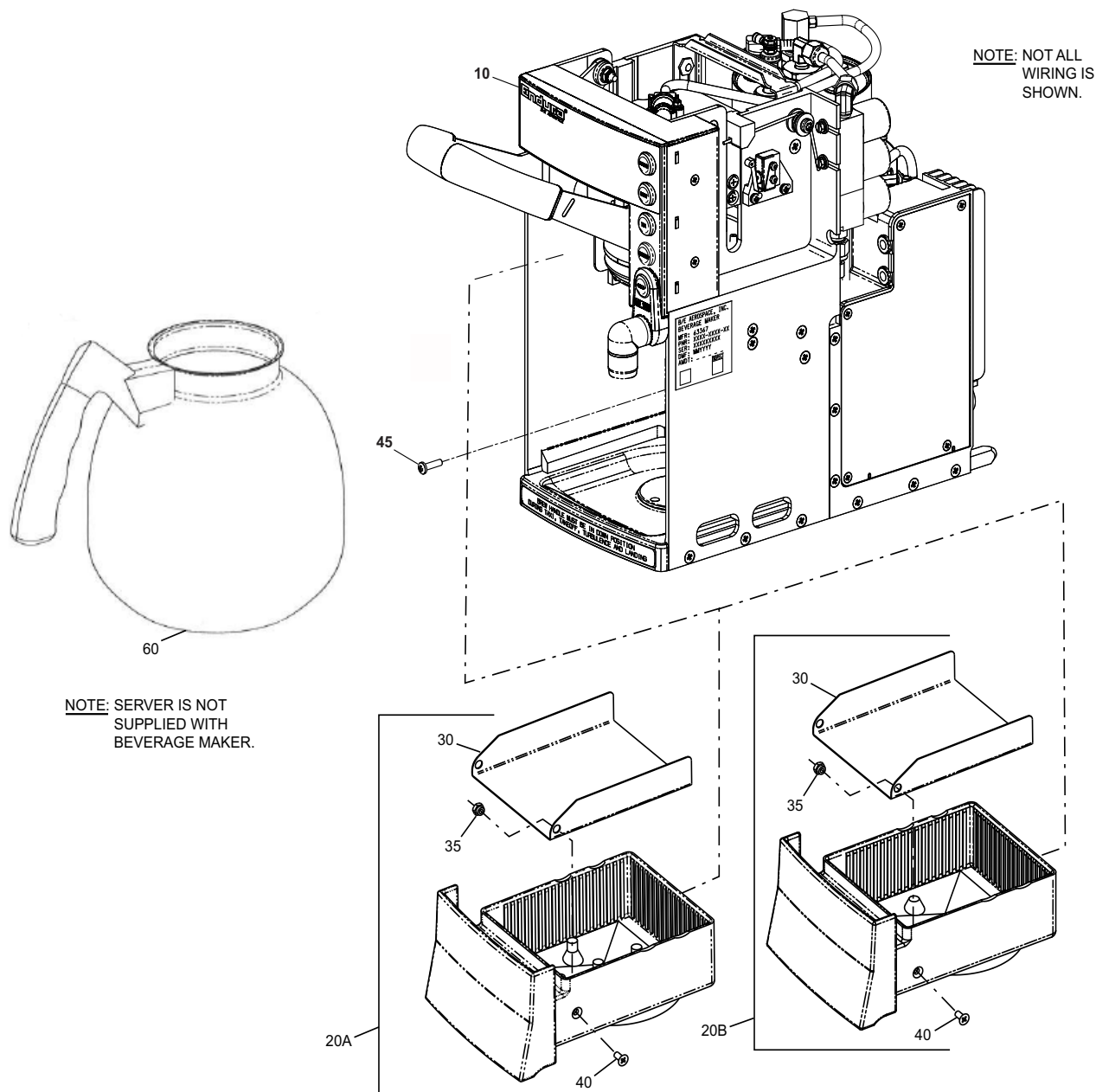
- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
62-0210-1		2B	35	1
		2C	35	1
		2	490	1
		2A	490	1
		2B	490	1
62-0210-2		2C	490	1
		2	505	1
		2A	505	1
		2B	505	1
		2C	505	1
62-0210-3		2	495	1
		2A	495	1
		2B	495	1
		2C	495	1
		2	480	1
62-0211-1		2A	480	1
		2B	480	1
		2C	480	1
		3	195	RF
		3	200	1
62-0223-1		3	215	RF
62-0223-3		3	-215A	DEL
62-0223-5		3	210	RF
62-0223-9		3	-210A	DEL
62-0223-11		3	220	RF
62-0223-11		3	-220A	DEL

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series



IPL Figure 1
Beverage Maker

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-1A	4510-11UF-00		BEVERAGE MAKER	A	RF
-1B	4510-11LF-00		BEVERAGE MAKER	B	RF
-1C	4510-12UF-00		BEVERAGE MAKER	C	RF
-1D	4510-12LF-00		BEVERAGE MAKER	D	RF
-1E	4510-13UF-00		BEVERAGE MAKER	E	RF
-1F	4510-13LF-00		BEVERAGE MAKER	F	RF
-1G	4510-14UF-00		BEVERAGE MAKER	G	RF
-1H	4510-14LF-00		BEVERAGE MAKER	H	RF
-1J	4510-15UF-00		BEVERAGE MAKER	J	RF
-1K	4510-15LF-00		BEVERAGE MAKER	K	RF
-1L	4510-16UF-00		BEVERAGE MAKER	L	RF
-1M	4510-16LF-00		BEVERAGE MAKER	M	RF
-1N	4510-17UF-00		BEVERAGE MAKER	N	RF
-1P	4510-17LF-00		BEVERAGE MAKER	P	RF
-1R	4510-18UF-00		BEVERAGE MAKER	R	RF
-1T	4510-18LF-00		BEVERAGE MAKER	T	RF
10A	3520-0526-11		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	A,B	1
10B	3520-0526-12		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	C,D	1
10C	3520-0526-13		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	E,F	1
10D	3520-0526-14		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	G,H	1
10E	3520-0526-15		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	J,K	1
10F	3520-0526-16		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	L,M	1
10G	3520-0526-17		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	N,P	1
10H	3520-0526-18		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	R,T	1
20A	3520-0540-01		• BREW CUP ASSY (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

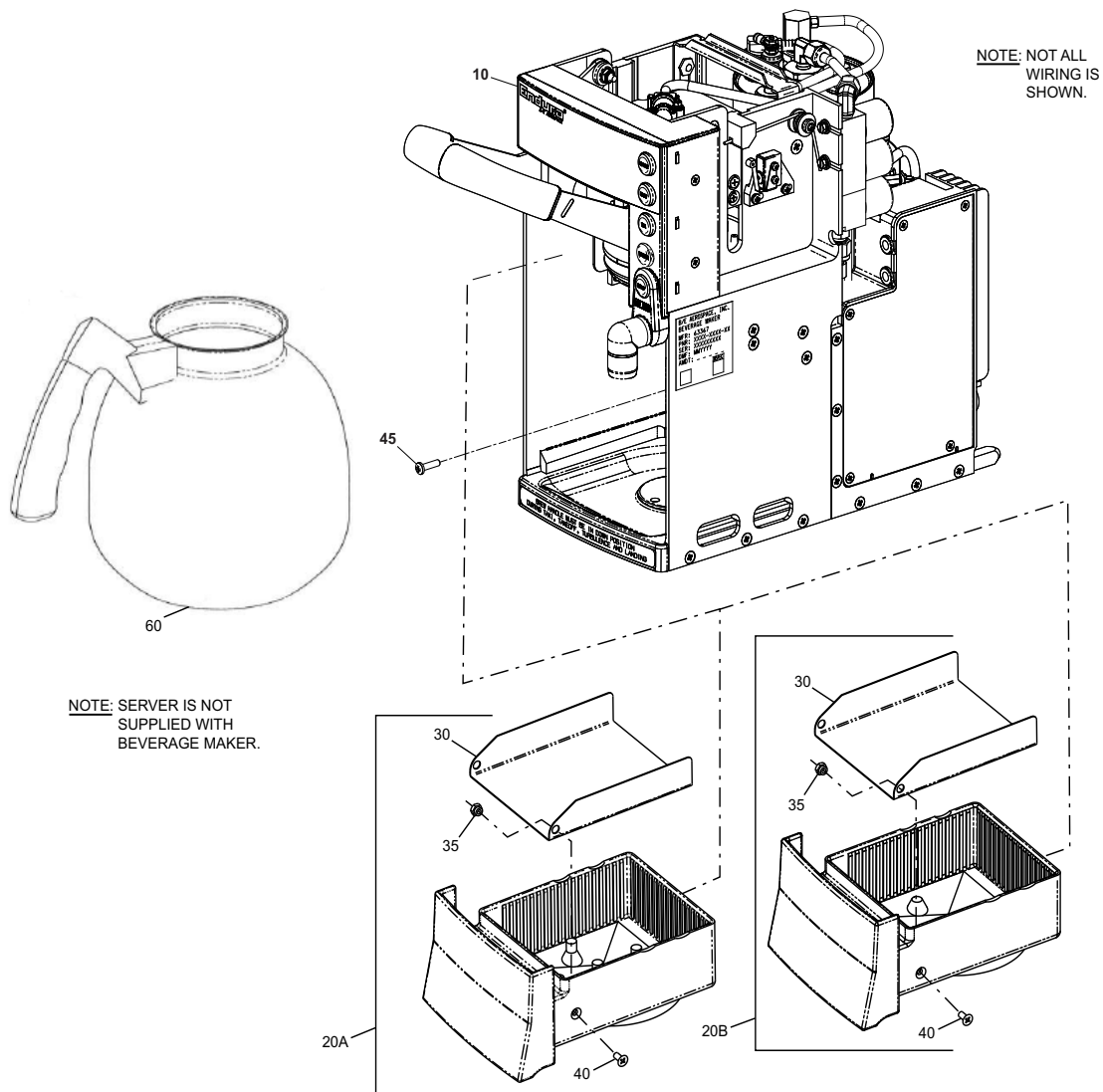
B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
20B	3520-0540-03		• BREW CUP ASSY (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • BREW CUP (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	RF
-25B	3530-0254-03		• • BREW CUP (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	RF
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 8-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER		AR
60*	MISC3201		• SERVER (TPN MISC-3201)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-72*	3530-0311-01		• • GASKET, SERVER COVER		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			<u>*NOTE:</u> NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series



IPL Figure 1A
Beverage Maker

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1A					
-1A	4510-21UF-00		BEVERAGE MAKER	A	RF
-1B	4510-21LF-00		BEVERAGE MAKER	B	RF
-1C	4510-22UF-00		BEVERAGE MAKER	C	RF
-1D	4510-22LF-00		BEVERAGE MAKER	D	RF
-1E	4510-23UF-00		BEVERAGE MAKER	E	RF
-1F	4510-23LF-00		BEVERAGE MAKER	F	RF
-1G	4510-24UF-00		BEVERAGE MAKER	G	RF
-1H	4510-24LF-00		BEVERAGE MAKER	H	RF
-1J	4510-25UF-00		BEVERAGE MAKER	J	RF
-1K	4510-25LF-00		BEVERAGE MAKER	K	RF
-1L	4510-26UF-00		BEVERAGE MAKER	L	RF
-1M	4510-26LF-00		BEVERAGE MAKER	M	RF
-1N	4510-27UF-00		BEVERAGE MAKER	N	RF
-1P	4510-27LF-00		BEVERAGE MAKER	P	RF
-1R	4510-28UF-00		BEVERAGE MAKER	R	RF
-1T	4510-28LF-00		BEVERAGE MAKER	T	RF
10A	3520-0526-21		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	A,B	1
10B	3520-0526-22		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	C,D	1
10C	3520-0526-23		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	E,F	1
10D	3520-0526-24		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	G,H	1
10E	3520-0526-25		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	J,K	1
10F	3520-0526-26		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	L,M	1
10G	3520-0526-27		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	N,P	1
10H	3520-0526-28		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	R,T	1
20A	3520-0540-01		• BREW CUP ASSY (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

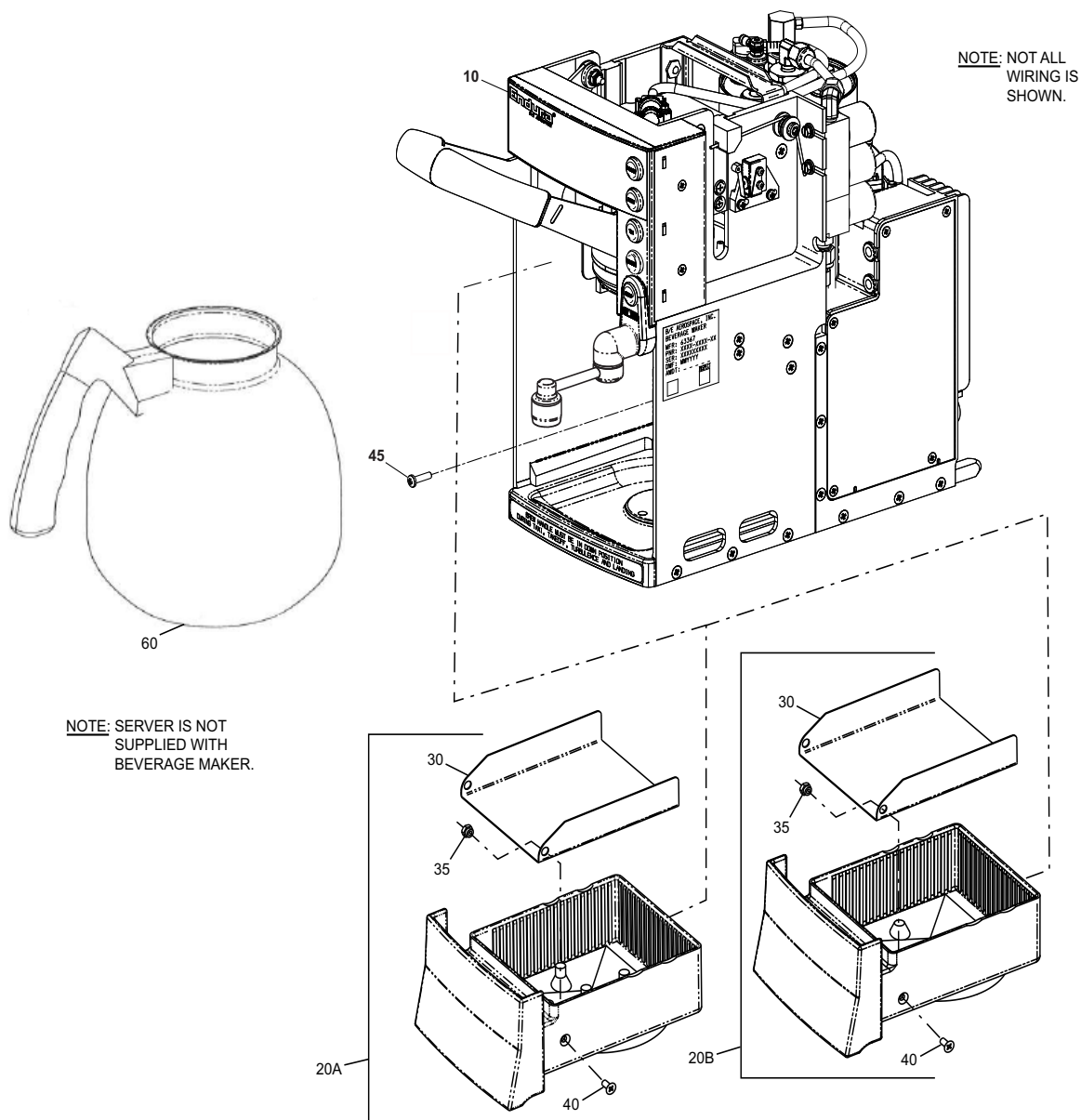
B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1A					
20B	3520-0540-03		• BREW CUP ASSY (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • BREW CUP (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	RF
-25B	3530-0254-03		• • BREW CUP (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	RF
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 8-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER		AR
60*	MISC3201		• SERVER (TPN MISC-3201)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-72*	3530-0311-01		• • GASKET, SERVER COVER		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			<u>*NOTE:</u> NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series



IPL Figure 1B
Beverage Maker

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
-1A	4510-11UG-00		• BEVERAGE MAKER	A	RF
-1B	4510-11LG-00		• BEVERAGE MAKER	B	RF
-1C	4510-12UG-00		• BEVERAGE MAKER	C	RF
-1D	4510-12LG-00		• BEVERAGE MAKER	D	RF
-1E	4510-13UG-00		• BEVERAGE MAKER	E	RF
-1F	4510-13LG-00		• BEVERAGE MAKER	F	RF
-1G	4510-14UG-00		• BEVERAGE MAKER	G	RF
-1H	4510-14LG-00		• BEVERAGE MAKER	H	RF
-1J	4510-15UG-00		• BEVERAGE MAKER	J	RF
-1K	4510-15LG-00		• BEVERAGE MAKER	K	RF
-1L	4510-16UG-00		• BEVERAGE MAKER	L	RF
-1M	4510-16LG-00		• BEVERAGE MAKER	M	RF
-1N	4510-17UG-00		• BEVERAGE MAKER	N	RF
-1P	4510-17LG-00		• BEVERAGE MAKER	P	RF
-1R	4510-18UG-00		• BEVERAGE MAKER	R	RF
-1T	4510-18LG-00		• BEVERAGE MAKER	T	RF
10A	3520-0526-31		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	A,B	1
10B	3520-0526-32		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	C,D	1
10C	3520-0526-33		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	E,F	1
10D	3520-0526-34		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	G,H	1
10E	3520-0526-35		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	J,K	1
10F	3520-0526-36		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	L,M	1
10G	3520-0526-37		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	N,P	1
10H	3520-0526-38		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	R,T	1
20A	3520-0540-01		• BREW CUP ASSY (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

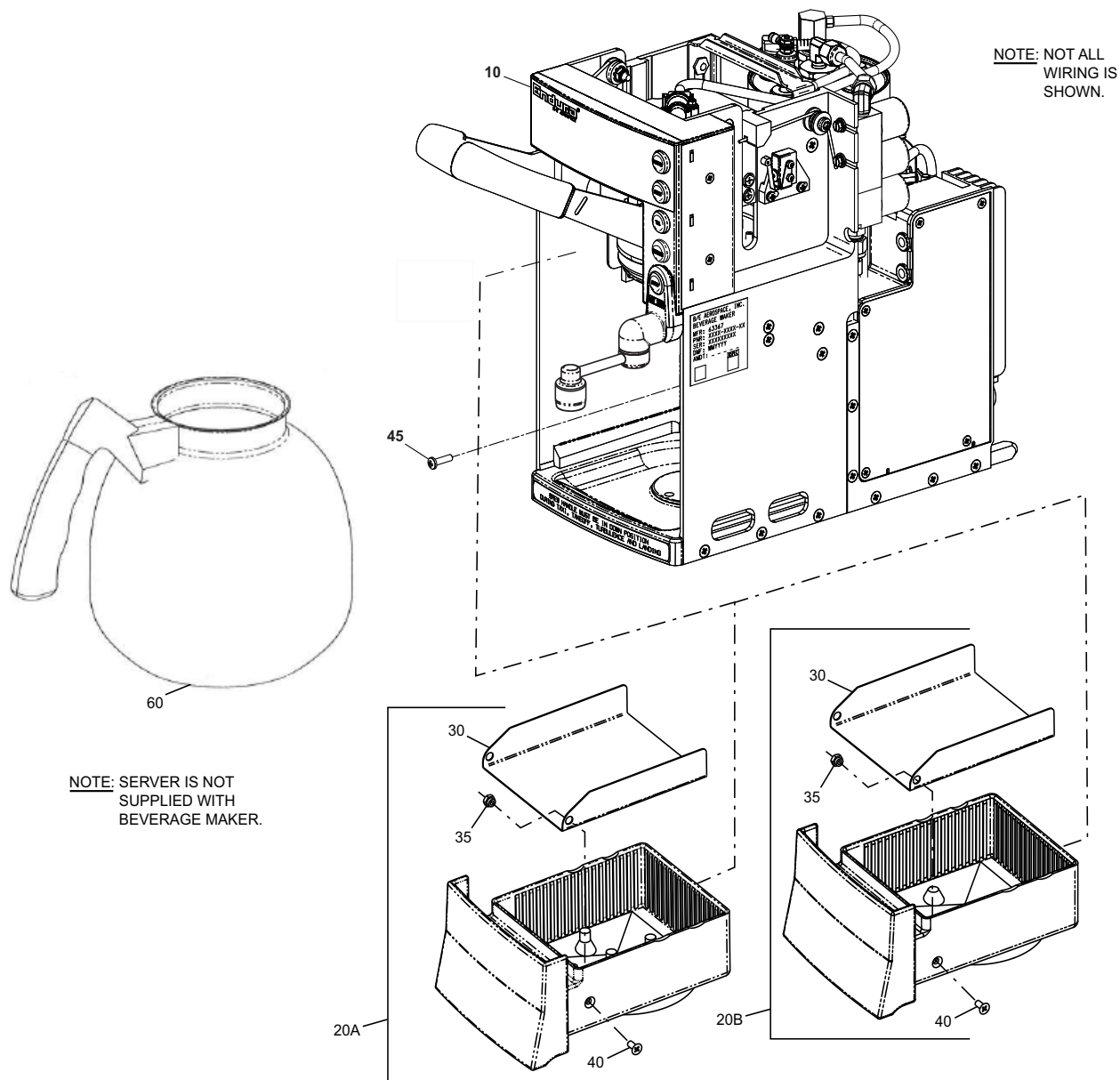
B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
20B	3520-0540-03		• BREW CUP ASSY (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • BREW CUP (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	RF
-25B	3530-0254-03		• • BREW CUP (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	RF
30	3575-0010-01		• • PLATE, SPACER ATTACHING PARTS		1
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG * * *		2
45	MS51957-46		• SCREW, PH, 8-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER		AR
60*	MISC3201		• SERVER (TPN MISC-3201)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-72*	3530-0311-01		• • GASKET, SERVER COVER		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			<u>*NOTE:</u> NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series



IPL Figure 1C
Beverage Maker

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1C					
-1A	4510-21UG-00		BEVERAGE MAKER	A	RF
-1B	4510-21LG-00		BEVERAGE MAKER	B	RF
-1C	4510-22UG-00		BEVERAGE MAKER	C	RF
-1D	4510-22LG-00		BEVERAGE MAKER	D	RF
-1E	4510-23UG-00		BEVERAGE MAKER	E	RF
-1F	4510-23LG-00		BEVERAGE MAKER	F	RF
-1G	4510-24UG-00		BEVERAGE MAKER	G	RF
-1H	4510-24LG-00		BEVERAGE MAKER	H	RF
-1J	4510-25UG-00		BEVERAGE MAKER	J	RF
-1K	4510-25LG-00		BEVERAGE MAKER	K	RF
-1L	4510-26UG-00		BEVERAGE MAKER	L	RF
-1M	4510-26LG-00		BEVERAGE MAKER	M	RF
-1N	4510-27UG-00		BEVERAGE MAKER	N	RF
-1P	4510-27LG-00		BEVERAGE MAKER	P	RF
-1R	4510-28UG-00		BEVERAGE MAKER	R	RF
-1T	4510-28LG-00		BEVERAGE MAKER	T	RF
10A	3520-0526-41		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	A,B	1
10B	3520-0526-42		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	C,D	1
10C	3520-0526-43		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	E,F	1
10D	3520-0526-44		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	G,H	1
10E	3520-0526-45		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	J,K	1
10F	3520-0526-46		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	L,M	1
10G	3520-0526-47		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	N,P	1
10H	3520-0526-48		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	R,T	1
20A	3520-0540-01		• BREW CUP ASSY (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1C					
20B	3520-0540-03		• BREW CUP ASSY (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • BREW CUP (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	RF
-25B	3530-0254-03		• • BREW CUP (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	RF
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 8-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER		AR
60*	MISC3201		• SERVER (TPN MISC-3201)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-72*	3530-0311-01		• • GASKET, SERVER COVER		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			<u>*NOTE:</u> NOT SUPPLIED WITH BEVERAGE MAKER		

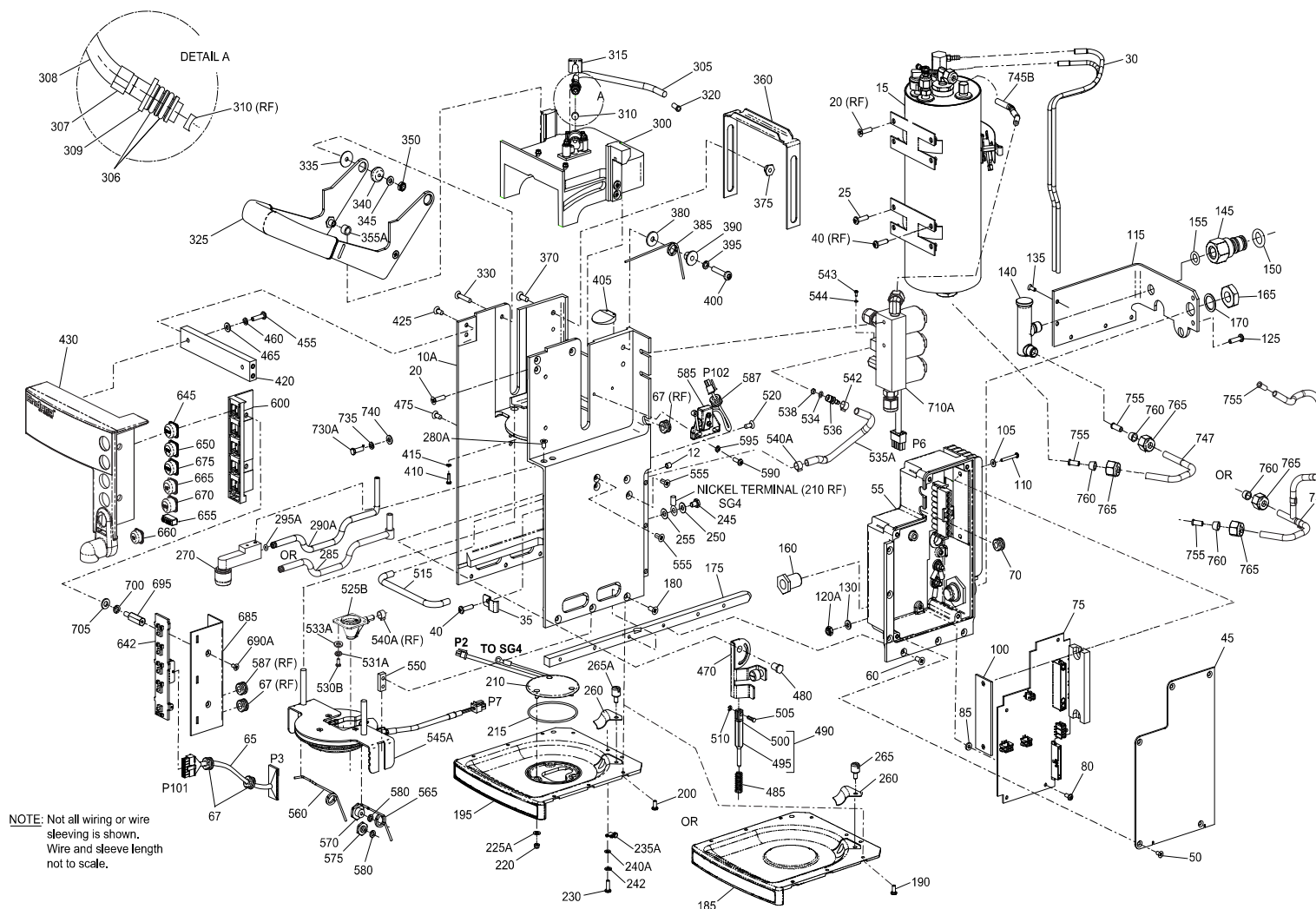
- ITEM NOT ILLUSTRATED

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the IPL Figure 2.

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

Temporary
Revision No. 8A

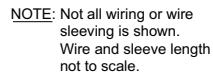


IPL Figure 2
Beverage Maker Main Subassembly

U.S. Export Classification: EAR99.
COLLINS AEROSPACE PROPRIETARY
Subject to the restriction on the cover page.

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4510-1 Series and 4510-2 Series



Beverage Maker Main Subassembly

U.S. Export Classification: EAR99
COLLINS AEROSPACE PROPRIETARY
Subject to the restriction on the cover page.

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-1A	3520-0526-11		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	A	RF
-1B	3520-0526-12		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	B	RF
-1C	3520-0526-13		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	C	RF
-1D	3520-0526-14		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	D	RF
-1E	3520-0526-15		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	E	RF
-1F	3520-0526-16		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	F	RF
-1G	3520-0526-17		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	G	RF
-1H	3520-0526-18		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	H	RF
-10	3520-0548-01		• CHASSIS ASSY (SUPSD BY ITEM 10A AFTER 20/01/2006)		1
10A	3520-0548-01		• CHASSIS ASSY (SUPSDS ITEM 10 AFTER 20/01/2006, REQUIRING FH SCREWS ITEM 370)		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• TANK ASSY (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
25	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		2
35	62-0144-1		• HOSE CLIP		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
45	3575-1223-01		• POWER MODULE COVER		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		
55	3520-0530-01		• POWER MODULE ASSY (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• HARNESS ASSY		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• CONTROLLER BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
160	62-0037-1		ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		NUT, JAM, HEX, CRS (SUPSDS ITEM 165)		1
170	LW163		WASHER, LK, 9/16 (TPN LW-163)		1
185	3520-0561-01		PLATEN ASSY	A, B, C, G	1
			ATTACHING PARTS		
190	MS51957-27		SCREW, PH, 6-32 X .31	A, B, C, G	8
			* * *		
195	3520-0561-03		PLATEN ASSY (USED WITH PLATEN HEATER)	D, E, F, H	1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		
140	62-0014-1		• DRAIN VALVE ASSY (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• WATER SHUTOFF PLUG ASSY		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-11		• RAIL GUIDE		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-01		• PLATEN ASSY	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	9
			* * *		
195	3520-0561-03		• PLATEN ASSY (USED WITH PLATEN HEATER)	D,E,F,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44		1
-235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201) (SUPSD BY ITEM 235A)	D,E,H	1
235A	CL50032		• CLAMP, NYLON (TPN CL-50032) (SUPSDS ITEM 235)		1
-240	WA120		• WASHER, CLAMP (TPN WA-120) (SUPSD BY ITEM 240A)	D,E,F,H	1
240A	NAS620C6		• WASHER, FLAT, REDUCED-OD (SUPSDS ITEM 240)		1
242	MS35338-136		• WASHER, LOCK (USED WITH ITEM 235A)		1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	8
			* * *		
-205	3575-1269-01		• • INLAY, PLATEN		1
210	3530-0261-01		• PLATEN HEATER	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A) (PRE SIL 4510-SIL-03)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1214-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
-265	3575-1215-01		• FASTENER, SCREEN, DRAIN (ALUMINUM) (SUPSD BY ITEM 265A AFTER 2004/05/05)		1
265A	3575-1215-01		• FASTENER, SCREEN, DRAIN (PLASTIC) (SUPSDS ITEM 265 AFTER 2004/05/05)		1
			* * *		
270	3520-0525-01		• FAUCET ASSY (SEE IPL FIGURE 7 FOR DET BKDN)		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, PH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 280)		2
			* * *		
285	3520-0535-01		• FAUCET TUBE ASSY	B,C,E,F G,H	1
-290	3520-0535-03		• FAUCET TUBE ASSY (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• BREW HEAD ASSY (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• BREW TUBE ASSY		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±.2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• FLOW WASHER		1
315	467-0912-1		• CLIP		1
320	PF202		• INSERT, TUBE (TPN PF-202)		1
			* * *		
325	3520-0533-01		• BREW HANDLE ASSY		1
-327	430531		• DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
335	411-0724-1		• HANDLE WASHER		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
-355	411-0692-1		• BUSHING, HANDLE (PLASTIC) (SUPSD BY ITEM 355A AFTER 2005/10/07)		2
355A	411-0692-1		• BUSHING, HANDLE (METAL) (SUPSDS ITEM 355 AFTER 2005/10/07)		2
-356	425-0702-1		• • GUIDE BUSHING ASSEMBLY		2
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375 AFTER 2006/01/20)		
370	MS24693C272		• SCREW, FH, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
420	3520-0542-01		• CHASSIS BRACE ASSY ATTACHING PARTS		1
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) * * *		4
430	3520-0560-01		• FRONT PANEL ASSY	F,H	1
-435	3520-0560-03		• FRONT PANEL ASSY	B,D,G	1
-440	3520-0560-05		• FRONT PANEL ASSY	A	1
-445	3520-0560-07		• FRONT PANEL ASSY	E	1
-450	3520-0560-09		• FRONT PANEL ASSY ATTACHING PARTS	C	1
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8 * * *		2
470	3575-0650-01		• LOCATING PIN LEVER ATTACHING PARTS		1
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• PIVOT PIN ASSY * * *		1
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• LOCATING PIN ASSY		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING, .375 DIAMETER (TPN OR-0571) ATTACHING PARTS		1
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING * * *		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3520-0236-01		• FUNNEL, TUBE (SUPSD BY ITEM 525A AFTER 2006/01/15)	B,F,G,H	1
-525A	3530-0289-01		• TUBE, FUNNEL (SUPSDS ITEM 525 AFTER 2006/01/15) (SUPSD BY ITEM 525B) (PRE SB 4510-25-14)	B,F,G,H	1
525B	3530-0879-01		• FUNNEL, TEA (SUPSDS ITEM 525A) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
-530	MS24693C1		• SCREW, 4-40 X .19 (TPN MS24693-C1) (SUPSD BY ITEMS 530A, 531, AND 533 AFTER 2006/01/15)	B,F,G,H	2
-530A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 530B)	B,F,G,H	2
530B	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530A)		2
-531	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 531A)	B,F,G,H	2
531A	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 531)		2
-533	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 533A)	B,F,G,H	2
533A	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 533)		2
534	OR0571		• O-RING, .375 DIAMETER (TPN OR-0571) (AFTER 2005/02/25)	B,F,G,H	1
-535	TU50242		• TUBE (5/16 IN OD X 3/16 IN ID) 80 IN LONG, (TPN TU-50242) (SUPSD BY ITEM 535A PER SB 4510-25-02)	B,F,G,H	1
535A	3575-1312-01		• TUBE, TEA, HEAT FORMED (SUPSDS ITEM 535 PER SB 4510-25-02)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSD BY ITEM 555A)		4
-555A	MS24693C49		• SCREW, CSK, 8-32 X.437, CRES-CLND (TPN MS24693-C49) (SUPSDS ITEM 555)		4
560	3575-0050-01		• SERVER RETAINER SPRING, LEFT HAND		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
536	3575-1283-01		• CARTRIDGE, REGULATOR (AFTER 2005/02/25)	B,F,G,H	1
538	62-0066-7		• FLOW WASHER (AFTER 2005/02/25)	B,F,G,H	1
-540	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSD BY ITEMS 540A AND 542) (PRE SB 4510-25-02)	B,F,G,H	2
540A	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
542	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
543	MS51957-13		• SCREW, PNH, 4-40 X .25 (AFTER 2005/02/25)	B,F,G,H	1
544	MS35338-135		• WASHER, LK, #4 (AFTER 2005/02/25)	B,F,G,H	1
			* * *		
-545	3520-0539-01		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSD BY ITEM 545A) (PRE SB 4510-25-03)		1
545A	3520-0580-03		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSDS ITEM 545) (POST SB 4510-25-03)		1
			ATTACHING PARTS		
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		8
560	3575-0050-01		• SERVER RETAINER SPRING, LEFT HAND		1
565	3575-0050-02		• SERVER RETAINER SPRING, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• SWITCH ASSY		1
587	GR098		• • GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A			DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A AFTER 2007/01/25)	F,G	1
710A	3530-0232-01		• ASSY, MANIFOLD (SUPSDS ITEM 710 AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN) (SUPSD BY ITEM 710B)	F,G	1
-710B	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0233-01		• ASSY, MANIFOLD (SUPSD BY ITEM 715A AFTER 2004/05/12)	C,E	1
-715A	3530-0232-05		• MANIFOLD ASSY (SUPSDS ITEM 715 AFTER 2004/05/12) (SUPSD BY ITEM 715B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715B	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0236-01		• ASSY, MANIFOLD (SUPSD BY ITEM 720A AFTER 2004/05/12)	A,D	1
-720A	3530-0232-07		• MANIFOLD ASSY (SUPSDS ITEM 720 AFTER 2004/05/12) (SUPSD BY ITEM 720B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720B	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0265-01		• ASSY, MANIFOLD (SUPSD BY ITEM 725A AFTER 2004/05/12)	B,H	1
-725A	3530-0232-03		• MANIFOLD ASSY (SUPSDS ITEM 725 AFTER 2004/05/12) (SUPSD BY ITEM 725B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725B	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
745B	TU50020		TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17) (SUPSD BY ITEM 747A)	A,B,D,H	1
-747A	62-0196-3		TUBE, FORMED (SUPSDS ITEM 747)	A,B,D,H	1
750	3520-0554-01		WATER INLET TUBE ASSY	C,E,F,G	1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
-745	TU50020		TUBING, 8.5 IN (TPN TU-50020) (SUPSD BY ITEM 745A)		2
-745A	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 7.5 IN (TPN TU-50020) (SUPSDS ITEM 745) (SUPSD BY ITEM 745B AFTER 2006/08/17)		2
745B	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		• TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17)	A,B,D,H	1
750	3520-0554-01		• WATER INLET TUBE ASSY	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		

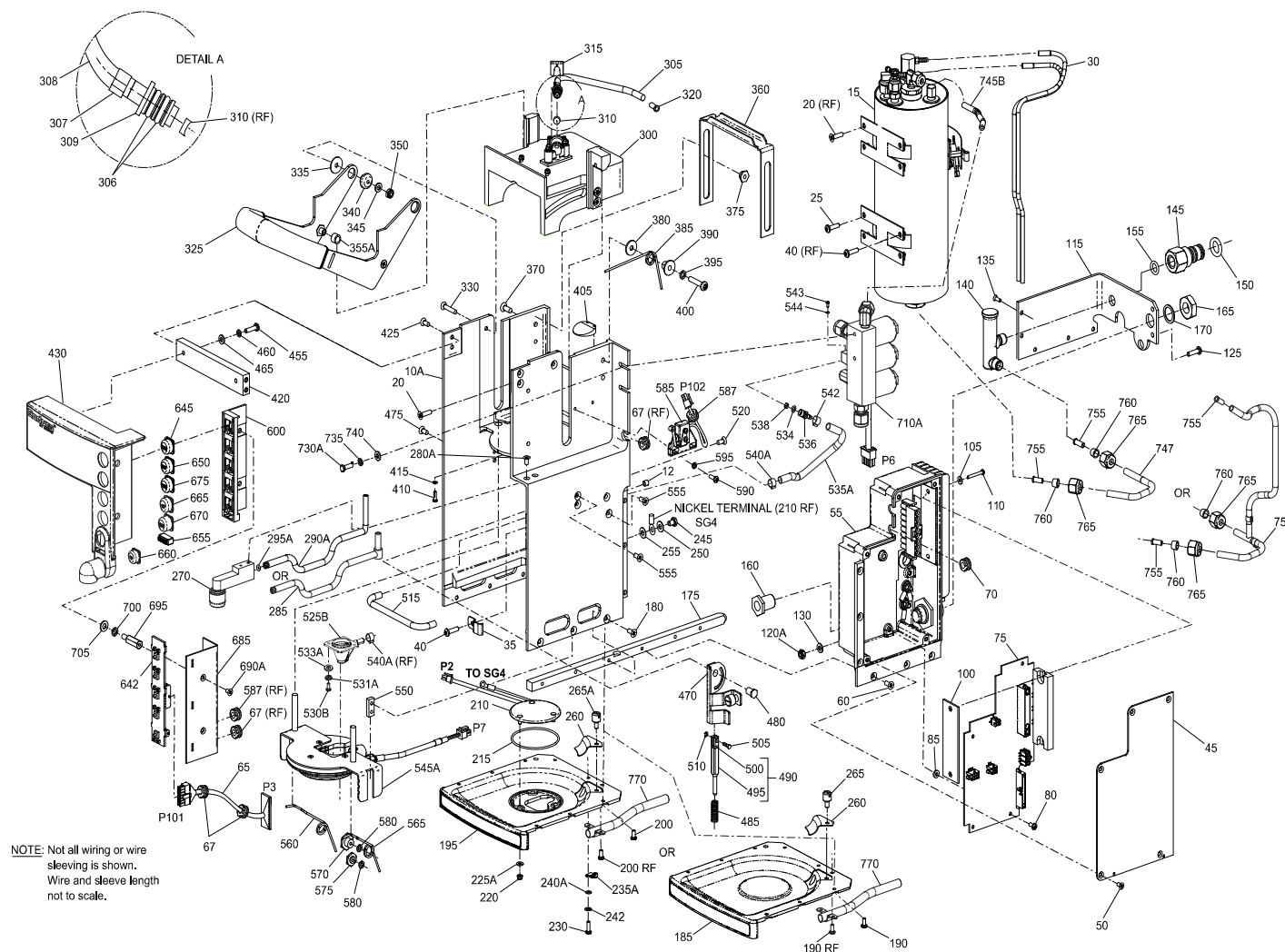
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the IPL Figure 2A.

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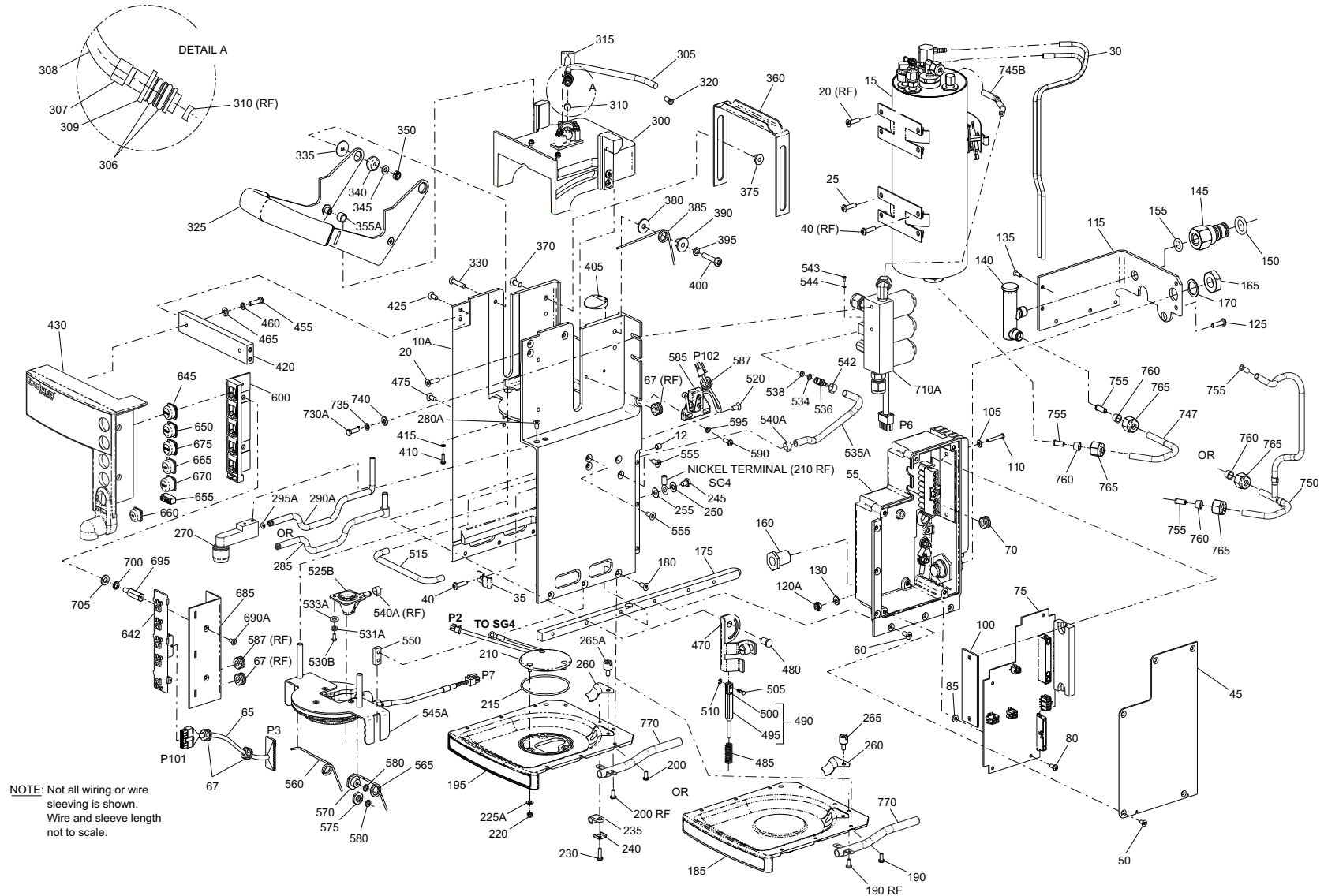


IPL Figure 2A
Beverage Maker Main Subassembly

U.S. Export Classification: EAR99.
COLLINS AEROSPACE PROPRIETARY
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IPL Figure 2A
 Beverage Maker Main Subassembly

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
-1A	3520-0526-21		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	A	RF
-1B	3520-0526-22		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	B	RF
-1C	3520-0526-23		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	C	RF
-1D	3520-0526-24		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	D	RF
-1E	3520-0526-25		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	E	RF
-1F	3520-0526-26		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	F	RF
-1G	3520-0526-27		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	G	RF
-1H	3520-0526-28		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	H	RF
-10	3520-0548-01		• CHASSIS ASSY (SUPSD BY ITEM 10A AFTER 2006/01/20)		1
10A	3520-0548-01		• CHASSIS ASSY (SUPSDS ITEM 10 AFTER 2006/01/20, REQUIRING FH SCREWS ITEM 370)		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• TANK ASSY (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
25	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		2
35	62-0144-1		• HOSE CLIP		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
45	3575-1223-01		• POWER MODULE COVER ATTACHING PARTS		1
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) * * *		5
55	3520-0530-01		• POWER MODULE ASSY (SEE IPL FIGURE 4 FOR DET BKDN) ATTACHING PARTS		1
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) * * *		6
65	3520-0553-01		• HARNESS ASSY		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098) * * *		2
75	3530-0121-01		• CONTROLLER BOARD ASSY (REFER TO CMM 25-33-32) ATTACHING PARTS		1
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875 * * *		2
115	62-0018-1		• CONNECTOR, SUPPORT		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
160	62-0037-1		ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		NUT, JAM, HEX, CRS (SUPSDS ITEM 165)		1
170	LW163		WASHER, LK, 9/16 (TPN LW-163)		1
185	3520-0561-01		PLATEN ASSY	A, B, C, G	1
			ATTACHING PARTS		
190	MS51957-27		SCREW, PH, 6-32 X .31	A, B, C, G	8
			* * *		

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (MS24693-C48)		5
			* * *		
140	62-0014-1		• DRAIN VALVE ASSY (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• WATER SHUTOFF PLUG ASSY		1
150	OR067		• O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-11		• RAIL GUIDE		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-01		• PLATEN ASSY	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	9
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44		1
-235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201) (SUPSD BY ITEM 235A)	D,E,H	1
235A	CL50032		• CLAMP, NYLON (TPN CL-50032) (SUPSDS ITEM 235)		1
-240	WA120		• WASHER, CLAMP (TPN WA-120) (SUPSD BY ITEM 240A)	D,E,F,H	1
240A	NAS620C6		• WASHER, FLAT, REDUCED-OD (SUPSDS ITEM 240)		1
242	MS35338-136		• WASHER, LOCK (USED WITH ITEM 235A)		1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
195	3520-0561-03		• PLATEN ASSY	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	8
			* * *		
-205	3575-1269-01		• • INLAY, PLATEN		1
210	3530-0261-01		• PLATEN HEATER	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A) (PRE SIL 4510-SIL-03)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1214-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
-265	3575-1215-01		• FASTENER, SCREEN, DRAIN (ALUMINUM) (SUPSD BY ITEM 265A AFTER 2004/05/05)		1
265A	3575-1215-01		• FASTENER, SCREEN, DRAIN (PLASTIC) (SUPSDS ITEM 265 AFTER 2004/05/05)		1
			* * *		
270	3520-0525-01		• FAUCET ASSY (SEE IPL FIGURE 7 FOR DET BKDN)		1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (SUPSD BY ITEM 280A) (TPN MS24693-C26)		2
280A	MS24693C24		• SCREW, FH, 6-32 X .25 (SUPSDS ITEM 280) (TPN MS24693-C24)		2
			* * *		
285	3520-0535-01		• FAUCET TUBE ASSY	B,C,E,F G,H	1
-290	3520-0535-03		• FAUCET TUBE ASSY (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• ASSY, BREWHEAD (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• BREW TUBE ASSY		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±.2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• FLOW WASHER		1
315	467-0912-1		• CLIP		1
320	PF202		• INSERT, TUBE (TPN PF-202)		1
			* * *		
325	3520-0533-01		• BREW HANDLE ASSY		1
-327	430531		• DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
-355	411-0692-1		• BUSHING, HANDLE (PLASTIC) (SUPSD BY ITEM 355A AFTER 2005/10/07)		2
355A	411-0692-1		• BUSHING, HANDLE (METAL) (SUPSDS ITEM 355 AFTER 2005/10/07)		2
-356	425-0702-1		• • GUIDE BUSHING ASSEMBLY		2
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE		
370	MS24693C272		• SCREW, FH, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
420	3520-0542-01		• CHASSIS BRACE ASSY ATTACHING PARTS		1
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) * * *		4
430	3520-0560-01		• FRONT PANEL ASSY	F,H	1
-435	3520-0560-03		• FRONT PANEL ASSY	B,D,G	1
-440	3520-0560-05		• FRONT PANEL ASSY	A	1
-445	3520-0560-07		• FRONT PANEL ASSY	E	1
-450	3520-0560-09		• FRONT PANEL ASSY ATTACHING PARTS	C	1
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8 * * *		2
470	3575-0650-01		• LOCATING PIN LEVER ATTACHING PARTS		1
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• PIVOT PIN ASSY * * *		1
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• LOCATING PIN ASSY		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING, .375 DIAMETER (TPN OR-0571) ATTACHING PARTS		1
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING * * *		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3520-0236-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A AFTER 2006/01/15)	B,F,G,H	1
-525A	3530-0289-01		• TUBE, FUNNEL (SUPSDS ITEM 525 AFTER 2006/01/15) (SUPSD BY ITEM 525B) (PRE SB 4510-25-14)	B,F,G,H	1
525B	3530-0879-01		• FUNNEL, TEA (SUPSDS ITEM 525A) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
-530	MS24693C1		• SCREW 4-40 X .19 (TPN MS24693-C1) (SUPSD BY ITEMS 530A, 531, AND 533 AFTER 2006/01/15)	B,F,G,H	2
-530A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 530B)	B,F,G,H	2
530B	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530A)		2
-531	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 531A)	B,F,G,H	2
531A	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 531)		2
-533	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 533A)	B,F,G,H	2
533A	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 533)		2
534	OR0571		• O-RING, .375 DIAMETER (TPN OR-0571) (AFTER 2005/02/25)	B,F,G,H	1
-535	TU50242		• TUBE (5/16 IN OD X 3/16 IN ID), 8.0 IN LONG (TPN TU-50242) (SUPSD BY ITEM 535A PER SB 4510-25-02)	B,F,G,H	1
535A	3575-1312-01		• TUBE, TEA, HEAT FORMED (SUPSDS ITEM 535 PER SB 4510-25-02)	B,F,G,H	1

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSD BY ITEM 555A)		4
-555A	MS24693C49		• SCREW, CSK, 8-32 X.437, CRES-CLND (TPN MS24693-C49) (SUPSDS ITEM 555)		4
560	3575-0050-01		• SERVER RETAINER SPRING, LEFT HAND		1

- ITEM NOT ILLUSTRATED

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
536	3575-1283-01		• CARTRIDGE, REGULATOR (AFTER 2005/02/25)	B,F,G,H	1
538	62-0066-7		• FLOW WASHER (AFTER 2005/02/25)	B,F,G,H	1
-540	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSD BY ITEMS 540A AND 542) (PRE SB 4510-25-02)	B,F,G,H	2
540A	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
542	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
543	MS51957-13		• SCREW, PNH, 4-40 X .25 (AFTER 2005/02/25)	B,F,G,H	1
544	MS35338-135		• WASHER, LK, #4 (AFTER 2005/02/25)	B,F,G,H	1
			* * *		
-545	3520-0539-01		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSD BY ITEM 545A)		1
545A	3520-0580-03		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSDS ITEM 545)		1
			ATTACHING PARTS		
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		8
560	3575-0050-01		• SERVER RETAINER SPRING, LEFT HAND		1
565	3575-0050-02		• SERVER RETAINER SPRING, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• SWITCH ASSY		1
587	GR098		• • GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2
			* * *		

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A			DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A AFTER 2007/01/25)	F,G	1
710A	3530-0232-01		• ASSY, MANIFOLD (SUPSDS ITEM 710 AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN) (SUPSD BY ITEM 710B)	F,G	1
-710B	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0233-01		• ASSY, MANIFOLD (SUPSD BY ITEM 715A AFTER 2004/05/12)	C,E	1
-715A	3530-0232-05		• ASSY, MANIFOLD (SUPSDS ITEM 715 AFTER 2004/05/12) (SUPSD BY ITEM 715B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715B	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0236-01		• ASSY, MANIFOLD (SUPSD BY ITEM 720A AFTER 2004/05/12)	A,D	1
-720A	3530-0232-07		• ASSY, MANIFOLD (SUPSDS ITEM 720 AFTER 2004/05/12) (SUPSD BY ITEM 720B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720B	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0265-01		• ASSY, MANIFOLD (SUPSD BY ITEM 725A AFTER 2004/05/12)	B,H	1
-725A	3530-0232-03		• ASSY, MANIFOLD (SUPSDS ITEM 725 AFTER 2004/05/12) (SUPSD BY ITEM 725B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725B	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
745B	TU50020		TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17) (SUPSD BY ITEM 747A)	A, B, D, H	1
-747A	62-0196-3		TUBE, FORMED (SUPSDS ITEM 747)	A, B, D, H	1
750	3520-0554-01		WATER INLET TUBE ASSY	C,E,F,G	1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
-745	TU50020		• TUBING, 8.5 IN (SUPSD BY ITEM 745A) (TPN TU-50020)		2
-745A	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 7.5 IN (TPN TU-50020) (SUPSDS ITEM 745) (SUPSD BY ITEM 745B AFTER 2006/08/17)		2
745B	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		• TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17)	A,B,D,H	1
750	3520-0554-01		• WATER INLET TUBE ASSY	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		
770	3520-0528-01		• DRAIN TUBE ASSY		1
			ATTACHING PARTS		
-775	MS51957-27		• SCREW, PH, 6-32 X .31		1
			* * *		

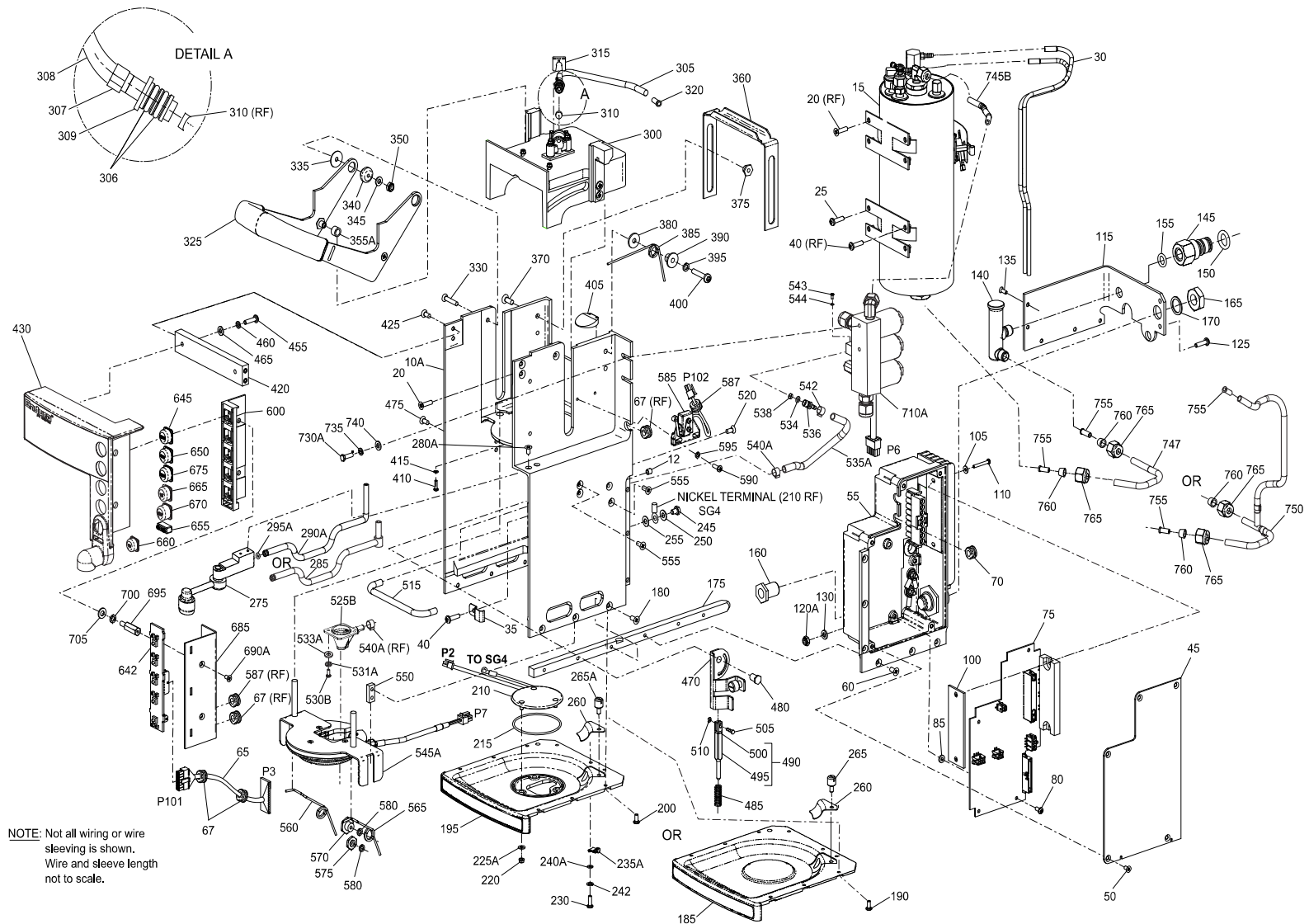
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COMPONENT MAINTENANCE MANUAL
 4510-1 Series and 4510-2 Series

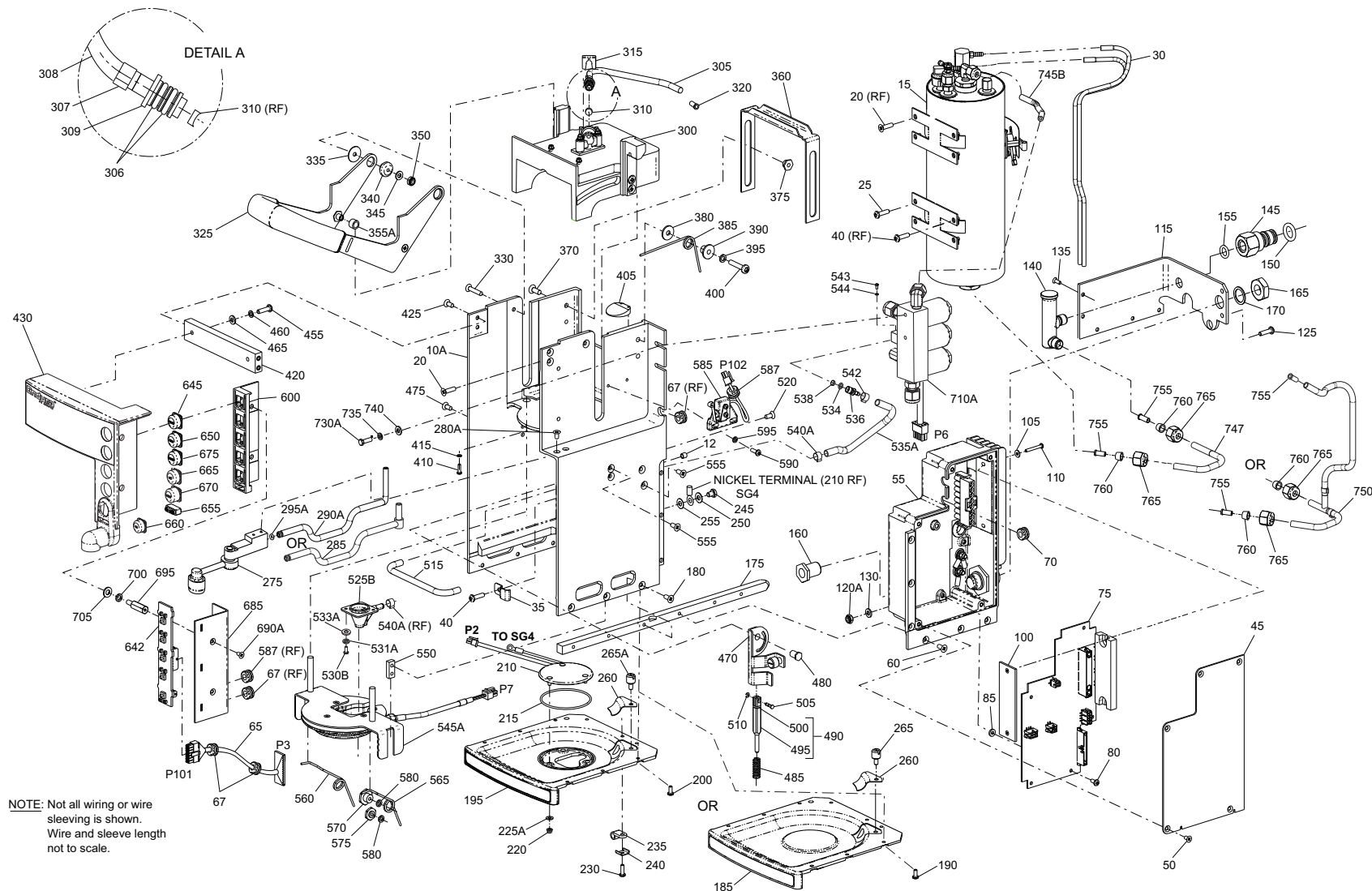
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IPL Figure 2B
Beverage Maker Subassembly

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IPL Figure 2B
 Beverage Maker Subassembly

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-1A	3520-0526-31		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	A	RF
-1B	3520-0526-32		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	B	RF
-1C	3520-0526-33		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	C	RF
-1D	3520-0526-34		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	D	RF
-1E	3520-0526-35		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	E	RF
-1F	3520-0526-36		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	F	RF
-1G	3520-0526-37		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	G	RF
-1H	3520-0526-38		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	H	RF
-10	3520-0548-01		• CHASSIS ASSY (SUPSD BY ITEM 10A AFTER 2006/01/20)		1
10A	3520-0548-01		• CHASSIS ASSY (SUPSDS ITEM 10 AFTER 2006/01/20, REQUIRING FH SCREWS ITEM 370)		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• TANK ASSY (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
25	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		2
35	62-0144-1		• HOSE CLIP		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
45	3575-1223-01		• POWER MODULE COVER		1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		
55	3520-0530-01		• POWER MODULE ASSY (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• HARNESS ASSY		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
			* * *		
75	3530-0121-01		• CONTROLLER BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
160	62-0037-1		ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		NUT, JAM, HEX, CRS (SUPSDS ITEM 165)		1
170	LW163		WASHER, LK, 9/16 (TPN LW-163)		1
185	3520-0561-01		PLATEN ASSY	A, B, C, G	1
			ATTACHING PARTS		
190	MS51957-27		SCREW, PH, 6-32 X .31	A, B, C, G	8

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		
140	62-0014-1		• DRAIN VALVE ASSY (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• WATER SHUTOFF PLUG ASSY		1
150	OR067		• O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-11		• RAIL GUIDE		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-01		• PLATEN ASSY	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	9
			* * *		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44		1
-235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201) (SUPSD BY ITEM 235A)	D,E,H	1
235A	CL50032		• CLAMP, NYLON (TPN CL-50032) (SUPSDS ITEM 235)		1
-240	WA120		• WASHER, CLAMP (TPN WA-120) (SUPSD BY ITEM 240A)	D,E,F,H	1
240A	NAS620C6		• WASHER, FLAT, REDUCED-OD (SUPSDS ITEM 240)		1
242	MS35338-136		• WASHER, LOCK (USED WITH ITEM 235A)		1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
195	3520-0561-03		• PLATEN ASSY	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	8
			* * *		
-205	3575-1269-01		• • INLAY, PLATEN		1
210	3530-0261-01		• PLATEN HEATER	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A) (PRE SIL 4510-SIL-03)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1214-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
-265	3575-1215-01		• FASTENER, SCREEN, DRAIN (ALUMINUM) (SUPSD BY ITEM 265A AFTER 2004/05/05)		1
265A	3575-1215-01		• FASTENER, SCREEN, DRAIN (PLASTIC) (SUPSDS ITEM 265 AFTER 2004/05/05)		1
			* * *		
275	3520-0559-01		• SWIVEL FAUCET ASSY (SEE IPL FIGURE 8 FOR DET BKDN)		1

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (SUPSD BY ITEM 280A) (TPN MS24693-C26)		2
280A	MS24693C24		• SCREW, FH, 6-32 X .25 (SUPSDS ITEM 280) (TPN MS24693-C24)		2
			* * *		
285	3520-0535-01		• FAUCET TUBE ASSY	B,C,E, F,G,H	1
-290	3520-0535-03		• FAUCET TUBE ASSY (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (SUPSDS ITEM 295) (TPN OR-0381)		1
300	3520-0534-01		• BREW HEAD ASSY (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• BREW TUBE ASSY		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±.2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• FLOW WASHER		1
315	467-0912-1		• CLIP		1
320	PF202		• INSERT, TUBE (TPN PF-202)		1
			* * *		
325	3520-0533-01		• BREW HANDLE ASSY		1
-327	430531		• DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
-355	411-0692-1		• BUSHING, HANDLE (PLASTIC) (SUPSD BY ITEM 355A AFTER 2005/10/07)		2
355A	411-0692-1		• BUSHING, HANDLE (METAL) (SUPSDS ITEM 355 AFTER 2005/10/07)		2
-356	425-0702-1		• • GUIDE BUSHING ASSEMBLY		2
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 365 AFTER 2006/01/20)		2
370	MS24693C272		• SCREW, FH, 10-32 X.50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		

- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
420	3520-0542-01		• CHASSIS BRACE ASSY ATTACHING PARTS		1
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) * * *		4
430	3520-0560-01		• FRONT PANEL ASSY	F,H	1
-435	3520-0560-03		• FRONT PANEL ASSY	B,D,G	1
-440	3520-0560-05		• FRONT PANEL ASSY	A	1
-445	3520-0560-07		• FRONT PANEL ASSY	E	1
-450	3520-0560-09		• FRONT PANEL ASSY ATTACHING PARTS	C	1
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8 * * *		2
470	3575-0650-01		• LOCATING PIN LEVER ATTACHING PARTS		1
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• PIVOT PIN ASSY * * *		1
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• LOCATING PIN ASSY		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING, .375 DIAMETER (TPN OR-0571) ATTACHING PARTS		1
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING * * *		1

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3520-0236-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A AFTER 2006/01/15)	B,F,G,H	1
-525A	3530-0289-01		• TUBE, FUNNEL (SUPSDS ITEM 525 AFTER 2006/01/15) (SUPSD BY ITEM 525B) (PRE SB 4510-25-14)	B,F,G,H	1
525B	3530-0879-01		• FUNNEL, TEA (SUPSDS ITEM 525A) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
-530	MS24693C1		• SCREW 4-40 X .19 (TPN MS24693-C1) (SUPSD BY ITEMS 530A, 531, AND 533 AFTER 2006/01/15)	B,F,G,H	2
-530A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 530B)	B,F,G,H	2
530B	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530A)		2
-531	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 531A)	B,F,G,H	2
531A	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 531)		2
-533	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 533A)	B,F,G,H	2
533A	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 533)		2
534	OR0571		• O-RING, .375 DIAMETER (TPN OR-0571) (AFTER 2005/02/25)	B,F,G,H	1
-535	TU50242		• TUBE (5/16 IN OD X 3/16 IN ID), 8.0 IN LONG (TPN TU-50242) (SUPSD BY ITEM 535A PER SB 4510-25-02)	B,F,G,H	1
535A	3575-1312-01		• TUBE, TEA, HEAT FORMED (SUPSDS ITEM 535 PER SB 4510-25-02)	B,F,G,H	1

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

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Revision No. 8A

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSD BY ITEM 555A)		4
-555A	MS24693C49		• SCREW, CSK, 8-32 X.437, CRES-CLND (TPN MS24693-C49) (SUPSDS ITEM 555)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
536	3575-1283-01		• CARTRIDGE, REGULATOR (AFTER 2005/02/25)	B,F,G,H	1
538	62-0066-7		• FLOW WASHER (AFTER 2005/02/25)	B,F,G,H	1
-540	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSD BY ITEMS 540A AND 542) (PRE SB 4510-25-02)	B,F,G,H	2
540A	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
542	CL50010		• CLAMP, TUBE, 9/32 INCH (CL-50010) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
543	MS51957-13		• SCREW, PNH, 4-40 X .25 (AFTER 2005/02/25)	B,F,G,H	1
544	MS35338-135		• WASHER, LK, #4 (AFTER 2005/02/25)	B,F,G,H	1
			* * *		
-545	3520-0539-01		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSD BY ITEM 545A) (PRE SB 4510-25-03)		1
545A	3520-0580-03		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSDS ITEM 545) (POST SB 4510-25-03)		1
			ATTACHING PARTS		
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		8
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• SWITCH ASSY		1
587	GR098		• • GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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4510-1 Series and 4510-2 Series

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A			DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (SUPSDS ITEM 690) (TPN MS24693-C24)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2

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B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A AFTER 2007/01/25)	F,G	1
710A	3530-0232-01		• ASSY, MANIFOLD (SUPSDS ITEM 710 AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN) (SUPSD BY ITEM 710B)	F,G	1
-710B	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0233-01		• ASSY, MANIFOLD (SUPSD BY ITEM 715A AFTER 2004/05/12)	C,E	1
-715A	3530-0232-05		• ASSY, MANIFOLD (SUPSDS ITEM 715 AFTER 2004/05/12) (SUPSD BY ITEM 715B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715B	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0236-01		• ASSY, MANIFOLD (SUPSD BY ITEM 720A AFTER 2004/05/12)	A,D	1
-720A	3530-0232-07		• ASSY, MANIFOLD (SUPSDS ITEM 720 AFTER 2004/05/12) (SUPSD BY ITEM 720B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720B	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0265-01		• ASSY, MANIFOLD (SUPSD BY ITEM 725A AFTER 2004/05/12)	B,H	1
-725A	3530-0232-03		• ASSY, MANIFOLD (SUPSDS ITEM 725 AFTER 2004/05/12) (SUPSD BY ITEM 725B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725B	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
745B	TU50020		TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17) (SUPSD BY ITEM 747A)	A,B,D,H	1
-747A	62-0196-3		TUBE, FORMED (SUPSDS ITEM 747)	A,B,D,H	1
750	3520-0554-01		WATER INLET TUBE ASSY	C,E,F,G	1

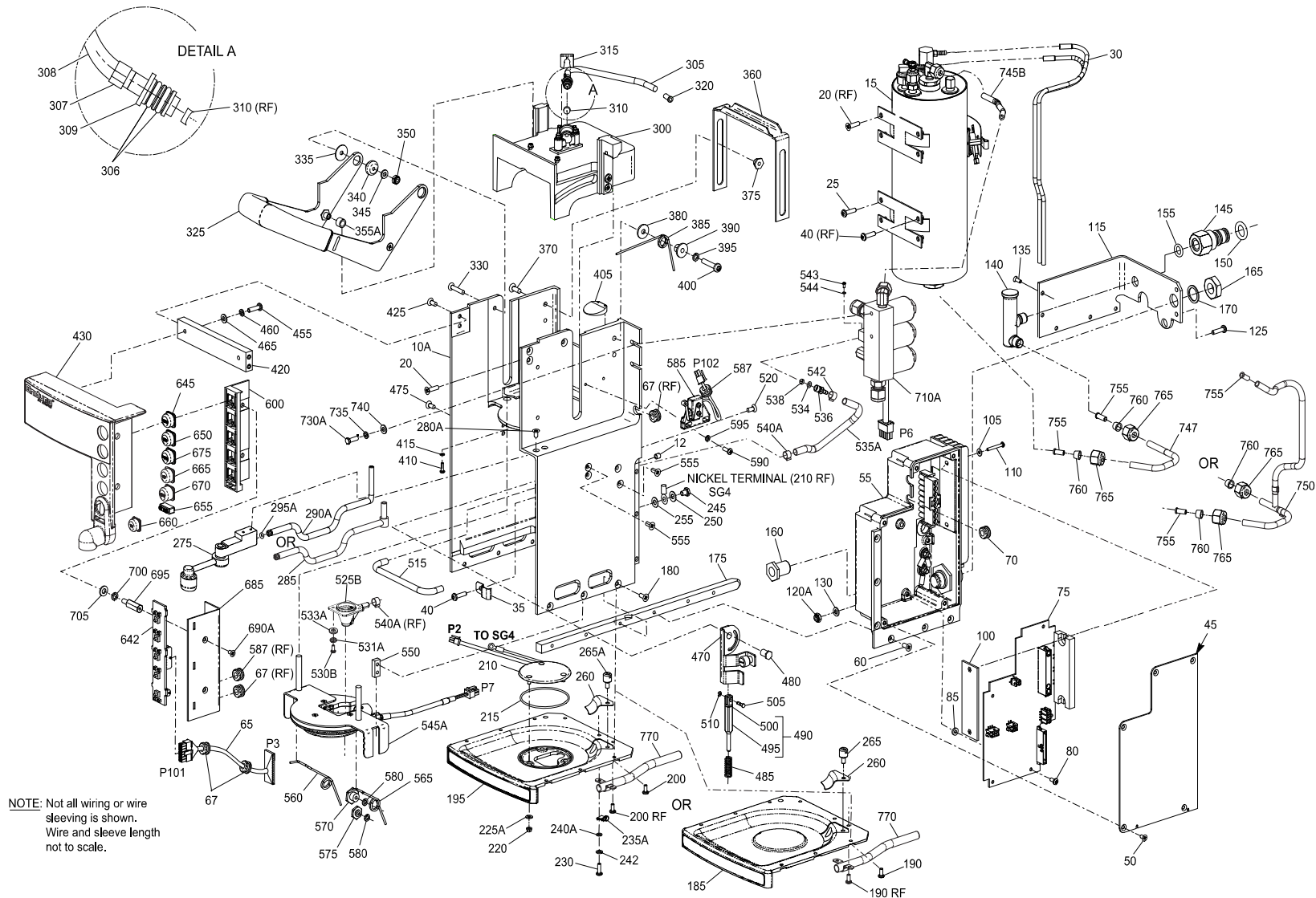
- ITEM NOT ILLUSTRATED

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
-745	TU50020		• TUBING, 8.5 IN (TPN TU-50020) (SUPSD BY ITEM 745A)		2
-745A	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 7.5 IN (TPN TU-50020) (SUPSDS ITEM 745) (SUPSD BY ITEM 745B AFTER 2006/08/17)		2
745B	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		• TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17)	A,B,D,H	1
750	3520-0554-01		• WATER INLET TUBE ASSY	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		

- ITEM NOT ILLUSTRATED

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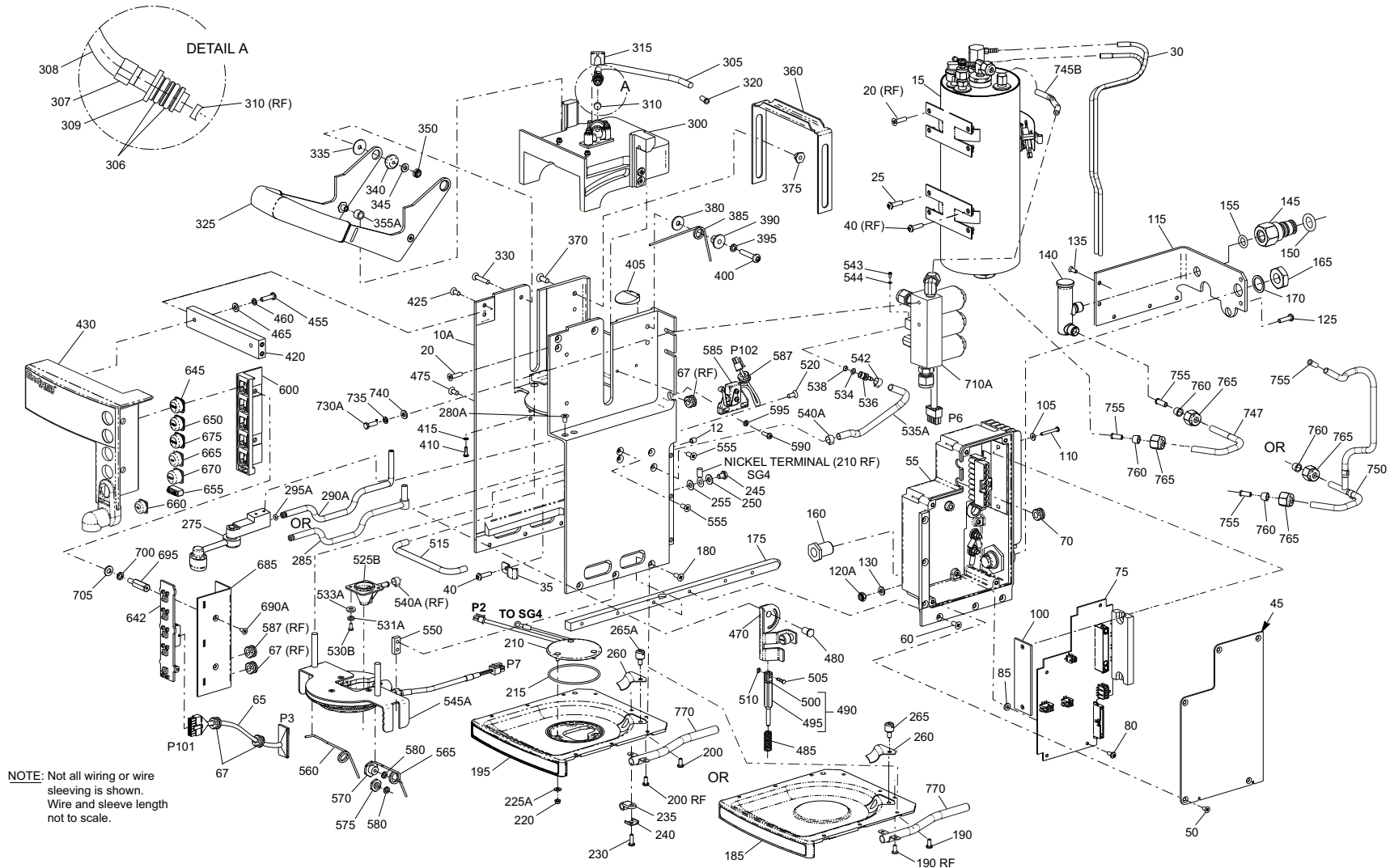


IPL Figure 2C
Beverage Maker Main Subassembly

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IPL Figure 2C
Beverage Maker Main Subassembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-1A	3520-0526-41		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	A	RF
-1B	3520-0526-42		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	B	RF
-1C	3520-0526-43		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	C	RF
-1D	3520-0526-44		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	D	RF
-1E	3520-0526-45		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	E	RF
-1F	3520-0526-46		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	F	RF
-1G	3520-0526-47		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	G	RF
-1H	3520-0526-48		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	H	RF
-10	3520-0548-01		• CHASSIS ASSY (SUPSD BY ITEM 10A AFTER 2006/01/20)		1
10A	3520-0548-01		• CHASSIS ASSY (SUPSDS ITEM 10 AFTER 2006/01/20, REQUIRING FH SCREWS ITEM 370)		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• TANK ASSY (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
25	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		2
35	62-0144-1		• HOSE CLIP		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		1
			* * *		
45	3575-1223-01		• POWER MODULE COVER		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		
55	3520-0530-01		• POWER MODULE ASSY (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• HARNESS ASSY		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• CONTROLLER BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
160	62-0037-1		ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		NUT, JAM, HEX, CRS (SUPSDS ITEM 165)		1
170	LW163		WASHER, LK, 9/16 (TPN LW-163)		1
185	3520-0561-01		PLATEN ASSY	A, B, C, G	1
			ATTACHING PARTS		
190	MS51957-27		SCREW, PH, 6-32 X .31	A, B, C, G	8
			* * *		
195	3520-0561-03		PLATEN ASSY	D, E, F, H	1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		
140	62-0014-1		• DRAIN VALVE ASSY (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• WATER SHUTOFF PLUG ASSY		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• ALIGNMENT PIN GUIDE		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-11		• RAIL GUIDE		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-01		• PLATEN ASSY	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	9
			* * *		
195	3520-0561-03		• PLATEN ASSY	D,E,F,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
230	MS51957-29		• SCREW, PH, 6-32 X .44	D,E,F,H	1
-235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201) (SUPSD BY ITEM 235A)	D,E,H	1
235A	CL50032		• CLAMP, NYLON (TPN CL-50032) (SUPSDS ITEM 235)		1
-240	WA120		• WASHER, CLAMP (TPN WA-120) (SUPSD BY ITEM 240A)	D,E,F,H	1
240A	NAS620C6		• WASHER, FLAT, REDUCED-OD (SUPSDS ITEM 240)		1
242	MS35338-136		• WASHER, LOCK (USED WITH ITEM 235A)		1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	8
			* * *		
-205	3575-1269-01		• • INLAY, PLATEN		1
210	3530-0261-01		• PLATEN HEATER	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A) (PRE SIL 4510-SIL-03)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225) (POST SIL 4510-SIL-03)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1214-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
-265	3575-1215-01		• FASTENER, SCREEN, DRAIN (ALUMINUM) (SUPSD BY ITEM 265A AFTER 2004/05/05)		
265A	3575-1215-01		• FASTENER, SCREEN, DRAIN (PLASTIC) (SUPSDS ITEM 265 AFTER 2004/05/05)		1
			* * *		
275	3520-0559-01		• SWIVEL FAUCET ASSY (SEE IPL FIGURE 8 FOR DET BKDN)		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, FH, 6-32 X .25 (SUPSDS ITEM 280) (TPN MS24693-C24)		2
			* * *		
285	3520-0535-01		• FAUCET TUBE ASSY	B,C,E,F G,H	1
-290	3520-0535-03		• FAUCET TUBE ASSY (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• BREWHEAD ASSY (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• BREW TUBE ASSY		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±.2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• FLOW WASHER		1
315	467-0912-1		• CLIP		1
320	PF202		• INSERT, TUBE (TPN PF-202)		1
			* * *		
325	3520-0533-01		• BREW HANDLE ASSY		1
-327	430531		• DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
-355	411-0692-1		• BUSHING, HANDLE (PLASTIC) (SUPSD BY ITEM 355A AFTER 2005/10/07)		2
355A	411-0692-1		• BUSHING, HANDLE (METAL) (SUPSDS ITEM 355 AFTER 2005/10/07)		2
-356	425-0702-1		• • GUIDE BUSHING ASSEMBLY		2
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375 AFTER 2006/01/20)		
370	MS24693C272		• SCREW, FH, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A AND AN ITEM 370 SCREW)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
420	3520-0542-01		• CHASSIS BRACE ASSEMBLY ATTACHING PARTS		1
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) * * *		4
430	3520-0560-01		• FRONT PANEL ASSY	F,H	1
-435	3520-0560-03		• FRONT PANEL ASSY	B,D,G	1
-440	3520-0560-05		• FRONT PANEL ASSY	A	1
-445	3520-0560-07		• FRONT PANEL ASSY	E	1
-450	3520-0560-09		• FRONT PANEL ASSY ATTACHING PARTS	C	1
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8 * * *		2
470	3575-0650-01		• LEVER, PIN, LOCATING ATTACHING PARTS		1
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• PIVOT PIN ASSY * * *		1
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• LOCATING PIN ASSY		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING, .375 DIAMETER (TPN OR-0571) ATTACHING PARTS		1
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING * * *		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3520-0236-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A AFTER 2006/01/15)	B,F,G,H	1
-525A	3530-0289-01		• TUBE, FUNNEL (SUPSDS ITEM 525 AFTER 2006/01/15) (SUPSD BY ITEM 525B) (PRE SB 4510-25-14)	B,F,G,H	1
525B	3530-0879-01		• FUNNEL, TEA (SUPSDS ITEM 525A) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
-530	MS24693C1		• SCREW 4-40 X .19 (TPN MS24693-C1) (SUPSD BY ITEMS 530A, 531, AND 533 AFTER 2006/01/15)	B,F,G,H	2
-530A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 530B)	B,F,G,H	2
530B	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 530A)		2
-531	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 531A)	B,F,G,H	2
531A	MS35338-135		• WASHER, LK, #4 (SUPSDS ITEM 531)		2
-533	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 530 AFTER 2006/01/15) (SUPSD BY ITEM 533A)	B,F,G,H	2
533A	MS15795-804		• WASHER, FL, 4-40 (SUPSDS ITEM 533)		2
534	OR0571		• O-RING, .375 DIAMETER (TPN OR-0571) (AFTER 2005/02/25)	B,F,G,H	1
-535	TU50242		• TUBE (5/16 IN OD X 3/16 IN ID), 8.0 IN LONG (TPN TU-50242) (SUPSD BY ITEM 535A PER SB 4510-25-02)	B,F,G,H	1
535A	3575-1312-01		• TUBE, TEA, HEAT FORMED (SUPSDS ITEM 535 PER SB 4510-25-02)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSD BY ITEM 555A)		4
-555A	MS24693C49		• SCREW, CSK, 8-32 X.437, CRES-CLND (TPN MS24693-C49) (SUPSDS ITEM 555)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
536	3575-1283-01		• CARTRIDGE, REGULATOR (AFTER 2005/02/25)	B,F,G,H	1
538	62-0066-7		• FLOW WASHER (AFTER 2005/02/25)	B,F,G,H	1
-540	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSD BY ITEMS 540A AND 542) (PRE SB 4510-25-02)	B,F,G,H	2
540A	CL50034		• CLAMP, TUBE (TPN CL-50034) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
542	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010) (SUPSDS ITEM 540) (POST SB 4510-25-02)	B,F,G,H	1
543	MS51957-13		• SCREW, PNH, 4-40 X .25 (AFTER 2005/02/25)	B,F,G,H	1
544	MS35338-135		• WASHER, LK, #4 (AFTER 2005/02/25)	B,F,G,H	1
			* * *		
-545	3520-0539-01		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSD BY ITEM 545A) (PRE SB 4510-25-03)		1
545A	3520-0580-03		• SERVER RETAINER ASSY (SEE IPL FIGURE 10 FOR DET BKDN) (SUPSDS ITEM 545) (POST SB 4510-25-03)		1
			ATTACHING PARTS		
550	3520-0537-01		• GUIDE LLS ASSY		2
555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		8
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• SWITCH ASSY		1
587	GR098		• • GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A			DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C, D,E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (SUPSDS ITEM 690) (TPN MS24693-C24)		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A AFTER 2007/01/25)	F,G	1
710A	3530-0232-01		• ASSY, MANIFOLD (SUPSDS ITEM 710 AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN) (SUPSD BY ITEM 710B)	F,G	1
-710B	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0233-01		• ASSY, MANIFOLD (SUPSD BY ITEM 715A AFTER 2004/05/12)	C,E	1
-715A	3530-0232-05		• ASSY, MANIFOLD (SUPSDS ITEM 715 AFTER 2004/05/12) (SUPSD BY ITEM 715B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715B	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0236-01		• ASSY, MANIFOLD (SUPSD BY ITEM 720A AFTER 2004/05/12)	A,D	1
-720A	3530-0232-07		• ASSY, MANIFOLD (SUPSDS ITEM 720 AFTER 2004/05/12) (SUPSD BY ITEM 720B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720B	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0265-01		• ASSY, MANIFOLD (SUPSD BY ITEM 725A AFTER 2004/05/12)	B,H	1
-725A	3530-0232-03		• ASSY, MANIFOLD (SUPSDS ITEM 725 AFTER 2004/05/12) (SUPSD BY ITEM 725B AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725B	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725A AFTER 2007/01/25) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
745B	TU50020		TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17) (SUPSD BY ITEM 747A)	A,B,D,H	1
-747A	62-0196-3		TUBE, FORMED (SUPSDS ITEM 747)	A,B,D,H	1
750	3520-0554-01		ASSY, TUBE, INLET, WATER	C,E,F,G	1

- ITEM NOT ILLUSTRATED

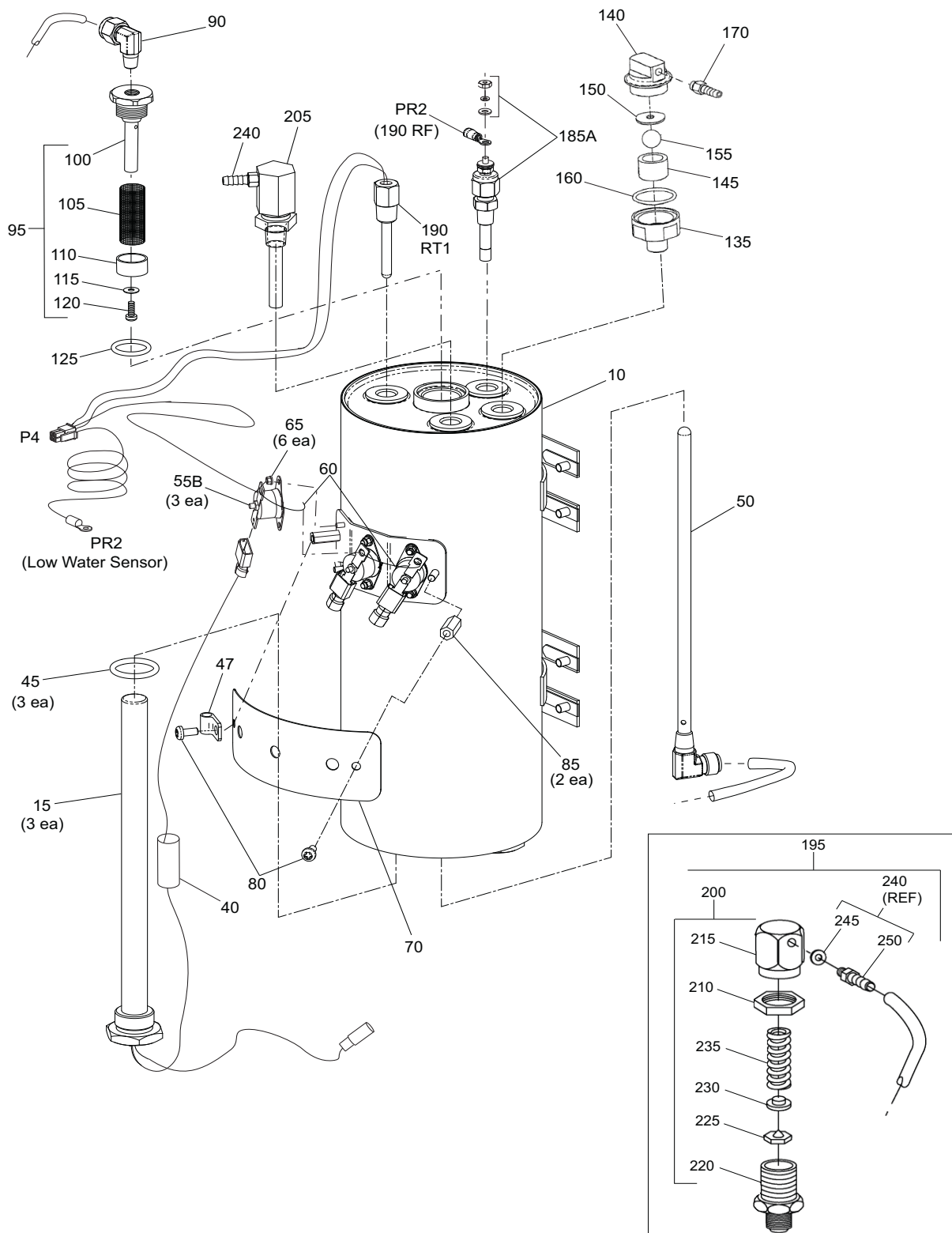
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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
-745	TU50020		• TUBING, 8.5 IN (TPN TU-50020) (SUPSD BY ITEM 745A)		2
-745A	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 7.5 IN (TPN TU-50020) (SUPSDS ITEM 745) (SUPSD BY ITEM 745B AFTER 2006/08/17)		2
745B	TU50020		• TUBING (1/4 IN OD X 0.63 IN WALL), 3.5 IN (TPN TU-50020) (SUPSDS ITEM 745A AFTER 2006/08/17)		1
747	TU4505		• TUBING TEFLON (1/4 IN OD X 0.030 IN WALL), 4.0 IN (TPN TU-4505) (AFTER 2006/08/17)	A,B,D,H	1
750	3520-0554-01		• ASSY, TUBE, INLET, WATER	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		
770	3520-0528-01		• DRAIN TUBE ASSY		1
			ATTACHING PARTS		
-775	MS51957-27		• SCREW, PH, 6-32 X .31		1
			* * *		

- ITEM NOT ILLUSTRATED

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IPL Figure 3
 Tank Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-1	3520-0529-01		TANK ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)	A	RF
10	3520-0555-01		• TANK WELDED ASSY		1
15	3520-0551-01		• TANK HEATER ASSY (HR1, HR2, AND HR3)		3
-20	TE50432		• • TERMINAL, FAST-ON (TPN TE-50432)		1
-25	CO50313		• • CONNECTOR (TPN CO-50313)		1
-30	CO50314		• • CONNECTOR PIN (TPN CO-50314)		1
-35	3575-1480-07		• • HEATER ASSY TANK		1
40	TU50039		• SLEEVING, VARGLAS, BLACK, 8 IN (TPN TU-50039) (AFTER 2006/07/19)		1
45	OR0941		• O-RING (TPN OR-0941)		3
50	3520-0509-01		• TUBE ASSY		1
-55	3530-0268-01		• THERMOSTAT, OVERTEMPERATURE (TS1, TS2, TS3) (SUPSD BY ITEM 55A)		3
-55A	3530-0280-01		• THERMOSTAT, OVERTEMPERATURE (TS1, TS2, TS3) (SUPSDS ITEM 55) (SUPSD BY ITEM 55B)		3
55B	352-0983-3		• THERMOSTAT, OVERTEMPERATURE (TS1, TS2, TS3) (SUPSDS ITEM 55A) (PER SIL 4510-SIL-02)		3
			ATTACHING PARTS		
60	WI156		• WIRE, BUSS, SOLID, 3 IN, 16 GA (TPN WI-156)		1
65	MS21043-04		• NUT, LK, 4-40		6
			* * *		
70	3575-0630-01		• THERMAL SWITCH COVER		1
75	CL315		• CLAMP (TPN CL-315)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		2
85	NS50113		• SPACER, HEX, 1/4 IN DIA, 6-32 X .625 IN (TPN NS-50113)		2
			* * *		

- ITEM NOT ILLUSTRATED

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
90	PF102		• FITTING, ELBOW (TPN PF-102)		1
95	62-0020-1		• FILTER ASSY,		1
100	3575-0647-01		• • WATER JET		1
105	62-0007-1		• • FILTER SCREEN		1
110	62-0011-1		• • RETAINER		1
115	MS35338-136		• • WASHER, LK, #6		1
120	MS51957-26		• • SCREW, PH, 6-32 X .25		1
125	OR0941		• O-RING (TPN OR-0941)		1
-130	3520-0523-01		• VENT VALVE ASSY (SUPSD BY ITEM 130A) (PRE SB 4510-25-07)		1
-130A	3520-0594-03		• VENT VALVE ASSY (SUPSDS ITEM 130) (POST SB 4510-25-07)	A	1
-130B	3520-0594-01		• VENT VALVE ASSY (POST SB 4510-25-07)	B	1
135	3575-1324-01		• • ADAPTER, VENT VALVE		1
140	3575-1325-01		• • BODY, VENT VALVE		1
145	3575-1327-01		• • LINER, VENT VALVE (SUPSD BY ITEM 145A) (PRE SB 3520-0594-25-003)		1
-145A	3575-2785-01		• • LINER, VENT VALVE (SUPSDS ITEM 145) (POST SB 3520-0594-25-003)		1
150	3575-1329-01		• • WASHER, VENT VALVE (SUPSD BY ITEM 150A) (PRE SB 3520-0594-25-003)		1
-150A	3575-2791-01		• • WASHER, VENT VALVE (SUPSDS ITEM 150) (POST SB 3520-0594-25-003)		1
155	MISC50411		• • BALL, VENT VALVE (TPN MISC-50411) (SUPSD BY ITEM 165) (PRE SB 3520-0594-25-001)		1
160	OR50019		• • O-RING (TPN OR-50019)		1
-165	MISC50470		• • BALL, VENT VALVE (TPN MISC-50470) (SUPSDS ITEM 155) (SUPSD BY ITEM 165A) (POST SB 3520-0594-25-001) (PRE SB 3520-0594-25-003)		1
-165A	MISC50509		• • BALL, VENT VALVE (TPN MISC-50509) (SUPSDS ITEM 165) (POST SB 3520-0594-25-003)		1

- ITEM NOT ILLUSTRATED

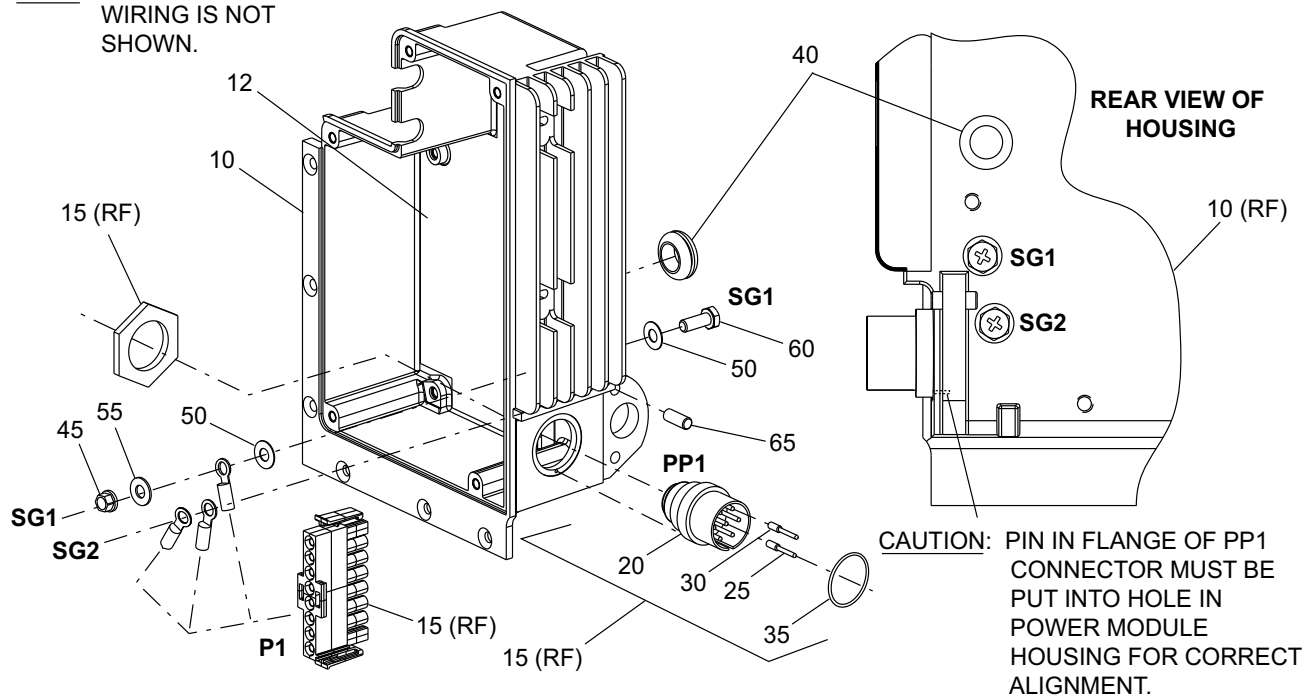
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
170	62-0136-7		• • FITTING, TUBE		1
-175	62-0136-9		• • • GASKET, OD .24 X ID .15 X .022 THICK		1
-180	62-0136-11		• • • FITTING, BARBED		1
-185	MISC50389		• SENSOR, WATER, LOW (SUPSD BY ITEM 185A) (TPN MISC-50389) (PRE SB 4510-25-08)		1
185A	3520-0581-01		• LOW WATER SENSOR ASSY (SUPSDS ITEM 185) (POST SB 4510-25-08)		1
190	3520-0550-01		• ASSY, RTD		1
195	62-0223-1		• PRESSURE RELIEF VALVE ASSY (SUPSD BY ITEM 205) (PRE SB 4510-25-009)		RF
200	62-0223-3		• • RELIEF VALVE SUBASSY (SUPSD BY ITEM 205) (PRE SB 4510-25-009)		1
205	VA50107		• VALVE, PRESSURE RELIEF (TPN VA-50107) (SUPSDS ITEMS 195 AND 200) (POST SB 4510-25-009)		1
210	62-0223-9		• • • NUT, LOCK (SUPSD BY ITEM 210A)		RF
-210A	62-0223-9		DELETED (SUPSDS ITEM 210)		DEL
215	62-0223-5		• • • CAP, ADJUSTMENT (SUPSD BY ITEM 215A)		RF
-215A	62-0223-5		DELETED (SUPSDS ITEM 215)		DEL
220	62-0223-11		• • • BODY, VALVE (SUPSD BY ITEM 220A)		RF
-220A	62-0223-11		DELETED (SUPSDS ITEM 220)		DEL
225	VA1031		• • • ASSY, VALVE (TPN VA-1031) (SUPSD BY ITEM 225A)		RF
-225A	VA1031		DELETED (SUPSDS ITEM 225)		DEL
230	VA1003		• • • REST, SPRING (TPN VA-1003) (SUPSD BY ITEM 230A)		RF
-230A	VA1003		DELETED (SUPSDS ITEM 230)		DEL
235	VA1004		• • • SPRING (TPN VA-1004) (SUPSD BY ITEM 235A)		RF
-235A	VA1004		DELETED (SUPSDS ITEM 235)		DEL
240	62-0136-7		• FITTING, BARBED	B	1
245	62-0136-9		• • GASKET, OD .24 X ID .15 X .022 THICK		1
250	62-0136-11		• • FITTING, NICKEL PLATE		1

- ITEM NOT ILLUSTRATED

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NOTE: FOR CLARITY,
 WIRING IS NOT
 SHOWN.



IPL Figure 4
 Power Module Assembly

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-1	3520-0530-01		POWER MODULE ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
10	3520-0531-01		• POWER MODULE HOUSING ASSY		1
12	3575-1250-01		• • PC BOARD INSULATOR		1
15	3520-0544-01		• HARNESS ASSY		1
-16	CO50311		• • RECEPTACLE, CONNECTOR (TPN CO-50311) (CONNECTORS J8, J9, J10. REFER TO FIGURE 2002)		3
-17	CO50312		• • CONTACT SOCKET (TPN CO-50312) (CONNECTORS J8, J9, J10. REFER TO FIGURE 2002)		3
20	CO6505		• • CONNECTOR, 7-PIN (TPN CO-6505)		1
25	CO6505-01		• • • PIN (TPN CO-6505-01)		1
30	CO6505-03		• • • PIN (TPN CO-6505-03)		6
35	CO6505-05		• • • O-RING (TPN CO-6505-05)		1
40	GR50053		• GROMMET (TPN GR-50053)		1
45	MS21042L3		• NUT, LK, 10-32		2
50	NAS1149D0316H		• WASHER, FL, #10		4
55	NAS1149F0332P		• WASHER, FL, #10		2
60	NAS1801-3-8		• SCREW, HEX, 10-32 X .50 (SG1 AND SG2)		2
65	PN5855		• PIN (TPN PN-5855)		1

- ITEM NOT ILLUSTRATED

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IPL Figure 5 (DELETED)
Controller Board Assy (Refer to CMM 25-33-32)

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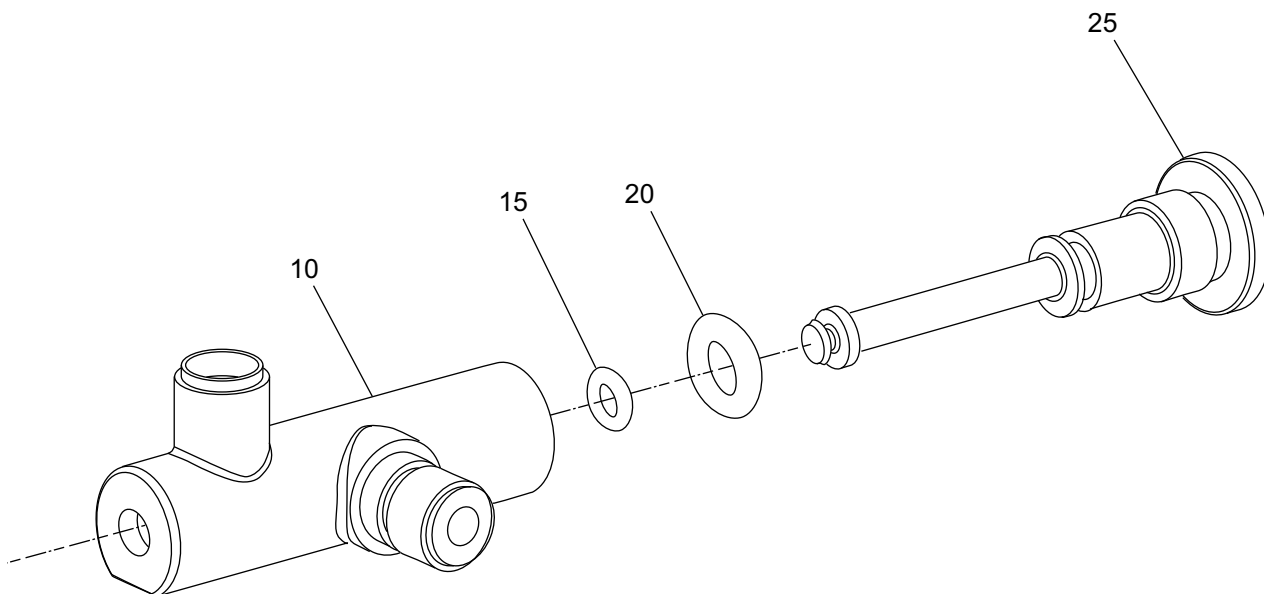
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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE							EFF CODE	UNITS/ ASSY
			1	2	3	4	5	6	7		
5 -1	3530-0121-01		DELETED (REFER TO CMM 25-33-32)								DEL

- ITEM NOT ILLUSTRATED

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4510-1 Series and 4510-2 Series



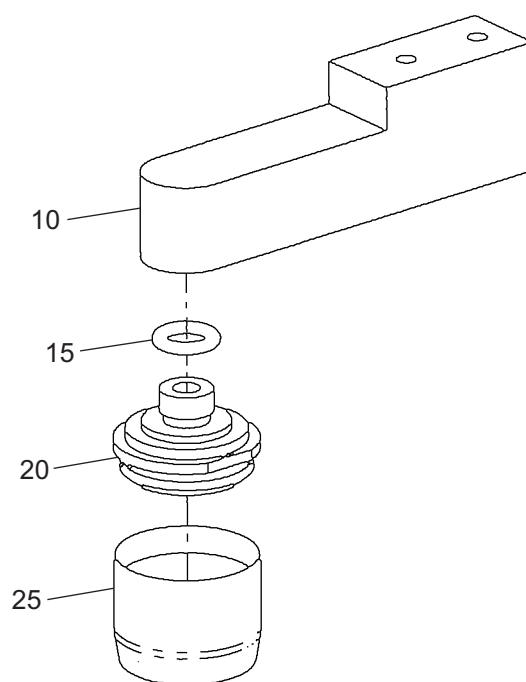
IPL Figure 6
Drain Valve Assembly

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
-1	62-0014-1		DRAIN VALVE ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
10	62-0008-1		• BODY		1
15	OR0341		• O-RING (TPN OR-0341)		1
20	OR0571		• O-RING (TPN OR-0571)		1
25	62-0013-1		• STEM ASSY		1

- ITEM NOT ILLUSTRATED

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4510-1 Series and 4510-2 Series



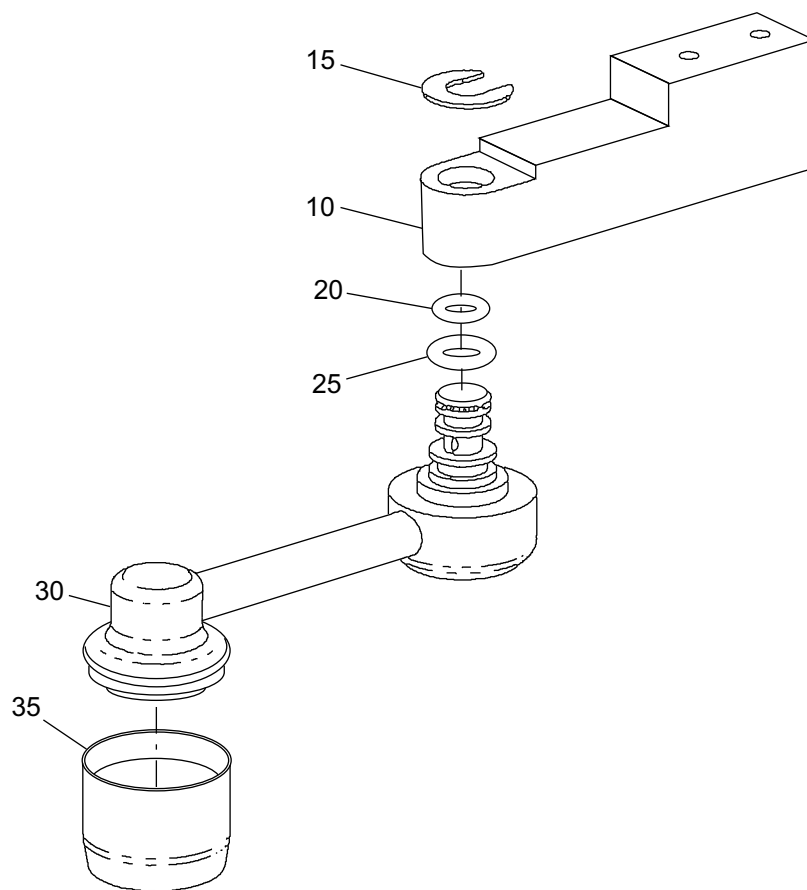
IPL Figure 7
Faucet Assembly

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
-1	3520-0525-01		FAUCET ASSY (SEE IPL FIGURES 2, AND 2A FOR NHA)		RF
10	3575-1227-01		• BAR, FAUCET		1
15	OR0571		• O-RING (TPN OR-0571)		1
20	3575-1225-01		• ADAPTER, AERATOR		1
25	MISC013		• AERATOR (TPN MISC-013)		1

- ITEM NOT ILLUSTRATED

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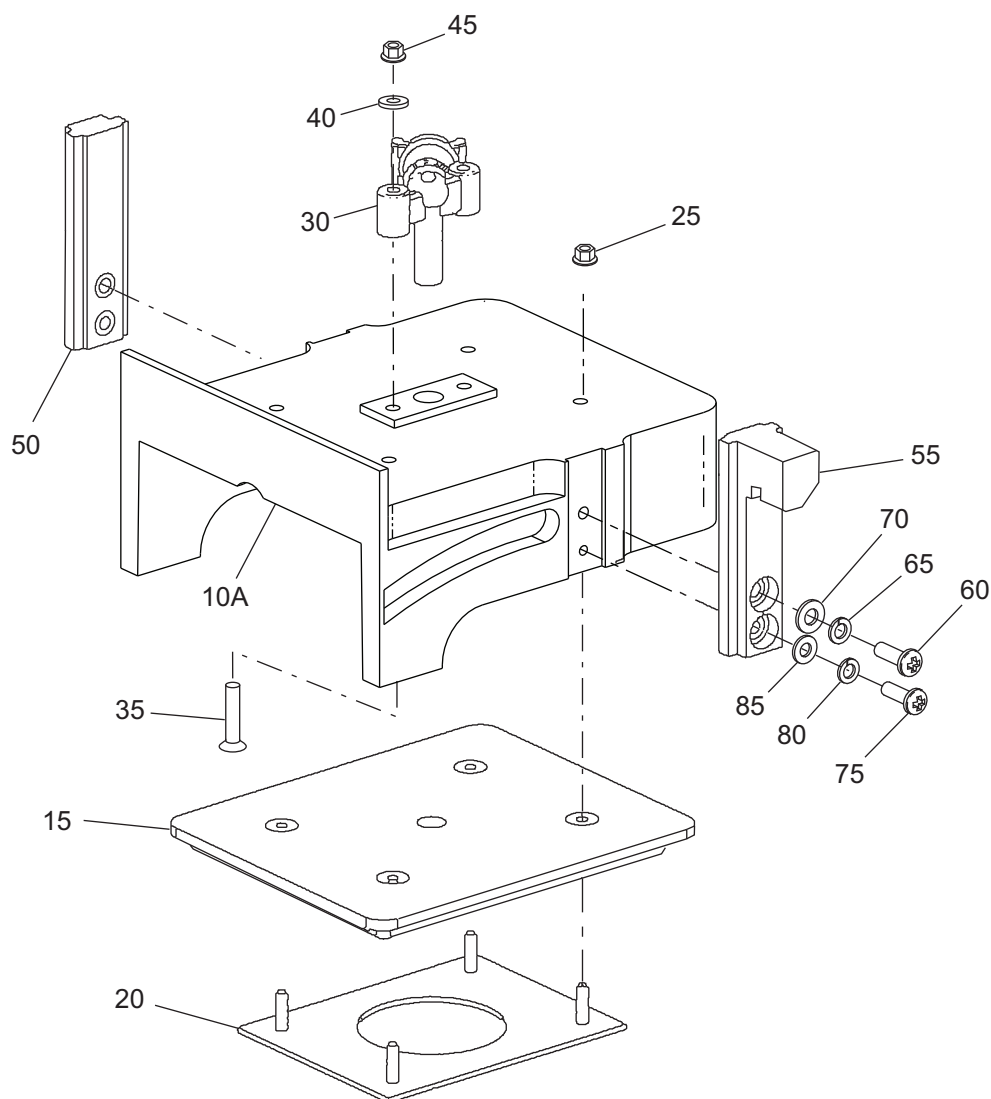
IPL Figure 8
Swivel Faucet Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-1	3520-0559-01		ASSY, SWIVEL FAUCET (SEE IPL FIGURES 2B AND 2C FOR NHA)		RF
10	3575-1246-01		• SWIVEL FAUCET BAR		1
15	MS16633-4031		• RING, RETAINING		1
20	OR50001		• O-RING (TPN OR-50001)		1
25	OR0571		• O-RING (TPN OR-0571)		1
30	3520-0558-01		• WELDED FAUCET ASSY		1
35	MISC013		• AERATOR (TPN MISC-013)		1

- ITEM NOT ILLUSTRATED

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IPL Figure 9
Brew Head Assembly

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
9					
-1	3520-0534-01		ASSY, BREWHEAD (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
-10	3575-1230-01		• BREW HEAD (SUPSD BY ITEM 10A)		1
-10A	3575-1230-01		• BREW HEAD (SUPSDS ITEM 10 AFTER 2006/04/13, PART WAS REDESIGNED) (OPT FOR ITEM 10B)		1
10B	3530-0272-01		• BREW HEAD (OPT FOR ITEM 10A) (PRFD OVER ITEM 10A)		1
15	411-0693-1		• GASKET		1
			ATTACHING PARTS		
20	411-0709-1		• RETAINER, GASKET		1
25	MS21043-06		• NUT, LK, 6-32		4
			* * *		
30	3575-0189-01		• FITTING, BREW		1
			ATTACHING PARTS		
35	MS24693C31		• SCREW, FH, 6-32 X .87 (TPN MS24693-C31)		2
40	MS15795-805		• WASHER, FL, #6		2
45	MS21043-06		• NUT, LK, 6-32		2
			* * *		
50	411-0748-1		• GUIDE ASSY (LEFT HAND)		1
55	411-0748-3		• GUIDE ASSY (RIGHT HAND)		1
			ATTACHING PARTS		
60	MS51957-45		• SCREW, PH, 8-32 X .50 (SUPSD BY ITEM 60A)		2
-60A	SC50280-45		• SCREW, PH, 8-32 X .50 (SUPSDS ITEM 60)		2
65	MS35338-137		• WASHER, LK, #8		2
70	MS15795-807		• WASHER, FL, #8		2
75	MS51957-29		• SCREW, PH, 6-32 X .44 (SUPSD BY ITEM 75A)		2
-75A	SC50280-29		• SCREW, PH, 6-32 X .44 (TPN SC-50280-29) (SUPSDS ITEM 75)		2

- ITEM NOT ILLUSTRATED

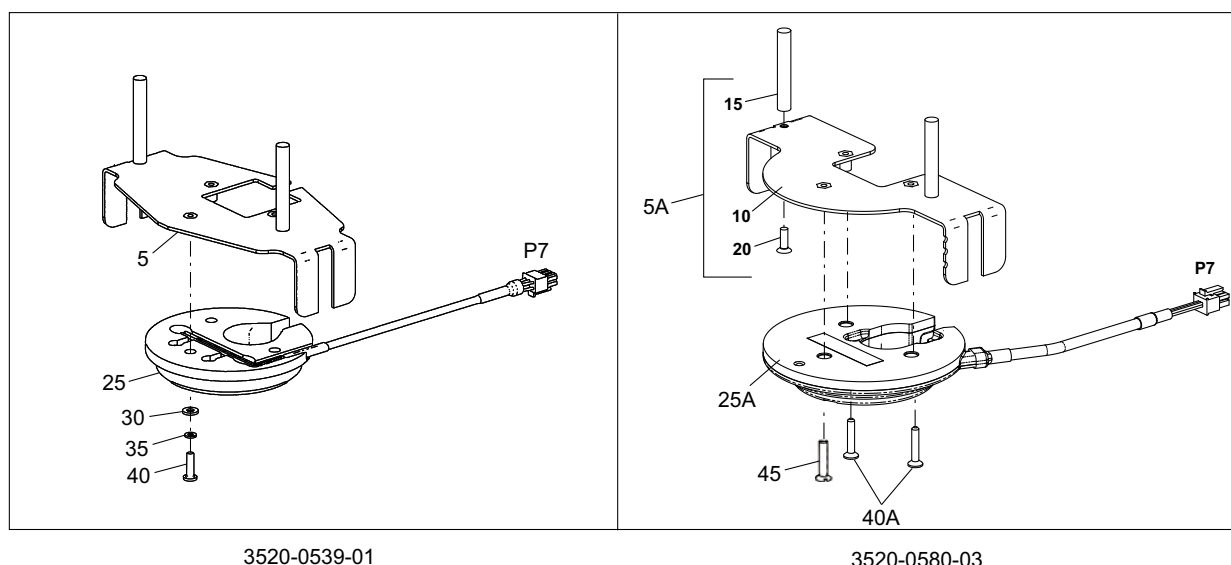
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 4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
9					
80	MS35338-136		• WASHER, LK, #6		2
85	MS15795-805		• WASHER, FL, #6		2
			* * *		

- ITEM NOT ILLUSTRATED

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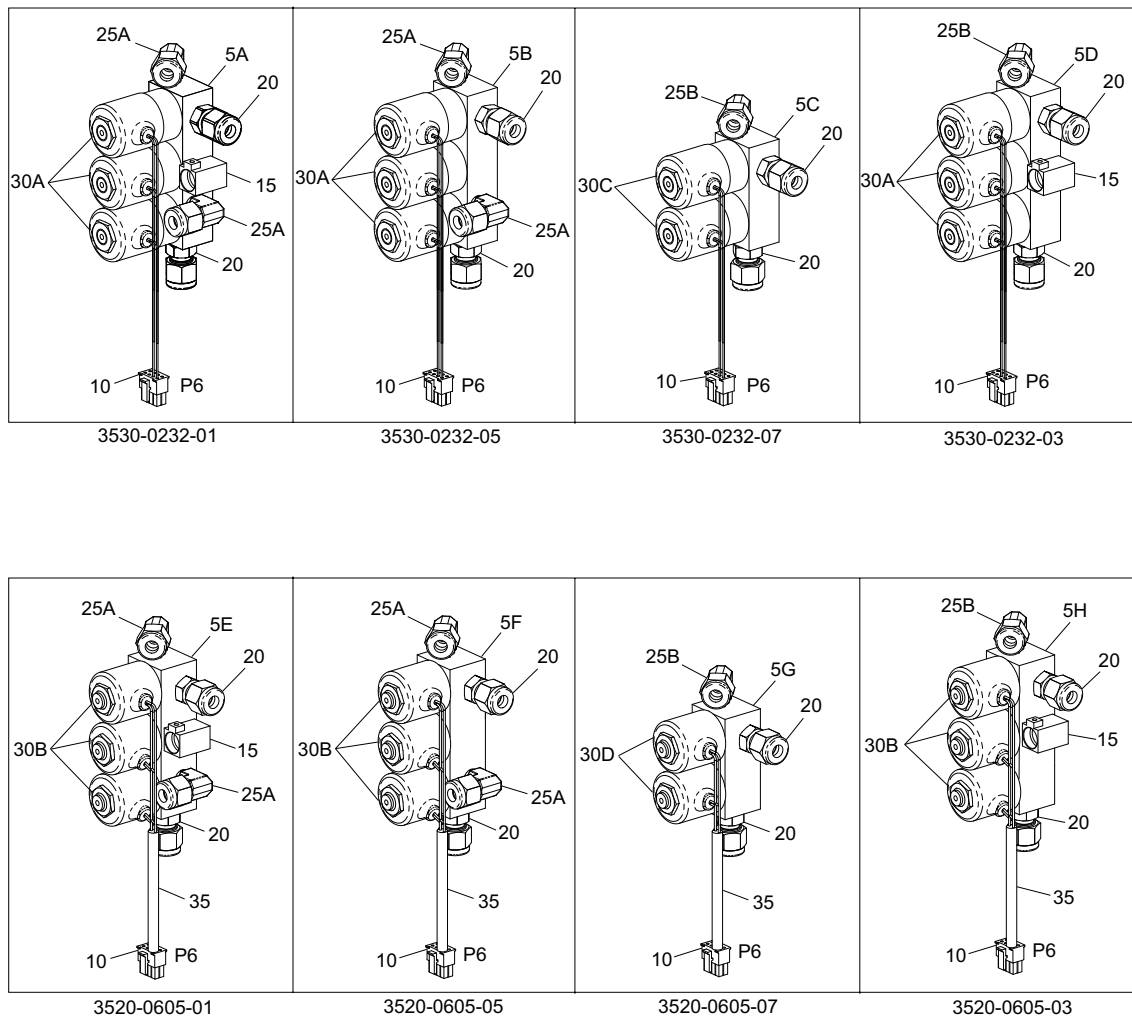
IPL Figure 10
Retainer Server Assembly

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
10					
-1	3520-0539-01		SERVER RETAINER ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1A PER SB 4510-25-03)		RF
-1A	3520-0580-03		SERVER RETAINER ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1 PER SB 4510-25-03)		RF
5	3520-0543-01		• LLS RETAINER ASSY (SUPSD BY ITEM 5A PER SB 4510-25-03)		1
5A	3520-0595-03		• RETAINER PLATE AND ROD ASSY (SUPSDS ITEM 5 PER SB 4510-25-03)		1
10	3520-0578-03		• • RETAINER PLATE ASSY		1
15	3575-1236-05		• • LLS RETAINER ROD		2
			ATTACHING PARTS		
20	MS24693C27		• • SCREW, FH, 6-32 X .44 * * *		2
25	3520-0557-01		• LLS POTTED ASSY (SUPSD BY ITEM 25A PER SB 4510-25-03)		1
25A	3520-0579-01		• INFRARED SENSOR ASSY (SUPSDS ITEM 25 PER SB 4510-25-03) ATTACHING PARTS		1
30	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 40A PER SB 4510-25-03)		3
35	MS35338-136		• WASHER, LK, #6 (SUPSD BY ITEM 40A PER SB 4510-25-03)		3
40	MS51957-30		• SCREW, PH, 6-32 X .50 (SUPSD BY ITEM 40A) (PRE SB 4510-25-03)		3
40A	SC50268-30		• SCREW, FH, NYLOC THREADS, 6-32 X .75 (TPN SC-50268-30) (SUPSDS ITEMS 30, 35, AND 40 AFTER 2008/06/16) (POST SB 4510-25-03)		2
45	SC50270		• SCREW, FH, NYLON, 6-32 X .625 (TPN SC-50270) * * *		1

- ITEM NOT ILLUSTRATED

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 4510-1 Series and 4510-2 Series



IPL Figure 11
 Manifold Assembly

B/E AEROSPACE, A PART OF COLLINS AEROSPACE
COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
11					
-1A	3530-0232-01		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1E AFTER 2007/01/05)	A	RF
-1B	3530-0232-05		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1F AFTER 2007/01/05)	B	RF
-1C	3530-0232-07		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1G AFTER 2007/01/05)	C	RF
-1D	3530-0232-03		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1H AFTER 2007/01/05)	D	RF
-1E	3520-0605-01		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1A AFTER 2007/01/05)	E	RF
-1F	3520-0605-05		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1B AFTER 2007/01/05)	F	RF
-1G	3520-0605-07		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1C AFTER 2007/01/05)	G	RF
-1H	3520-0605-03		MANIFOLD ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1D AFTER 2007/01/05)	H	RF
5A	3575-1211-01		• BODY, MANIFOLD	A	1
5B	3575-1211-05		• BODY, MANIFOLD	B	1
5C	3575-1211-07		• BODY, MANIFOLD	C	1
5D	3575-1211-03		• BODY, MANIFOLD	D	1
5E	3575-1351-01		• MANIFOLD, SOLENOID	E	1
5F	3575-1351-05		• MANIFOLD, SOLENOID	F	1
5G	3575-1351-07		• MANIFOLD, SOLENOID	G	1
5H	3575-1351-03		• MANIFOLD, SOLENOID	H	1
10	CO50409		• CONNECTOR, EIGHT PIN, DUAL ROW (P6) (TPN CO-50409)		1
15	3575-1284-01		• FITTING, REGULATOR	A,D,E, H	1
20	PF110		• FITTING, STRAIGHT (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-110)		2

- ITEM NOT ILLUSTRATED

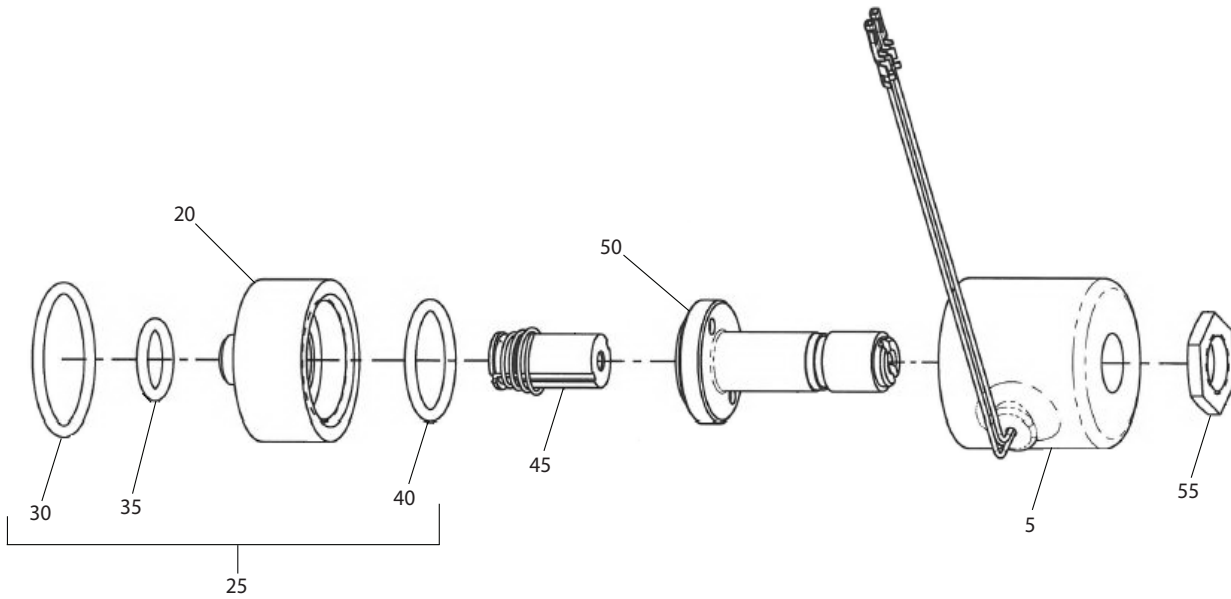
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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
11					
25A	PF102		• FITTING, 90 DEGREE, ELBOW (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-102)	A,B,E,F	2
25B	PF102		• FITTING, 90 DEGREE, ELBOW (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-102)	C,D,E, G,H,J	1
30A	3530-0279-01		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	A,B,D	3
30B	3530-0279-23		• SOLENOID VALVE ASSY (REPLD BY ITEM 30E) (PRE SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 12 FOR DET BKDN)	E,F,H	3
30C	3530-0279-01		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	C	2
30D	3530-0279-23		• SOLENOID VALVE ASSY (REPLD BY ITEM 30F) (PRE SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 12 FOR DET BKDN)	G	2
-30E	3520-1529-01		• 2-WAY NORMALLY CLOSED VALVE (REPLS ITEM 30B) (POST SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 12A FOR DET BKDN)	E,F,H	3
-30F	3520-1529-01		• 2-WAY NORMALLY CLOSED VALVE (REPLS ITEM 30D) (POST SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 12A FOR DET BKDN)	G	2
-32	TE50442		• TERMINAL, FEMALE CRIMP (TPN TE-50442) (USED WITH ITEM 30E, 30F)		AR
35	TU50052		• VARGLAS, NON-FRAY SLEEVING (TPN TU-50052)		1

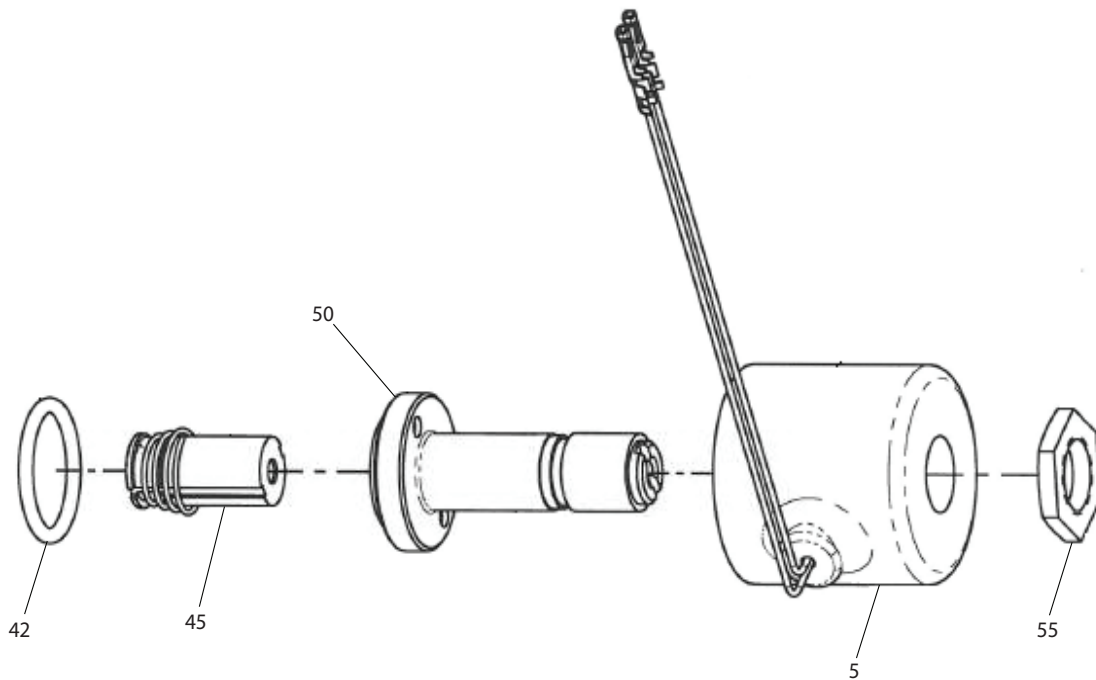
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 4510-1 Series and 4510-2 Series



3530-0279-01



3530-0279-23

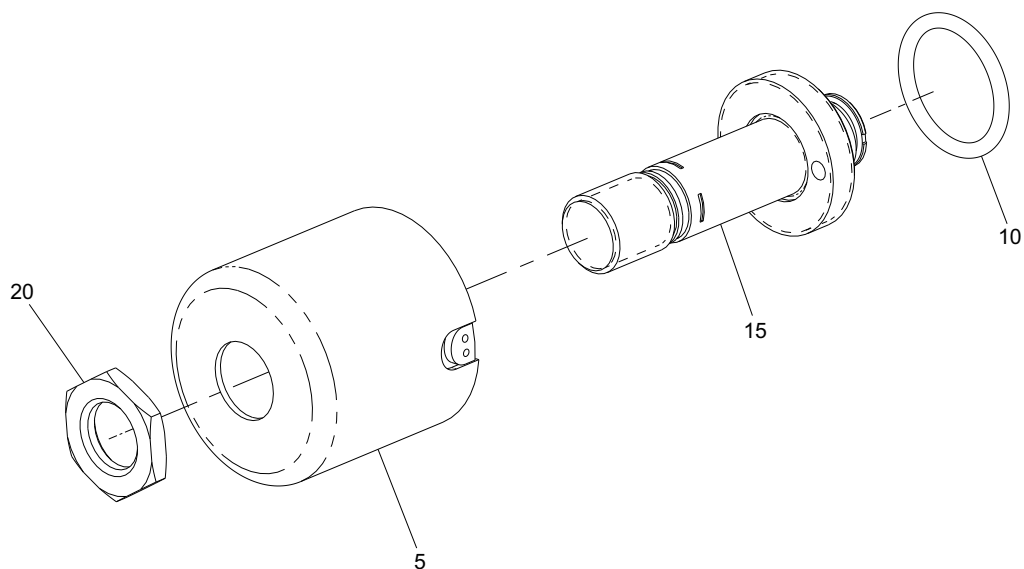
IPL Figure 12
 Solenoid Valve Assembly

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4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
12					
-1A	3530-0279-01		SOLENOID VALVE ASSY (SEE IPL FIGURE 11 FOR NHA)	A	RF
-1B	3530-0279-23		SOLENOID VALVE ASSY (PRE SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 11 FOR NHA)	B	RF
5	3530-0279-11		• COIL ASSY		1
-10	3530-0279-21		• • COIL		1
-15	TE50442		• • CRIMP TERMINAL (TPN TE-50442)		2
20	3530-0279-05		• BODY	A	1
25	3530-0279-03		• O-RING KIT	A	1
30	3530-0279-15		• • O-RING	A	1
35	3530-0279-17		• • O-RING	A	1
40	3530-0279-19		• • O-RING	A	1
42	3530-0279-25		• O-RING	B	1
45	3530-0279-09		• PLUNGER ASSY		1
			ATTACHING PARTS		
50	3530-0279-07		• GUIDE ASSY		1
55	3530-0279-13		• NUT		1
			* * *		

- ITEM NOT ILLUSTRATED

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4510-1 Series and 4510-2 Series



IPL Figure 12A
2-Way Normally Closed Valve

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COMPONENT MAINTENANCE MANUAL
4510-1 Series and 4510-2 Series

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
12A					
-1	3520-1529-01		VALVE, 2-WAY NORMALLY CLOSED (POST SIL 4X1X-EN-SIL-02) (SEE IPL FIGURE 11 FOR NHA)		RF
5	3520-1616-01		• COIL ASSY		1
10	0R50099		• O-RING (TPN OR-50099)		1
15	3520-1620-01		• PLUNGER GUIDE ASSY		1
20	3575-3037-01		• NUT		1

- ITEM NOT ILLUSTRATED

